On compositions of graphs, signed graphs, and gain graphs

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Abstract and publications arising from this thesis

The theory of signed and gain graphs extends classical graph theory by equipping edges with signs or group elements, enabling the modelling of interactions more nuanced than simple adjacency. A fundamental equivalence relation in this setting, known as switching, preserves many spectral properties yet can change the underlying signature or gain structure. This thesis develops a systematic framework for extending classical graph operations—such as joins, corona products, and subdivision-based constructions—to signed and gain graphs in a manner that respects switching equivalence.

We introduce the $\mathcal{T}$-corona product for signed graphs, defined using pseudo-potential functions with respect to a fixed spanning tree, and prove that it satisfies identity preservation, structural compatibility, and switching stability. This construction yields infinite families of non-switching-isomorphic signed graphs that are cospectral with respect to both the adjacency and net Laplacian matrices. We also extend the H-join operation to complex unit gain graphs, providing both a direct variant and a switching-stable variant that achieves full stability through pseudo-potentials.

Using the framework of G-phases and group representations, we introduce gain analogues of the classical Q and R graphs. For an r-regular G-gain graph and an irreducible unitary representation π, we determine the π-spectra of the represented adjacency and Laplacian matrices, recovering known classical and signed results while providing new spectral formulas for signed Q graphs. We further introduce six subdivision-based corona products for gain graphs, derive their π-characteristic polynomials, and develop pseudo-potentials over abelian groups to obtain switching-stable versions in which balanced components yield balanced products.

The pseudo-potential technique emerges as a unifying concept throughout this thesis, consistently providing switching-stable generalisations of classical graph operations. Our results recover and extend numerous known spectral formulas, and the explicit constructions demonstrate the utility of switching-stable operations in spectral graph theory.

This thesis is based on the following publications and works-in-progress:

- Chapter 2: C. Huntington, F. Belardo, S. Barik. Graphs with integer reciprocal eigenvalue properties. Under review. [60]
- Chapter 3: C. Huntington. Spectra and balance of signed graphs obtained from vertex splittings. Discuss. Math. Graph Theory. Online (2026) DOI: 10.7151/dmgt.2643 [58]
- Chapter 4: F. Belardo, A. Donno, C. Huntington, S. Spessato. A switching-stable corona product of signed graphs. Under review. [20]
- Chapter 5: C. Huntington, S. Spessato, F. Belardo. On H-joins of complex unit gain graphs and their stability. Discrete Math. **349** (2026) 115069. DOI: 10.1016/j.disc.2026.115069 [61]
- Chapter 6: C. Huntington, O. Waka. Spectra of gain Q and R graphs via G-phases and group representations. Under review. [63]
- Chapter 7: C. Huntington. Products derived from gain subdivision graphs via group representations. In preparation. [59]

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Chapter 1

Introduction and preliminaries

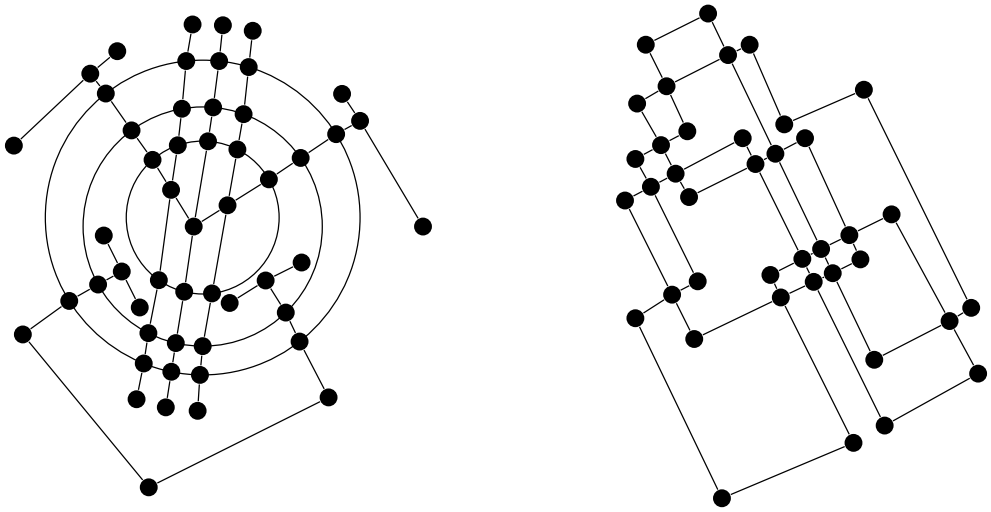


Figure 1.1: TikZ reproduction of Pablo Picasso, *Constellations*, 1924.

The eigenvalues of the matrices associated with a graph encode its principal structural properties. Just as astronomers study stellar spectra to determine the composition of distant stars, graph theorists consider eigenvalues to deduce the structure of a graph [36]. Eigenvalues are closely related to almost all major invariants of a graph, linking one extremal property to another, and play a central role in our fundamental understanding of graphs.

In the summer of 1924, working from a sketchbook at Juan-les-Pins on the Côte d'Azur in southern France, Pablo Picasso produced a series of drawings known as the Constellations [84]: ink dots connected by fine pen lines, mapping stars onto guitars, mandolins, and musical staves.

Astronomers read stellar spectra. Graph theorists read eigenvalues. Picasso read the night sky and produced a combinatorial encoding of it, dots (vertices) connected by lines (edges) that reveal hidden structure. Each asks the same question: what can you deduce about an invisible underlying reality from a compressed, schematic representation?

1.1 Motivation and overview

Spectral graph theory is a rich field that explores the connections between the combinatorial structure of graphs and the eigenvalues and eigenvectors of associated matrices. A key advantage of this approach is that spectra can be computed in polynomial time, making eigenvalues and eigenspaces powerful tools for studying graphs.

In recent decades, increasing attention has been devoted to the study of signed graphs, that is, graphs in which each edge is labelled by a positive or negative sign [100]. Classical graphs can be regarded as special cases of signed graphs by assigning to each edge a positive sign. Signed graphs admit a natural equivalence relation, called switching equivalence, whereby equivalent signed graphs share many spectral properties and can be regarded as essentially the same object in certain respects. More generally, in a gain graph each oriented edge is labelled by an element of a group, with reversed orientations receiving inverse elements; signed graphs correspond to the case where the group is $\{+1, -1\}$.

In graph theory there are various types of products and operations whose purpose is to produce a new graph Γ from given graphs $\Gamma_1, \Gamma_2, \ldots, \Gamma_k$. Among the most classical of these constructions are the Cartesian, Kronecker (or tensor), strong, and lexicographic products, whose spectra are well understood for ordinary graphs, see [38] and [47].

Alongside these sit operations of a different character: the join of two graphs and its generalisation the H-join (first introduced by Schwenk in 1974 [87], and then rediscovered by Cardoso et al. in 2011 [28]), the corona product (introduced by Frucht and Harary in 1970 [40]), the subdivision graph (used implicitly by Kuratowski in 1930 [66]; reformulated by Wagner in 1937 [97], with the formal definition of the nth subdivision graph later introduced by Harary and Nash-Williams in 1965 [52]), the line graph (studied by Whitney in 1932 [98] and characterised by Krausz in 1943 [65], with the term itself coined by Harary and Norman in 1960 [53]), the total graph (introduced by Behzad in 1965 [14]), and the Q and R graphs (introduced by Cvetković in 1975 [37]), whose interrelations underpin much of spectral graph theory.

A topic of active research interest is the extension of these operations to signed graphs [22, 24, 42, 45, 57, 68, 88, 80, 81, 104], their relations to oriented graphs [62, 83, 92], the extension to complex unit gain graphs [6, 16, 17, 19, 23], and, more generally, extension to arbitrary gain graphs [31, 32, 34, 35]. In other words, the goal is to define a desirable signature or gain function on the composed graph Γ when the factors $\Gamma_1, \Gamma_2, \ldots, \Gamma_k$ are already equipped with such structures.

When generalising a graph product to the setting of gain graphs, one would like the construction to satisfy the following three properties:

- **Identity preservation.** If all edge gains are the identity element, the resulting gain graph also has all edges labelled by the identity.
- **Structural compatibility.** Geometric and spectral properties are preserved or suitably generalised.
- **Switching stability.** The operation is invariant under switching: replacing any of the input gain graphs with a switching-equivalent gain graph does not change the switching class of the product.

Identity preservation allows us to regard the new operation as a genuine extension of the classical one for ordinary graphs. Structural compatibility is important to ensure that the new product remains connected to the classical results concerning its non-gain version. Switching stability, as one might expect, guarantees stability under switching, which is a fundamental operation in the theory of signed and gain graphs.

The thesis is structured as follows. In Chapter 2, we study graphs with integer reciprocal eigenvalue properties, first determining when these properties are satisfied for known classes and products of graphs, and then introducing two new classes, based on [60]. In Chapter 3, we introduce two new products of signed graphs and, as an application, show how they can be implemented to obtain infinite families of non-isomorphic cospectral pairs of signed graphs, based on [58]. In Chapter 4, we introduce and study a new notion of the corona product of signed graphs, with our construction having the benefit of being switching stable via fixing a spanning tree and using pseudo-potentials, based on [20]. In Chapter 5, we extend the H-join to the realm of complex unit gain graphs, proposing two new variants, with the second satisfying all three desired properties, based on [61]. In Chapter 6, we introduce gain Q and R graphs as generalisations of their classical and signed counterparts, constructed via G-phases and group representations, then show that switching-equivalent gain graphs give rise to switching-equivalent gain Q and R graphs, and provide spectral results under certain conditions, based on [63]. In Chapter 7, we continue by introducing gain variants of the subdivision-vertex, Q-vertex, R-vertex, subdivision-edge, Q-edge, and R-edge corona products, constructed via G-phases and group representations, and then develop the notion of pseudo-potentials for G-gain graphs, enabling us to define switching-stable versions of all six products, based on [59].

For any results or concepts concerning graph spectra, signed graphs, or gain graphs used in this thesis which have not been explicitly stated, the reader is directed to [38, 100, 101].

1.2 General notation

We shall use the following notations throughout this thesis. The identity matrix is denoted by I_n when it is of order n or simply by I if the relevant size is clear. We denote the all-one matrix by $J_{m\times n}$ if it is an $m \times n$ matrix, by J_n if it is square of order n, or just by J. Likewise we denote the all-zero matrix by $O_{m\times n}$ if it is an $m \times n$ matrix, or by O_n, or just O. In addition to this, the all-one vector $J_{n\times 1}$ may be denoted by $\mathbf{1}_n$ and the all-zero vector $O_{n\times 1}$ may be denoted by $\mathbf{0}_n$.

Throughout, we distinguish notationally between graph operations and graph matrices: operations producing a new graph are written with parentheses, as in the line graph $L(\Gamma)$ or the subdivision graph $S(\Gamma)$, while matrices associated with a graph carry the graph as a subscript, as in the adjacency matrix $A_{(\Gamma,\sigma)}$ or the Laplacian matrix $L_{(\Gamma,\sigma)}$.

Furthermore, we adopt the standard notations of P_n, K_n, and C_n to denote the path, complete, and cycle graphs on n vertices, respectively.

1.3 Graphs, signed graphs, and gain graphs

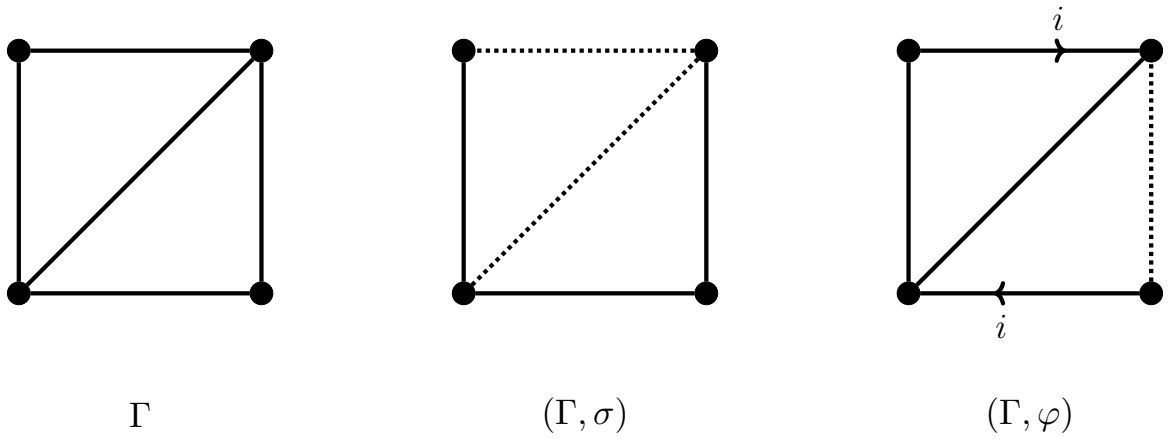


Figure 1.2: A graph Γ, a signed graph (Γ, σ), and a (complex unit) gain graph (Γ, φ). Here and throughout, solid lines represent positive edges, dashed lines represent negative edges, and lines directed from a to b and labelled x represent edges with gain $\varphi(a, b) = x$.

All graphs considered in this thesis are finite, simple, and undirected; where connectedness is required, it is assumed explicitly.

1.3.1 Classical graphs

A graph Γ is a pair (V_Γ, E_Γ), where V_Γ is the vertex set and $E_\Gamma \subseteq \{\{x, y\} \subseteq V_\Gamma : x \neq y\}$ is the edge set. Two vertices x and y are called adjacent if and only if $\{x, y\} \in E_\Gamma$. Given a vertex v, its degree $d_\Gamma(v)$, or simply $d(v)$ when the graph is clear from the context, is the number of vertices adjacent to v.

For $v \in V_\Gamma$ and $e \in E_\Gamma$ we write $v \in e$ if the edge e is incident to v. If $e, f \in E_\Gamma$ are both incident to a vertex we say that e and f are incident and we denote that vertex by $e \cap f$; if e and f are not incident then we write $e \cap f = \varnothing$.

We will denote by $O(E_\Gamma)$ the set of oriented edges of Γ, that is, the set of ordered pairs of adjacent vertices, so that $O(E_\Gamma) = \{(u, v) : \{u, v\} \in E_\Gamma\}$.

For brevity, we shall often write uv for the edge $\{u, v\}$, using the two notations interchangeably; we favour $\{u, v\}$ where the edge is treated as a set, and the juxtaposition uv within set-builder expressions and as the argument of signature and gain functions. Here $uv = vu$, in contrast to the arc (u, v).

The line graph of Γ is the graph $L(\Gamma) = (V_{L(\Gamma)}, E_{L(\Gamma)})$ where $V_{L(\Gamma)} = E_\Gamma$, and two vertices $e, f \in V_{L(\Gamma)}$ $(e \neq f)$ are adjacent in $L(\Gamma)$ if and only if $e \cap f \neq \varnothing$.

For an unsigned graph Γ, its complement Γ^{c} is the graph on the same vertex set in which two distinct vertices are adjacent if and only if they are not adjacent in Γ.

Given two graphs Γ_1 and Γ_2, an isomorphism is a bijective map $f\colon V_{\Gamma_1} \to V_{\Gamma_2}$ such that $\{x, y\} \in E_{\Gamma_1}$ if and only if $\{f(x), f(y)\} \in E_{\Gamma_2}$.

A walk W of length k in a graph Γ is an ordered sequence of vertices $v_1 v_2 \dots v_{k+1}$ such that v_i is adjacent to v_{i+1} for each $1 \leq i \leq k$; the walk is closed if $v_{k+1} = v_1$.

1.3.2 Signed graphs and gain graphs

A signed graph (Γ, σ) is a pair consisting of an ordinary graph $\Gamma = (V_\Gamma, E_\Gamma)$, called the underlying graph, together with a map $\sigma\colon E_\Gamma \to \{+1, -1\}$, referred to as the signature. The edge set of a signed graph is thus partitioned into positive and negative edges. We denote the sign of the edge $\{u, v\} \in E_\Gamma$ by $\sigma(u, v)$; with this notation we have $\sigma(u, v) = \sigma(v, u)$. We regard an ordinary graph Γ as the all-positive signed graph $(\Gamma, +)$ whose signature assigns the value $+1$ to all of its edges.

Let G be a group with identity element 1_G. A G-gain graph is a pair (Γ, φ) consisting of an underlying graph $\Gamma = (V_\Gamma, E_\Gamma)$ and a gain function $\varphi\colon O(E_\Gamma) \to G$ satisfying $\varphi(u, v) = \varphi(v, u)^{-1}$ for every $(u, v) \in O(E_\Gamma)$.

Let $\mathbb{T}$ be the group of all complex numbers z with $|z| = 1$. A complex unit gain graph, or simply a $\mathbb{T}$-gain graph, is a G-gain graph with $G = \mathbb{T}$.

Since every element of $\{+1, -1\}$ is its own inverse, a signature may equally be regarded as a gain function with values in the group $\{+1, -1\}$: signed graphs are precisely the $\{+1, -1\}$-gain graphs, and an ordinary graph is the gain graph whose gains all equal the identity. This observation allows us to define the notions of switching, balance, and their relatives just once, in the setting of G-gain graphs, and to recover the corresponding notions for signed and ordinary graphs by specialisation.

Following [32], we denote by $G(\Gamma)$ the set of all G-gain functions on Γ. Note that, despite its appearance, $G(\Gamma)$ does not denote the image of Γ under a map; it is a single symbol for the set of gain functions, which shall be retained for consistency with the literature.

1.4 Graph matrices, characteristic polynomials, and spectra

Having defined the underlying graph Γ together with its gain function φ, we can now record this combinatorial data in a variety of matrices.

The adjacency matrix $A_{(\Gamma,\varphi)} = (a_{ij})_{n\times n}$ of a G-gain graph (Γ, φ) with vertex set $V_\Gamma = \{v_1, v_2, \dots, v_n\}$ is defined by

$$a_{ij} = \begin{cases} \varphi(v_i, v_j), & \text{if } \{v_i, v_j\} \in E_\Gamma, \\ 0, & \text{otherwise.} \end{cases}$$

For a signed graph (Γ, σ) the entries of $A_{(\Gamma,\sigma)}$ lie in $\{-1, 0, +1\}$ and the adjacency matrix is real and symmetric; for a complex unit gain graph (Γ, φ) the relation $\varphi(v_j, v_i) = \varphi(v_i, v_j)^{-1} = \overline{\varphi(v_i, v_j)}$ makes $A_{(\Gamma,\varphi)}$ a Hermitian matrix. In both cases the spectral theorem guarantees that all eigenvalues are real. For an arbitrary group G, however, the entries of the adjacency matrix are group elements rather than numbers, and one cannot speak of a spectrum until a unitary representation $\pi\colon G \to U(d)$ is chosen, where $U(d)$ denotes the group of $d \times d$ unitary matrices, and each entry is replaced by the block $\pi(\varphi(v_i, v_j))$; the resulting spectrum genuinely depends on the choice of π. This representation-theoretic point of view is developed in Chapter 6.

Example 1.4.1. Consider the graphs displayed in Figure 1.2. With a suitable vertex ordering, their adjacency matrices are

$$A_\Gamma = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}, \qquad A_{(\Gamma,\sigma)} = \begin{pmatrix} 0 & -1 & 0 & 1 \\ -1 & 0 & 1 & -1 \\ 0 & 1 & 0 & 1 \\ 1 & -1 & 1 & 0 \end{pmatrix}, \qquad A_{(\Gamma,\varphi)} = \begin{pmatrix} 0 & i & 0 & 1 \\ -i & 0 & -1 & 1 \\ 0 & -1 & 0 & i \\ 1 & 1 & -i & 0 \end{pmatrix}.$$

For a graph Γ with n vertices, let $d_i = d_\Gamma(v_i)$ be the degree of vertex v_i, and let D_Γ denote the degree matrix of Γ, the $n \times n$ diagonal matrix with entries $d_1, d_2, \ldots, d_n$. The Laplacian matrix $L_{(\Gamma,\varphi)}$ of a G-gain graph (Γ, φ) is defined by $L_{(\Gamma,\varphi)} = D_\Gamma - A_{(\Gamma,\varphi)}$. In particular, for a signed graph (Γ, σ) we have $L_{(\Gamma,\sigma)} = D_\Gamma - A_{(\Gamma,\sigma)}$.

Example 1.4.2. Consider the complex unit gain graph (Γ, φ) displayed in Figure 1.2. With a suitable vertex ordering, its degree and Laplacian matrices are

$$D_\Gamma = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}, \qquad L_{(\Gamma,\varphi)} = \begin{pmatrix} 2 & -i & 0 & -1 \\ i & 3 & 1 & -1 \\ 0 & 1 & 2 & -i \\ -1 & -1 & i & 3 \end{pmatrix}.$$

The standard notion of the Laplacian involves the degrees of the vertices but does not take into account the signs of the edges. For this reason, it is natural to introduce an alternative notion of degree, specifically tailored to signed graphs, which will in turn lead to an alternative definition of the Laplacian matrix. Given a vertex v in (Γ, σ), the net degree of v is the integer $d^{\pm}_\sigma(v) = d^+(v) - d^-(v)$, where $d^+(v)$ is the number of vertices w such that $\sigma(v, w) = +1$ and $d^-(v)$ is the number of vertices w such that $\sigma(v, w) = -1$. In other words, $d^+(v)$ is the number of positive neighbours of v, and $d^-(v)$ is the number of negative neighbours of v. The net Laplacian matrix is then defined as

$$L^{\pm}_{(\Gamma,\sigma)} = D^{\pm}_{(\Gamma,\sigma)} - A_{(\Gamma,\sigma)}, \tag{1.4.1}$$

where $D^{\pm}_{(\Gamma,\sigma)}$ is the diagonal matrix of net degrees $d^{\pm}_\sigma(v)$. The original definition and the main properties of the net Laplacian of a signed graph can be found in [90].

Example 1.4.3. Consider the signed graph (Γ, σ) displayed in Figure 1.2. With a suitable vertex ordering, its net degree and net Laplacian matrices are

$$D^{\pm}_{(\Gamma,\sigma)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad L^{\pm}_{(\Gamma,\sigma)} = \begin{pmatrix} 0 & 1 & 0 & -1 \\ 1 & -1 & -1 & 1 \\ 0 & -1 & 2 & -1 \\ -1 & 1 & -1 & 1 \end{pmatrix}.$$

Let M be an $n \times n$ matrix. Its characteristic polynomial, $\det(\lambda I_n - M)$, will be denoted by $P(M; \lambda)$. For a signed graph (Γ, σ), we define $P(M_{(\Gamma,\sigma)}; \lambda)$ to be the M-characteristic polynomial of (Γ, σ) for a graph matrix M. The roots of the M-characteristic polynomial of (Γ, σ) are the M-eigenvalues of (Γ, σ), and the multiset of these M-eigenvalues, including multiplicities, is called the M-spectrum of (Γ, σ). By the characteristic polynomial, the eigenvalues, and the spectrum of a signed graph (without any reference to the matrix), we mean the A-characteristic polynomial, the A-eigenvalues, and the A-spectrum, respectively. The same conventions apply to complex unit gain graphs, whose characteristic polynomial is $P(A_{(\Gamma,\varphi)}; \lambda) = \det\big(\lambda I - A_{(\Gamma,\varphi)}\big)$. When no confusion can arise, we may write $P((\Gamma, \sigma); \lambda)$, $P((\Gamma, \varphi); \lambda)$, and $P(\Gamma; \lambda)$ in place of $P(A_{(\Gamma,\sigma)}; \lambda)$, $P(A_{(\Gamma,\varphi)}; \lambda)$, and $P(A_\Gamma; \lambda)$, respectively.

The spectrum of (Γ, φ) is denoted by $\mathrm{spec}(\Gamma, \varphi)$, where we write $\lambda^{(k)}$ to indicate that the eigenvalue λ occurs with multiplicity k. A signed or gain graph is said to have a symmetric spectrum if $-\lambda$ is an eigenvalue whenever λ is. Two graphs are said to be cospectral if they have precisely the same spectrum. For non-abelian groups, however, caution is required: eigenvalues must be understood in the appropriate sense, i.e. left or right. In the case of quaternion unit graphs, the subtleties of this distinction are discussed in detail in [18]. In later chapters we avoid this issue entirely by working with a representation π of the group, which sends non-commutative group elements to complex matrices, where the usual notion of eigenvalue is unambiguous.

Many combinatorial properties of a classical graph can be determined from its adjacency and Laplacian spectra. From the adjacency spectrum:

- Basic parameters: The number of vertices and number of edges.
- Bipartiteness: A graph is bipartite if and only if its adjacency spectrum is symmetric about the origin.

- Closed walk information: The spectral moments $\sum \lambda_i^k$ give the total number of closed walks of length k; in particular, $\frac{1}{6}\sum \lambda_i^3$ gives the number of triangles.
- Regularity: A connected graph is regular if and only if the largest eigenvalue equals the maximum degree.

From the Laplacian spectrum:

- Number of connected components: The multiplicity of the zero eigenvalue equals the number of connected components.
- Spanning trees: For a connected graph, the number of spanning trees is $\frac{1}{n}\prod_{i=1}^{n-1} \mu_i$, where μ_i are the non-zero Laplacian eigenvalues.
- Sum of squared degrees: the trace of the Laplacian gives $\mathrm{Tr}(L) = 2m$, thus $\mathrm{Tr}(L^2) = \sum_i d_i^2 + 2m$, hence $\sum_i d_i^2 = \mathrm{Tr}(L^2) - \mathrm{Tr}(L)$.
- Connectivity: The second smallest eigenvalue (algebraic connectivity) is positive if and only if the graph is connected.

For signed and gain graphs, the spectral picture changes in several respects. Unlike classical graphs, non-bipartite signed and gain graphs can have symmetric spectra. Additionally, the adjacency moments reflect signed structure: for a signed graph, the k-th moment is the difference between the numbers of positive and negative closed walks of length k, while for a complex unit gain graph it is the sum of the gains of the closed walks. In the Laplacian spectrum, the multiplicity of zero no longer counts connected components, but instead counts the number of balanced connected components.

Example 1.4.4. Again consider the graphs displayed in Figure 1.2. Their spectra are

$$\mathrm{spec}(\Gamma) = \{2.5616, 0, -1, -1.5616\}, \quad \mathrm{spec}(\Gamma,\sigma) = \{2, 1, -1, -2\}, \quad \mathrm{spec}(\Gamma,\varphi) = \{2.2361, 0, 0, -2.2361\}.$$

Notice that the spectrum of Γ is not symmetric, which is expected since Γ is not a bipartite graph. However (Γ,σ) and (Γ,φ) do both have symmetric spectra, which is possible for a non-bipartite underlying graph with a suitable signature or gain function.

The rank of a graph, signed graph, or complex unit gain graph is the rank of its adjacency matrix; since the adjacency matrix is symmetric or Hermitian, this equals the number of non-zero eigenvalues, counted with multiplicity.

Given an $m \times n$ matrix $A = (a_{ij})$ and a $p \times q$ matrix B, the Kronecker product $A \otimes B$ is the $mp \times nq$ block matrix whose (i,j) block is $a_{ij}B$. We shall use two standard facts: the mixed-product property $(A \otimes B)(C \otimes D) = AC \otimes BD$, valid whenever the products AC and BD are defined, and the fact that if A and B are square with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_m$ and $\mu_1, \mu_2, \ldots, \mu_p$ respectively, then the eigenvalues of $A \otimes B$ are the mp products $\lambda_i\mu_j$, see [55].

1.5 Degrees and regularities

A graph Γ is r-regular if every vertex has degree r. A signed graph (Γ,σ) is k-net-regular if every vertex has the same net degree $d_\sigma^{\pm}(v) = k$. The notion of co-regular signed graphs was proposed in 2015 by Hameed et al. [46]: a signed graph (Γ,σ) is said to be co-regular if its underlying graph Γ is r-regular for some positive integer r and (Γ,σ) is k-net-regular for some integer k. The corresponding co-regularity pair is then (r,k).

The neighbourhood $N_\Gamma(v)$ of a vertex v, written $N(v)$ when the graph is clear, is the set of vertices adjacent to v, so that $d_\Gamma(v) = |N_\Gamma(v)|$.

In [23], Brunetti gave definitions of outgain, ingain, and regularity for complex unit gain graphs. Here, we extend these concepts to gain graphs over arbitrary groups, interpreting them in the group ring $\mathbb{C}G$. In essence, the G-outgain of a vertex v in (Γ,φ) is the sum of the gains of all the arcs originating in v. A formal definition follows.

Definition 1.5.1. Let u be a vertex of a G-gain graph (Γ,φ). The values

$$d^{\rightarrow}_{(\Gamma,\varphi)}(u) = \sum_{uv \in E_\Gamma} \varphi(uv) \qquad \text{and} \qquad d^{\leftarrow}_{(\Gamma,\varphi)}(u) = \sum_{uv \in E_\Gamma} \varphi(vu)$$

are, respectively, called the G-outgain and the G-ingain of the vertex u.

These sums are to be understood as formal sums in the group ring $\mathbb{C}G$. If u is an isolated vertex, then $d^{\rightarrow}_{(\Gamma,\varphi)}(u)$ and $d^{\leftarrow}_{(\Gamma,\varphi)}(u)$ are assumed to be equal to zero.

Here, we extend Brunetti's notion of regularity for complex unit gain graphs to gain graphs over arbitrary groups.

Definition 1.5.2. A G-gain graph (Γ, φ) is g-G-regular if there exists a fixed element $g \in \mathbb{C}G$ such that, for every vertex $u \in V_\Gamma$,

$$d^{\rightarrow}_{(\Gamma,\varphi)}(u) = \sum_{uv \in E_\Gamma} \varphi(uv) = g.$$

Recall that π extends linearly to an algebra homomorphism $\mathbb{C}G \to M_d(\mathbb{C})$. If $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$, then the represented adjacency matrix $\pi(A_{(\Gamma,\varphi)})$ has constant row sum γ_π. For $G = \mathbb{T}$ and $d = 1$, the scalar $\pi(g)$ is precisely the complex outgain, and this recovers Brunetti's notion of regularity for complex unit gain graphs.

Let $A_{(\Gamma,\varphi)}$ be the adjacency matrix of a G-gain graph (Γ, φ). It is immediately clear from the definitions that $d^{\rightarrow}_{(\Gamma,\varphi)}(u_j)$ gives the sum of the elements in the j-th row of $A_{(\Gamma,\varphi)}$ and $d^{\leftarrow}_{(\Gamma,\varphi)}(u_j)$ gives the sum of the elements in the j-th column of $A_{(\Gamma,\varphi)}$.

1.6 Switching and balance

Although signed graphs appear implicitly in earlier works, such as by Dénes Kőnig in 1936 [64], the explicit notion of balance in triangles originated in social psychology with Fritz Heider in 1946 [54]. The term *signed graph* and the foundational graph-theoretic characterisations of balance were later formalised by Frank Harary in 1953 [49]. There, he provided the following theorem.

Theorem 1.6.1 ([49, Theorem 3])**.** *A signed graph (Γ, σ) is balanced if and only if its vertex set V_Γ can be partitioned into two disjoint subsets in such a way that each positive edge of (Γ, σ) joins two vertices of the same subset and each negative edge joins two vertices of different subsets.*

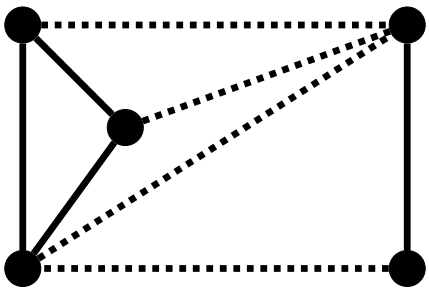


Figure 1.3: A balanced signed graph, which can be seen from Theorem 1.6.1 by partitioning the vertices into two subsets: one consisting of the three on the left, the other of the two on the right.

When Thomas Zaslavsky introduced gain graphs in 1989 [101, 102], he was similarly interested in balance. Accordingly, a fundamental concept in the theory of gain graphs, inherited from the theory of signed graphs, is switching equivalence. Two gain functions φ and φ' on the same underlying graph Γ are said to be switching equivalent, denoted by $\varphi \sim \varphi'$, if there exists a switching function $s\colon V_\Gamma \to G$ such that

$$\varphi'(u,v) = s(u)^{-1}\varphi(u,v)s(v) \quad \text{for all } (u,v) \in O(E_\Gamma). \tag{1.6.1}$$

In this case we may write $(\Gamma, \varphi) \sim (\Gamma, \varphi')$. Throughout this thesis the symbol $\sim$ is reserved for switching equivalence. For signed graphs, the switching function takes values in $\{+1, -1\}$, thus $s(v)^{-1} = s(v)$, and for two signatures σ and σ', condition (1.6.1) takes the familiar form

$$\sigma'(u,v) = s(u)\sigma(u,v)s(v).$$

The switching relation can also be expressed in terms of matrices. For complex unit gain graphs, (Γ, φ) and (Γ, φ') are switching equivalent if and only if there exists a diagonal matrix S with diagonal entries in $\mathbb{T}$ such that $A_{(\Gamma,\varphi')} = S^* A_{(\Gamma,\varphi)} S$, where S^* is the conjugate transpose of S. For signed graphs, S is real with diagonal entries in $\{+1, -1\}$, so that $S^* = S^{-1} = S$ and the condition reads $A_{(\Gamma,\sigma')} = S^{-1} A_{(\Gamma,\sigma)} S$. Such a matrix S is known as a switching matrix.

Switching does not alter the adjacency spectrum, since matrix conjugation preserves eigenvalues. Hence, switching-equivalent signed and complex unit gain graphs share the same characteristic polynomials and eigenvalues. For an arbitrary group G, the analogous statement holds once a representation is fixed: if (Γ, φ) and (Γ, φ') are switching equivalent via the switching function s, then for every unitary representation π of G the associated block matrices are conjugate by the block-diagonal unitary matrix with blocks $\pi(s(v_1)), \pi(s(v_2)), \ldots, \pi(s(v_n))$, and hence share the same spectrum.

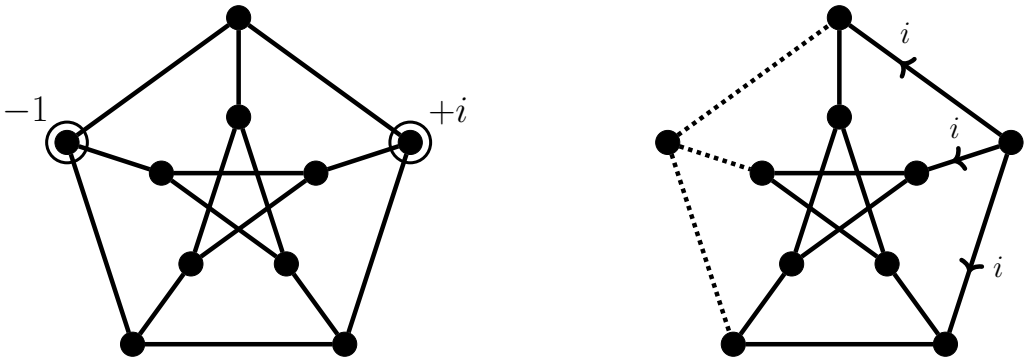


Figure 1.4: A complex unit switching performed on the Petersen graph. The actions are -1 on the vertex furthest left and $+i$ on the vertex furthest right.

Two switching-equivalent gain graphs (Γ, φ) and (Γ, φ') are said to belong to the same switching class, denoted by $[\varphi]$. Clearly, all members of a switching class share the same adjacency characteristic polynomial and spectrum. Moreover, since switching does not change the underlying graph Γ, the degree matrix D_Γ also remains unchanged, so the Laplacian spectra of switching-equivalent graphs coincide. However, the same does not hold for the net Laplacian spectrum, since $D^{\pm}_{(\Gamma,\sigma)}$ and $D^{\pm}_{(\Gamma,\sigma')}$ can differ even if (Γ, σ) and (Γ, σ') are switching equivalent.

More generally, two gain graphs (Γ_1, φ_1) and (Γ_2, φ_2) are said to be switching isomorphic if there exists a graph isomorphism $f\colon V_{\Gamma_1} \to V_{\Gamma_2}$ such that φ_1 is switching equivalent to the gain function $\varphi_2 \circ f$ on Γ_1, defined as $(\varphi_2 \circ f)(u, v) = \varphi_2(f(u), f(v))$. For signed graphs this amounts to the existence of a switching matrix S and a permutation matrix P such that

$$A_{(\Gamma_2,\sigma_2)} = (PS)^{-1} A_{(\Gamma_1,\sigma_1)} (PS),$$

with the matrix PS being a signed permutation matrix, also referred to as a $\{1, 0, -1\}$-monomial matrix. Since isomorphic graphs clearly share the same adjacency and Laplacian spectra, switching-isomorphic signed and complex unit gain graphs are cospectral; as before, the statement extends to general G-gain graphs once a representation is fixed.

When Γ is endowed with a gain function φ, the gain of a closed walk $W = v_1 v_2 \ldots v_{k+1}$ is

$$\varphi(W) = \prod_{i=1}^{k} \varphi(v_i, v_{i+1}),$$

and for a signature σ, the sign of W is $\sigma(W) = \prod_{i=1}^{k} \sigma(v_i, v_{i+1})$. In particular, the sign of a cycle is the product of the signs of its edges, so a cycle in a signed graph is positive precisely when it contains an even number of negative edges.

A G-gain graph (Γ, φ) is said to be balanced if $\varphi(W) = 1_G$ for every closed walk W in Γ. Since the gain of a reversed walk is the inverse of the original gain, balance is orientation-independent. For a signed graph (Γ, σ) this equates to requiring $\sigma(W) = +1$ for every closed walk W, or equivalently, that all of its cycles, if any exist, are positive. Balance admits the following characterisation in terms of switching.

Proposition 1.6.2 ([101, Lemma 5.3]). *A G-gain graph (Γ, φ) is balanced if and only if φ is switching equivalent to the identity gain function. In particular, a signed graph (Γ, σ) is balanced if and only if it is switching equivalent to the all-positive $(\Gamma, +)$.*

1.7 Pseudo-potentials

Pseudo-potentials play a central role in the switching-stable constructions of later chapters. For this reason, we record their basic theory here in greater detail than the other preliminaries.

Pseudo-potential functions on signed graphs were introduced by Cavaleri, Donno, and Spessato [33, Definition 2.1], and we extended the notion to complex unit gain graphs in [61, Definition 4.1].

In keeping with the conventions of this chapter, we state the definition and its basic properties once, for G-gain graphs, and recover the signed and complex unit versions by specialisation. Pseudo-potentials are a fundamental ingredient in the switching-stable constructions of Chapters 4 and 7.

Definition 1.7.1. Let (Γ, φ) be a connected G-gain graph. Fix a spanning tree $\mathcal{T}$ of Γ and fix a vertex v in V_Γ. The pseudo-potential $U_{\mathcal{T},v,\varphi} \colon V_\Gamma \to G$ is the function defined, for each w in V_Γ, as

$$U_{\mathcal{T},v,\varphi}(w) = \varphi(\gamma_{vw}),$$

where γ_{vw} is the unique path in $\mathcal{T}$ from v to w.

Note that $U_{\mathcal{T},v,\varphi}(v) = 1_G$. Moreover, since $\varphi(u,w)\varphi(w,u) = 1_G$ for any pair of adjacent vertices, any backtracking in a walk contributes trivially to its gain; consequently $U_{\mathcal{T},v,\varphi}(w)$ equals the gain of any walk contained in $\mathcal{T}$ from v to w, not only that of the unique path.

Throughout this thesis, we adopt the convention that any construction defined for a connected graph using a spanning tree extends to the disconnected case by applying the same construction componentwise using a spanning forest, with one root per connected component. In particular, for a disconnected G-gain graph (Γ, φ), the pseudo-potential is defined by replacing the spanning tree with a maximal spanning forest and the fixed vertex v with a list $\overline{v} = \{v_1, \ldots, v_r\}$ of vertices, one chosen from each connected component. The definition then applies componentwise, so all results stated for connected pseudo-potentials extend verbatim to the disconnected case.

Remark 1.7.2. The prefix *pseudo-* distinguishes the notion of Definition 1.7.1 from that of a potential. A potential of a G-gain graph (Γ, φ) is a function $U \colon V_\Gamma \to G$ such that $\varphi(u,w) = U(u)^{-1}U(w)$ for every $(u,w) \in O(E_\Gamma)$; for a signed graph this reads $\sigma(u,w) = U(u)U(w)$, recovering the classical notion. By Proposition 1.6.2, a G-gain graph is balanced if and only if it admits a potential. The pseudo-potential satisfies the defining property of a potential on the edges of $\mathcal{T}$, and is a genuine potential precisely when (Γ, φ) is balanced: if (Γ, φ) is balanced and $(u,w) \in O(E_\Gamma)$, then the walk obtained by following γ_{vu}, the edge (u,w), and the reverse of γ_{vw} is closed, so

$$U_{\mathcal{T},v,\varphi}(u)\, \varphi(u,w)\, U_{\mathcal{T},v,\varphi}(w)^{-1} = 1_G,$$

that is, $\varphi(u,w) = U_{\mathcal{T},v,\varphi}(u)^{-1} U_{\mathcal{T},v,\varphi}(w)$.

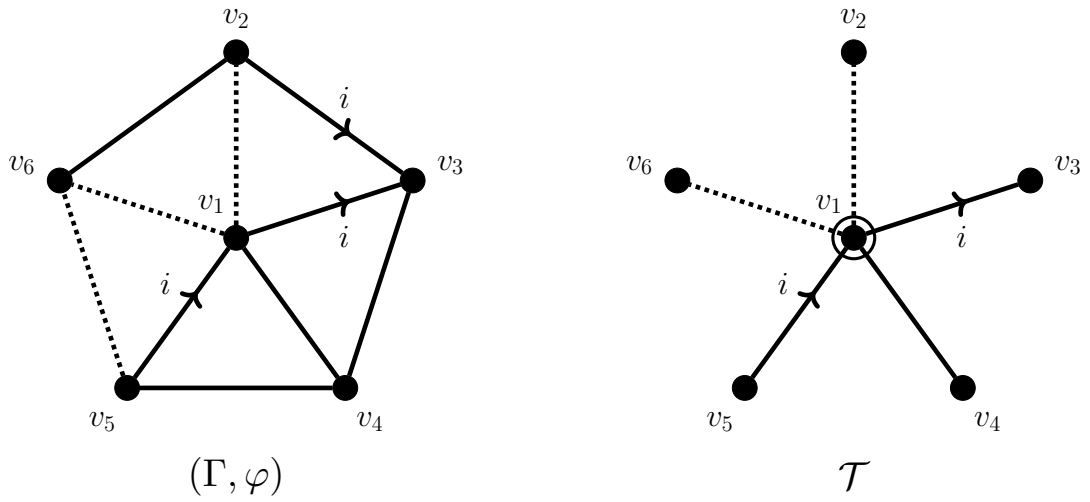


Figure 1.5: A complex unit gain graph (Γ, φ), a spanning tree $\mathcal{T}$, and a fixed vertex v_1.

Example 1.7.3. Consider the complex unit gain graph (Γ, φ), spanning tree $\mathcal{T}$, and fixed vertex v_1 of Figure 1.5. The corresponding pseudo-potential gives:

$$\begin{aligned} U_{\mathcal{T},v_1,\varphi}(v_1) &= +1, \\ U_{\mathcal{T},v_1,\varphi}(v_2) &= -1, \\ U_{\mathcal{T},v_1,\varphi}(v_3) &= +i, \\ U_{\mathcal{T},v_1,\varphi}(v_4) &= +1, \\ U_{\mathcal{T},v_1,\varphi}(v_5) &= -i, \\ U_{\mathcal{T},v_1,\varphi}(v_6) &= -1. \end{aligned}$$

With a slight abuse of notation, we denote by the same symbol $U_{\mathcal{T},v,\varphi}$ the diagonal matrix with entries $(U_{\mathcal{T},v,\varphi})_{ww} = U_{\mathcal{T},v,\varphi}(w)$.

We now record how the pseudo-potential depends on its three parameters: the base vertex, the gain function within a switching class, and the spanning tree.

Lemma 1.7.4. *Let (Γ, φ) be a connected G-gain graph and let $\mathcal{T}$ be a spanning tree of Γ. Then for all $v, v' \in V_\Gamma$,*

$$U_{\mathcal{T},v',\varphi} = \varphi(\gamma_{v'v})\, U_{\mathcal{T},v,\varphi},$$

where $\gamma_{v'v}$ is the unique path in $\mathcal{T}$ from v' to v. In particular, for a signed graph either $U_{\mathcal{T},v',\sigma} = U_{\mathcal{T},v,\sigma}$ or $U_{\mathcal{T},v',\sigma} = -U_{\mathcal{T},v,\sigma}$, and for a complex unit gain graph $U_{\mathcal{T},v',\varphi} = z\, U_{\mathcal{T},v,\varphi}$ for some $z \in \mathbb{T}$.

Proof. For $w \in V_\Gamma$, the path $\gamma_{v'v}$ followed by the path γ_{vw} is a walk in $\mathcal{T}$ from v' to w, whose gain is $\varphi(\gamma_{v'v})\, U_{\mathcal{T},v,\varphi}(w)$. By the walk-independence observed after Definition 1.7.1, this gain equals $U_{\mathcal{T},v',\varphi}(w)$. □

Lemma 1.7.5. *Let (Γ, φ) and (Γ, φ') be two switching-equivalent G-gain graphs, with $\varphi'(u,w) = s(u)^{-1}\varphi(u,w)s(w)$ for some switching function $s\colon V_\Gamma \to G$. Fix a spanning tree $\mathcal{T}$ of Γ and a vertex $v \in V_\Gamma$. Then for all $y \in V_\Gamma$,*

$$U_{\mathcal{T},v,\varphi'}(y) = s(v)^{-1}\, U_{\mathcal{T},v,\varphi}(y)\, s(y).$$

Proof. Let $\gamma_{vy} = v, u_1, u_2, \ldots, u_{k-1}, y$ be the unique path in $\mathcal{T}$ from v to y. Then

$$\begin{aligned} U_{\mathcal{T},v,\varphi'}(y) &= \varphi'(v,u_1)\varphi'(u_1,u_2)\cdots\varphi'(u_{k-1},y) \\ &= s(v)^{-1}\varphi(v,u_1)s(u_1)\, s(u_1)^{-1}\varphi(u_1,u_2)s(u_2)\cdots s(u_{k-1})^{-1}\varphi(u_{k-1},y)s(y) \\ &= s(v)^{-1}\, \varphi(v,u_1)\varphi(u_1,u_2)\cdots\varphi(u_{k-1},y)\, s(y) \\ &= s(v)^{-1}\, U_{\mathcal{T},v,\varphi}(y)\, s(y). \end{aligned}$$

□

Corollary 1.7.6. *Under the assumptions of Lemma 1.7.5, the following hold:*

(i) if $s(v) = 1_G$, then $U_{\mathcal{T},v,\varphi'}(y) = U_{\mathcal{T},v,\varphi}(y)\, s(y)$ for all $y \in V_\Gamma$;

(ii) for each $y \in V_\Gamma$ with $s(y) = 1_G$, we have $U_{\mathcal{T},v,\varphi'}(y) = s(v)^{-1}\, U_{\mathcal{T},v,\varphi}(y)$.

Remark 1.7.7. The normalisation $s(v) = 1_G$ in Corollary 1.7.6(i) may be assumed without loss of generality whenever only the switching class of (Γ, φ') matters. Indeed, given any switching function $s\colon V_\Gamma \to G$, define $\tilde{s}\colon V_\Gamma \to G$ by $\tilde{s}(y) = s(v)^{-1}s(y)$, so that $\tilde{s}(v) = 1_G$. The gain functions obtained by switching φ with s and with $\tilde{s}$ are both switching equivalent to φ, hence to each other.

Lemma 1.7.8. *Let (Γ, φ) be a connected balanced G-gain graph. Fix a vertex $v \in V_\Gamma$. Then for any two spanning trees $\mathcal{T}$ and $\mathcal{T}'$ of Γ and all $y \in V_\Gamma$,*

$$U_{\mathcal{T},v,\varphi}(y) = U_{\mathcal{T}',v,\varphi}(y).$$

Proof. Let γ_{vy} and γ'_{vy} be the unique paths in $\mathcal{T}$ and $\mathcal{T}'$ from v to y, respectively. Then γ_{vy} followed by the reverse of γ'_{vy} is a closed walk based at v. Since (Γ, φ) is balanced, every closed walk has gain 1_G, so

$$U_{\mathcal{T},v,\varphi}(y)\, U_{\mathcal{T}',v,\varphi}(y)^{-1} = \varphi(\gamma_{vy})\, \varphi(\gamma'_{vy})^{-1} = 1_G,$$

and hence $U_{\mathcal{T},v,\varphi}(y) = U_{\mathcal{T}',v,\varphi}(y)$ for each $y \in V_\Gamma$. □

For signed and complex unit gain graphs, where G is abelian and the gains are matrix entries, Lemma 1.7.5 takes the following matrix form.

Corollary 1.7.9. *Let φ and φ' be two switching-equivalent $\mathbb{T}$-gain functions on a connected graph Γ, let $\mathcal{T}$ be a spanning tree of Γ, and let $v \in V_\Gamma$. Then there exists a diagonal matrix S with diagonal entries in $\mathbb{T}$ such that*

$$A_{(\Gamma,\varphi')} = S^* A_{(\Gamma,\varphi)} S \quad \text{and} \quad U_{\mathcal{T},v,\varphi'} = U_{\mathcal{T},v,\varphi}\, S = S\, U_{\mathcal{T},v,\varphi}.$$

In particular, for two switching-equivalent signatures σ and σ', the matrix S is real with diagonal entries in $\{+1, -1\}$, and $A_{(\Gamma,\sigma')} = S^{-1} A_{(\Gamma,\sigma)} S$ and $U_{\mathcal{T},v,\sigma'} = S U_{\mathcal{T},v,\sigma} = U_{\mathcal{T},v,\sigma} S$.

Proof. Let s be a switching function with $\varphi'(u,w) = s(u)^{-1}\varphi(u,w)s(w)$. Set $S_0 = \operatorname{diag}(s(v_1), s(v_2), \ldots, s(v_n))$. Then $A_{(\Gamma,\varphi')} = S_0^* A_{(\Gamma,\varphi)} S_0$ and, by Lemma 1.7.5, $U_{\mathcal{T},v,\varphi'} = s(v)^{-1} U_{\mathcal{T},v,\varphi} S_0$.

Now define $S = s(v)^{-1} S_0$. Because $\mathbb{T}$ is abelian, $s(v)^{-1}$ commutes with all matrices, so $S^* A_{(\Gamma,\varphi)} S = S_0^* A_{(\Gamma,\varphi)} S_0 = A_{(\Gamma,\varphi')}$, and $U_{\mathcal{T},v,\varphi'} = U_{\mathcal{T},v,\varphi} S = S U_{\mathcal{T},v,\varphi}$. □

Finally, the pseudo-potential gives a short proof of the existence of distinguished representatives for switching classes, generalising a classical result of Zaslavsky for signed graphs [100, Lemma 3.1].

Definition 1.7.10. Let (Γ, φ) be a G-gain graph and let $\mathcal{T}$ be a spanning tree (or maximal spanning forest) of Γ. Then (Γ, φ) is $\mathcal{T}$-positive if $\varphi(u, w) = 1_G$ for each edge $\{u, w\}$ in $\mathcal{T}$.

Lemma 1.7.11. *Let Γ be a connected graph and let $\mathcal{T}$ be a spanning tree of Γ. Every switching class of G-gain functions on Γ contains a $\mathcal{T}$-positive representative, and any two $\mathcal{T}$-positive representatives of the same class are obtained from one another by conjugating every gain by a single fixed element of G. In particular, if G is abelian, each switching class contains exactly one $\mathcal{T}$-positive representative.*

Proof. For existence, let φ be a gain function on Γ, fix $v \in V_\Gamma$, and switch φ by the function $s(y) = U_{\mathcal{T},v,\varphi}(y)^{-1}$. For an edge $\{u, w\}$ of $\mathcal{T}$, oriented so that u lies on the path γ_{vw}, we have $U_{\mathcal{T},v,\varphi}(w) = U_{\mathcal{T},v,\varphi}(u)\, \varphi(u, w)$, and therefore

$$s(u)^{-1}\varphi(u,w)s(w) = U_{\mathcal{T},v,\varphi}(u)\, \varphi(u,w)\, \varphi(u,w)^{-1}\, U_{\mathcal{T},v,\varphi}(u)^{-1} = 1_G.$$

For uniqueness, suppose φ_1 and φ_2 are both $\mathcal{T}$-positive with $\varphi_2(u,w) = s(u)^{-1}\varphi_1(u,w)s(w)$ for a switching function s. On each edge of $\mathcal{T}$ this gives $1_G = s(u)^{-1}s(w)$, so $s(u) = s(w)$; since $\mathcal{T}$ spans Γ, the function s is constant, say $s \equiv c$. Hence $\varphi_2(u,w) = c^{-1}\varphi_1(u,w)c$ for every oriented edge. If G is abelian, this yields $\varphi_2 = \varphi_1$. □

Definition 1.7.12. Let Γ be a graph, let $\mathcal{T}$ be a spanning tree (or maximal spanning forest) of Γ, let G be abelian, and let $[\varphi]$ be a switching class of G-gain functions on Γ. The $\mathcal{T}$-representative of $[\varphi]$ is the unique $\mathcal{T}$-positive gain function $\varphi_{\mathcal{T}} \in [\varphi]$.

1.8 Auxiliary results

The following is the very important result known as Schur's complement formula. This was first introduced in 1917 by Issai Schur [86, pp. 215–216] and appears in many textbooks, see [38].

Lemma 1.8.1 (Schur's complement formula)**.** *Let* $Q = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ *be a block matrix. Let* A *and* D *be square matrices. It follows that*

(a) if A *is an invertible matrix,* $\det(Q) = \det(A)\det\big(D - CA^{-1}B\big)$,

(b) if D *is an invertible matrix,* $\det(Q) = \det(D)\det\big(A - BD^{-1}C\big)$.

The matrix $D - CA^{-1}B$ ($A - BD^{-1}C$, respectively) is usually called the Schur complement of the matrix A (the matrix D, respectively) in the matrix Q and is denoted Q/A (Q/D, respectively). Both expressions can be obtained by using block Gaussian elimination. For instance, Lemma 1.8.1(a) follows from the factorisation

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & O \\ CA^{-1} & I \end{pmatrix} \begin{pmatrix} A & B \\ O & D - CA^{-1}B \end{pmatrix}.$$

When applying Schur's complement to a characteristic matrix, we typically assume that the relevant diagonal block is invertible, i.e., that λ avoids the spectrum of that block. The resulting expression is a rational function of λ. Since the final determinant is a polynomial in λ, the identity extends to all λ by clearing denominators. Consequently, when applying Lemma 1.8.1 to a characteristic matrix, the invertibility assumption is only needed to define the resolvent; the final polynomial identity follows for all λ. We will use this standard polynomial-continuity argument implicitly throughout.

The subsequent result comes immediately from applying Schur's complement formula to particular matrices.

Lemma 1.8.2 ([41, Lemma 3])**.** *Let* X *and* Y *be* $n \times m$ *matrices and let* M *be an* $n \times n$ *invertible matrix. Then*

(a) $\det\big(I_n + XY^T\big) = \det\big(I_m + Y^T X\big)$.

(b) $\det\big(M + XY^T\big) = \det(M)\det\big(I_m + Y^T M^{-1} X\big)$.

For further applications and historical notes pertaining to Schur's result, see [103].

The graph invariant known as the coronal was introduced in 2011 by McLeman and McNicholas in [75]. In the following we adapt their definition directly to a matrix.

Definition 1.8.3 ([75, Definition 1])**.** Let M be an $n \times n$ matrix. The coronal $\chi_M(\lambda)$ of M is the rational function

$$\chi_M(\lambda) = \mathbf{1}_n^T (\lambda I_n - M)^{-1} \mathbf{1}_n,$$

defined for $\lambda \notin \operatorname{spec}(M)$; as a rational function it may have poles at spectral values.

We also have the following useful result.

Lemma 1.8.4 ([75, Proposition 6])**.** *Let* M *be an* $n \times n$ *matrix with all row sums equal to* r*. Then*

$$\chi_M(\lambda) = \frac{n}{\lambda - r}.$$

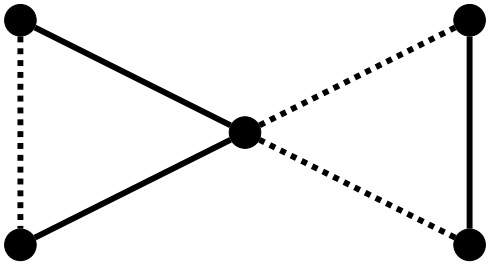


Figure 1.6: A net-regular signed graph (Γ, σ).

Example 1.8.5. Consider the signed graph (Γ, σ) of order 5 depicted in Figure 1.6. With a suitable vertex ordering, its adjacency matrix is

$$A_{(\Gamma,\sigma)} = \begin{pmatrix} 0 & 1 & 1 & -1 & -1 \\ 1 & 0 & -1 & 0 & 0 \\ 1 & -1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 0 \end{pmatrix},$$

which, since the signed graph is 0-net-regular, is a 5×5 matrix with all row sums equal to 0.

By Definition 1.8.3, the coronal $\chi_{A_{(\Gamma,\sigma)}}(\lambda)$ of $A_{(\Gamma,\sigma)}$ can be calculated as

$$\chi_{A_{(\Gamma,\sigma)}}(\lambda) = \mathbf{1}_5^T (\lambda I_5 - A_{(\Gamma,\sigma)})^{-1} \mathbf{1}_5,$$

and by Lemma 1.8.4, this is equal to $5/\lambda$.

Chapter 2

Graphs with integer reciprocal eigenvalue properties

As natural companions to the integer reciprocal eigenvalue properties m-(R) and m-(SR) recently introduced by Barik, Pati, and Verma [13], we introduce the integer anti-reciprocal eigenvalue properties m-($-$R) and m-($-$SR) for graphs and signed graphs. We show that these properties are compatible with the Kronecker product for signed graphs [42]; that simple m-coronae of signed graphs always satisfy m-($-$SR), with m-(SR) holding whenever the root graph has symmetric spectrum; and that the unsigned hypercube Q_n satisfies integer reciprocal eigenvalue properties for $n \leq 4$ but not for $n \geq 5$, whereas a suitable signing from [70] always yields n-(SR) and n-($-$SR). Finally, we construct explicit infinite families of graphs satisfying m-(R) but not m-(SR), together with non-corona graphs satisfying both m-(SR) and m-($-$SR), in both the classical and signed settings.

This chapter is based on [60], joint work with Francesco Belardo and Sasmita Barik.

2.1 Introduction and preliminaries

Since the reciprocal eigenvalue property was first introduced in 2006 [10], graphs satisfying reciprocal and anti-reciprocal eigenvalue properties have been studied in a number of recent papers, see [4, 7, 8, 9, 11, 12, 44, 67, 77, 79]. These properties have also been extended to weighted graphs [3, 78] and to signed graphs [5, 21]. More recently, Barik et al. introduced the notion of the integer reciprocal eigenvalue property [13].

Definition 2.1.1 ([13, Definition 3])**.** Let m be a positive integer. Given a graph Γ, we say that Γ satisfies the integer reciprocal eigenvalue property m-(R) if whenever λ is a non-zero eigenvalue of Γ, m/λ is also an eigenvalue of Γ. Further to this, if λ and m/λ have the same multiplicity in each instance, we say that Γ satisfies the strong integer reciprocal eigenvalue property m-(SR).

Here we introduce the natural integer analogue of the anti-reciprocal eigenvalue property.

Definition 2.1.2. Let m be a positive integer. Given a graph Γ, we say that Γ satisfies the integer anti-reciprocal eigenvalue property m-($-$R) if whenever λ is a non-zero eigenvalue of Γ, $-m/\lambda$ is also an eigenvalue of Γ. Further to this, if λ and $-m/\lambda$ have the same multiplicity in each instance, we say that Γ satisfies the strong integer anti-reciprocal eigenvalue property m-($-$SR).

Throughout this chapter we follow the notation and conventions of Chapter 1; recall in particular that graph operations are written with parentheses, while matrices carry subscripts.

A simple corona graph, often denoted as $\Gamma \circ K_1$, is a graph constructed by taking a root graph Γ and attaching a new pendant vertex to every existing vertex in Γ. In the following we define a simple m-corona graph.

Definition 2.1.3. Take any graph Γ. Then the simple m-corona graph of Γ is obtained by attaching m new pendant vertices to every existing vertex in Γ.

Let $\cup_m K_1$ denote the disjoint union of m copies of K_1, the graph with a single vertex. Then we can see that the simple m-corona of any given graph Γ is precisely $\Gamma \circ \cup_m K_1$.

Note that the simple m-corona of the graph K_1 is the complete bipartite graph $K_{1,m}$, also known as the star graph S_m.

Lemma 2.1.4 ([13, Lemma 8])**.** *Let Γ_1 be a graph on $n \geq 2$ vertices and $m \geq 2$ be a positive integer. Let $\Gamma = \Gamma_1 \circ \cup_m K_1$. Then Γ satisfies property m-$(-R)$. Furthermore, if Γ_1 is bipartite, then Γ satisfies properties m-(SR) and m-$(-SR)$.*

We note that the second conclusion of this lemma is stated for bipartite graphs since, in the unsigned setting, a graph has symmetric spectrum if and only if it is bipartite. In the signed setting, however, there exist signed graphs with symmetric spectrum that are not bipartite, and so the upcoming Proposition 2.2.4 gives a strictly more general result.

Example 2.1.5. The spectrum of the star graph $S_m = K_{1,m}$ consists of $\sqrt{m}$ once, $-\sqrt{m}$ once, and 0 with multiplicity $m-1$. Since $\sqrt{m}$ is its own m-reciprocal, it is easy to see that S_m satisfies properties m-(SR) and m-$(-$SR).

The following result relates the coefficients of the characteristic polynomial of a graph satisfying m-(SR).

Lemma 2.1.6 ([13, Lemma 18])**.** *Let $m \geq 2$ and Γ be a graph on n vertices with rank k. If Γ satisfies property m-(SR), then*

$$|a_{k-i}| = m^{\frac{k}{2}-i}|a_i|,$$

for $1 \leq i \leq k$, where a_i is the coefficient of λ^{k-i} in $P(\Gamma; \lambda)$.

Here we provide the analogous result relating the coefficients of the characteristic polynomial of a signed graph satisfying m-$(-$SR), and note that it recovers the previous result when σ is the all-positive signature.

Lemma 2.1.7. *Let $m \geq 2$ and (Γ, σ) be a signed graph on n vertices with rank k. If (Γ, σ) satisfies property m-$(-SR)$, then*

$$a_{k-i} = (-1)^{\frac{k}{2}-i} m^{\frac{k}{2}-i}\, a_i,$$

for $0 \leq i \leq k$, where a_i is the coefficient of λ^{k-i} in $P((\Gamma, \sigma); \lambda)$. In particular, k is even, $a_0 = 1$, and $a_k = (-1)^{k/2} m^{k/2}$.

Proof. Let the non-zero eigenvalues of (Γ, σ) be $\lambda_1, \lambda_2, \ldots, \lambda_k$, so that $P((\Gamma, \sigma); \lambda) = \lambda^{n-k} \prod_{i=1}^{k} (\lambda - \lambda_i)$. Define $Q(\lambda) = \prod_{i=1}^{k} (\lambda - \lambda_i) = \sum_{i=0}^{k} a_i \lambda^{k-i}$ (so $a_0 = 1$). The property m-$(-$SR) pairs each λ_i with $-m/\lambda_i$, hence k is even and

$$Q(\lambda) = \prod_{i=1}^{k} \left(\lambda + \frac{m}{\lambda_i}\right) = (-1)^{k/2} m^{-k/2} \lambda^k Q\left(-\frac{m}{\lambda}\right).$$

After substituting $Q(\lambda) = \sum_{i=0}^{k} a_i \lambda^{k-i}$ and $Q(-m/\lambda) = \sum_{i=0}^{k} a_i (-m)^{k-i} \lambda^{i-k}$, comparing coefficients of λ^{k-i} yields

$$a_{k-i} = (-1)^{\frac{k}{2}-i} m^{\frac{k}{2}-i} a_i$$

for $0 \leq i \leq k$. In particular $a_k = (-1)^{k/2} m^{k/2}$. □

The remainder of this chapter is structured as follows. In Section 2.2, we show when Kronecker products, m-corona products, and hypercubes satisfy integer reciprocal eigenvalue properties. In Section 2.3, we provide explicit constructions for classes of graphs, both signed and unsigned, which satisfy integer reciprocal eigenvalue properties. In Section 2.4, we offer concluding remarks.

2.2 Eigenvalue properties of Kronecker products, signed m-coronae, and hypercubes

In this section, we begin by extending the notions of integer eigenvalue reciprocity for Kronecker products and simple m-coronae to the theory of signed graphs. We then provide a full classification of when hypercubes satisfy these properties, considering both the all-positive case and the case in which every 4-cycle is negative.

2.2.1 Kronecker products

Graph products are useful in constructing many important classes of graphs. Following [42], we define the Kronecker product of two signed graphs as follows. Let (Γ_1, σ_1) and (Γ_2, σ_2) be two signed graphs with disjoint vertex sets $V_{\Gamma_1} = \{u_1, u_2, \ldots, u_p\}$ and $V_{\Gamma_2} = \{v_1, v_2, \ldots, v_q\}$, respectively. The Kronecker product of (Γ_1, σ_1) and (Γ_2, σ_2), denoted by $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$, is the signed graph whose vertex set is $V_{\Gamma_1} \times V_{\Gamma_2}$. Two vertices (u_i, v_j) and (u_r, v_s) are adjacent in $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ if u_i and u_r are adjacent in (Γ_1, σ_1) and v_j and v_s are adjacent in (Γ_2, σ_2), in which case the sign of the edge is $\sigma_1(u_i, u_r)\sigma_2(v_j, v_s)$.

Moreover, it was proved in [42, Theorem 3.1] that if $\lambda_1, \lambda_2, \ldots, \lambda_p$ are eigenvalues of (Γ_1, σ_1) and $\mu_1, \mu_2, \ldots, \mu_q$ are eigenvalues of (Γ_2, σ_2), then the eigenvalues of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ are

$$\lambda_i \mu_j, \quad \text{for } 1 \le i \le p;\ 1 \le j \le q.$$

We show that the m-(R) and m-($-$R) properties are compatible with the Kronecker product in a precise sense: if two graphs satisfy such properties, their Kronecker product satisfies a related property with its parameter equal to the product of the two original parameters.

Proposition 2.2.1. *Let (Γ_1, σ_1) and (Γ_2, σ_2) be signed graphs with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$ and $\mu_1, \mu_2, \ldots, \mu_q$ respectively. If (Γ_1, σ_1) satisfies m_1-(R) and (Γ_2, σ_2) satisfies m_2-(R), then $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$-(R).*

Proof. Let $\lambda_i \mu_j$ be a non-zero eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$. Since (Γ_1, σ_1) satisfies m_1-(R), m_1/λ_i is an eigenvalue of (Γ_1, σ_1), and since (Γ_2, σ_2) satisfies m_2-(R), m_2/μ_j is an eigenvalue of (Γ_2, σ_2). Therefore

$$\left(\frac{m_1}{\lambda_i}\right)\left(\frac{m_2}{\mu_j}\right) = \frac{m_1 m_2}{\lambda_i \mu_j}$$

is an eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$. Since $\lambda_i \mu_j$ was arbitrary, every eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ has its $m_1 m_2$-reciprocal also as an eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$, and hence $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ has the $m_1 m_2$-(R) property. □

Corollary 2.2.2. *Let (Γ_1, σ_1) and (Γ_2, σ_2) be signed graphs with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$ and $\mu_1, \mu_2, \ldots, \mu_q$ respectively.*

(i) If (Γ_1, σ_1) satisfies m_1-(R) and (Γ_2, σ_2) satisfies m_2-($-$R), then $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$-($-$R).

(ii) If (Γ_1, σ_1) satisfies m_1-($-$R) and (Γ_2, σ_2) satisfies m_2-($-$R), then $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$-(R).

Proof. Let $\lambda_i \mu_j$ be a non-zero eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$; if the eigenvalue is zero, there is nothing to prove.

(i) Since (Γ_1, σ_1) satisfies m_1-(R), m_1/λ_i is an eigenvalue of (Γ_1, σ_1), and since (Γ_2, σ_2) satisfies m_2-($-$R), $-m_2/\mu_j$ is an eigenvalue of (Γ_2, σ_2). Therefore

$$\left(\frac{m_1}{\lambda_i}\right)\left(\frac{-m_2}{\mu_j}\right) = \frac{-m_1 m_2}{\lambda_i \mu_j}$$

is an eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$, and hence $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$-($-$R).

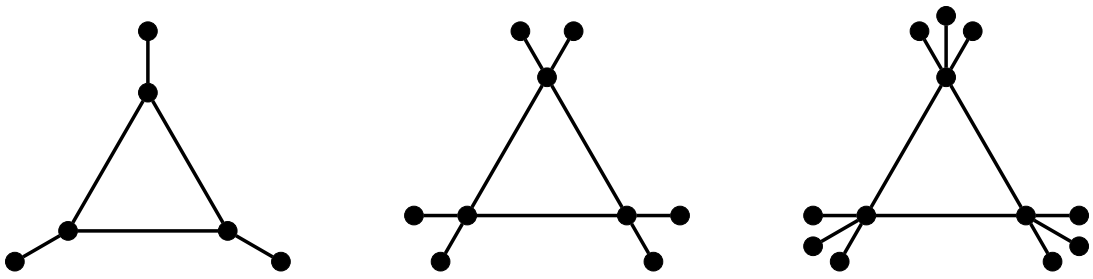
(ii) Since (Γ_1, σ_1) satisfies m_1-$(-\mathrm{R})$, $-m_1/\lambda_i$ is an eigenvalue of (Γ_1, σ_1), and since (Γ_2, σ_2) satisfies m_2-$(-\mathrm{R})$, $-m_2/\mu_j$ is an eigenvalue of (Γ_2, σ_2). Therefore

$$\left(\frac{-m_1}{\lambda_i}\right)\left(\frac{-m_2}{\mu_j}\right) = \frac{m_1 m_2}{\lambda_i \mu_j}$$

is an eigenvalue of $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$, and hence $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$-(R). □

Remark 2.2.3. In fact, the strong properties are also preserved under the Kronecker product. If (Γ_1, σ_1) satisfies m_1-(SR) and (Γ_2, σ_2) satisfies m_2-(SR), then $(\Gamma_1, \sigma_1) \otimes (\Gamma_2, \sigma_2)$ satisfies $m_1 m_2$-(SR), and similarly for the mixed cases of the corollary. This follows because, for non-zero eigenvalues, the strong reciprocal pairings in each factor induce a multiplicity-preserving bijection between the pairs contributing to the multiplicity of $\lambda_i \mu_j$ and those contributing to the multiplicity of $m_1 m_2/(\lambda_i \mu_j)$.

2.2.2 Signed m-coronae



$K_3 \circ K_1$ $K_3 \circ \cup_2 K_1$ $K_3 \circ \cup_3 K_1$

Figure 2.1: The simple 1-corona, 2-corona, and 3-corona of K_3.

As we know from [13, Lemma 8], taking the simple m-corona of a graph gives a product satisfying property m-$(-\mathrm{R})$. Here we extend this to signed graphs and show that the stronger property m-$(-\mathrm{SR})$ is always satisfied, with m-(SR) also satisfied whenever the root signed graph has symmetric spectrum.

Proposition 2.2.4. *Let (Γ, σ) be a signed graph on $n \geq 2$ vertices and let m be any positive integer. Then $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ satisfies the strong integer anti-reciprocal eigenvalue property m-$(-SR)$. Moreover, if (Γ, σ) has symmetric spectrum, then $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ also satisfies the strong integer reciprocal eigenvalue property m-(SR).*

Proof. With a suitable vertex ordering, the adjacency matrix of $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ has the following blockwise form

$$A_{((\Gamma,\sigma)\circ\cup_m(K_1,+))} = \begin{pmatrix} A_{(\Gamma,\sigma)} & I_n \otimes \mathbf{1}_m^T \\ I_n \otimes \mathbf{1}_m & O_{mn} \end{pmatrix}.$$

By passing to the characteristic polynomial and applying Schur's complement formula, we have

$$\begin{aligned} P((\Gamma,\sigma)\circ\cup_m(K_1,+);\,\lambda) &= \det\begin{pmatrix} \lambda I_n - A_{(\Gamma,\sigma)} & -I_n \otimes \mathbf{1}_m^T \\ -I_n \otimes \mathbf{1}_m & \lambda I_{mn} \end{pmatrix} \\ &= \det(\lambda I_{mn}) \det\big(\lambda I_n - A_{(\Gamma,\sigma)} - (I_n \otimes \mathbf{1}_m^T)(\lambda I_{mn})^{-1}(I_n \otimes \mathbf{1}_m)\big) \\ &= \lambda^{mn} \det\Big(\lambda I_n - A_{(\Gamma,\sigma)} - \frac{m}{\lambda} I_n\Big) \\ &= \lambda^{mn} \det\Big(\Big(\lambda - \frac{m}{\lambda}\Big) I_n - A_{(\Gamma,\sigma)}\Big) \\ &= \lambda^{mn} P\left((\Gamma,\sigma);\,\lambda - \frac{m}{\lambda}\right). \end{aligned}$$

Hence, a non-zero λ is an eigenvalue of $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ if and only if $\lambda - m/\lambda$ is an eigenvalue of (Γ, σ). If μ is an eigenvalue of (Γ, σ), then the corresponding eigenvalues of $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ satisfy

$\lambda - m/\lambda = \mu$, giving $\lambda^2 - \mu\lambda - m = 0$, with roots $\lambda^{\pm} = (\mu \pm \sqrt{\mu^2 + 4m})/2$. Since

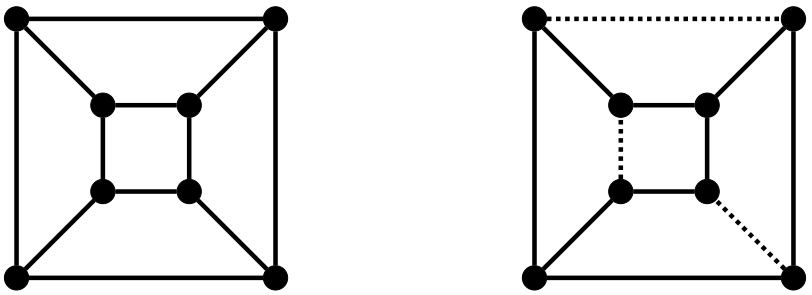
$$\lambda^+\lambda^- = \left(\frac{\mu + \sqrt{\mu^2+4m}}{2}\right)\left(\frac{\mu - \sqrt{\mu^2+4m}}{2}\right) = -m,$$

we have $\lambda^- = -m/\lambda^+$. Moreover, since $\mu = \lambda - m/\lambda$ is uniquely determined by λ, distinct eigenvalues of (Γ, σ) give rise to disjoint root pairs, and the multiplicity of each non-zero eigenvalue λ of $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ equals the multiplicity of μ in (Γ, σ). As λ and $-m/\lambda$ arise from the same μ, they have equal multiplicities, and so $(\Gamma, \sigma) \circ \cup_m (K_1, +)$ satisfies m-(−SR).

If (Γ, σ) has symmetric spectrum, meaning that $-\mu$ is an eigenvalue whenever μ is, then the roots corresponding to $-\mu$ are $-\lambda^{\mp}$. Since $m/\lambda^{\pm} = -\lambda^{\mp}$, whenever λ is an eigenvalue of $(\Gamma, \sigma) \circ \cup_m (K_1, +)$, so is m/λ, with the same multiplicity, giving m-(SR). □

2.2.3 Hypercubes

The hypercube of dimension n, denoted Q_n, is the graph whose vertex set consists of all binary strings of length n, with two vertices adjacent if and only if they differ in exactly one coordinate. Equivalently, Q_n is the n-fold Cartesian product of K_2 with itself, and may be decomposed into two copies of Q_{n-1} joined by a perfect matching.



$(Q_3, +)$ (Q_3, σ_3)

Figure 2.2: The hypercubes $(Q_3, +)$ and (Q_3, σ_3). Here and throughout, a solid line between two vertices represents a positive edge and a dashed line represents a negative edge.

A classical or all-positive hypercube has spectrum

$$\operatorname{spec}(Q_n, +) = \{(n-2k)^{\binom{n}{k}} : 0 \le k \le n\}.$$

In particular, we have

$$\begin{aligned}
\operatorname{spec}(Q_1, +) &= \{1, -1\},\\
\operatorname{spec}(Q_2, +) &= \{2, 0^{(2)}, -2\},\\
\operatorname{spec}(Q_3, +) &= \{3, 1^{(3)}, -1^{(3)}, -3\},\\
\operatorname{spec}(Q_4, +) &= \{4, 2^{(4)}, 0^{(6)}, -2^{(4)}, -4\},
\end{aligned}$$

and one can verify that Q_1 satisfies 1-(SR) and 1-(−SR), Q_2 satisfies 4-(SR) and 4-(−SR), Q_3 satisfies 3-(R) and 3-(−R), and Q_4 satisfies 8-(R) and 8-(−R).

However, as n increases and the spectrum of Q_n acquires more distinct eigenvalues, the graph no longer satisfies any such property.

Proposition 2.2.5. *For $n \ge 5$, the hypercube Q_n does not satisfy m-(R) or m-(−R) for any positive integer m.*

Proof. Since the spectrum of Q_n is symmetric about 0, λ is an eigenvalue of Q_n if and only if $-\lambda$ is an eigenvalue of Q_n, so m-(−R) holds if and only if m-(R) holds. Thus without loss of generality we consider only m-(R).

Suppose that Q_n satisfies m-(R). The spectrum of Q_n is made up exclusively of integers and so for every non-zero eigenvalue λ, m/λ is also an eigenvalue and hence an integer. Therefore λ must divide m for all non-zero eigenvalues λ, and the least common multiple L_n of the distinct non-zero eigenvalues divides m. In particular, since n is the largest eigenvalue, m/n must also be an eigenvalue, must be an integer, and must satisfy $m/n \leq n$, which implies $m \leq n^2$. Thus if $L_n > n^2$ we have a contradiction.

For odd $n \geq 7$, the non-zero eigenvalues of Q_n include n, $n-2$, and $n-4$, which are pairwise coprime: $\gcd(n, n-2)$ and $\gcd(n-2, n-4)$ divide 2, $\gcd(n, n-4)$ divides 4, and all three numbers are odd. Hence

$$L_n \geq \operatorname{lcm}(n, n-2, n-4) = n(n-2)(n-4).$$

This exceeds n^2 when $(n-2)(n-4) > n$, i.e. $n^2 - 7n + 8 > 0$, which holds for all $n \geq 7$.

For even $n \geq 14$, the non-zero eigenvalues of Q_n include n, $n-2$, and $n-4$, all even. Writing $n = 2r$, we have $\operatorname{lcm}(n, n-2, n-4) = 2 \cdot \operatorname{lcm}(r, r-1, r-2)$. Since r, $r-1$, $r-2$ are three consecutive integers, $\gcd(r, r-1) = \gcd(r-1, r-2) = 1$ and $\gcd(r, r-2)$ divides 2, so $\operatorname{lcm}(r, r-1, r-2) \geq r(r-1)(r-2)/2$, and hence

$$L_n \geq 2 \cdot \frac{r(r-1)(r-2)}{2} = r(r-1)(r-2) = \frac{n(n-2)(n-4)}{8}.$$

This exceeds n^2 when $(n-2)(n-4) > 8n$, i.e. $n^2 - 14n + 8 > 0$, which holds for all $n \geq 14$.

It remains to consider $n \in \{5, 6, 8, 10, 12\}$. For $n = 10$, one verifies directly that $L_{10} = 120 > 100 = 10^2$, so this case is also handled by the lcm argument. For $n \in \{5, 6, 8, 12\}$, the candidates for m with $L_n \mid m \leq n^2$ are as follows:

n	L_n	candidates for m	reason for failure
5	15	15	$15/1 = 15 \notin \operatorname{spec}(Q_5)$
6	12	12, 24, 36	$12/4 = 3,\ 24/2 = 12,\ 36/4 = 9 \notin \operatorname{spec}(Q_6)$
8	24	24, 48	$24/8 = 3,\ 48/4 = 12 \notin \operatorname{spec}(Q_8)$
12	120	120	$120/8 = 15 \notin \operatorname{spec}(Q_{12})$

In each case, no valid m exists. □

In [56], Huang proved that every subgraph of Q_n induced on more than half the vertices has maximum degree at least $\sqrt{n}$, thus proving the sensitivity conjecture. In [70, Lemma 2.2], Laplante et al. show that Huang also implicitly proved that for each $n \geq 2$, there exists a signature σ_n on Q_n such that in (Q_n, σ_n) all 4-cycles are negative. The spectrum of this signed hypercube is well-known, leading directly to our next result.

Proposition 2.2.6. *For $n \geq 2$, the signed hypercube (Q_n, σ_n) in which all 4-cycles are negative satisfies both n-(SR) and n-($-$SR).*

Proof. By [70, Lemma 2.2], such a signed hypercube (Q_n, σ_n) has exactly two distinct eigenvalues, namely $\sqrt{n}$ and $-\sqrt{n}$, each with multiplicity 2^{n-1}. It follows that the adjacency matrix $A_{(Q_n,\sigma_n)}$ satisfies $A^2_{(Q_n,\sigma_n)} = nI$. Moreover, since the spectrum is symmetric and consists of only these two values, n-(SR) and n-($-$SR) are satisfied and the result follows immediately. □

2.3 Classes of graphs satisfying eigenvalue properties

In this section, we provide explicit constructions for several classes of graphs that satisfy integer reciprocal eigenvalue properties. First, we introduce a class of graphs that satisfy m-(R) but fail to satisfy m-(SR) due to mismatched multiplicities in the m-reciprocal eigenvalue pairs. Second, we present a class of non-corona graphs that satisfy both m-(SR) and m-($-$SR). In each case, we analyse the all-positive signatures first before constructing signed variants, which render the graphs unbalanced while preserving the eigenvalue properties of their unsigned counterparts.

2.3.1 Graphs satisfying m-(R) but not m-(SR)

It is clear from the definition that when a graph satisfies m-(SR), it also satisfies m-(R). The converse, however, is not necessarily true. Inspired by [77, Definition 2.1], we provide a method for constructing an infinite class of graphs to demonstrate this.

Classical variant

Construction 2.3.1. Set $R = R_1 \circ \cup_m K_1$ with $R_1 = K_1$, $V_{R_1} = \{u\}$, and $V_R = \{u, v_1, v_2, \ldots, v_m\}$. Let $\Gamma_{m,n}$, with $m \geq 1$ and $n \geq 2$, be the graph obtained by taking n copies of $K_2 \circ \cup_m K_1$ and one copy of R, and then joining the vertex $u \in V_{R_1}$ to every vertex in the first copy of $K_2 \circ \cup_m K_1$ and to the vertices of degree $m+1$ in the remaining $n-1$ copies of $K_2 \circ \cup_m K_1$. We denote such a family of graphs by

$$\Omega_1^m = \{\Gamma_{m,n} : m, n \in \mathbb{N}, m \geq 1, n \geq 2\}.$$

Remark 2.3.2. Note that by setting $m = 1$ we recover precisely the construction provided in [77, Definition 2.1].

Label the vertices in the first copy of $K_2 \circ \cup_m K_1$ as $\{x_1, x_2, \ldots, x_{2m+2}\}$, with edges $\{x_1, x_2\}$, $\{x_1, x_o\}$ for $o \in \{3, 5, \ldots, 2m+1\}$, and $\{x_2, x_e\}$ for $e \in \{4, 6, \ldots, 2m+2\}$. Similarly, for $1 \leq i \leq n-1$, label the vertices in each remaining copy of $K_2 \circ \cup_m K_1$ as $\{y_1^i, y_2^i, \ldots, y_{2m+2}^i\}$, with edges $\{y_1^i, y_2^i\}$, $\{y_1^i, y_o^i\}$ for $o \in \{3, 5, \ldots, 2m+1\}$, and $\{y_2^i, y_e^i\}$ for $e \in \{4, 6, \ldots, 2m+2\}$.

The graph $\Gamma_{2,2} \in \Omega_1^2$, as well as the vertex labelling convention, is shown in Figure 2.3.

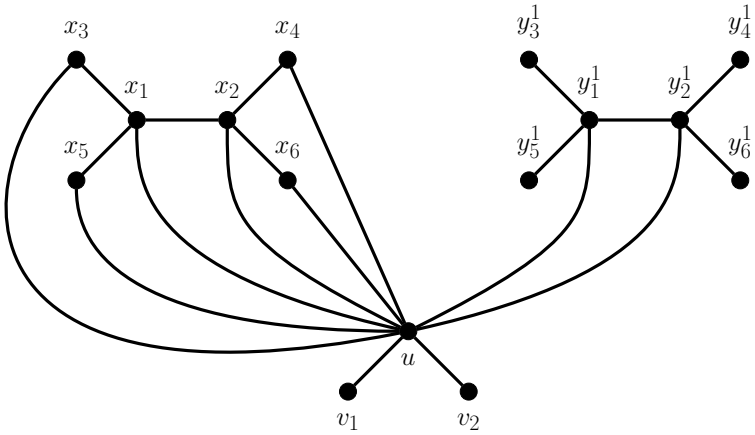


Figure 2.3: The depicted graph $\Gamma_{2,2} \in \Omega_1^2$ from Construction 2.3.1 satisfies 2-(R) but not 2-(SR).

The cycles present in the graph $\Gamma_{m,n} \in \Omega_1^m$ can be partitioned into six distinct types. We have

- Type (i): ux_1x_2u, of which there is only one. Removing the vertices of the cycle of this type from $\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\}$ leaves $2m$ copies of K_1 and $n-1$ copies of $K_2 \circ \cup_m K_1$.
- Type (ii): ux_1x_ou or ux_2x_eu, of which there are $2m$. Removing the vertices of a cycle of this type from $\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\}$ leaves $m-1$ copies of K_1, one copy of $K_1 \circ \cup_m K_1$, and $n-1$ copies of $K_2 \circ \cup_m K_1$.
- Type (iii): $ux_1x_2x_eu$ or $ux_2x_1x_ou$, of which there are $2m$. Removing the vertices of a cycle of this type from $\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\}$ leaves $2m-1$ copies of K_1 and $n-1$ copies of $K_2 \circ \cup_m K_1$.
- Type (iv): $ux_ox_1x_2x_eu$, of which there are m^2. Removing the vertices of a cycle of this type from $\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\}$ leaves $2m-2$ copies of K_1 and $n-1$ copies of $K_2 \circ \cup_m K_1$.
- Type (v): $ux_ox_1x_{o'}u$ with $o \neq o'$, or $ux_ex_2x_{e'}u$ with $e \neq e'$, of which there are $2\binom{m}{2} = m(m-1)$. Removing the vertices of a cycle of this type from $\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\}$ leaves $m-2$ copies of K_1, one copy of $K_1 \circ \cup_m K_1$, and $n-1$ copies of $K_2 \circ \cup_m K_1$.
- Type (vi): $uy_1^iy_2^iu$, of which there are $n-1$. Removing the vertices of a cycle of this type from $\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\}$ leaves $2m$ copies of K_1 and $n-1$ copies of $K_2 \circ \cup_m K_1$.

An example of each cycle type is shown in Figure 2.4.

The two following lemmas are classical results on the characteristic polynomial of a graph, due to Schwenk [87]; see also [38].

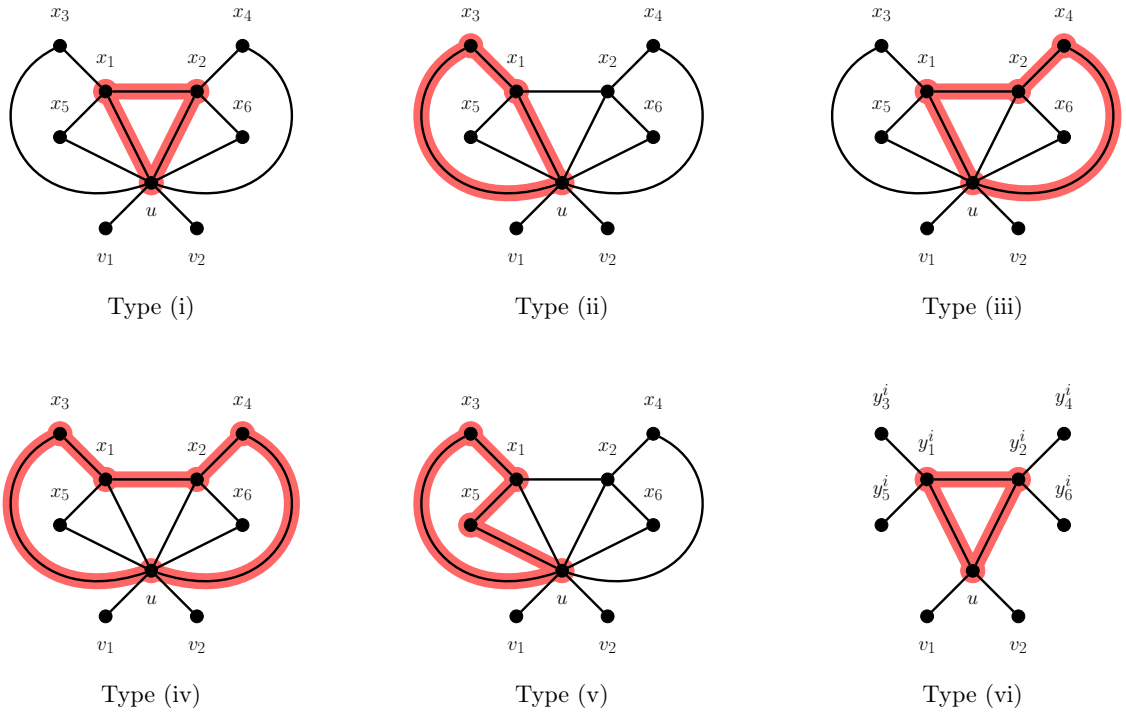


Figure 2.4: Examples of each of the cycle types (i)–(vi) in $\Gamma_{2,2} \in \Omega_1^2$ from Construction 2.3.1.

Lemma 2.3.3. *Let v be a vertex in the graph Γ and $C(v)$ be the set of all cycles in Γ containing v. Then*

$$P(\Gamma;\lambda) = \lambda P(\Gamma - v;\lambda) - \sum_{u\in N(v)} P(\Gamma - u - v;\lambda) - 2\sum_{Z\in C(v)} P(\Gamma - Z;\lambda).$$

Lemma 2.3.4. *Let v be a vertex of degree 1 in the graph Γ and u be the vertex adjacent to v. Then*

$$P(\Gamma;\lambda) = \lambda P(\Gamma - v;\lambda) - P(\Gamma - u - v;\lambda).$$

Remark 2.3.5. Let $A = P(K_2 \circ \cup_m K_1;\lambda)$, $B = P(K_1;\lambda)$, $C = P(K_1 \circ \cup_m K_1;\lambda)$, and $D = P(K_2 \circ \cup_m K_1 - x;\lambda)$, where x is a pendant vertex.

To obtain an explicit expression for A, we begin with $K_2 \circ \cup_m K_1$ and strip one side of K_2 of its pendant vertices by applying Lemma 2.3.4 in m steps, this gives

$$\begin{aligned} A &= \lambda^m P(K_{1,m+1};\lambda) - m\lambda^{m-1}P(K_{1,m};\lambda) \\ &= \lambda^m\Big[\lambda^m(\lambda^2-(m+1))\Big] - m\lambda^{m-1}\Big[\lambda^{m-1}(\lambda^2-m)\Big] \\ &= \lambda^{2m-2}(\lambda^4-(2m+1)\lambda^2+m^2). \end{aligned}$$

Clearly $B = P(K_1;\lambda) = \lambda$ and $C = P(K_{1,m};\lambda) = \lambda^{m-1}(\lambda^2-m)$.

With D we can follow the same pattern as with A, but since we begin with one fewer pendant vertex, we only need $m-1$ iterations. This results in

$$\begin{aligned} D &= \lambda^{m-1}P(K_{1,m+1};\lambda) - (m-1)\lambda^{m-2}P(K_{1,m};\lambda) \\ &= \lambda^{2m-3}(\lambda^4 - 2m\lambda^2 + m(m-1)). \end{aligned}$$

Theorem 2.3.6. *Let $\Gamma_{m,n} \in \Omega_1^m$. Then $\Gamma_{m,n}$ satisfies m-(R) but does not satisfy m-(SR). The characteristic polynomial is*

$$\begin{aligned} P(\Gamma_{m,n};\lambda) = (\lambda^{2m+2} - (2m+1)\lambda^{2m} + m^2\lambda^{2m-2})^{n-1}\Big\{&\lambda^{3m+3} - (2n+5m+1)\lambda^{3m+1} \\ &- (2n+4m)\lambda^{3m} + m(2n+5m-1)\lambda^{3m-1} + 2m^2\lambda^{3m-2} - m^3\lambda^{3m-3}\Big\}. \end{aligned}$$

Proof. We shall use A, B, C, and D from Remark 2.3.5.

By iterating Lemma 2.3.4, we see that

$$\begin{aligned}
P(\Gamma_{1,n};\,\lambda) &= \lambda P(\Gamma_{1,n}-v_1;\,\lambda) - P(\Gamma_{1,n}-v_1-u;\,\lambda),\\
P(\Gamma_{2,n};\,\lambda) &= \lambda^2 P(\Gamma_{2,n}-\{v_1,v_2\};\,\lambda) - \lambda P(\Gamma_{2,n}-\{v_1,v_2\}-u;\,\lambda) - P(\Gamma_{2,n}-v_1-u;\,\lambda),\\
P(\Gamma_{3,n};\,\lambda) &= \lambda^3 P(\Gamma_{3,n}-\{v_1,v_2,v_3\};\,\lambda) - \lambda^2 P(\Gamma_{3,n}-\{v_1,v_2,v_3\}-u;\,\lambda)\\
&\quad - \lambda P(\Gamma_{3,n}-\{v_1,v_2\}-u;\,\lambda) - P(\Gamma_{3,n}-v_1-u;\,\lambda),\\
&\vdots\\
P(\Gamma_{m,n};\,\lambda) &= \lambda^m P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\};\,\lambda) - \lambda^{m-1} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-u;\,\lambda)\\
&\quad - \cdots - \lambda P(\Gamma_{m,n}-\{v_1,v_2\}-u;\,\lambda) - P(\Gamma_{m,n}-v_1-u;\,\lambda).
\end{aligned}$$

It can be seen that

$$\begin{aligned}
\lambda^{m-1} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-u;\,\lambda) &= \lambda^{m-1}A^n,\\
\lambda^{m-2} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_{m-1}\}-u;\,\lambda) &= \lambda^{m-2}A^nB\\
&= \lambda^{m-1}A^n,\\
&\vdots\\
\lambda P(\Gamma_{m,n}-\{v_1,v_2\}-u;\,\lambda) &= \lambda A^nB^{m-2}\\
&= \lambda^{m-1}A^n,\\
P(\Gamma_{m,n}-v_1-u;\,\lambda) &= A^nB^{m-1}\\
&= \lambda^{m-1}A^n.
\end{aligned}$$

Hence the sum of all of these parts gives

$$P(\Gamma_{m,n};\,\lambda) = \lambda^m P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\};\,\lambda) - m\lambda^{m-1}A^n. \tag{2.3.1}$$

In the graph $\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}$, all of the cycles contain the vertex u. So, by Lemma 2.3.3, we see that

$$\begin{aligned}
&P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\};\,\lambda)\\
&\quad= \lambda P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-u;\,\lambda) - \sum_{s\in N(u)} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-u-s;\,\lambda)\\
&\qquad - 2\sum_{Z\in C(u)} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-Z;\,\lambda).
\end{aligned}$$

Then, we have

$$\begin{aligned}
&\sum_{s\in N(u)} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-u-s;\,\lambda)\\
&\qquad= 2\Big[A^{n-1}B^mC\Big] + 2m\Big[A^{n-1}D\Big] + 2(n-1)\Big[A^{n-1}B^mC\Big]\\
&\qquad= 2\lambda^m\Big[A^{n-1}C\Big] + 2m\Big[A^{n-1}D\Big] + 2(n-1)\lambda^m\Big[A^{n-1}C\Big]\\
&\qquad= 2mA^{n-1}D + 2n\lambda^mA^{n-1}C,
\end{aligned} \tag{2.3.2}$$

and

$$\begin{aligned}
&\sum_{Z\in C(u)} P(\Gamma_{m,n}-\{v_1,v_2,\dots,v_m\}-Z;\,\lambda)\\
&\qquad= 1\Big[A^{n-1}B^{2m}\Big] + 2m\Big[A^{n-1}B^{m-1}C\Big] + 2m\Big[A^{n-1}B^{2m-1}\Big]\\
&\qquad\quad + m^2\Big[A^{n-1}B^{2m-2}\Big] + m(m-1)\Big[A^{n-1}B^{m-2}C\Big] + (n-1)\Big[A^{n-1}B^{2m}\Big]
\end{aligned}$$

$$
\begin{aligned}
&= \lambda^{2m}\Big[A^{n-1}\Big] + 2m\lambda^{m-1}\Big[A^{n-1}C\Big] + 2m\lambda^{2m-1}\Big[A^{n-1}\Big] \\
&\quad + m^2\lambda^{2m-2}\Big[A^{n-1}\Big] + m(m-1)\lambda^{m-2}\Big[A^{n-1}C\Big] + (n-1)\lambda^{2m}\Big[A^{n-1}\Big] \\
&= (n\lambda^{2m} + 2m\lambda^{2m-1} + m^2\lambda^{2m-2})A^{n-1} + m\lambda^{m-2}(2\lambda + m - 1)A^{n-1}C. \qquad (2.3.3)
\end{aligned}
$$

So, putting Equations (2.3.2) and (2.3.3) into the application of Lemma 2.3.3, we have

$$
\begin{aligned}
&P(\Gamma_{m,n} - \{v_1, v_2, \ldots, v_m\};\, \lambda) \\
&\quad = \lambda A^n - \Big[2mA^{n-1}D + 2n\lambda^m A^{n-1}C\Big] \\
&\qquad - 2\Big[(n\lambda^{2m} + 2m\lambda^{2m-1} + m^2\lambda^{2m-2})A^{n-1} + m\lambda^{m-2}(2\lambda + m - 1)A^{n-1}C\Big] \\
&\quad = \lambda A^n - \Big[2mD + 2n\lambda^{2m} + 4m\lambda^{2m-1} + 2m^2\lambda^{2m-2}\Big]A^{n-1} \\
&\qquad - \Big[2n\lambda^m + 2m\lambda^{m-2}(2\lambda + m - 1)\Big]A^{n-1}C. \qquad (2.3.4)
\end{aligned}
$$

Now we shall combine Equations (2.3.1) and (2.3.4) and then substitute in the expressions for A, B, C, and D that are given in Remark 2.3.5. This yields

$$
\begin{aligned}
P(\Gamma_{m,n};\, \lambda) &= \lambda^m\Big\{\lambda A^n - \Big[2mD + 2n\lambda^{2m} + 4m\lambda^{2m-1} + 2m^2\lambda^{2m-2}\Big]A^{n-1} \\
&\qquad - \Big[2n\lambda^m + 2m\lambda^{m-2}(2\lambda + m - 1)\Big]A^{n-1}C\Big\} - m\lambda^{m-1}A^n \\
&= \lambda^{m+1}A^n - \lambda^m\Big[2mD + 2n\lambda^{2m} + 4m\lambda^{2m-1} + 2m^2\lambda^{2m-2}\Big]A^{n-1} \\
&\qquad - \lambda^m\Big[2n\lambda^m + 2m\lambda^{m-2}(2\lambda + m - 1)\Big]A^{n-1}C - m\lambda^{m-1}A^n \\
&= A^{n-1}\Big\{\lambda^{m+1}A - \lambda^m\Big[2mD + 2n\lambda^{2m} + 4m\lambda^{2m-1} + 2m^2\lambda^{2m-2}\Big] \\
&\qquad - \lambda^m\Big[2n\lambda^m + 2m\lambda^{m-2}(2\lambda + m - 1)\Big]C - m\lambda^{m-1}A\Big\} \\
&= A^{n-1}\Big\{\lambda^{3m-1}(\lambda^4 - (2m+1)\lambda^2 + m^2) - 2m\lambda^{3m-3}(\lambda^4 - 2m\lambda^2 + m(m-1)) \\
&\qquad - 2n\lambda^{3m} - 4m\lambda^{3m-1} - 2m^2\lambda^{3m-2} \\
&\qquad - \Big[2n\lambda^m + 2m\lambda^{m-2}(2\lambda + m - 1)\Big](\lambda^{2m+1} - m\lambda^{2m-1}) \\
&\qquad - m\lambda^{3m-3}(\lambda^4 - (2m+1)\lambda^2 + m^2)\Big\} \\
&= A^{n-1}\Big\{\lambda^{3m+3} - (5m + 2n + 1)\lambda^{3m+1} - (4m + 2n)\lambda^{3m} \\
&\qquad + m(5m + 2n - 1)\lambda^{3m-1} + 2m^2\lambda^{3m-2} - m^3\lambda^{3m-3}\Big\} \\
&= (\lambda^{2m+2} - (2m+1)\lambda^{2m} + m^2\lambda^{2m-2})^{n-1}\Big\{\lambda^{3m+3} - (5m + 2n + 1)\lambda^{3m+1} \\
&\qquad - (4m + 2n)\lambda^{3m} + m(5m + 2n - 1)\lambda^{3m-1} + 2m^2\lambda^{3m-2} - m^3\lambda^{3m-3}\Big\}.
\end{aligned}
$$

The roots of

$$
\lambda^{2m+2} - (2m+1)\lambda^{2m} + m^2\lambda^{2m-2} \qquad (2.3.5)
$$

are given by

$$
\lambda_0 = 0, \qquad \lambda_{(1,\pm)} = \frac{1}{2}\left(1 \pm \sqrt{4m+1}\right), \qquad \lambda_{(2,\pm)} = \frac{1}{2}\left(-1 \pm \sqrt{4m+1}\right),
$$

and the roots of

$$
\lambda^{3m+3} - (5m + 2n + 1)\lambda^{3m+1} - (4m + 2n)\lambda^{3m} + m(5m + 2n - 1)\lambda^{3m-1} + 2m^2\lambda^{3m-2} - m^3\lambda^{3m-3} \qquad (2.3.6)
$$

are given by

$$
\lambda_0 = 0, \qquad \lambda_{(3,\pm)} = \frac{1}{2}\left(-1 \pm \sqrt{4m+1}\right),
$$

$$\lambda_{(4,\pm)} = \frac{1}{4} + \frac{1}{4}\sqrt{24m+8n+1} \pm \frac{1}{2}\sqrt{2(2m+n) + \frac{8(2m+n)+8m+1}{2\sqrt{24m+8n+1}} - 2m + \frac{1}{2}},$$

$$\lambda_{(5,\pm)} = \frac{1}{4} - \frac{1}{4}\sqrt{24m+8n+1} \pm \frac{1}{2}\sqrt{2(2m+n) - \frac{8(2m+n)+8m+1}{2\sqrt{24m+8n+1}} - 2m + \frac{1}{2}}.$$

It can be checked that $\lambda_{(1,+)}\lambda_{(2,+)} = m$, $\lambda_{(1,-)}\lambda_{(2,-)} = m$, $\lambda_{(1,+)}\lambda_{(3,+)} = m$, $\lambda_{(1,-)}\lambda_{(3,-)} = m$, $\lambda_{(4,+)}\lambda_{(4,-)} = m$ and $\lambda_{(5,+)}\lambda_{(5,-)} = m$.

Therefore, for each non-zero eigenvalue λ of $\Gamma_{m,n}$, we also have m/λ as an eigenvalue of $\Gamma_{m,n}$, and property m-(R) is satisfied.

Since $\lambda_{(2,\pm)} = \lambda_{(3,\pm)}$, these eigenvalues appear as roots of both Equations (2.3.5) and (2.3.6). Specifically, they appear with multiplicity $n-1$ in (2.3.5) and, being distinct from $\lambda_{(4,\pm)}$ and $\lambda_{(5,\pm)}$, as a simple root of (2.3.6), giving a total multiplicity of n in $P(\Gamma_{m,n};\lambda)$. However, the m-reciprocal partner $\lambda_{(1,\pm)}$, which is distinct from every root of (2.3.6), appears only as a root of (2.3.5), with multiplicity $n-1$. Clearly $n \neq n-1$, and so the multiplicities of these m-reciprocal eigenvalues differ, and property m-(SR) is not satisfied. □

Example 2.3.7. The graph $\Gamma_{2,2} \in \Omega_1^2$ is depicted in Figure 2.3. Its spectrum is

$$\operatorname{spec}(\Gamma_{2,2}) = \{4.036, 2, 1^{(2)}, 0.495, 0^{(5)}, -0.709, -1, -2^{(2)}, -2.823\}.$$

The pairs $\{4.036, 0.495\}$, $\{2,1\}$, $\{-0.709, -2.823\}$, $\{-1,-2\}$ demonstrate how $\Gamma_{2,2}$ satisfies property 2-(R), but the difference in the multiplicities of 2 and 1 (and also -1 and -2) show that property 2-(SR) is not satisfied.

Furthermore, its characteristic polynomial is

$$P(\Gamma_{2,2};\lambda) = \lambda^{15} - 20\lambda^{13} - 12\lambda^{12} + 105\lambda^{11} + 68\lambda^{10} - 198\lambda^9 - 88\lambda^8 + 144\lambda^7 + 32\lambda^6 - 32\lambda^5,$$

which matches the expression provided in Theorem 2.3.6 after making the relevant substitutions.

Signed variant

Taking the same construction but with every edge negative gives a signed graph and since each $\Gamma_{m,n}$ contains cycles of odd length, these signed graphs are unbalanced. We denote such a family of graphs by

$$(\Omega_1^m, -) = \{(\Gamma_{m,n}, -) : m, n \in \mathbb{N}, m \geq 1, n \geq 2\}.$$

Theorem 2.3.8. *Let $(\Gamma_{m,n}, -) \in (\Omega_1^m, -)$. Then $(\Gamma_{m,n}, -)$ satisfies m-(R) but does not satisfy m-(SR).*

Proof. This follows immediately from Theorem 2.3.6 and the fact that the adjacency spectrum of an all-negative signed graph $(\Gamma, -)$ is the negation of the spectrum of the underlying unsigned graph Γ, since the adjacency matrix of $(\Gamma, -)$ is equal to $-A_\Gamma$. □

2.3.2 Non-corona graphs satisfying m-(SR) and m-($-$SR)

We can construct classes of signed graphs that are not corona graphs having the properties m-(SR) and m-($-$SR). This can be achieved by performing alternating joins between various vertices of m-coronae.

We can follow the constructions in [79] for the unsigned cases and in [21] for the signed cases. In each appropriate instance, we take the simple m-corona instead of the simple corona, and we obtain the m-(SR) and m-($-$SR) properties rather than (SR) and ($-$SR).

Classical variant

Construction 2.3.9. Let $\Gamma_{k,m,n}$ with $k \geq 2$, and $m, n \geq 1$, be the graph obtained from the following construction. Begin with a vertex u and form the simple m-corona, so that u is adjacent to vertices labelled $v_1, v_2, \ldots, v_m$. Next take n copies of the cycle C_{2k}, with vertices labelled bipartitely so that the edges are $\{x_1, x_{k+1}\}$ and $\{x_1, x_{2k}\}$, $\{x_2, x_{k+1}\}$ and $\{x_2, x_{k+2}\}$, and so on. Now form the simple m-corona for each copy of C_{2k}, so that vertex x_1 is adjacent to vertices labelled $y_1, y_2, \ldots, y_m$, vertex x_2 is adjacent

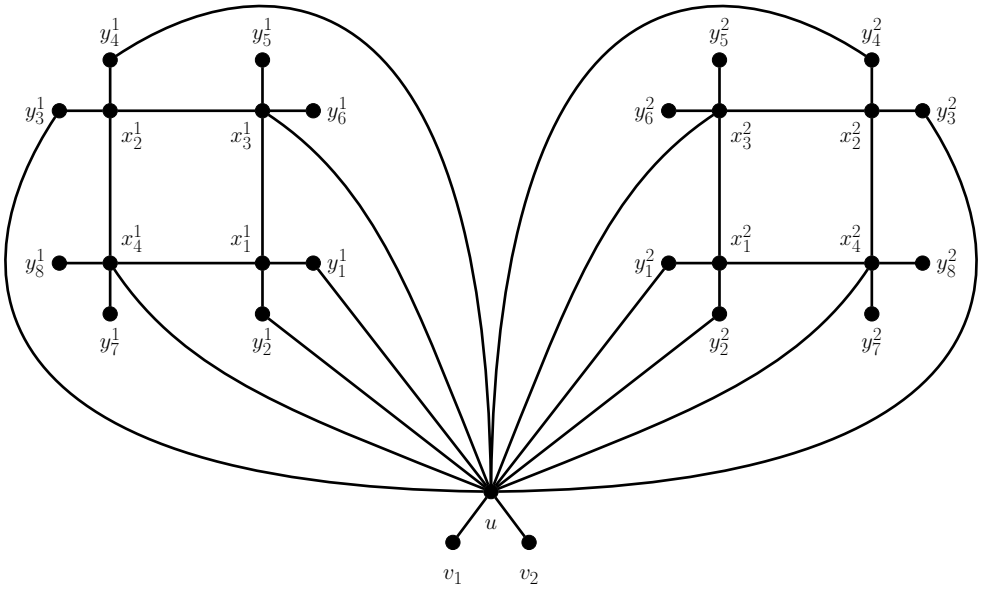


Figure 2.5: The depicted non-corona graph $\Gamma_{2,2,2} \in \Omega_2^2$ from Construction 2.3.9 has properties 2-(SR) and 2-(−SR).

to vertices labelled $y_{m+1}, y_{m+2}, \ldots, y_{2m}$, and so on. Lastly, we make adjacencies between u and the first km pendant vertices in each copy of the m-corona cycle, i.e. the vertices $y_1, y_2, \ldots, y_{km}$, and between u and the last k cyclic vertices in each copy of the m-corona cycle, i.e. the vertices $x_{k+1}, x_{k+2}, \ldots, x_{2k}$. We denote such a family of graphs by

$$\Omega_2^m = \{\Gamma_{k,m,n} : k, m, n \in \mathbb{N}, k \geq 2, m, n \geq 1\}.$$

We note that the graphs in Ω_2^m are not simple m-corona graphs. Indeed, if $\Gamma_{k,m,n}$ were a simple m-corona, then every vertex of degree greater than 1 would have exactly m neighbours of degree 1. To contradict this, consider the vertex y_1^i, which is adjacent to precisely two vertices: x_1^i and u. Since $k \geq 2$ and $m \geq 1$, both x_1^i and u have degree greater than 1, while y_1^i itself has degree 2. Thus y_1^i has no neighbours of degree 1, proving that $\Gamma_{k,m,n}$ is not a simple m-corona graph.

Remark 2.3.10. Note that by setting $m = 1$ we recover precisely the construction provided as Class $\mathcal{NC}_1^1$ in [79].

Theorem 2.3.11. *Let $\Gamma_{k,m,n} \in \Omega_2^m$. Then $\Gamma_{k,m,n}$ satisfies properties m-(SR) and m-(−SR).*

Proof. With a suitable vertex ordering of such a graph, the adjacency matrix may be written as

$$A_{\Gamma_{k,m,n}} = \left(\begin{array}{c|c|c} I_n \otimes \left(\begin{array}{c|cc} O_{2km} & I_{2k} \otimes \mathbf{1}_m & \\ \hline I_{2k} \otimes \mathbf{1}_m^T & O_k & C \\ & C^T & O_k \end{array}\right) & \mathbf{1}_n \otimes \begin{pmatrix} \mathbf{1}_{km} \\ \mathbf{0}_{km} \\ \mathbf{0}_k \\ \mathbf{1}_k \end{pmatrix} & O_{2kn(m+1)\times m} \\ \hline \mathbf{1}_n^T \otimes \begin{pmatrix} \mathbf{1}_{km}^T & \mathbf{0}_{km}^T & \mathbf{0}_k^T & \mathbf{1}_k^T \end{pmatrix} & 0 & \mathbf{1}_m^T \\ \hline O_{m\times 2kn(m+1)} & \mathbf{1}_m & O_m \end{array}\right),$$

where $\begin{pmatrix} O_k & C \\ C^T & O_k \end{pmatrix}$ is the adjacency matrix of the cycle C_{2k}.

Let a_i and b_i be the singular vector pairs corresponding to the singular values λ_i of C for $1 \leq i \leq k$. Without loss of generality, we can assume that the sets $\{a_1, a_2, \ldots, a_k\}$ and $\{b_1, b_2, \ldots, b_k\}$ are orthonormal.

Moreover, since $C\mathbf{1}_k = 2\mathbf{1}_k$ and $C^T\mathbf{1}_k = 2\mathbf{1}_k$ by the 2-regularity of cycles, we can further assume that $a_1 = b_1 = (1/\sqrt{k})\mathbf{1}_k$.

Let

$$Z_i^+ = \begin{pmatrix} a_i \otimes \mathbf{1}_m \\ b_i \otimes \mathbf{1}_m \\ c_i a_i \\ c_i b_i \end{pmatrix} \quad \text{and} \quad Z_i^- = \begin{pmatrix} a_i \otimes \mathbf{1}_m \\ -b_i \otimes \mathbf{1}_m \\ d_i a_i \\ -d_i b_i \end{pmatrix},$$

where c_i and d_i are non-zero scalars. Then, for $2 \le i \le k$ and $1 \le j \le n$, we obtain

$$A_{\Gamma_{k,m,n}} \begin{pmatrix} e_j \otimes Z_i^+ \\ \mathbf{0}_{m+1} \end{pmatrix} = \begin{pmatrix} e_j \otimes \begin{pmatrix} c_i(a_i \otimes \mathbf{1}_m) \\ c_i(b_i \otimes \mathbf{1}_m) \\ (m + \lambda_i c_i)a_i \\ (m + \lambda_i c_i)b_i \end{pmatrix} \\ \mathbf{0}_{m+1} \end{pmatrix} = c_i \begin{pmatrix} e_j \otimes \begin{pmatrix} a_i \otimes \mathbf{1}_m \\ b_i \otimes \mathbf{1}_m \\ ((m + \lambda_i c_i)/c_i)a_i \\ ((m + \lambda_i c_i)/c_i)b_i \end{pmatrix} \\ \mathbf{0}_{m+1} \end{pmatrix},$$

since each pendant vertex receives a contribution only from its unique cyclic neighbour, the vertex u having entry zero where adjacent, giving $c_i(a_i \otimes \mathbf{1}_m)$ and $c_i(b_i \otimes \mathbf{1}_m)$, respectively, and each cyclic vertex receives a contribution of mc_i from its m pendants and $\lambda_i c_i$ from its cyclic neighbours via $Cb_i = \lambda_i a_i$ and $C^T a_i = \lambda_i b_i$, giving $(m + \lambda_i c_i)a_i$ and $(m + \lambda_i c_i)b_i$. The entry at u is

$$\mathbf{1}_{km}^T(a_i \otimes \mathbf{1}_m) + c_i \mathbf{1}_k^T b_i = m(\mathbf{1}_k^T a_i) + c_i(\mathbf{1}_k^T b_i) = 0,$$

since a_i and b_i are orthogonal to $a_1 = b_1 = (1/\sqrt{k})\mathbf{1}_k$ for $i \ge 2$, and the entry at each pendant of u is zero; this is precisely why the range $2 \le i \le k$ is required here.

Therefore $(e_j \otimes Z_i^+, \mathbf{0}_{m+1})^T$ is an eigenvector corresponding to the eigenvalue c_i of $A_{\Gamma_{k,m,n}}$ if and only if $c_i = (m + \lambda_i c_i)/c_i$, that is, if and only if $c_i^2 - \lambda_i c_i - m = 0$.

The two roots of $c_i^2 - \lambda_i c_i - m = 0$ are given explicitly by

$$c_i^{\pm} = \frac{\lambda_i \pm \sqrt{\lambda_i^2 + 4m}}{2},$$

and their product satisfies $c_i^+ c_i^- = -m$, so the two eigenvalues contributed by Z_i^+ are m-anti-reciprocal partners of one another.

Similarly for Z_i^-, it can be seen by an analogous computation that the eigenvector condition becomes $d_i^2 + \lambda_i d_i - m = 0$, with roots

$$d_i^{\pm} = \frac{-\lambda_i \pm \sqrt{\lambda_i^2 + 4m}}{2},$$

again with the product $d_i^+ d_i^- = -m$, so the two eigenvalues contributed by Z_i^- are also m-anti-reciprocal partners of one another. Moreover, the eigenvalues contributed by Z_i^- are the negatives of those contributed by Z_i^+, giving simultaneously the m-(SR) and m-($-$SR) pairing for each $2 \le i \le k$.

The eigenvalues c_i^+, c_i^-, d_i^+, and d_i^- each appear with multiplicity n, and so with Z_i^+ and Z_i^- for $2 \le i \le k$ and $1 \le j \le n$, we have found $4n(k-1)$ eigenvalues of $A_{\Gamma_{k,m,n}}$.

Now, let

$$Z_1^+ = \begin{pmatrix} \mathbf{1}_{km} \\ \mathbf{1}_{km} \\ c_1\mathbf{1}_k \\ c_1\mathbf{1}_k \end{pmatrix} \quad \text{and} \quad Z_1^- = \begin{pmatrix} \mathbf{1}_{km} \\ -\mathbf{1}_{km} \\ d_1\mathbf{1}_k \\ -d_1\mathbf{1}_k \end{pmatrix},$$

where c_1 and d_1 are some non-zero scalars. Then, for $2 \le j \le n$, denoting $A_{\Gamma_{k,m,n}}$ simply by A, we have

$$A\begin{pmatrix} (e_1 - e_j) \otimes Z_1^+ \\ \mathbf{0}_{m+1} \end{pmatrix} = c_1 \begin{pmatrix} (e_1 - e_j) \otimes Z_1^+ \\ \mathbf{0}_{m+1} \end{pmatrix} \iff c_1 = \frac{m + 2c_1}{c_1} \iff c_1^{\pm} = 1 \pm \sqrt{1+m},$$
$$A\begin{pmatrix} (e_1 - e_j) \otimes Z_1^- \\ \mathbf{0}_{m+1} \end{pmatrix} = d_1 \begin{pmatrix} (e_1 - e_j) \otimes Z_1^- \\ \mathbf{0}_{m+1} \end{pmatrix} \iff d_1 = \frac{m - 2d_1}{d_1} \iff d_1^{\pm} = -1 \pm \sqrt{1+m}.$$

Hence $1 \pm \sqrt{1+m}$ and $-1 \pm \sqrt{1+m}$ are eigenvalues of $A_{\Gamma_{k,m,n}}$, each with multiplicity $n-1$. Furthermore, we have $c_1^{\pm} = -m/c_1^{\mp}$, $d_1^{\pm} = -m/d_1^{\mp}$, $c_1^{+} = m/d_1^{+}$, and $c_1^{-} = m/d_1^{-}$, meaning these eigenvalues satisfy both m-(SR) and m-($-$SR).

So far we have listed $4(kn-1)$ eigenvalues, corresponding to the eigenvectors as described above. Since there are $(1+m)(1+2kn)$ vertices in total, we have $2kn(m-1)+m+5$ eigenvalues left to find.

When multiplying $A_{\Gamma_{k,m,n}}$ by a vector $\mathbf{x}$, the entry of $A_{\Gamma_{k,m,n}}\mathbf{x}$ at vertex v is the sum of the entries of $\mathbf{x}$ at all neighbours of v. Consider a vector $\mathbf{x}$ which is zero everywhere except on the m pendants of a cyclic vertex v, and whose entries on the m pendants of v sum to zero. Now we can check the entry of $A_{\Gamma_{k,m,n}}\mathbf{x}$ at each vertex:

- a pendant p of v: its only neighbour is the cyclic vertex v, which has entry zero in $\mathbf{x}$, and so $(A_{\Gamma_{k,m,n}}\mathbf{x})_p = 0$.
- cyclic vertex v: its pendant neighbours contribute a sum of zero by construction, and all other neighbours have entry zero, and so $(A_{\Gamma_{k,m,n}}\mathbf{x})_v = 0$.
- the vertex u: if $v \in \{x_1, \ldots, x_k\}$ then u is adjacent to the m pendants of v, whose entries sum to zero by construction, while every other neighbour of u has entry zero in $\mathbf{x}$; if $v \in \{x_{k+1}, \ldots, x_{2k}\}$ then u is adjacent to no pendant of v. In either case $(A_{\Gamma_{k,m,n}}\mathbf{x})_u = 0$.
- any other vertex w: the pendants of v are adjacent only to v and possibly to u, so no neighbour of w has a non-zero entry in $\mathbf{x}$, and so $(A_{\Gamma_{k,m,n}}\mathbf{x})_w = 0$.

Hence $A_{\Gamma_{k,m,n}}\mathbf{x} = 0\mathbf{x}$, and zero is an eigenvalue of $A_{\Gamma_{k,m,n}}$. For each of the $2kn$ cyclic vertices we have m pendants, so there are $m-1$ independent choices for making these pendant entries sum to zero, and so we get $m-1$ zero eigenvalues per cyclic vertex. Then vertex u also has m pendants, and by the same argument we have a further $m-1$ zero eigenvalues. Thus we have 0 appearing as an eigenvalue of $A_{\Gamma_{k,m,n}}$ with multiplicity $(2kn+1)(m-1)$.

This means that there are $2kn(m-1)+m+5-(2kn+1)(m-1) = 6$ non-trivial eigenvalues remaining to be found.

Let $\mu_1, \mu_2, \ldots, \mu_6$ denote the six remaining eigenvalues of $A_{\Gamma_{k,m,n}}$, corresponding to eigenvectors $X_1, X_2, \ldots, X_6$, respectively. Let

$$Y_1 = \begin{pmatrix} \mathbf{1}_n \otimes \begin{pmatrix} \mathbf{1}_{km} \\ \mathbf{0}_{km} \\ \mathbf{0}_k \\ \mathbf{0}_k \end{pmatrix} \\ \mathbf{0}_{m+1} \end{pmatrix}, \quad Y_2 = \begin{pmatrix} \mathbf{1}_n \otimes \begin{pmatrix} \mathbf{0}_{km} \\ \mathbf{1}_{km} \\ \mathbf{0}_k \\ \mathbf{0}_k \end{pmatrix} \\ \mathbf{0}_{m+1} \end{pmatrix}, \quad Y_3 = \begin{pmatrix} \mathbf{1}_n \otimes \begin{pmatrix} \mathbf{0}_{km} \\ \mathbf{0}_{km} \\ \mathbf{1}_k \\ \mathbf{0}_k \end{pmatrix} \\ \mathbf{0}_{m+1} \end{pmatrix},$$

$$Y_4 = \begin{pmatrix} \mathbf{1}_n \otimes \begin{pmatrix} \mathbf{0}_{km} \\ \mathbf{0}_{km} \\ \mathbf{0}_k \\ \mathbf{1}_k \end{pmatrix} \\ \mathbf{0}_{m+1} \end{pmatrix}, \quad Y_5 = \begin{pmatrix} \mathbf{0}_{2kn(m+1)} \\ 1 \\ \mathbf{0}_m \end{pmatrix}, \quad Y_6 = \begin{pmatrix} \mathbf{0}_{2kn(m+1)} \\ 0 \\ \mathbf{1}_m \end{pmatrix}.$$

The vectors $Y_1, \ldots, Y_6$ are the characteristic vectors of the six cells of the partition of $V_{\Gamma_{k,m,n}}$, consisting of: the pendants of $x_1, \ldots, x_k$ (equivalently, those pendants adjacent to u), the pendants of $x_{k+1}, \ldots, x_{2k}$, the cyclic vertices $x_1, \ldots, x_k$, and the cyclic vertices $x_{k+1}, \ldots, x_{2k}$, each taken over all n copies; and, additionally, the lone vertex u and the pendants of vertex u. A direct computation of the neighbourhood counts shows that this partition is equitable; consequently $W = \operatorname{span}\{Y_1, \ldots, Y_6\}$ is invariant under $A_{\Gamma_{k,m,n}}$, and the Y_i, having pairwise disjoint supports, are linearly independent, so $\dim W = 6$.

Let U denote the span of the eigenvectors exhibited above, so that $\dim U = 4(kn-1)+(m-1)(2kn+1)$. Each of these eigenvectors sums to zero on every cell of the partition, and therefore lies in $W^{\perp}$; hence $U \subseteq W^{\perp}$. Since the total dimension is $2kn(m+1)+m+1$, we have

$$\dim W^{\perp} = 2kn(m+1)+m+1-6 = 4(kn-1)+(m-1)(2kn+1) = \dim U,$$

so $U = W^{\perp}$ and thus $W = U^{\perp}$. As $A_{\Gamma_{k,m,n}}$ is symmetric and U is spanned by eigenvectors, the six remaining eigenvectors lie in $U^{\perp} = W$, and so $X_i = \sum_{j=1}^{6} s_j Y_j$ for some scalars s_j not all zero.

Therefore, the equation $A_{\Gamma_{k,m,n}}X_i = \mu_i X_i$ holds if and only if

$$\begin{aligned} s_1\mu_i - s_3 - s_5 &= 0,\\ s_2\mu_i - s_4 &= 0,\\ ms_1 - s_3\mu_i + 2s_4 &= 0,\\ ms_2 + 2s_3 - s_4\mu_i + s_5 &= 0,\\ kmn\, s_1 + kn\, s_4 - s_5\mu_i + ms_6 &= 0,\\ s_5 - s_6\mu_i &= 0. \end{aligned}$$

Evidently, this means that the equation $A_{\Gamma_{k,m,n}}X_i = \mu_i X_i$ holds if and only if

$$\det\begin{pmatrix} -\mu_i & 0 & 1 & 0 & 1 & 0\\ 0 & -\mu_i & 0 & 1 & 0 & 0\\ m & 0 & -\mu_i & 2 & 0 & 0\\ 0 & m & 2 & -\mu_i & 1 & 0\\ kmn & 0 & 0 & kn & -\mu_i & m\\ 0 & 0 & 0 & 0 & 1 & -\mu_i \end{pmatrix} = 0.$$

So, the six remaining eigenvalues are the solutions to

$$(\mu^2 - m)(\mu^4 - (kn(m+1) + 2m + 4)\mu^2 + m^2) = 0,$$

which gives $\mu = \pm\sqrt{m}$ from the first factor, along with $\mu = \pm\sqrt{t_1}$ and $\mu = \pm\sqrt{t_2}$ from the second factor, where

$$\begin{aligned} t_1 &= \frac{1}{2}\Big(kn(m+1) + 2m + 4 + \sqrt{(kn(m+1) + 2m + 4)^2 - 4m^2}\Big),\\ t_2 &= \frac{1}{2}\Big(kn(m+1) + 2m + 4 - \sqrt{(kn(m+1) + 2m + 4)^2 - 4m^2}\Big). \end{aligned}$$

Clearly $\sqrt{m}$ is its own m-reciprocal partner, and $-\sqrt{m}$ is its m-anti-reciprocal partner. Since t_1 and t_2 are the roots of $t^2 - (kn(m+1) + 2m + 4)t + m^2 = 0$, we have $t_1 t_2 = m^2$, and so $\sqrt{t_1}\sqrt{t_2} = m$, meaning $\sqrt{t_1}$ and $\sqrt{t_2}$ are m-reciprocal partners and $\sqrt{t_1}$ and $-\sqrt{t_2}$ are m-anti-reciprocal partners. Hence all six remaining eigenvalues satisfy both m-(SR) and m-($-$SR), and the proof is complete. □

Example 2.3.12. The graph $\Gamma_{2,2,2} \in \Omega_2^2$ is depicted in Figure 2.5. Computed directly, its spectrum is

$$\begin{aligned} \operatorname{spec}(\Gamma_{2,2,2}) = \{&4.449, 2.732, 1.414^{(5)}, 0.732, 0.449, 0^{(9)},\\ &-0.449, -0.732, -1.414^{(5)}, -2.732, -4.449\}. \end{aligned}$$

It is easy to check that $\Gamma_{2,2,2}$ satisfies properties 2-(SR) and 2-($-$SR).

We can also compute the spectrum via the method presented in Theorem 2.3.11. With $k = 2$, we have $C = J_2$, which has singular values $\lambda_1 = 2$ and $\lambda_2 = 0$. So, first, we have

$$\begin{aligned} c_2^+ &= \frac{\lambda_2 + \sqrt{\lambda_2^2 + 4m}}{2} = \frac{0 + \sqrt{0^2 + 4\cdot 2}}{2} = 1.414,\\ c_2^- &= \frac{\lambda_2 - \sqrt{\lambda_2^2 + 4m}}{2} = \frac{0 - \sqrt{0^2 + 4\cdot 2}}{2} = -1.414,\\ d_2^+ &= \frac{-\lambda_2 + \sqrt{\lambda_2^2 + 4m}}{2} = \frac{-0 + \sqrt{0^2 + 4\cdot 2}}{2} = 1.414,\\ d_2^- &= \frac{-\lambda_2 - \sqrt{\lambda_2^2 + 4m}}{2} = \frac{-0 - \sqrt{0^2 + 4\cdot 2}}{2} = -1.414, \end{aligned}$$

each appearing with multiplicity $n = 2$.

Next, we have

$$c_1^+ = 1 + \sqrt{1+m} = 1 + \sqrt{1+2} = 2.732, \quad c_1^- = 1 - \sqrt{1+m} = 1 - \sqrt{1+2} = -0.732,$$

$$d_1^+ = -1 + \sqrt{1+m} = -1 + \sqrt{1+2} = 0.732, \quad d_1^- = -1 - \sqrt{1+m} = -1 - \sqrt{1+2} = -2.732,$$

each appearing with multiplicity $n - 1 = 1$.

For the remainder we have 0 appearing with multiplicity $(2kn + 1)(m - 1) = 9$, along with the final six eigenvalues:

$$\sqrt{2} = 1.414,\ -\sqrt{2} = -1.414,\ \sqrt{t_1} = 4.449,\ -\sqrt{t_1} = -4.449,\ \sqrt{t_2} = 0.449,\ -\sqrt{t_2} = -0.449.$$

Collecting these, we see that they give precisely the same spec($\Gamma_{2,2,2}$) as above.

Signed variant

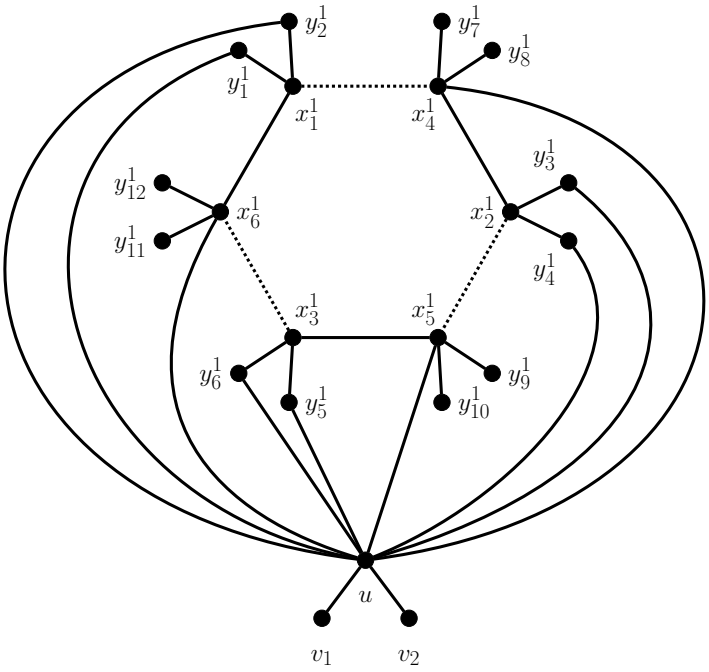


Figure 2.6: The depicted non-corona, unbalanced signed graph $(\Gamma_{3,2,1}, \sigma) \in (\Omega_2^2, \Sigma)$ from Construction 2.3.13 has properties 2-(SR) and 2-(−SR).

Let $k \geq 2$ and take (C_{2k}, τ) to be the signed graph such that τ assigns all edges of one perfect matching of C_{2k} to be positive, and all edges of the other to be negative.

Construction 2.3.13. Let $(\Gamma_{k,m,n}, \sigma)$ with $k \geq 2$, and $m, n \geq 1$, be the signed graph obtained from the following construction. Begin with a vertex u and form the simple positive m-corona, so that u is adjacent to vertices labelled $v_1, v_2, \ldots, v_m$ with each edge having positive signature. Next take n copies of the cycle (C_{2k}, τ), with vertices labelled bipartitely so that the edges are $\{x_1, x_{k+1}\}$ and $\{x_1, x_{2k}\}$, $\{x_2, x_{k+1}\}$ and $\{x_2, x_{k+2}\}$, and so on. Now form the simple positive m-corona for each copy of (C_{2k}, τ), so that vertex x_1 is adjacent to vertices labelled $y_1, y_2, \ldots, y_m$, vertex x_2 is adjacent to vertices labelled $y_{m+1}, y_{m+2}, \ldots, y_{2m}$, and so on, with all such edges having positive signature. Lastly, we make positive adjacencies between u and the first km pendant vertices in each copy of the m-corona cycle, i.e. the vertices $y_1, y_2, \ldots, y_{km}$, and between u and the last k cyclic vertices in each copy of the m-corona cycle, i.e. the vertices $x_{k+1}, x_{k+2}, \ldots, x_{2k}$. We denote such a family of graphs by

$$(\Omega_2^m, \Sigma) = \{(\Gamma_{k,m,n}, \sigma) : k, m, n \in \mathbb{N}, k \geq 2, m, n \geq 1\}.$$

Since the negative perfect matching has k edges, (C_{2k}, τ) is unbalanced if and only if k is odd, and since (C_{2k}, τ) appears as a subgraph of $(\Gamma_{k,m,n}, \sigma)$, this immediately means that $(\Gamma_{k,m,n}, \sigma)$ is unbalanced when k is odd. When k is even, (C_{2k}, τ) is balanced, however $(\Gamma_{k,m,n}, \sigma)$ is still unbalanced. To see this, observe that the cycle $uy_1^1x_1^1x_{k+1}^1u$ contains a single negative edge, namely $\{x_1^1, x_{k+1}^1\}$, and so is negative, making $(\Gamma_{k,m,n}, \sigma)$ unbalanced.

Theorem 2.3.14. *Let $(\Gamma_{k,m,n},\sigma)\in(\Omega_2^m,\Sigma)$. Then $(\Gamma_{k,m,n},\sigma)$ satisfies properties m-(SR) and m-($-$SR).*

Proof. The proof follows a similar strategy to that of Theorem 2.3.11 along with the approach of [21, Theorem 3.2], with the main difference being that we now have a 0-net-regular cycle rather than a 2-regular all-positive cycle.

As before, with a suitable vertex ordering of such a graph, the adjacency matrix may be written as

$$A_{(\Gamma_{k,m,n},\sigma)}=\left(\begin{array}{c|c|c} I_n\otimes\left(\begin{array}{c|cc} O_{2km} & I_{2k}\otimes\mathbf{1}_m \\ \hline I_{2k}\otimes\mathbf{1}_m^T & \begin{array}{cc} O_k & C \\ C^T & O_k\end{array}\end{array}\right) & \mathbf{1}_n\otimes\begin{pmatrix}\mathbf{1}_{km}\\ \mathbf{0}_{km}\\ \mathbf{0}_k\\ \mathbf{1}_k\end{pmatrix} & O_{2kn(m+1)\times m} \\ \hline \mathbf{1}_n^T\otimes\begin{pmatrix}\mathbf{1}_{km}^T & \mathbf{0}_{km}^T & \mathbf{0}_k^T & \mathbf{1}_k^T\end{pmatrix} & 0 & \mathbf{1}_m^T \\ \hline O_{m\times 2kn(m+1)} & \mathbf{1}_m & O_m\end{array}\right),$$

where $\begin{pmatrix}O_k & C\\ C^T & O_k\end{pmatrix}$ is the adjacency matrix of the cycle (C_{2k},τ) and the block C is negative on its diagonal.

Let a_i and b_i be the singular vector pairs corresponding to the singular values λ_i of C for $1\le i\le k$. Without loss of generality, we can assume that the sets $\{a_1,a_2,\dots,a_k\}$ and $\{b_1,b_2,\dots,b_k\}$ are orthonormal. Moreover, since $C\mathbf{1}_k=\mathbf{0}$ and $C^T\mathbf{1}_k=\mathbf{0}$ by the 0-net-regularity of the cycle (C_{2k},τ), the vector $\frac{1}{\sqrt{k}}\mathbf{1}_k$ lies in the null spaces of both C and C^T, and so we may choose $a_1=b_1=\frac{1}{\sqrt{k}}\mathbf{1}_k$ as the singular vectors corresponding to the singular value $\lambda_1=0$.

Taking the same Z_i^+ and Z_i^- with non-zero scalars c_i and d_i as in the classical case and following the same logic, we obtain the same eigenvalues

$$c_i^{\pm}=\frac{\lambda_i\pm\sqrt{\lambda_i^2+4m}}{2}\quad\text{and}\quad d_i^{\pm}=\frac{-\lambda_i\pm\sqrt{\lambda_i^2+4m}}{2}.$$

These each appear with multiplicity n, and so with Z_i^+ and Z_i^- for $2\le i\le k$ and $1\le j\le n$, we have found $4n(k-1)$ eigenvalues of $A_{(\Gamma_{k,m,n},\sigma)}$.

Now, let

$$Z_1^+=\begin{pmatrix}\mathbf{1}_{km}\\ \mathbf{1}_{km}\\ c_1\mathbf{1}_k\\ c_1\mathbf{1}_k\end{pmatrix}\quad\text{and}\quad Z_1^-=\begin{pmatrix}\mathbf{1}_{km}\\ -\mathbf{1}_{km}\\ d_1\mathbf{1}_k\\ -d_1\mathbf{1}_k\end{pmatrix},$$

where c_1 and d_1 are some non-zero scalars. Then, for $2\le j\le n$, denoting $A_{(\Gamma_{k,m,n},\sigma)}$ simply by A, we have

$$A\begin{pmatrix}(e_1-e_j)\otimes Z_1^+\\ \mathbf{0}_{m+1}\end{pmatrix}=c_1\begin{pmatrix}(e_1-e_j)\otimes Z_1^+\\ \mathbf{0}_{m+1}\end{pmatrix}\iff c_1=\frac{m}{c_1}\iff c_1^{\pm}=\pm\sqrt{m},$$
$$A\begin{pmatrix}(e_1-e_j)\otimes Z_1^-\\ \mathbf{0}_{m+1}\end{pmatrix}=d_1\begin{pmatrix}(e_1-e_j)\otimes Z_1^-\\ \mathbf{0}_{m+1}\end{pmatrix}\iff d_1=\frac{m}{d_1}\iff d_1^{\pm}=\pm\sqrt{m}.$$

Hence $+\sqrt{m}$ and $-\sqrt{m}$ are eigenvalues of $A_{(\Gamma_{k,m,n},\sigma)}$, each with multiplicity $2(n-1)$. Furthermore, we have $c_1^{\pm}=-m/c_1^{\mp}$, $d_1^{\pm}=-m/d_1^{\mp}$, $c_1^{+}=m/d_1^{+}$, and $c_1^{-}=m/d_1^{-}$, meaning these eigenvalues satisfy both m-(SR) and m-($-$SR).

So far we have listed $4(kn-1)$ eigenvalues, corresponding to the eigenvectors as described above. Since there are $(1+m)(1+2kn)$ vertices in total, we have $2kn(m-1)+m+5$ eigenvalues left to find.

For precisely the same reasons as in the unsigned construction, $A_{(\Gamma_{k,m,n},\sigma)}\mathbf{x}=0\mathbf{x}$, and zero is an eigenvalue of $A_{(\Gamma_{k,m,n},\sigma)}$ with multiplicity $(2kn+1)(m-1)$.

This means that there are $2kn(m-1)+m+5-(2kn+1)(m-1)=6$ non-trivial eigenvalues remaining to be found.

Let $\mu_1, \mu_2, \ldots, \mu_6$ denote the six remaining eigenvalues of $A_{(\Gamma_{k,m,n},\sigma)}$, corresponding to eigenvectors $X_1, X_2, \ldots, X_6$, respectively, and let $Y_1, Y_2, \ldots, Y_6$ be the same as before.

As in the proof of Theorem 2.3.11, the Y_i are the characteristic vectors of the cells of a partition of $V_{\Gamma_{k,m,n}}$. Every edge outside the cycles is positive, and each cyclic vertex meets exactly one positive and one negative edge of (C_{2k}, τ), so the row sums of $A_{(\Gamma_{k,m,n},\sigma)}$ over each cell are constant on each cell; that is, the partition is equitable for the signed adjacency matrix. Consequently $W = \operatorname{span}\{Y_1, \ldots, Y_6\}$ is invariant under $A_{(\Gamma_{k,m,n},\sigma)}$, and the Y_i, having pairwise disjoint supports, are linearly independent, so $\dim W = 6$.

Let U denote the span of the eigenvectors exhibited above, so that $\dim U = 4(kn-1)+(m-1)(2kn+1)$. Each of these eigenvectors sums to zero on every cell of the partition, and therefore lies in $W^\perp$; hence $U \subseteq W^\perp$. Since the total dimension is $2kn(m+1)+m+1$, we have

$$\dim W^\perp = 2kn(m+1)+m+1-6 = 4(kn-1)+(m-1)(2kn+1) = \dim U,$$

so $U = W^\perp$ and thus $W = U^\perp$. As $A_{(\Gamma_{k,m,n},\sigma)}$ is symmetric and U is spanned by eigenvectors, the six remaining eigenvectors lie in $U^\perp = W$, and so $X_i = \sum_{j=1}^{6} s_j Y_j$ for some scalars s_j not all zero.

This time, however, the equation $A_{(\Gamma_{k,m,n},\sigma)} X_i = \mu_i X_i$ holds if and only if

$$\begin{aligned} s_1\mu_i - s_3 - s_5 &= 0,\\ s_2\mu_i - s_4 &= 0,\\ ms_1 - s_3\mu_i &= 0,\\ ms_2 - s_4\mu_i + s_5 &= 0,\\ kmn\, s_1 + kn\, s_4 - s_5\mu_i + ms_6 &= 0,\\ s_5 - s_6\mu_i &= 0. \end{aligned}$$

Evidently, this means that the equation $A_{(\Gamma_{k,m,n},\sigma)} X_i = \mu_i X_i$ holds if and only if

$$\det \begin{pmatrix} -\mu_i & 0 & 1 & 0 & 1 & 0 \\ 0 & -\mu_i & 0 & 1 & 0 & 0 \\ m & 0 & -\mu_i & 0 & 0 & 0 \\ 0 & m & 0 & -\mu_i & 1 & 0 \\ kmn & 0 & 0 & kn & -\mu_i & m \\ 0 & 0 & 0 & 0 & 1 & -\mu_i \end{pmatrix} = 0.$$

So, the six remaining eigenvalues are the solutions to

$$(\mu^2 - m)(\mu^4 - (kn(m+1)+2m)\mu^2 + m^2) = 0,$$

which gives $\mu = \pm\sqrt{m}$ from the first factor, along with $\mu = \pm\sqrt{t_1}$ and $\mu = \pm\sqrt{t_2}$ from the second factor, where

$$\begin{aligned} t_1 &= \frac{1}{2}\Big(kn(m+1)+2m+\sqrt{(kn(m+1)+2m)^2-4m^2}\Big),\\ t_2 &= \frac{1}{2}\Big(kn(m+1)+2m-\sqrt{(kn(m+1)+2m)^2-4m^2}\Big). \end{aligned}$$

Clearly $\sqrt{m}$ is its own m-reciprocal partner, and $-\sqrt{m}$ is its m-anti-reciprocal partner. Combined with the $\sqrt{m}$ and $-\sqrt{m}$ of multiplicity $2(n-1)$ found earlier, $\sqrt{m}$ and $-\sqrt{m}$ each appear with total multiplicity $2n-1$ in the full spectrum. Since t_1 and t_2 are the roots of $t^2-(kn(m+1)+2m)t+m^2=0$, we have $t_1 t_2 = m^2$, and so $\sqrt{t_1}\sqrt{t_2} = m$, meaning $\sqrt{t_1}$ and $\sqrt{t_2}$ are m-reciprocal partners and $\sqrt{t_1}$ and $-\sqrt{t_2}$ are m-anti-reciprocal partners. Hence all six remaining eigenvalues satisfy both m-(SR) and m-($-$SR), and the proof is complete. □

Example 2.3.15. The graph $(\Gamma_{3,2,1}, \sigma) \in (\Omega_2^2, \Sigma)$ is depicted in Figure 2.6. Computed directly, its spectrum is

$$\operatorname{spec}(\Gamma_{3,2,1}, \sigma) = \{3.562, 2.524^{(2)}, 1.414, 0.792^{(2)}, 0.562, 0^{(7)},$$

$$-0.562, -0.792^{(2)}, -1.414, -2.524^{(2)}, -3.562\}.$$

It is easy to check that $(\Gamma_{3,2,1}, \sigma)$ satisfies properties 2-(SR) and 2-(−SR).

We can also compute the spectrum via the method presented in Theorem 2.3.14. With $k = 3$, we have

$$C = \begin{pmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{pmatrix},$$

which has singular values $\lambda_1 = 0$ and $\lambda_2 = \lambda_3 = \sqrt{3}$.

So, first, we have

$$\begin{aligned}
c_2^+ = c_3^+ &= \frac{\lambda_2 + \sqrt{\lambda_2^2 + 4m}}{2} = \frac{\lambda_3 + \sqrt{\lambda_3^2 + 4m}}{2} = \frac{\sqrt{3} + \sqrt{3 + 4 \cdot 2}}{2} = 2.524, \\
c_2^- = c_3^- &= \frac{\lambda_2 - \sqrt{\lambda_2^2 + 4m}}{2} = \frac{\lambda_3 - \sqrt{\lambda_3^2 + 4m}}{2} = \frac{\sqrt{3} - \sqrt{3 + 4 \cdot 2}}{2} = -0.792, \\
d_2^+ = d_3^+ &= \frac{-\lambda_2 + \sqrt{\lambda_2^2 + 4m}}{2} = \frac{-\lambda_3 + \sqrt{\lambda_3^2 + 4m}}{2} = \frac{-\sqrt{3} + \sqrt{3 + 4 \cdot 2}}{2} = 0.792, \\
d_2^- = d_3^- &= \frac{-\lambda_2 - \sqrt{\lambda_2^2 + 4m}}{2} = \frac{-\lambda_3 - \sqrt{\lambda_3^2 + 4m}}{2} = \frac{-\sqrt{3} - \sqrt{3 + 4 \cdot 2}}{2} = -2.524,
\end{aligned}$$

each appearing with multiplicity $n = 1$.

Since $n = 1$, the eigenvalues $\pm\sqrt{m}$ each appear with multiplicity $2(n-1) = 0$ and so do not contribute to the spectrum in this example.

For the remainder we have 0 appearing with multiplicity $(2kn + 1)(m - 1) = 7$, along with the final six eigenvalues:

$$\sqrt{2} = 1.414, \ -\sqrt{2} = -1.414, \ \sqrt{t_1} = 3.562, \ -\sqrt{t_1} = -3.562, \ \sqrt{t_2} = 0.562, \ -\sqrt{t_2} = -0.562.$$

Collecting these, we see that they give precisely the same $\mathrm{spec}(\Gamma_{3,2,1}, \sigma)$ as above.

2.4 Summary and outlook

In this chapter, we have introduced the integer anti-reciprocal eigenvalue properties m-(−R) and m-(−SR) and investigated graphs satisfying these alongside the integer reciprocal eigenvalue properties m-(R) and m-(SR) introduced by Barik et al. [13]. It is interesting to observe how results for unsigned graphs can be extended to signed graphs with suitable settings.

We have shown that the m-(R) and m-(−R) properties interact naturally with the Kronecker product, and that m-coronae of signed graphs always satisfy m-(−SR), with m-(SR) following when the root graph has symmetric spectrum. We have also established that for $n \geq 5$, the classical hypercube Q_n satisfies no integer reciprocal eigenvalue property, while a suitable signing always yields n-(SR) and n-(−SR). Finally, we have constructed explicit infinite families of graphs satisfying m-(R) but not m-(SR), and non-corona graphs satisfying both m-(SR) and m-(−SR), in both the unsigned and signed settings.

In each case the signed result required no fundamentally new machinery, only the right choice of signature on the underlying construction. What is perhaps notable is that the signed constructions yield the same eigenvalue properties as their unsigned counterparts while being genuinely unbalanced.

A natural next question is whether these properties persist under the gain graph generalisations studied in later chapters. The constructions here are far from exhaustive, and we expect that many further families satisfying these properties remain to be found.

Chapter 3

Spectra and balance of signed graphs obtained from vertex splittings

This chapter introduces new constructions of non-regular cospectral signed graphs via two operations: the neighbours splitting (NS) join and the non-neighbours splitting (NNS) join. We compute the adjacency and Laplacian characteristic polynomials for each join, allowing spectral analysis for arbitrary signed graphs and explicit eigenvalue calculations for co-regular signed graphs. A second approach employs pseudo-potential functions to define more robust switching-stable versions of these joins, preserving spectral properties under switching equivalence. As an application of these techniques, we construct infinite families of non-isomorphic signed graphs exhibiting cospectrality for both adjacency and Laplacian matrices. We also characterise balancedness conditions for each of the constructions.

This chapter is based on [58].

3.1 Introduction and preliminaries

For a graph matrix M, two signed graphs (Γ, σ) and (Γ', σ') are said to be M-cospectral if they have the same M-spectrum. In this chapter, if two or more M-cospectral signed graphs are not switching isomorphic, they are called M-cospectral mates.

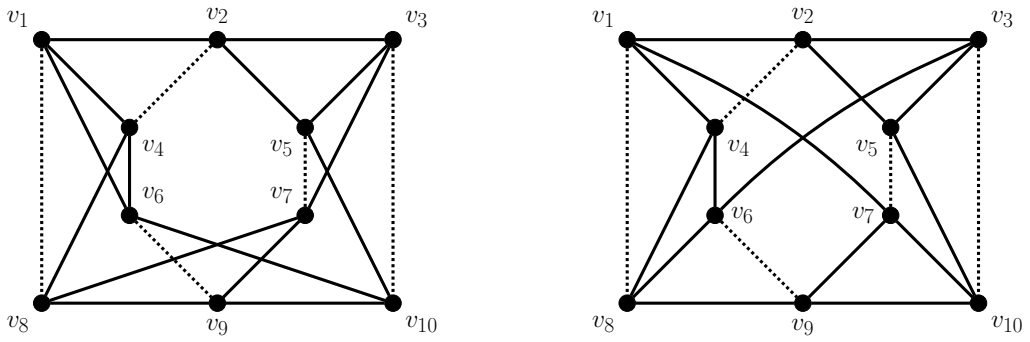


Figure 3.1: A pair of $(4, 2)$-co-regular non-isomorphic cospectral signed graphs (Ψ, σ_Ψ) and (Φ, σ_Φ) on ten vertices. Here and throughout, solid lines represent positive edges and dashed lines represent negative edges.

It is known (see [38, p. 398]) that all regular unsigned graphs on fewer than ten vertices are determined by their spectrum, that is, any unsigned graph cospectral with a regular graph on fewer than ten vertices is isomorphic to it. Furthermore, there are four pairs of cospectral regular unsigned graphs on ten vertices, as shown by van Dam and Haemers [39]. By identifying a perfect matching present in both unsigned graphs forming such a pair, that is, a subset of the edge set in which every vertex is incident with

exactly one edge, and turning precisely those edges negative, we obtain a non-trivial pair of co-regular, non-isomorphic, cospectral signed graphs.

Recently, several papers have focused on constructions of cospectral signed and gain graphs that are not switching isomorphic [1, 15, 17, 95]. An example of a pair of non-isomorphic cospectral signed graphs is given in Figure 3.1. One can verify that the graphs are non-isomorphic by seeing that in Ψ there exist edges, such as the edge v_1v_4, that lie in three triangles, whereas no edge of Φ has this property.

As discussed in Chapter 1, a well-designed generalisation of a product or join operation on signed graphs should aim to be switching stable, meaning that replacing the signatures of the components with switching-equivalent ones does not change the switching class of the product. This is particularly useful in spectral computations due to the fact that switching-equivalent signed graphs are cospectral.

In [33], the authors introduce the concept of pseudo-potential functions on signed graphs and provide a solution to an open problem regarding the switching stability of the lexicographic product of signed graphs. Following the approach of [48] and [73], we extend two join operations to the setting of signed graphs. Subsequently, we apply a result from [82], the pseudo-potential functions of [33], and a method developed in [61] to define distinct and more robust constructions that avoid the stability and balance issues present in the simpler versions.

For any results or concepts concerning graph spectra or signed graphs used in this chapter which have not been clearly stated the reader is directed to [38] and [100].

We now provide two remarks detailing how we shall denote the adjacency and Laplacian eigenvalues of co-regular signed graphs.

Remark 3.1.1. Let (Γ, σ) be an (r, k)-co-regular signed graph on n vertices. We shall denote its adjacency eigenvalues by setting $\lambda_1(\Gamma, \sigma) = k$ and ordering the remainder so that $\lambda_2(\Gamma, \sigma) \geq \lambda_3(\Gamma, \sigma) \geq \cdots \geq \lambda_n(\Gamma, \sigma)$. Note that k is not necessarily the largest adjacency eigenvalue of (Γ, σ) in general, unlike the unsigned case where $\lambda_1(\Gamma) = r$ is always the spectral radius. For the underlying graph Γ, the adjacency eigenvalues can be written as $\lambda_1(\Gamma) = r \geq \lambda_2(\Gamma) \geq \lambda_3(\Gamma) \geq \cdots \geq \lambda_n(\Gamma)$.

Remark 3.1.2. Let (Γ, σ) be an (r, k)-co-regular signed graph on n vertices. We shall denote its Laplacian eigenvalues by setting $\mu_1(\Gamma, \sigma) = r - k$ and ordering the remainder so that $\mu_2(\Gamma, \sigma) \leq \mu_3(\Gamma, \sigma) \leq \cdots \leq \mu_n(\Gamma, \sigma)$. Note that $r - k$ is not necessarily the smallest Laplacian eigenvalue of (Γ, σ) in general, unlike the unsigned case where $\mu_1(\Gamma) = 0$ always. For the underlying graph Γ, the Laplacian eigenvalues can be written as $\mu_1(\Gamma) = 0 \leq \mu_2(\Gamma) \leq \mu_3(\Gamma) \leq \cdots \leq \mu_n(\Gamma)$.

Recall Schur's complement formula, Lemma 1.8.1. This can be manipulated to simplify the computation of determinants of block matrices with various structures. The following extends [48, Lemma 2.6] by introducing a scalar parameter α.

Proposition 3.1.3. *Let M be a block matrix*

$$M = \begin{pmatrix} A & B & \alpha J_{n_1 \times n_2} \\ B & C & O_{n_1 \times n_2} \\ \alpha J_{n_2 \times n_1} & O_{n_2 \times n_1} & D \end{pmatrix}$$

where A, B, C are square matrices of order n_1, D is a square matrix of order n_2, and α is any real number. Then for the determinant of the characteristic matrix of M we have

$$\det(\lambda I_{2n_1+n_2} - M) = \det(\lambda I_{n_2} - D) \det \begin{pmatrix} \lambda I_{n_1} - A - \alpha^2 \chi_D(\lambda) J_{n_1} & -B \\ -B & \lambda I_{n_1} - C \end{pmatrix}.$$

Proof. The characteristic matrix of M is

$$\lambda I_{2n_1+n_2} - M = \begin{pmatrix} \lambda I_{n_1} - A & -B & -\alpha J_{n_1 \times n_2} \\ -B & \lambda I_{n_1} - C & O_{n_1 \times n_2} \\ -\alpha J_{n_2 \times n_1} & O_{n_2 \times n_1} & \lambda I_{n_2} - D \end{pmatrix}.$$

Suppose that λ is not an eigenvalue of D. Then $\lambda I_{n_2} - D$ is invertible and by Schur's complement formula we obtain the determinant of the characteristic matrix of M as

$$\det(\lambda I_{2n_1+n_2} - M) = \det(\lambda I_{n_2} - D) \det \left(\begin{pmatrix} \lambda I_{n_1} - A & -B \\ -B & \lambda I_{n_1} - C \end{pmatrix} \right.$$

$$
\begin{aligned}
&\quad - \begin{pmatrix} -\alpha J_{n_1\times n_2} \\ O_{n_1\times n_2} \end{pmatrix} (\lambda I_{n_2} - D)^{-1} \begin{pmatrix} -\alpha J_{n_2\times n_1} & O_{n_2\times n_1} \end{pmatrix} \Bigg) \\
&= \det(\lambda I_{n_2} - D) \det \Bigg(\begin{pmatrix} \lambda I_{n_1} - A & -B \\ -B & \lambda I_{n_1} - C \end{pmatrix} \\
&\quad - \alpha^2 \begin{pmatrix} \mathbf{1}_{n_1}\mathbf{1}_{n_2}^T \\ O_{n_1\times n_2} \end{pmatrix} (\lambda I_{n_2} - D)^{-1} \begin{pmatrix} \mathbf{1}_{n_2}\mathbf{1}_{n_1}^T & O_{n_2\times n_1} \end{pmatrix} \Bigg) \\
&= \det(\lambda I_{n_2} - D) \det \begin{pmatrix} \lambda I_{n_1} - A - \alpha^2 \chi_D(\lambda) J_{n_1} & -B \\ -B & \lambda I_{n_1} - C \end{pmatrix}.
\end{aligned}
$$

Since both sides are polynomials in λ that coincide for all λ that are not eigenvalues of D, and since the set of eigenvalues of D is finite, the two polynomials must coincide identically. □

The adjugate of a matrix A, defined as the transpose of its cofactor matrix, is denoted by $\mathrm{adj}(A)$, and satisfies the identity $A\,\mathrm{adj}(A) = \det(A)I$. Consequently, when A is invertible, we have $\mathrm{adj}(A) = \det(A)A^{-1}$. This final result is a special case of the matrix determinant lemma and can be found in textbooks such as [55].

Proposition 3.1.4. *If A is an $n \times n$ real matrix and α is a real number then*

$$\det(A + \alpha J_n) = \det(A) + \alpha \mathbf{1}_n^T \,\mathrm{adj}(A)\mathbf{1}_n.$$

The remainder of this chapter is structured as follows. In Section 3.2, we define the signed NS and NNS joins, derive their adjacency and Laplacian characteristic polynomials for two arbitrary signed graphs, compute their spectra for two co-regular signed graphs, and demonstrate a method of constructing cospectral mates. In Section 3.3, we use pseudo-potentials to provide alternative definitions for the signatures of the NS and NNS joins that are stable under switchings of their components. In Section 3.4, we offer concluding remarks and open problems which are beyond the scope of this chapter.

3.2 The signed NS and NNS joins

As described by Harary [51], the join of two ordinary graphs Γ_1 and Γ_2, denoted $\Gamma_1 \vee \Gamma_2$, is formed by connecting each vertex of Γ_1 to every vertex of Γ_2. In the case of signed graphs, however, there are multiple possible ways to define such a join, since the result depends on the signatures assigned to the newly added edges. We shall mainly focus on the join of two signed graphs (Γ_1, σ_1) and (Γ_2, σ_2) where each newly introduced edge is made to be positive. We will call this procedure the positive join of (Γ_1, σ_1) and (Γ_2, σ_2) and write it as $(\Gamma_1, \sigma_1) \vee_+ (\Gamma_2, \sigma_2)$.

A variation on the standard join operation was introduced by Lu et al. [73] and called the splitting V-vertex join. Throughout the remainder of this chapter, we will refer to this operation as the neighbours splitting (NS) join. A further variation, named the non-neighbours splitting (NNS) join, was defined by Hamud and Berman [48]. Both of these were introduced in the setting of ordinary unsigned graphs, and we extend them to signed graphs (Γ_1, σ_1) and (Γ_2, σ_2) by assigning all newly introduced edges to be positive.

Definition 3.2.1 (Signed NS join, adapted from [73])**.** Let (Γ_1, σ_1) and (Γ_2, σ_2) be two vertex-disjoint signed graphs of order n_1 and n_2, respectively, and write $V_{\Gamma_1} = \{u_1, u_2, \ldots, u_{n_1}\}$ and $V_{\Gamma_2} = \{v_1, v_2, \ldots, v_{n_2}\}$. The signed neighbours splitting (NS) join of (Γ_1, σ_1) and (Γ_2, σ_2), denoted $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$, is the signed graph obtained by forming the positive join $(\Gamma_1, \sigma_1) \vee_+ (\Gamma_2, \sigma_2)$, introducing the edgeless graph $(\Gamma_3, \sigma_3) = (K_{n_1}^c, \varnothing)$ on n_1 vertices with vertex set $V_{\Gamma_3} = \{w_1, w_2, \ldots, w_{n_1}\}$, where $\varnothing$ denotes the empty signature, and connecting u_i to w_j with a positive edge if and only if u_i is adjacent to u_j in Γ_1.

We can see that symbolically for the NS join of (Γ_1, σ_1) and (Γ_2, σ_2), we have vertex and edge sets

$$
\begin{aligned}
V_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)} &= V_{\Gamma_1} \cup V_{\Gamma_2} \cup V_{\Gamma_3}, \\
E_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)} &= E_{\Gamma_1} \cup E_{\Gamma_2} \cup \{u_i v_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\} \cup \{u_i w_j : u_i u_j \in E_{\Gamma_1}\},
\end{aligned}
$$

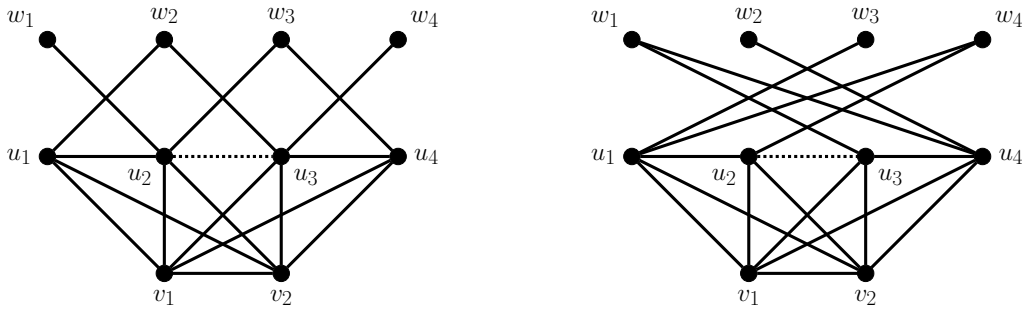


Figure 3.2: The NS and NNS joins $(P_4, \sigma_1) \veebar (P_2, \sigma_2)$ and $(P_4, \sigma_1) \barwedge (P_2, \sigma_2)$.

where the signature σ of $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$ is defined by

$$\sigma(e) = \begin{cases} \sigma_1(e), & \text{if } e \in E_{\Gamma_1}, \\ \sigma_2(e), & \text{if } e \in E_{\Gamma_2}, \\ +1, & \text{if } e \in \{u_i v_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\} \cup \{u_i w_j : u_i u_j \in E_{\Gamma_1}\}. \end{cases}$$

Definition 3.2.2 (Signed NNS join, adapted from [48])**.** Let (Γ_1, σ_1) and (Γ_2, σ_2) be two vertex-disjoint signed graphs of order n_1 and n_2, respectively, and write $V_{\Gamma_1} = \{u_1, u_2, \ldots, u_{n_1}\}$ and $V_{\Gamma_2} = \{v_1, v_2, \ldots, v_{n_2}\}$. The signed non-neighbours splitting (NNS) join of (Γ_1, σ_1) and (Γ_2, σ_2), denoted $(\Gamma_1, \sigma_1) \barwedge (\Gamma_2, \sigma_2)$, is the signed graph obtained by forming the positive join $(\Gamma_1, \sigma_1) \vee_+ (\Gamma_2, \sigma_2)$, introducing the edgeless graph $(\Gamma_3, \sigma_3) = (K_{n_1}^{\mathrm{c}}, \varnothing)$ on n_1 vertices with vertex set $V_{\Gamma_3} = \{w_1, w_2, \ldots, w_{n_1}\}$, where $\varnothing$ denotes the empty signature, and connecting u_i to w_j with a positive edge if and only if $i \neq j$ and u_i is not adjacent to u_j in Γ_1.

Now we see that for the NNS join of (Γ_1, σ_1) and (Γ_2, σ_2), we have the vertex and edge sets

$$\begin{aligned} V_{(\Gamma_1,\sigma_1)\barwedge(\Gamma_2,\sigma_2)} &= V_{\Gamma_1} \cup V_{\Gamma_2} \cup V_{\Gamma_3}, \\ E_{(\Gamma_1,\sigma_1)\barwedge(\Gamma_2,\sigma_2)} &= E_{\Gamma_1} \cup E_{\Gamma_2} \cup \{u_i v_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\} \cup \{u_i w_j : i \neq j \text{ and } u_i u_j \notin E_{\Gamma_1}\}, \end{aligned}$$

where the signature σ of $(\Gamma_1, \sigma_1) \barwedge (\Gamma_2, \sigma_2)$ is defined by

$$\sigma(e) = \begin{cases} \sigma_1(e), & \text{if } e \in E_{\Gamma_1}, \\ \sigma_2(e), & \text{if } e \in E_{\Gamma_2}, \\ +1, & \text{if } e \in \{u_i v_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\} \cup \{u_i w_j : i \neq j \text{ and } u_i u_j \notin E_{\Gamma_1}\}. \end{cases}$$

3.2.1 Characteristic polynomials of the signed NS and NNS joins

Here we compute the adjacency and Laplacian characteristic polynomials of both $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$ and $(\Gamma_1, \sigma_1) \barwedge (\Gamma_2, \sigma_2)$ for two arbitrary signed graphs (Γ_1, σ_1) and (Γ_2, σ_2). These results are then used to construct pairs of non-regular cospectral mates with respect to both the adjacency and Laplacian matrices.

Adjacency characteristic polynomials

Theorem 3.2.3. *Let (Γ_1, σ_1) and (Γ_2, σ_2) be vertex-disjoint signed graphs on n_1 and n_2 vertices, respectively. Then*

(a) The adjacency characteristic polynomial of the NS join $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$ is

$$P(A_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)}; \lambda) = \lambda^{n_1} P\big(A_{(\Gamma_2,\sigma_2)}; \lambda\big) \det\Big(\lambda I_{n_1} - N_\lambda\Big)\Big[1 - \chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)\chi_{N_\lambda}(\lambda)\Big],$$

where $N_\lambda = A_{(\Gamma_1,\sigma_1)} + \frac{1}{\lambda} A_{\Gamma_1}^2$.

(b) The adjacency characteristic polynomial of the NNS join $(\Gamma_1,\sigma_1)\,\overline{\wedge}\,(\Gamma_2,\sigma_2)$ *is*

$$P(A_{(\Gamma_1,\sigma_1)\overline{\wedge}(\Gamma_2,\sigma_2)};\lambda)=\lambda^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};\lambda\big)\det\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}\Big)\Big[1-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)\chi_{N_\lambda^{\mathrm{c}}}(\lambda)\Big],$$

where $N_\lambda^{\mathrm{c}}=A_{(\Gamma_1,\sigma_1)}+\frac{1}{\lambda}A^2_{\Gamma_1^{\mathrm{c}}}$.

Proof. By labelling the vertices of $(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)$ and $(\Gamma_1,\sigma_1)\,\overline{\wedge}\,(\Gamma_2,\sigma_2)$ in a suitable order, we obtain the adjacency matrices

$$A_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)}=\begin{pmatrix}A_{(\Gamma_1,\sigma_1)} & A_{\Gamma_1} & J_{n_1\times n_2}\\ A_{\Gamma_1} & O_{n_1} & O_{n_1\times n_2}\\ J_{n_2\times n_1} & O_{n_2\times n_1} & A_{(\Gamma_2,\sigma_2)}\end{pmatrix},\quad A_{(\Gamma_1,\sigma_1)\overline{\wedge}(\Gamma_2,\sigma_2)}=\begin{pmatrix}A_{(\Gamma_1,\sigma_1)} & A_{\Gamma_1^{\mathrm{c}}} & J_{n_1\times n_2}\\ A_{\Gamma_1^{\mathrm{c}}} & O_{n_1} & O_{n_1\times n_2}\\ J_{n_2\times n_1} & O_{n_2\times n_1} & A_{(\Gamma_2,\sigma_2)}\end{pmatrix}.$$

Owing to the resemblance between the corresponding matrices when the graph vertices are ordered appropriately, the two joins admit proofs of a similar structure. For this reason, we shall provide a full proof only for the NNS join.

Denote $P_A^{\overline{\wedge}}=P(A_{(\Gamma_1,\sigma_1)\overline{\wedge}(\Gamma_2,\sigma_2)};\,\lambda)$. Hence, by applying Proposition 3.1.3 with $|\alpha|=1$, we get the adjacency characteristic polynomial

$$P_A^{\overline{\wedge}}=\det\big(\lambda I_{n_2}-A_{(\Gamma_2,\sigma_2)}\big)\det\begin{pmatrix}\lambda I_{n_1}-A_{(\Gamma_1,\sigma_1)}-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)J_{n_1} & -A_{\Gamma_1^{\mathrm{c}}}\\ -A_{\Gamma_1^{\mathrm{c}}} & \lambda I_{n_1}\end{pmatrix}.$$

Now we can apply Lemma 1.8.1 and then make use of Proposition 3.1.4 to acquire

$$\begin{aligned}P_A^{\overline{\wedge}}&=\det\big(\lambda I_{n_2}-A_{(\Gamma_2,\sigma_2)}\big)\det\big(\lambda I_{n_1}\big)\det\big(\lambda I_{n_1}-A_{(\Gamma_1,\sigma_1)}-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)J_{n_1}-A_{\Gamma_1^{\mathrm{c}}}(\lambda I_{n_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}\big)\\
&=\lambda^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};\lambda\big)\det\Big(\lambda I_{n_1}-A_{(\Gamma_1,\sigma_1)}-\frac{1}{\lambda}A^2_{\Gamma_1^{\mathrm{c}}}-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)J_{n_1}\Big)\\
&=\lambda^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};\lambda\big)\det\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)J_{n_1}\Big)\\
&=\lambda^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};\lambda\big)\left[\det\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}\Big)-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)\mathbf{1}_{n_1}^T\operatorname{adj}\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}\Big)\mathbf{1}_{n_1}\right]\\
&=\lambda^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};\lambda\big)\det\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}\Big)\left[1-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)\mathbf{1}_{n_1}^T\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}\Big)^{-1}\mathbf{1}_{n_1}\right].\end{aligned}$$

Therefore

$$P(A_{(\Gamma_1,\sigma_1)\overline{\wedge}(\Gamma_2,\sigma_2)};\,\lambda)=\lambda^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};\lambda\big)\det\Big(\lambda I_{n_1}-N_\lambda^{\mathrm{c}}\Big)\Big[1-\chi_{A_{(\Gamma_2,\sigma_2)}}(\lambda)\chi_{N_\lambda^{\mathrm{c}}}(\lambda)\Big].$$

□

Laplacian characteristic polynomials

Theorem 3.2.4. *Let* (Γ_1,σ_1) *and* (Γ_2,σ_2) *be vertex-disjoint signed graphs on* n_1 *and* n_2 *vertices, respectively. Then*

(a) The Laplacian characteristic polynomial of the NS join $(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)$ *is*

$$\begin{aligned}P(L_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)};\mu)=\;&\det\Big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\Big)\det\Big(\mu I_{n_1}-D_{\Gamma_1}\Big)\\
&\det\Big((\mu-n_2)I_{n_1}-M_\mu\Big)\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)\chi_{M_\mu}(\mu-n_2)\Big],\end{aligned}$$

where $M_\mu=L_{(\Gamma_1,\sigma_1)}+D_{\Gamma_1}+A_{\Gamma_1}\big(\mu I_{n_1}-D_{\Gamma_1}\big)^{-1}A_{\Gamma_1}$.

(b) The Laplacian characteristic polynomial of the NNS join $(\Gamma_1,\sigma_1)\,\overline{\wedge}\,(\Gamma_2,\sigma_2)$ *is*

$$\begin{aligned}P(L_{(\Gamma_1,\sigma_1)\overline{\wedge}(\Gamma_2,\sigma_2)};\mu)=\;&\det\Big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\Big)\det\Big((\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\Big)\\
&\det\Big((\mu-n_1-n_2+1)I_{n_1}-M_\mu^{\mathrm{c}}\Big)\\
&\cdot\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)\chi_{M_\mu^{\mathrm{c}}}(\mu-n_1-n_2+1)\Big],\end{aligned}$$

where $M_\mu^{\mathrm{c}}=L_{(\Gamma_1,\sigma_1)}-D_{\Gamma_1}+A_{\Gamma_1^{\mathrm{c}}}((\mu-n_1+1)I_{n_1}+D_{\Gamma_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}$.

Proof. By labelling the vertices of $(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)$ and $(\Gamma_1,\sigma_1)\barwedge(\Gamma_2,\sigma_2)$ in a suitable order, we obtain the Laplacian matrices

$$L_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)}=\begin{pmatrix} n_2I_{n_1}+D_{\Gamma_1}+L_{(\Gamma_1,\sigma_1)} & -A_{\Gamma_1} & -J_{n_1\times n_2}\\ -A_{\Gamma_1} & D_{\Gamma_1} & O_{n_1\times n_2}\\ -J_{n_2\times n_1} & O_{n_2\times n_1} & n_1I_{n_2}+L_{(\Gamma_2,\sigma_2)}\end{pmatrix}$$

and

$$L_{(\Gamma_1,\sigma_1)\barwedge(\Gamma_2,\sigma_2)}=\begin{pmatrix} (n_1+n_2-1)I_{n_1}+L_{(\Gamma_1,\sigma_1)}-D_{\Gamma_1} & -A_{\Gamma_1^{\mathrm{c}}} & -J_{n_1\times n_2}\\ -A_{\Gamma_1^{\mathrm{c}}} & (n_1-1)I_{n_1}-D_{\Gamma_1} & O_{n_1\times n_2}\\ -J_{n_2\times n_1} & O_{n_2\times n_1} & n_1I_{n_2}+L_{(\Gamma_2,\sigma_2)}\end{pmatrix}.$$

As in the proof of Theorem 3.2.3, we shall provide full details only for the NNS join.

Denote $P_L^{\barwedge}=P(L_{(\Gamma_1,\sigma_1)\barwedge(\Gamma_2,\sigma_2)};\mu)$. Hence, by applying Proposition 3.1.3, we get the Laplacian characteristic polynomial

$$P_L^{\barwedge}=\det\big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\big)\det\begin{pmatrix}X & A_{\Gamma_1^{\mathrm{c}}}\\ A_{\Gamma_1^{\mathrm{c}}} & (\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\end{pmatrix},$$

where $X=(\mu-n_1-n_2+1)I_{n_1}-L_{(\Gamma_1,\sigma_1)}+D_{\Gamma_1}-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)J_{n_1}$.

Now we can apply Lemma 1.8.1 and make use of Proposition 3.1.4 to acquire

$$\begin{aligned}
P_L^{\barwedge}&=\det\big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\big)\det\big((\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\big)\\
&\quad\det\big((\mu-n_1-n_2+1)I_{n_1}-L_{(\Gamma_1,\sigma_1)}+D_{\Gamma_1}\\
&\qquad-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)J_{n_1}-A_{\Gamma_1^{\mathrm{c}}}((\mu-n_1+1)I_{n_1}+D_{\Gamma_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}\big)\\
&=\det\big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\big)\det\big((\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\big)\\
&\quad\Big[\det\big((\mu-n_1-n_2+1)I_{n_1}-L_{(\Gamma_1,\sigma_1)}+D_{\Gamma_1}-A_{\Gamma_1^{\mathrm{c}}}((\mu-n_1+1)I_{n_1}+D_{\Gamma_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}\big)\\
&\qquad-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)\mathbf{1}_{n_1}^T\operatorname{adj}\big((\mu-n_1-n_2+1)I_{n_1}\\
&\qquad\quad-L_{(\Gamma_1,\sigma_1)}+D_{\Gamma_1}-A_{\Gamma_1^{\mathrm{c}}}((\mu-n_1+1)I_{n_1}+D_{\Gamma_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}\big)\mathbf{1}_{n_1}\Big]\\
&=\det\big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\big)\det\big((\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\big)\\
&\quad\Big[\det\big((\mu-n_1-n_2+1)I_{n_1}-M_\mu^{\mathrm{c}}\big)-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)\mathbf{1}_{n_1}^T\operatorname{adj}\big((\mu-n_1-n_2+1)I_{n_1}-M_\mu^{\mathrm{c}}\big)\mathbf{1}_{n_1}\Big]\\
&=\det\big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\big)\det\big((\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\big)\\
&\quad\det\big((\mu-n_1-n_2+1)I_{n_1}-M_\mu^{\mathrm{c}}\big)\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)\mathbf{1}_{n_1}^T\big((\mu-n_1-n_2+1)I_{n_1}-M_\mu^{\mathrm{c}}\big)^{-1}\mathbf{1}_{n_1}\Big].
\end{aligned}$$

Therefore

$$\begin{aligned}
P(L_{(\Gamma_1,\sigma_1)\barwedge(\Gamma_2,\sigma_2)};\mu)=\ &\det\Big((\mu-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\Big)\det\Big((\mu-n_1+1)I_{n_1}+D_{\Gamma_1}\Big)\\
&\det\Big((\mu-n_1-n_2+1)I_{n_1}-M_\mu^{\mathrm{c}}\Big)\\
&\cdot\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(\mu-n_1)\chi_{M_\mu^{\mathrm{c}}}(\mu-n_1-n_2+1)\Big],
\end{aligned}$$

□

3.2.2 Constructing non-regular cospectral signed pairs

Now that we have computed the characteristic polynomials for the NS and NNS joins, we can use the results to construct non-regular cospectral pairs of signed graphs. The following corollaries show that, under suitable co-regularity assumptions on the factor graphs, the NS and NNS joins preserve cospectrality. A remark explains why the co-regularity assumption is necessary. While the corollaries establish cospectrality, we conjecture that the compound graphs are in fact non-switching-isomorphic (and hence cospectral mates) whenever the factor graphs are and a concrete example is provided in which this stronger property is verified directly.

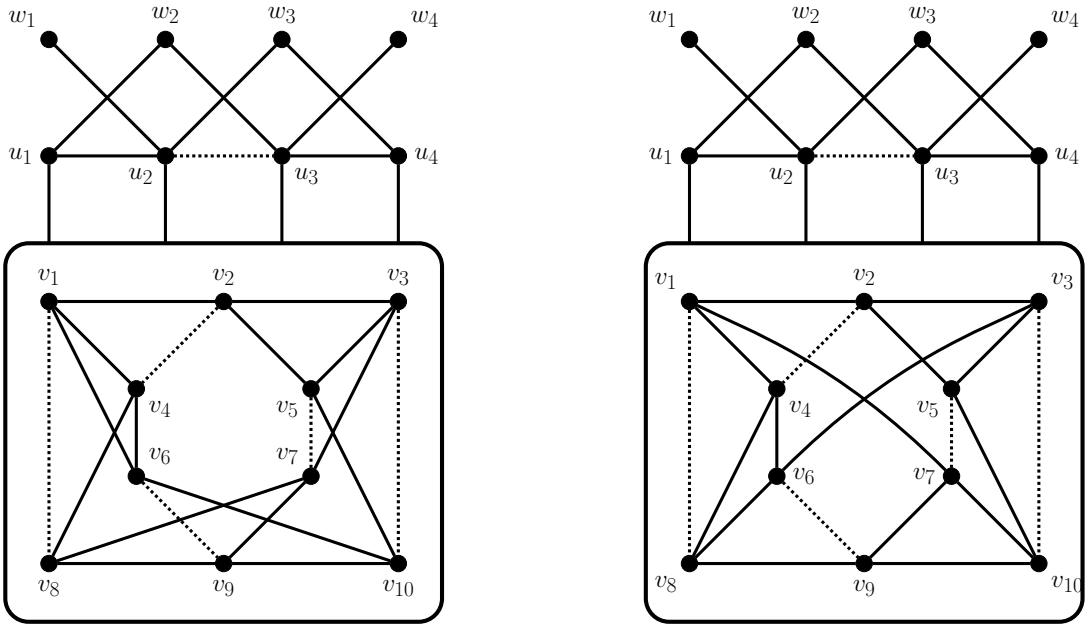


Figure 3.3: The NS joins $(\Gamma, \sigma) \veebar (\Psi, \sigma_\Psi)$ and $(\Gamma, \sigma) \veebar (\Phi, \sigma_\Phi)$, where $(\Gamma, \sigma) = (P_4, \sigma)$ and (Ψ, σ_Ψ), (Φ, σ_Φ) are from Figure 3.1. A line from a vertex to a box indicates edges connecting the single vertex to all vertices within the box.

Corollary 3.2.5. *Let (Ψ, σ_Ψ) and (Φ, σ_Φ) be (r, k)-co-regular signed graphs that are cospectral mates. Then for every signed graph (Γ, σ) we have*

(a) the graphs $(\Gamma, \sigma) \veebar (\Psi, \sigma_\Psi)$ and $(\Gamma, \sigma) \veebar (\Phi, \sigma_\Phi)$ are cospectral with respect to both the adjacency and Laplacian matrices,

(b) the graphs $(\Gamma, \sigma) \barwedge (\Psi, \sigma_\Psi)$ and $(\Gamma, \sigma) \barwedge (\Phi, \sigma_\Phi)$ are cospectral with respect to both the adjacency and Laplacian matrices.

Proof. We shall consider (a). By Theorems 3.2.3 and 3.2.4, we know that

$$P(A_{(\Gamma,\sigma)\veebar(\Psi,\sigma_\Psi)}; \lambda) = P(A_{(\Gamma,\sigma)\veebar(\Phi,\sigma_\Phi)}; \lambda) \qquad \text{and} \qquad P(L_{(\Gamma,\sigma)\veebar(\Psi,\sigma_\Psi)}; \mu) = P(L_{(\Gamma,\sigma)\veebar(\Phi,\sigma_\Phi)}; \mu).$$

Indeed, by Lemma 1.8.4, the adjacency matrices $A_{(\Psi,\sigma_\Psi)}$ and $A_{(\Phi,\sigma_\Phi)}$ share the same coronal by their k-net-regularity, and the Laplacian matrices $L_{(\Psi,\sigma_\Psi)}$ and $L_{(\Phi,\sigma_\Phi)}$ share the same coronal by their (r, k)-co-regularity. Combined with the cospectrality of (Ψ, σ_Ψ) and (Φ, σ_Φ), they also share the same characteristic polynomials. This completes the proof of (a) and the proof of (b) is similar. □

Corollary 3.2.6. *Let (Ψ, σ_Ψ) and (Φ, σ_Φ) be (r, k)-co-regular signed graphs that are cospectral mates. Then for every signed graph (Γ, σ) we have*

(a) the graphs $(\Psi, \sigma_\Psi) \veebar (\Gamma, \sigma)$ and $(\Phi, \sigma_\Phi) \veebar (\Gamma, \sigma)$ are cospectral with respect to both the adjacency and Laplacian matrices,

(b) the graphs $(\Psi, \sigma_\Psi) \barwedge (\Gamma, \sigma)$ and $(\Phi, \sigma_\Phi) \barwedge (\Gamma, \sigma)$ are cospectral with respect to both the adjacency and Laplacian matrices.

Proof. The proof is similar to that of the previous corollary. □

Remark 3.2.7. We note that the characteristic polynomial identities in Theorems 3.2.3 and 3.2.4 establish cospectrality of the compound graphs. It is natural to ask whether the compound graphs are also non-switching-isomorphic (and hence cospectral mates) whenever (Ψ, σ_Ψ) and (Φ, σ_Φ) are. This would follow from a recoverability lemma showing that any isomorphism of the NS or NNS join must preserve

the factor partition. We conjecture that such a lemma holds, and a targeted exhaustive search within a small net-regular signed K_5 family supports this conjecture. However, a full proof is beyond the scope of the present chapter.

Remark 3.2.8. It is necessary to restrict the graphs (Ψ, σ_Ψ) and (Φ, σ_Φ) in Corollaries 3.2.5 and 3.2.6 to being (r, k)-co-regular. To see this, consider the cospectral mates $(K_{1,4}, +)$ and $(C_4 \cup K_1, +)$, each with all-positive signature. By definition these share the same spectrum and hence their characteristic polynomials are equal. However, the values of their coronals differ. Thus, in general, the characteristic polynomials of the join graphs $(\Gamma, \sigma) \veebar (K_{1,4}, +)$ and $(\Gamma, \sigma) \veebar (C_4 \cup K_1, +)$ are not necessarily equal, and therefore they may not be adjacency or Laplacian cospectral.

Example 3.2.9. The signed graphs depicted in Figure 3.1 are both $(4, 2)$-co-regular and are not switching isomorphic: recall that Ψ contains edges lying in three triangles while Φ does not.

Let $(\Gamma, \sigma) = (P_4, \sigma)$, with $\sigma(u_1u_2) = \sigma(u_3u_4) = +1$ and $\sigma(u_2u_3) = -1$. Then, taking the NS joins $(\Gamma, \sigma) \veebar (\Psi, \sigma_\Psi)$ and $(\Gamma, \sigma) \veebar (\Phi, \sigma_\Phi)$ gives the two signed graphs depicted in Figure 3.3. These graphs share the following adjacency characteristic polynomial:

$$\begin{aligned} P(A_{(\Gamma,\sigma)\veebar(\Psi,\sigma_\Psi)}; \lambda) = P(A_{(\Gamma,\sigma)\veebar(\Phi,\sigma_\Phi)}; \lambda) = {} & \lambda^{18} - 69\lambda^{16} - 96\lambda^{15} + 1182\lambda^{14} + 1936\lambda^{13} \\ & - 8771\lambda^{12} - 14524\lambda^{11} + 32278\lambda^{10} + 49960\lambda^9 \\ & - 59603\lambda^8 - 77616\lambda^7 + 51914\lambda^6 + 45824\lambda^5 \\ & - 17185\lambda^4 - 7076\lambda^3 + 1569\lambda^2 + 312\lambda - 36. \end{aligned}$$

The two joins are therefore cospectral. Moreover, they are not switching isomorphic. Indeed, in either join the vertices of V_{Γ_1}, V_{Γ_2} and V_{Γ_3} have degrees $2d_{\Gamma_1}(u_i) + n_2$, $n_1 + d_{\Gamma_2}(v_k)$ and $d_{\Gamma_1}(u_j)$ respectively; with $\Gamma_1 = P_4$ and Γ_2 being 4-regular on ten vertices these are $\{12, 14\}$, $\{8\}$ and $\{1, 2\}$. As these three sets are pairwise disjoint, V_{Γ_2} is recovered in both joins as the set of vertices of degree 8, so every switching isomorphism between the joins maps V_{Γ_2} onto V_{Γ_2}. Since the join signature restricts to σ_Ψ (respectively σ_Φ) on E_{Γ_2}, and the restriction of a switching function is again a switching function, such a map would restrict to a switching isomorphism between (Ψ, σ_Ψ) and (Φ, σ_Φ), contradicting the above. Therefore $(\Gamma, \sigma) \veebar (\Psi, \sigma_\Psi)$ and $(\Gamma, \sigma) \veebar (\Phi, \sigma_\Phi)$ form a pair of cospectral mates.

3.2.3 Spectra of the signed NS and NNS joins

In this section we shall compute the adjacency and Laplacian spectra of graphs produced by the two join operations. Our analysis is restricted to joins of two (r, k)-co-regular signed graphs, as this allows for the application of Lemma 1.8.4, simplifying the computations. Without this restriction the calculations become considerably more difficult.

Adjacency spectrum of the signed NS join

Theorem 3.2.10. *Let (Γ_1, σ_1) be an (r_1, k_1)-co-regular signed graph on n_1 vertices and let (Γ_2, σ_2) be an (r_2, k_2)-co-regular signed graph on n_2 vertices. Then the adjacency spectrum of $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$ consists of*

(i) $\lambda_j(\Gamma_2, \sigma_2)$ for each $2 \le j \le n_2$,

(ii) the $2(n_1 - 1)$ roots of

$$\frac{\det\left(x^2 I_{n_1} - x A_{(\Gamma_1,\sigma_1)} - A_{\Gamma_1}^2\right)}{x^2 - k_1 x - r_1^2} = 0,$$

(iii) the three roots of the cubic equation $x^3 - (k_1 + k_2)x^2 - (r_1^2 - k_1k_2 + n_1n_2)x + k_2r_1^2 = 0$.

Proof. Denote $P_A^{\veebar} = P(A_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)}; x)$. Then, by Theorem 3.2.3, the adjacency characteristic polynomial of $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$ is

$$P_A^{\veebar} = x^{n_1} P\big(A_{(\Gamma_2,\sigma_2)}; x\big) \det\Big(x I_{n_1} - N_x\Big)\Big[1 - \chi_{A_{(\Gamma_2,\sigma_2)}}(x)\chi_{N_x}(x)\Big],$$

where $N_x = A_{(\Gamma_1,\sigma_1)} + \frac{1}{x}A_{\Gamma_1}^2$.

Since we are working with co-regular signed graphs, so that N_x has constant row sum $k_1 + \frac{1}{x}r_1^2$, we can apply Lemma 1.8.4 to obtain the adjacency characteristic polynomial as

$$\begin{aligned}
P_A^{\vee} &= x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det\Big(xI_{n_1}-N_x\Big)\Big[1-\frac{n_2}{x-k_2}\cdot\frac{n_1}{x-k_1-\frac{1}{x}r_1^2}\Big]\\
&= \prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det\Big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\Big)\Big[1-\frac{n_1n_2x}{\big(x-k_2\big)\big(x^2-k_1x-r_1^2\big)}\Big]\\
&= \prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det\Big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\Big)\Big[\frac{x^3-\big(k_1+k_2\big)x^2-\big(r_1^2+n_1n_2-k_1k_2\big)x+r_1^2k_2}{\big(x-k_2\big)\big(x^2-k_1x-r_1^2\big)}\Big],
\end{aligned}$$

where we have used $x^{n_1}\det(xI_{n_1}-N_x)=\det\big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\big)$.

Since $A_{(\Gamma_1,\sigma_1)}\mathbf{1}_{n_1}=k_1\mathbf{1}_{n_1}$ and $A_{\Gamma_1}\mathbf{1}_{n_1}=r_1\mathbf{1}_{n_1}$, the all-one vector contributes the factor $x^2-k_1x-r_1^2$ to this determinant; cancelling it together with the factor $x-k_2=x-\lambda_1(\Gamma_2,\sigma_2)$ of the product, by Remark 3.1.1, we get

$$P_A^{\vee}=\prod_{j=2}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\cdot\frac{\det\Big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\Big)}{x^2-k_1x-r_1^2}\cdot\Big[x^3-\big(k_1+k_2\big)x^2-\big(r_1^2-k_1k_2+n_1n_2\big)x+k_2r_1^2\Big].$$

Hence we have the desired result. □

Corollary 3.2.11. *When σ_1 is the all-positive signature, we have $A_{(\Gamma_1,\sigma_1)}=A_{\Gamma_1}$ and the determinant factors as $\prod_{i=1}^{n_1}\Big(x^2-\lambda_i(\Gamma_1)x-\lambda_i^2(\Gamma_1)\Big)$. Thus, Theorem 3.2.10(ii) becomes exactly the two roots of the quadratic equation $x^2-\lambda_i(\Gamma_1)x-\lambda_i^2(\Gamma_1)=0$ for each $2\le i\le n_1$.*

Adjacency spectrum of the signed NNS join

Theorem 3.2.12. *Let (Γ_1,σ_1) be an (r_1,k_1)-co-regular signed graph on n_1 vertices and let (Γ_2,σ_2) be an (r_2,k_2)-co-regular signed graph on n_2 vertices. Then the adjacency spectrum of $(\Gamma_1,\sigma_1)\,\overline{\wedge}\,(\Gamma_2,\sigma_2)$ consists of*

(i) $\lambda_j(\Gamma_2,\sigma_2)$ for each $2\le j\le n_2$,

(ii) the $2(n_1-1)$ roots of

$$\frac{\det\big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-(A_{\Gamma_1}+I_{n_1})^2\big)}{x^2-k_1x-(r_1+1)^2}=0,$$

(iii) the three roots of the cubic equation $x^3-\big(k_1+k_2\big)x^2+\big(k_1k_2-n_1n_2-(n_1-r_1-1)^2\big)x+k_2(n_1-r_1-1)^2=0$.

Proof. Denote $P_A^{\overline{\wedge}}=P(A_{(\Gamma_1,\sigma_1)\overline{\wedge}(\Gamma_2,\sigma_2)};x)$. Then, by Theorem 3.2.3, the adjacency characteristic polynomial of $(\Gamma_1,\sigma_1)\,\overline{\wedge}\,(\Gamma_2,\sigma_2)$ is

$$P_A^{\overline{\wedge}}=x^{n_1}P\big(A_{(\Gamma_2,\sigma_2)};x\big)\det\Big(xI_{n_1}-N_x^{\mathrm{c}}\Big)\Big[1-\chi_{A_{(\Gamma_2,\sigma_2)}}(x)\chi_{N_x^{\mathrm{c}}}(x)\Big],$$

where $N_x^{\mathrm{c}}=A_{(\Gamma_1,\sigma_1)}+\frac{1}{x}A_{\Gamma_1^{\mathrm{c}}}^2$.

We can use the substitution $A_{\Gamma_1^{\mathrm{c}}}=J_{n_1}-I_{n_1}-A_{\Gamma_1}$ and, since we are working with co-regular signed graphs, we can apply Lemma 1.8.4 to obtain the adjacency characteristic polynomial as

$$\begin{aligned}
P_A^{\overline{\wedge}} = {}& x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det\Big(xI_{n_1}-A_{(\Gamma_1,\sigma_1)}-\frac{1}{x}(J_{n_1}-I_{n_1}-A_{\Gamma_1})^2\Big)\\
&\Big[1-\chi_{A_{(\Gamma_2,\sigma_2)}}(x)\chi_{A_{(\Gamma_1,\sigma_1)}+\frac{1}{x}(J_{n_1}-I_{n_1}-A_{\Gamma_1})^2}(x)\Big]\\
= {}& x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)
\end{aligned}$$

$$\det\Big(xI_{n_1} - A_{(\Gamma_1,\sigma_1)} - \frac{1}{x}(J_{n_1}^2 - 2J_{n_1} + I_{n_1} - 2r_1J_{n_1} + 2A_{\Gamma_1} + A_{\Gamma_1}^2)\Big)$$
$$\left[1 - \frac{n_2}{x-k_2}\cdot\frac{n_1}{x-k_1-\frac{1}{x}(n_1-r_1-1)^2}\right]$$
$$= x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)$$
$$\det\Big(\Big(x-\frac{1}{x}\Big)I_{n_1} - A_{(\Gamma_1,\sigma_1)} - \frac{1}{x}(2A_{\Gamma_1} + A_{\Gamma_1}^2) - \frac{1}{x}(J_{n_1}^2 - 2J_{n_1} - 2r_1J_{n_1})\Big)$$
$$\left[1 - \frac{n_1n_2x}{(x-k_2)\big(x^2-k_1x-(n_1-r_1-1)^2\big)}\right].$$

Using Proposition 3.1.4, we can see that

$$\det\Big(\Big(x-\frac{1}{x}\Big)I_{n_1} - A_{(\Gamma_1,\sigma_1)} - \frac{1}{x}(2A_{\Gamma_1} + A_{\Gamma_1}^2) - \frac{1}{x}(J_{n_1}^2 - 2J_{n_1} - 2r_1J_{n_1})\Big)$$
$$= \det(B) - \frac{1}{x}(n_1-2-2r_1)\mathbf{1}_{n_1}^T\operatorname{adj}(B)\mathbf{1}_{n_1}$$

where

$$B = \Big(x-\frac{1}{x}\Big)I_{n_1} - A_{(\Gamma_1,\sigma_1)} - \frac{1}{x}(2A_{\Gamma_1} + A_{\Gamma_1}^2).$$

Making use of this along with Lemma 1.8.4 again, we have

$$P_A^{\bar{\wedge}} = x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det(B)$$
$$\left[1 - \frac{1}{x}(n_1-2-2r_1)\chi_{\frac{1}{x}I_{n_1}+A_{(\Gamma_1,\sigma_1)}+\frac{2}{x}A_{\Gamma_1}+\frac{1}{x}A_{\Gamma_1}^2}(x)\right]$$
$$\left[1 - \frac{n_1n_2x}{(x-k_2)\big(x^2-k_1x-(n_1-r_1-1)^2\big)}\right]$$
$$= x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det(B)$$
$$\left[1 - \frac{n_1-2-2r_1}{x}\cdot\frac{n_1}{x-\frac{1}{x}-k_1-\frac{2r_1}{x}-\frac{r_1^2}{x}}\right]$$
$$\left[1 - \frac{n_1n_2x}{(x-k_2)\big(x^2-k_1x-(n_1-r_1-1)^2\big)}\right]$$
$$= x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det(B)$$
$$\left[1 - \frac{n_1(n_1-2-2r_1)}{x^2-k_1x-r_1^2-2r_1-1}\right]$$
$$\left[1 - \frac{n_1n_2x}{(x-k_2)\big(x^2-k_1x-(n_1-r_1-1)^2\big)}\right]$$
$$= x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det(B)$$
$$\left[\frac{x^2-k_1x-r_1^2-2r_1-1-n_1(n_1-2-2r_1)}{x^2-k_1x-r_1^2-2r_1-1}\right]$$
$$\left[\frac{(x-k_2)\big(x^2-k_1x-(n_1-r_1-1)^2\big)-n_1n_2x}{(x-k_2)\big(x^2-k_1x-(n_1-r_1-1)^2\big)}\right]$$
$$= x^{n_1}\prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det(B)$$

$$\left[\frac{x^2-k_1x-(n_1-r_1-1)^2}{x^2-k_1x-r_1^2-2r_1-1}\cdot\frac{\big(x-k_2\big)\big(x^2-k_1x-(n_1-r_1-1)^2\big)-n_1n_2x}{\big(x-k_2\big)\big(x^2-k_1x-(n_1-r_1-1)^2\big)}\right]$$
$$= \prod_{j=1}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\det\big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-(A_{\Gamma_1}+I_{n_1})^2\big)$$
$$\left[\frac{\big(x-k_2\big)\big(x^2-k_1x-(n_1-r_1-1)^2\big)-n_1n_2x}{\big(x-k_2\big)\big(x^2-k_1x-r_1^2-2r_1-1\big)}\right],$$

where we have used $x^{n_1}\det(B)=\det\big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-(A_{\Gamma_1}+I_{n_1})^2\big)$.

Since $A_{(\Gamma_1,\sigma_1)}\mathbf{1}_{n_1}=k_1\mathbf{1}_{n_1}$ and $(A_{\Gamma_1}+I_{n_1})\mathbf{1}_{n_1}=(r_1+1)\mathbf{1}_{n_1}$, the all-one vector contributes the factor $x^2-k_1x-r_1^2-2r_1-1$ to this determinant; cancelling it together with the factor $x-k_2=x-\lambda_1(\Gamma_2,\sigma_2)$ of the product, by Remark 3.1.1, we get

$$P_A^{\bar{\wedge}}=\prod_{j=2}^{n_2}\Big(x-\lambda_j(\Gamma_2,\sigma_2)\Big)\cdot\frac{\det\big(x^2I_{n_1}-xA_{(\Gamma_1,\sigma_1)}-(A_{\Gamma_1}+I_{n_1})^2\big)}{x^2-k_1x-r_1^2-2r_1-1}$$
$$\cdot\Big[x^3-\big(k_1+k_2\big)x^2+\big(k_1k_2-n_1n_2-(n_1-r_1-1)^2\big)x+k_2(n_1-r_1-1)^2\Big].$$

Hence we have the desired result. □

Corollary 3.2.13. *When σ_1 is the all-positive signature, we have $A_{(\Gamma_1,\sigma_1)}=A_{\Gamma_1}$ and the determinant factors as $\prod_{i=1}^{n_1}\Big(x^2-\lambda_i(\Gamma_1)x-\big(\lambda_i(\Gamma_1)+1\big)^2\Big)$. Thus, Theorem 3.2.12(ii) becomes exactly the two roots of the quadratic equation $x^2-\lambda_i(\Gamma_1)x-\big(\lambda_i(\Gamma_1)+1\big)^2=0$ for each $2\le i\le n_1$.*

Laplacian spectrum of the signed NS join

Theorem 3.2.14. *Let (Γ_1,σ_1) be an (r_1,k_1)-co-regular signed graph on n_1 vertices and let (Γ_2,σ_2) be an (r_2,k_2)-co-regular signed graph on n_2 vertices. Then the Laplacian spectrum of $(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)$ consists of*

(i) $n_1+\mu_j(\Gamma_2,\sigma_2)$ for each $2\le j\le n_2$,

(ii) the $2(n_1-1)$ roots of

$$\frac{\det\big((x-r_1)(x-r_1-n_2)I_{n_1}-(x-r_1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\big)}{x^2-(2r_1+n_2+(r_1-k_1))x+n_2r_1+r_1(r_1-k_1)}=0,$$

(iii) the three roots of the cubic equation

$$x^3+\big(k_1+k_2-n_1-n_2-3r_1-r_2\big)x^2+\big(r_1^2+k_1k_2-k_1n_1-k_1r_1-k_1r_2-k_2n_2-3k_2r_1+3n_1r_1$$
$$+\,n_2r_1+n_2r_2+3r_1r_2\big)x+k_2r_1^2-n_1r_1^2-r_1^2r_2-k_1k_2r_1+k_1r_1r_2+k_2n_2r_1-n_2r_1r_2+k_1n_1r_1=0.$$

Proof. Denote $P_L^{\veebar}=P(L_{(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)};x)$. Then, by Theorem 3.2.4, the Laplacian characteristic polynomial of $(\Gamma_1,\sigma_1)\veebar(\Gamma_2,\sigma_2)$ is

$$P_L^{\veebar}=\det\Big((x-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\Big)\det\Big(xI_{n_1}-D_{\Gamma_1}\Big)$$
$$\det\Big((x-n_2)I_{n_1}-M_x\Big)\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(x-n_1)\chi_{M_x}(x-n_2)\Big],$$

where $M_x=L_{(\Gamma_1,\sigma_1)}+D_{\Gamma_1}+A_{\Gamma_1}\big(xI_{n_1}-D_{\Gamma_1}\big)^{-1}A_{\Gamma_1}$.

Since we are working with co-regular signed graphs, we can use the substitution $D_{\Gamma_1}=r_1I_{n_1}$ and then apply Lemma 1.8.4 to obtain the Laplacian characteristic polynomial as

$$P_L^{\veebar}=\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\det\Big((x-r_1)I_{n_1}\Big)$$
$$\det\Big((x-r_1-n_2)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}((x-r_1)I_{n_1})^{-1}A_{\Gamma_1}\Big)$$

$$
\begin{aligned}
&\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(x-n_1)\chi_{L_{(\Gamma_1,\sigma_1)}+r_1I_{n_1}+A_{\Gamma_1}((x-r_1)I_{n_1})^{-1}A_{\Gamma_1}}(x-n_2)\Big]\\
=\;&(x-r_1)^{n_1}\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\\
&\det\Big((x-r_1-n_2)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-\frac{1}{x-r_1}A_{\Gamma_1}^2\Big)\\
&\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(x-n_1)\chi_{L_{(\Gamma_1,\sigma_1)}+\frac{1}{x-r_1}A_{\Gamma_1}^2}(x-r_1-n_2)\Big]\\
=\;&(x-r_1)^{n_1}\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\det\Big((x-r_1-n_2)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-\frac{1}{x-r_1}A_{\Gamma_1}^2\Big)\\
&\left[1-\frac{n_2}{x-n_1-(r_2-k_2)}\cdot\frac{n_1}{x-r_1-n_2-(r_1-k_1)-\frac{r_1^2}{x-r_1}}\right]\\
=\;&\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\det\Big((x-r_1)(x-r_1-n_2)I_{n_1}-(x-r_1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\Big)\\
&\left[1-\frac{n_1n_2(x-r_1)}{\big(x-n_1-(r_2-k_2)\big)\big(x^2-(2r_1+n_2+(r_1-k_1))x+n_2r_1+r_1(r_1-k_1)\big)}\right],
\end{aligned}
$$

where we have used

$$(x-r_1)^{n_1}\det\Big((x-r_1-n_2)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-\frac{1}{x-r_1}A_{\Gamma_1}^2\Big)=\det\Big((x-r_1)(x-r_1-n_2)I_{n_1}-(x-r_1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\Big).$$

Since $L_{(\Gamma_1,\sigma_1)}\mathbf{1}_{n_1}=(r_1-k_1)\mathbf{1}_{n_1}$ and $A_{\Gamma_1}\mathbf{1}_{n_1}=r_1\mathbf{1}_{n_1}$, the all-one vector contributes the factor $x^2-(2r_1+n_2+(r_1-k_1))x+n_2r_1+r_1(r_1-k_1)$ to this determinant; cancelling it together with the factor $x-n_1-(r_2-k_2)=x-n_1-\mu_1(\Gamma_2,\sigma_2)$ of the product, by Remark 3.1.2, we get

$$
\begin{aligned}
P_L^{\vee}=\;&\prod_{j=2}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\cdot\frac{\det\big((x-r_1)(x-r_1-n_2)I_{n_1}-(x-r_1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\big)}{x^2-(2r_1+n_2+(r_1-k_1))x+n_2r_1+r_1(r_1-k_1)}\\
&\cdot\Big[\big(x-n_1-(r_2-k_2)\big)\big(x^2-(2r_1+n_2+(r_1-k_1))x+n_2r_1+r_1(r_1-k_1)\big)-n_1n_2(x-r_1)\Big]\\
=\;&\prod_{j=2}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\cdot\frac{\det\big((x-r_1)(x-r_1-n_2)I_{n_1}-(x-r_1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1}^2\big)}{x^2-(2r_1+n_2+(r_1-k_1))x+n_2r_1+r_1(r_1-k_1)}\\
&\cdot\Big[x^3+\big(k_1+k_2-n_1-n_2-3r_1-r_2\big)x^2\\
&\quad+\big(r_1^2+k_1k_2-k_1n_1-k_1r_1-k_1r_2-k_2n_2-3k_2r_1+3n_1r_1+n_2r_1+n_2r_2+3r_1r_2\big)x\\
&\quad+k_2r_1^2-n_1r_1^2-r_1^2r_2-k_1k_2r_1+k_1r_1r_2+k_2n_2r_1-n_2r_1r_2+k_1n_1r_1\Big].
\end{aligned}
$$

Hence we have the desired result. □

Corollary 3.2.15. *When σ_1 is the all-positive signature, we have $L_{(\Gamma_1,\sigma_1)}=L_{\Gamma_1}$ and the determinant factors as $\prod_{i=1}^{n_1}\Big(x^2-(n_2+2r_1+\mu_i(\Gamma_1))x+r_1\mu_i(\Gamma_1)-\mu_i^2(\Gamma_1)+2r_1\mu_i(\Gamma_1)+n_2r_1\Big)$. Thus, Theorem 3.2.14(ii) becomes exactly the two roots of the quadratic equation*

$$x^2-(n_2+2r_1+\mu_i(\Gamma_1))x+r_1\mu_i(\Gamma_1)-\mu_i^2(\Gamma_1)+2r_1\mu_i(\Gamma_1)+n_2r_1=0$$

for each $2\le i\le n_1$.

Laplacian spectrum of the signed NNS join

Theorem 3.2.16. *Let (Γ_1,σ_1) be an (r_1,k_1)-co-regular signed graph on n_1 vertices and let (Γ_2,σ_2) be an (r_2,k_2)-co-regular signed graph on n_2 vertices. Then the Laplacian spectrum of $(\Gamma_1,\sigma_1)\,\overline{\wedge}\,(\Gamma_2,\sigma_2)$ consists of*

(i) $n_1+\mu_j(\Gamma_2,\sigma_2)$ for each $2\le j\le n_2$,

(ii) the $2(n_1-1)$ *roots of*

$$\frac{\det\big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1}-(x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)}-(A_{\Gamma_1}+I_{n_1})^2\big)}{(x-n_1+r_1+1)(x-n_1-n_2+k_1+1)-(r_1+1)^2}=0,$$

(iii) the three roots of the cubic equation

$$\begin{aligned}x^3+\big(&k_1+k_2-3n_1-n_2+r_1-r_2+2\big)x^2+\big(2n_1^2-r_1^2+k_1+2k_2-2n_1-n_2-r_1-2r_2+k_1k_2\\&-2k_1n_1+k_1r_1-k_1r_2-2k_2n_1-k_2n_2+k_2r_1+n_1n_2+2n_1r_2-n_2r_1+n_2r_2-r_1r_2\big)x\\&+k_1n_1^2-k_2r_1^2-n_1^2r_1+n_1r_1^2+r_1^2r_2+k_1k_2-k_1n_1-k_1r_2-k_2n_2-k_2r_1+n_1r_1\\&+n_2r_2+r_1r_2-k_1k_2n_1+k_1k_2r_1-k_1n_1r_1+k_1n_1r_2-k_1r_1r_2+k_2n_1n_2\\&+k_2n_1r_1-k_2n_2r_1-n_1n_2r_2-n_1r_1r_2+n_2r_1r_2=0.\end{aligned}$$

Proof. Denote $P_L^{\bar\wedge}=P(L_{(\Gamma_1,\sigma_1)\bar\wedge(\Gamma_2,\sigma_2)};x)$. Then, by Theorem 3.2.4, the Laplacian characteristic polynomial of $(\Gamma_1,\sigma_1)\,\bar\wedge\,(\Gamma_2,\sigma_2)$ is

$$\begin{aligned}P_L^{\bar\wedge}=\;&\det\Big((x-n_1)I_{n_2}-L_{(\Gamma_2,\sigma_2)}\Big)\det\Big((x-n_1+1)I_{n_1}+D_{\Gamma_1}\Big)\\&\det\Big((x-n_1-n_2+1)I_{n_1}-M_x^{\mathrm{c}}\Big)\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(x-n_1)\chi_{M_x^{\mathrm{c}}}(x-n_1-n_2+1)\Big],\end{aligned}$$

where $M_x^{\mathrm{c}}=L_{(\Gamma_1,\sigma_1)}-D_{\Gamma_1}+A_{\Gamma_1^{\mathrm{c}}}((x-n_1+1)I_{n_1}+D_{\Gamma_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}$.

Since we are working with co-regular signed graphs, we can use the substitution $D_{\Gamma_1}=r_1I_{n_1}$ and then apply Lemma 1.8.4 to obtain the Laplacian characteristic polynomial as

$$\begin{aligned}P_L^{\bar\wedge}=\;&\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\det\Big((x-n_1+r_1+1)I_{n_1}\Big)\\&\det\Big((x-n_1-n_2+r_1+1)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1^{\mathrm{c}}}((x-n_1+r_1+1)I_{n_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}\Big)\\&\Big[1-\chi_{L_{(\Gamma_2,\sigma_2)}}(x-n_1)\chi_{L_{(\Gamma_1,\sigma_1)}+A_{\Gamma_1^{\mathrm{c}}}((x-n_1+r_1+1)I_{n_1})^{-1}A_{\Gamma_1^{\mathrm{c}}}}(x-n_1-n_2+r_1+1)\Big]\\=\;&(x-n_1+r_1+1)^{n_1}\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\\&\det\Big((x-n_1-n_2+r_1+1)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-\frac{1}{x-n_1+r_1+1}A_{\Gamma_1^{\mathrm{c}}}^2\Big)\\&\left[1-\frac{n_2}{x-n_1-(r_2-k_2)}\cdot\frac{n_1}{x-n_1-n_2+r_1+1-(r_1-k_1)-\frac{(n_1-r_1-1)^2}{x-n_1+r_1+1}}\right]\\=\;&\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\\&\det\Big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1}-(x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1^{\mathrm{c}}}^2\Big)\\&\left[1-\frac{n_1n_2(x-n_1+r_1+1)}{\big(x-n_1-(r_2-k_2)\big)\big((x-n_1-n_2+k_1+1)(x-n_1+r_1+1)-(n_1-r_1-1)^2\big)}\right],\end{aligned}$$

where we have used

$$\begin{aligned}&(x-n_1+r_1+1)^{n_1}\det\Big((x-n_1-n_2+r_1+1)I_{n_1}-L_{(\Gamma_1,\sigma_1)}-\frac{1}{x-n_1+r_1+1}A_{\Gamma_1^{\mathrm{c}}}^2\Big)\\&\qquad=\det\Big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1}-(x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)}-A_{\Gamma_1^{\mathrm{c}}}^2\Big),\end{aligned}$$

and in the bracket we have simplified $x-n_1-n_2+r_1+1-(r_1-k_1)=x-n_1-n_2+k_1+1$.

Now substitute $A_{\Gamma_1^{\mathrm{c}}}=J_{n_1}-I_{n_1}-A_{\Gamma_1}$. Then

$$A_{\Gamma_1^{\mathrm{c}}}^2=(J_{n_1}-I_{n_1}-A_{\Gamma_1})^2$$

$$= J_{n_1}^2 + I_{n_1} + A_{\Gamma_1}^2 - 2J_{n_1} - 2J_{n_1}A_{\Gamma_1} + 2A_{\Gamma_1}$$
$$= (n_1 - 2 - 2r_1)J_{n_1} + I_{n_1} + 2A_{\Gamma_1} + A_{\Gamma_1}^2.$$

Thus the determinant becomes

$$\begin{aligned}&\det\Big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1} - (x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)}\\ &\quad - (n_1-2-2r_1)J_{n_1} - I_{n_1} - 2A_{\Gamma_1} - A_{\Gamma_1}^2\Big)\\ =\ &\det\Big(\big[(x-n_1+r_1+1)(x-n_1-n_2+r_1+1)-1\big]I_{n_1} - (x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)}\\ &\quad - 2A_{\Gamma_1} - A_{\Gamma_1}^2 - (n_1-2-2r_1)J_{n_1}\Big).\end{aligned}$$

Now use $L_{(\Gamma_1,\sigma_1)} = r_1 I_{n_1} - A_{(\Gamma_1,\sigma_1)}$ and $(A_{\Gamma_1} + I_{n_1})^2 = I_{n_1} + 2A_{\Gamma_1} + A_{\Gamma_1}^2$. We can rewrite

$$\begin{aligned}&\big[(x-n_1+r_1+1)(x-n_1-n_2+r_1+1)-1\big]I_{n_1} - (x-n_1+r_1+1)(r_1 I_{n_1} - A_{(\Gamma_1,\sigma_1)})\\ &\quad - (A_{\Gamma_1}+I_{n_1})^2 + I_{n_1} - (n_1-2-2r_1)J_{n_1}.\end{aligned}$$

The I_{n_1} terms combine as

$$\begin{aligned}&\big[(x-n_1+r_1+1)(x-n_1-n_2+r_1+1)-1\big] - (x-n_1+r_1+1)r_1 + 1\\ &= (x-n_1+r_1+1)(x-n_1-n_2+r_1+1-r_1)\\ &= (x-n_1+r_1+1)(x-n_1-n_2+1).\end{aligned}$$

Thus the determinant simplifies to

$$\begin{aligned}&\det\Big((x-n_1+r_1+1)(x-n_1-n_2+1)I_{n_1} + (x-n_1+r_1+1)A_{(\Gamma_1,\sigma_1)}\\ &\quad - (A_{\Gamma_1}+I_{n_1})^2 - (n_1-2-2r_1)J_{n_1}\Big)\\ =\ &\det\Big(M - (n_1-2-2r_1)J_{n_1}\Big),\end{aligned}$$

where

$$M = (x-n_1+r_1+1)(x-n_1-n_2+1)I_{n_1} + (x-n_1+r_1+1)A_{(\Gamma_1,\sigma_1)} - (A_{\Gamma_1}+I_{n_1})^2.$$

Since $A_{(\Gamma_1,\sigma_1)}\mathbf{1}_{n_1} = k_1\mathbf{1}_{n_1}$ and $(A_{\Gamma_1}+I_{n_1})\mathbf{1}_{n_1} = (r_1+1)\mathbf{1}_{n_1}$, the matrix M has constant row sum

$$\delta(x) = (x-n_1+r_1+1)(x-n_1-n_2+1) + (x-n_1+r_1+1)k_1 - (r_1+1)^2.$$

By Proposition 3.1.4,

$$\begin{aligned}\det(M-(n_1-2-2r_1)J_{n_1}) &= \det(M) - (n_1-2-2r_1)\,\mathbf{1}_{n_1}^T \operatorname{adj}(M)\mathbf{1}_{n_1}\\ &= \det(M)\left[1 - \frac{n_1(n_1-2-2r_1)}{\delta(x)}\right]\\ &= \det(M)\cdot\frac{\delta(x) - n_1(n_1-2-2r_1)}{\delta(x)}.\end{aligned}$$

We simplify the numerator. First,

$$\delta(x) = (x-n_1+r_1+1)(x-n_1-n_2+1+k_1) - (r_1+1)^2 = (x-n_1+r_1+1)(x-n_1-n_2+k_1+1) - (r_1+1)^2.$$

Then

$$\begin{aligned}\delta(x) - n_1(n_1-2-2r_1) &= (x-n_1+r_1+1)(x-n_1-n_2+k_1+1) - (r_1+1)^2 - n_1^2 + 2n_1 + 2n_1r_1\\ &= (x-n_1+r_1+1)(x-n_1-n_2+k_1+1) - (n_1-r_1-1)^2.\end{aligned}$$

For the determinant $\det(M)$, we reverse the earlier simplification. Using $A_{(\Gamma_1,\sigma_1)} = r_1 I_{n_1} - L_{(\Gamma_1,\sigma_1)}$, we have

$$\begin{aligned} M &= (x-n_1+r_1+1)(x-n_1-n_2+1)I_{n_1} + (x-n_1+r_1+1)(r_1 I_{n_1} - L_{(\Gamma_1,\sigma_1)}) - (A_{\Gamma_1}+I_{n_1})^2 \\ &= (x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1} - (x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)} - (A_{\Gamma_1}+I_{n_1})^2. \end{aligned}$$

Thus $\det(M) = \det\big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1} - (x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)} - (A_{\Gamma_1}+I_{n_1})^2\big)$.

Substituting everything back,

$$\begin{aligned} P_L^{\bar{\wedge}} = &\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\cdot\det(M)\cdot\frac{(x-n_1+r_1+1)(x-n_1-n_2+k_1+1)-(n_1-r_1-1)^2}{\delta(x)} \\ &\cdot\left[1-\frac{n_1n_2(x-n_1+r_1+1)}{\big(x-n_1-(r_2-k_2)\big)\big((x-n_1-n_2+k_1+1)(x-n_1+r_1+1)-(n_1-r_1-1)^2\big)}\right] \\ = &\prod_{j=1}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\cdot\frac{\det(M)}{\delta(x)} \\ &\cdot\left[\frac{\big(x-n_1-(r_2-k_2)\big)\big((x-n_1-n_2+k_1+1)(x-n_1+r_1+1)-(n_1-r_1-1)^2\big)-n_1n_2(x-n_1+r_1+1)}{x-n_1-(r_2-k_2)}\right]. \end{aligned}$$

Since $L_{(\Gamma_1,\sigma_1)}\mathbf{1}_{n_1} = (r_1-k_1)\mathbf{1}_{n_1}$ and $(A_{\Gamma_1}+I_{n_1})\mathbf{1}_{n_1} = (r_1+1)\mathbf{1}_{n_1}$, the all-one vector contributes the factor $\delta(x)$ to $\det(M)$; cancelling it together with the factor $x-n_1-(r_2-k_2) = x-n_1-\mu_1(\Gamma_2,\sigma_2)$ of the product, by Remark 3.1.2, we get

$$\begin{aligned} P_L^{\bar{\wedge}} = &\prod_{j=2}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big)\cdot\frac{\det(M)}{\delta(x)} \\ &\cdot\Big[\big(x-n_1-(r_2-k_2)\big)\big((x-n_1-n_2+k_1+1)(x-n_1+r_1+1)-(n_1-r_1-1)^2\big) \\ &\qquad -n_1n_2(x-n_1+r_1+1)\Big] \\ = &\prod_{j=2}^{n_2}\Big(x-n_1-\mu_j(\Gamma_2,\sigma_2)\Big) \\ &\cdot\frac{\det\big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)I_{n_1}-(x-n_1+r_1+1)L_{(\Gamma_1,\sigma_1)}-(A_{\Gamma_1}+I_{n_1})^2\big)}{(x-n_1+r_1+1)(x-n_1-n_2+k_1+1)-(r_1+1)^2} \\ &\cdot\Big[x^3+\big(k_1+k_2-3n_1-n_2+r_1-r_2+2\big)x^2+\big(2n_1^2-r_1^2+k_1+2k_2-2n_1-n_2-r_1-2r_2 \\ &\quad+\ k_1k_2-2k_1n_1+k_1r_1-k_1r_2-2k_2n_1-k_2n_2+k_2r_1+n_1n_2+2n_1r_2-n_2r_1+n_2r_2 \\ &\quad-\ r_1r_2\big)x+k_1n_1^2-k_2r_1^2-n_1^2r_1+n_1r_1^2+r_1^2r_2+k_1k_2-k_1n_1-k_1r_2-k_2n_2-k_2r_1+n_1r_1 \\ &\quad+\ n_2r_2+r_1r_2-k_1k_2n_1+k_1k_2r_1-k_1n_1r_1+k_1n_1r_2-k_1r_1r_2+k_2n_1n_2+k_2n_1r_1 \\ &\quad-\ k_2n_2r_1-n_1n_2r_2-n_1r_1r_2+n_2r_1r_2\Big]. \end{aligned}$$

Hence we have the desired result. □

Corollary 3.2.17. *When σ_1 is the all-positive signature, we have $L_{(\Gamma_1,\sigma_1)} = L_{\Gamma_1}$ and the determinant factors as*

$$\prod_{i=1}^{n_1}\Big((x-n_1+r_1+1)(x-n_1-n_2+r_1+1)-(x-n_1+r_1+1)\mu_i(\Gamma_1)-(\lambda_i(\Gamma_1)+1)^2\Big).$$

Thus, Theorem 3.2.16(ii) becomes exactly the two roots of the quadratic equation

$$(x-n_1+r_1+1)(x-n_1-n_2+r_1+1)-(x-n_1+r_1+1)\mu_i(\Gamma_1)-(\lambda_i(\Gamma_1)+1)^2=0$$

for each $2 \le i \le n_1$.

3.3 Switching-stable signed NS and NNS joins

It can be shown, see, for example, [82], where the result is established in the more general setting of complex unit gain graphs, that a signed graph (Γ, σ) is balanced if and only if it admits a potential function, that is, a function $U\colon V_\Gamma \to \{+1, -1\}$ satisfying $\sigma(uv) = U(u)U(v)$ for every pair of adjacent vertices $u, v \in V_\Gamma$.

While the joins studied in the previous section achieve the goal of enabling constructions of non-regular cospectral signed graphs, the operations themselves are not stable under switchings of their component parts. In this section we use pseudo-potential functions defined by Cavaleri et al. [33] and discussed in Chapter 1 to define versions of the NS and NNS joins that are switching stable.

3.3.1 Switching-stable definitions

Following the work done in [33], where pseudo-potentials are applied to build a definition of the lexicographic product of signed graphs which is stable under switching equivalence, we can develop a method for defining switching-stable versions of the signed NS and NNS joins.

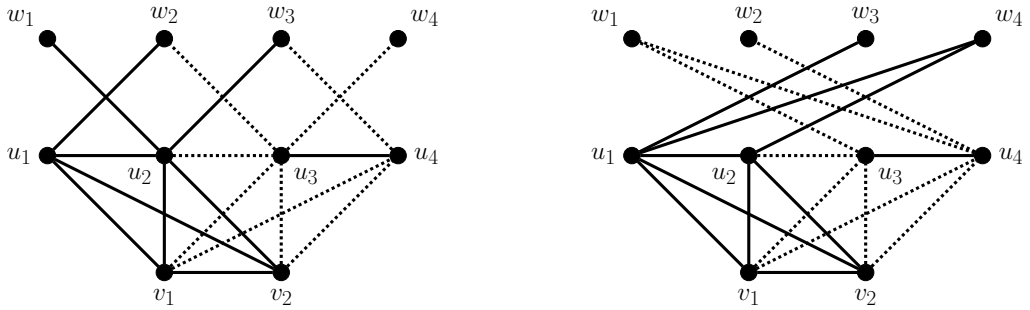


Figure 3.4: The switching-stable NS and NNS joins $(P_4, \sigma_1) \veebar_{\mathcal{T},\mathbf{v}} (P_2, \sigma_2)$ and $(P_4, \sigma_1) \barwedge_{\mathcal{T},\mathbf{v}} (P_2, \sigma_2)$, where $\mathcal{T}_1 = P_4$, $\mathcal{T}_2 = P_2$, and $\mathbf{v} = \{u_1, v_1\}$.

Let (Γ_1, σ_1) and (Γ_2, σ_2) be two vertex-disjoint signed graphs of order n_1 and n_2, respectively, and write $V_{\Gamma_1} = \{u_1, u_2, \ldots, u_{n_1}\}$ and $V_{\Gamma_2} = \{v_1, v_2, \ldots, v_{n_2}\}$.

Fix spanning trees $\mathcal{T}_1$ of Γ_1 and $\mathcal{T}_2$ of Γ_2 (with the usual convention that disconnected graphs are handled by taking spanning forests, one root per component) and set $\mathcal{T} = \mathcal{T}_1 \cup \mathcal{T}_2$, fix vertices $\mathbf{v} = \{u_i, v_j\}$ with u_i in V_{Γ_1} and v_j in V_{Γ_2}, and take the pseudo-potential functions $U_1(u_k) = \sigma_1(\gamma_{u_i u_k})$ for each u_k in V_{Γ_1} and $U_2(v_l) = \sigma_2(\gamma_{v_j v_l})$ for each v_l in V_{Γ_2}.

Introduce the edgeless graph on n_1 vertices, which we shall label $(\Gamma_3, \sigma_3) = (K_{n_1}^{\mathrm{c}}, \varnothing)$, where $\varnothing$ is the empty function and $V_{\Gamma_3} = \{w_1, w_2, \ldots, w_{n_1}\}$. As there are no edges in this graph, we shall take the pseudo-potential values to be $U_3(w_i) = 1$ for each $i = 1, 2, \ldots, n_1$.

Definition 3.3.1. The switching-stable signed neighbours splitting (ss-NS) join of (Γ_1, σ_1) and (Γ_2, σ_2), denoted $(\Gamma_1, \sigma_1) \veebar_{\mathcal{T},\mathbf{v}} (\Gamma_2, \sigma_2)$, is the signed graph obtained by taking the same underlying graph as the standard NS join and applying the signature

$$\sigma_{\mathcal{T},\mathbf{v}}(e) = \begin{cases} \sigma_1(e), & \text{if } e \in E_{\Gamma_1}, \\ \sigma_2(e), & \text{if } e \in E_{\Gamma_2}, \\ U_1(u_i)U_2(v_j), & \text{if } e \in \{u_i v_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\}, \\ U_1(u_i)U_3(w_j), & \text{if } e \in \{u_i w_j : u_i u_j \in E_{\Gamma_1}\}. \end{cases}$$

Definition 3.3.2. The switching-stable signed non-neighbours splitting (ss-NNS) join of (Γ_1, σ_1) and (Γ_2, σ_2), denoted $(\Gamma_1, \sigma_1) \barwedge_{\mathcal{T},\mathbf{v}} (\Gamma_2, \sigma_2)$, is the signed graph obtained by taking the same underlying graph

as the standard NNS join and applying the signature

$$\sigma_{\mathcal{T},\mathbf{v}}(e) = \begin{cases} \sigma_1(e), & \text{if } e \in E_{\Gamma_1}, \\ \sigma_2(e), & \text{if } e \in E_{\Gamma_2}, \\ U_1(u_i)U_2(v_j), & \text{if } e \in \{u_iv_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\}, \\ U_1(u_i)U_3(w_j), & \text{if } e \in \{u_iw_j : i \neq j \text{ and } u_iu_j \notin E_{\Gamma_1}\}. \end{cases}$$

Now that we have the definitions, we can present the main theorem of this section.

Theorem 3.3.3. *Let (Γ_1, σ_1) and (Γ_2, σ_2) be two vertex-disjoint signed graphs on n_1 and n_2 vertices, respectively. Then the ss-NS and ss-NNS join operations of (Γ_1, σ_1) and (Γ_2, σ_2) are switching stable. That is, replacing either σ_1 or σ_2 with some switching-equivalent σ_1' or σ_2' leaves the switching classes of $(\Gamma_1, \sigma_1) \veebar_{\mathcal{T},\mathbf{v}} (\Gamma_2, \sigma_2)$ and $(\Gamma_1, \sigma_1) \barwedge_{\mathcal{T},\mathbf{v}} (\Gamma_2, \sigma_2)$ unchanged.*

Proof. Since the proofs for the two joins are quite similar, we shall present a full proof only for the ss-NS join.

Suppose that we replace σ_1 and σ_2 with switching-equivalent signatures σ_1' and σ_2', respectively. Then there exist functions $s_1 \colon V_{\Gamma_1} \to \{+1, -1\}$ and $s_2 \colon V_{\Gamma_2} \to \{+1, -1\}$ such that

$$\sigma_1'(u_iu_j) = s_1(u_i)\sigma_1(u_iu_j)s_1(u_j) \quad \text{and} \quad \sigma_2'(v_iv_j) = s_2(v_i)\sigma_2(v_iv_j)s_2(v_j).$$

Since there are no edges to consider in $(\Gamma_3, \sigma_3) = (K_{n_1}^{\mathrm{c}}, \varnothing)$, we may choose an arbitrary function $s_3 \colon V_{\Gamma_3} \to \{+1, -1\}$; for definiteness, take $s_3 \equiv 1$.

By Remark 1.7.7, we may normalise the switching functions so that $s_1(u_i) = 1$ and $s_2(v_j) = 1$ without changing the switching classes of σ_1' and σ_2'. With this normalisation, Corollary 1.7.6(i) gives

$$U_1'(u) = U_1(u)s_1(u), \qquad U_2'(v) = U_2(v)s_2(v).$$

Since Γ_3 is edgeless, its pseudo-potential is identically 1 and is unaffected by switching; we may choose $s_3 \equiv 1$, so that $U_3'(w) = U_3(w) = 1$.

With this, the new signature of the ss-NS join is

$$\sigma'_{\mathcal{T},\mathbf{v}}(e) = \begin{cases} s_1(u_i)\sigma_1(u_iu_j)s_1(u_j), & \text{if } e = u_iu_j \in E_{\Gamma_1}, \\ s_2(v_i)\sigma_2(v_iv_j)s_2(v_j), & \text{if } e = v_iv_j \in E_{\Gamma_2}, \\ U_1(u_i)s_1(u_i) \cdot s_2(v_j)U_2(v_j), & \text{if } e \in \{u_iv_j : u_i \in V_{\Gamma_1}, v_j \in V_{\Gamma_2}\}, \\ U_1(u_i)s_1(u_i) \cdot s_3(w_j)U_3(w_j), & \text{if } e \in \{u_iw_j : u_iu_j \in E_{\Gamma_1}\}. \end{cases}$$

Note that all factors belong to $\{+1, -1\}$, so they commute. This allows us to group the s factors on the outside.

Comparing with the original signature $\sigma_{\mathcal{T},\mathbf{v}}$ from Definition 3.3.1, we see that

$$\sigma'_{\mathcal{T},\mathbf{v}}(xy) = s(x)\sigma_{\mathcal{T},\mathbf{v}}(xy)s(y)$$

for every edge xy in the ss-NS join, where

$$s(x) = \begin{cases} s_1(x), & \text{if } x \in V_{\Gamma_1}, \\ s_2(x), & \text{if } x \in V_{\Gamma_2}, \\ s_3(x), & \text{if } x \in V_{\Gamma_3}. \end{cases}$$

Thus $\sigma'_{\mathcal{T},\mathbf{v}}$ is switching equivalent to $\sigma_{\mathcal{T},\mathbf{v}}$, so the switching class of the join is unchanged. Hence the ss-NS join operation is switching stable. The proof for the ss-NNS join follows the same argument, using Definition 3.3.2 in place of Definition 3.3.1. □

Since the operations are switching stable, to compute the spectra of the ss-NS and ss-NNS joins, one can, without loss of generality, consider the graphs obtained by replacing (Γ_1, σ_1) and (Γ_2, σ_2) with the $\mathcal{T}$-positive representatives of their switching classes, which we know exist (and are unique) thanks to

Lemma 1.7.11. When (Γ_1, σ_1) and (Γ_2, σ_2) are $\mathcal{T}$-positive, every edge in $\mathcal{T}_1$ and $\mathcal{T}_2$ has signature $+1$, so the pseudo-potentials satisfy $U_1(u) = 1$ for all $u \in V_{\Gamma_1}$ and $U_2(v) = 1$ for all $v \in V_{\Gamma_2}$. Hence the signature on every newly added edge reduces to $+1$, and the ss-NS and ss-NNS joins coincide with the standard NS and NNS joins of Section 3.2. Since switching equivalence preserves spectra, all spectral results from that section apply unchanged.

3.3.2 Balancedness of the various joins

We now present characterisations of balancedness for the neighbours splitting and non-neighbours splitting joins, considering both the standard and switching-stable variations. Consequently, we see that the conditions are less restrictive in the cases of the switching-stable signatures.

Theorem 3.3.4. *Let (Γ_1, σ_1) and (Γ_2, σ_2) be two vertex-disjoint signed graphs. Then the join graphs $(\Gamma_1, \sigma_1) \veebar (\Gamma_2, \sigma_2)$ and $(\Gamma_1, \sigma_1) \barwedge (\Gamma_2, \sigma_2)$ are balanced if and only if the signatures on (Γ_1, σ_1) and (Γ_2, σ_2) are both all-positive.*

Proof. If both signatures are all-positive, then all edges in both joins are positive, so both joins are balanced.

Conversely, suppose that one of the joins is balanced. Let $u_iu_j \in E_{\Gamma_1}$ and $v_k \in V_{\Gamma_2}$. Both joins contain the triangle $u_iu_jv_k$, whose sign is

$$\sigma(u_iu_j)\sigma(u_jv_k)\sigma(v_ku_i) = \sigma_1(u_iu_j)(+1)(+1) = \sigma_1(u_iu_j),$$

so balancedness forces $\sigma_1(u_iu_j) = +1$ for every edge of Γ_1. Symmetrically, let $v_kv_l \in E_{\Gamma_2}$ and $u_i \in V_{\Gamma_1}$. Now both joins contain the triangle $u_iv_kv_l$, whose sign is $\sigma_2(v_kv_l)(+1)(+1) = \sigma_2(v_kv_l)$, and so for balance we must have $\sigma_2(v_kv_l) = +1$ for every edge of Γ_2. Hence both signatures are all-positive, this being vacuously true for an edgeless factor. □

Theorem 3.3.5. *The join graphs $(\Gamma_1, \sigma_1) \veebar_{\mathcal{T},\mathbf{v}} (\Gamma_2, \sigma_2)$ and $(\Gamma_1, \sigma_1) \barwedge_{\mathcal{T},\mathbf{v}} (\Gamma_2, \sigma_2)$ are balanced if and only if (Γ_1, σ_1) and (Γ_2, σ_2) are balanced. Moreover, when the joins are balanced, their signatures are independent of the choice of $\mathcal{T}$.*

Proof. We prove the result for the ss-NS join; the proof for the ss-NNS join is identical.

Suppose first that (Γ_1, σ_1) and (Γ_2, σ_2) are balanced. Then each is switching equivalent to an all-positive signature. By Theorem 3.3.3, replacing σ_1 and σ_2 with these all-positive representatives does not change the switching class of the join. When both signatures are all-positive, every edge in $\mathcal{T}_1$ and $\mathcal{T}_2$ has signature $+1$, so the pseudo-potentials satisfy $U_1(u) = 1$ for all $u \in V_{\Gamma_1}$ and $U_2(v) = 1$ for all $v \in V_{\Gamma_2}$. Hence the signature on every newly added edge is $+1$, and the join itself is all-positive, hence balanced. Since switching equivalence preserves balancedness, the original join is also balanced.

Conversely, suppose that the join is balanced. Since E_{Γ_1} is a subset of the edge set of the join, any closed walk in (Γ_1, σ_1) is also a closed walk in the join and must therefore have positive sign. Hence (Γ_1, σ_1) is balanced, and the same argument applies to (Γ_2, σ_2).

For the $\mathcal{T}$-independence, note that by Lemma 1.7.8, the pseudo-potentials of a balanced signed graph are independent of the choice of spanning tree. Since the join signature on the newly added edges is defined solely in terms of these pseudo-potentials, it too is independent of $\mathcal{T}$. □

Consider the graphs depicted in Figures 3.2 and 3.4. The same constituent parts $(\Gamma_1, \sigma_1) = (P_4, \sigma_1)$ and $(\Gamma_2, \sigma_2) = (P_2, \sigma_2)$ are used in each case, and they are both balanced, though (Γ_1, σ_1) does not have an all-positive signature. In Figure 3.2 the two products are unbalanced, conforming with Theorem 3.3.4, whereas the products in Figure 3.4 are balanced, as predicted by Theorem 3.3.5.

3.4 Summary and outlook

In this chapter, we have introduced two new signed graph operations: the neighbours splitting (NS) and non-neighbours splitting (NNS) joins. We have derived the adjacency and Laplacian characteristic polynomials and spectra for graphs with suitable properties, enabling the construction of infinite families

of non-isomorphic cospectral signed pairs. By further developing pseudo-potential techniques, we defined switching-stable variants of both joins that preserve spectral properties under switching equivalence while relaxing the conditions for balancedness.

Taking the definition provided by Chung [36] to signed graphs, the normalised Laplacian matrix $\mathcal{L}_{(\Gamma,\sigma)}$ of a signed graph (Γ,σ) is defined by $\mathcal{L}_{(\Gamma,\sigma)} = D_\Gamma^{-\frac{1}{2}} L_{(\Gamma,\sigma)} D_\Gamma^{-\frac{1}{2}}$ with the convention that the ith diagonal entry of $D_\Gamma^{-\frac{1}{2}}$ is $(d_i)^{-\frac{1}{2}}$ and if the degree of vertex v_i in Γ is zero then $(d_i)^{-\frac{1}{2}} = 0$. In other words,

$$(\mathcal{L}_{(\Gamma,\sigma)})_{ij} = \begin{cases} 1 & \text{if } i = j \text{ and } d_i \neq 0, \\ -\sigma(e_{ij})\frac{1}{\sqrt{d_i d_j}} & \text{if } i \neq j \text{ and } v_i \text{ is adjacent to } v_j, \\ 0 & \text{otherwise.} \end{cases}$$

If the signed graph (Γ,σ) in question is without isolated vertices then $\mathcal{L}_{(\Gamma,\sigma)} = I - D_\Gamma^{-\frac{1}{2}} A_{(\Gamma,\sigma)} D_\Gamma^{-\frac{1}{2}}$.

Applying the same methods used in Section 3.2, it is possible to compute the normalised Laplacian characteristic polynomials of the NS and NNS joins when (Γ_1,σ_1) and (Γ_2,σ_2) are on regular underlying graphs. When considering two arbitrary signed graphs, the calculations become much more involved.

Open Problem 3.4.1. Compute the characteristic polynomials of the normalised Laplacian matrices of the signed NS and NNS joins of any two arbitrary signed graphs.

The spectra provided in this chapter are restricted to joins of (r,k)-co-regular signed graphs. This allows for the use of Lemma 1.8.4 when doing the computations and thus makes the work simpler to carry out. It would be ideal if these, along with their normalised Laplacian analogues, could be generalised completely.

Open Problem 3.4.2. Compute the adjacency, Laplacian, and normalised Laplacian spectra of the signed NS and NNS joins of any two arbitrary signed graphs.

Chapter 4

A switching-stable corona product of signed graphs

The aim of this chapter is to introduce a notion of the corona product for signed graphs that is switching stable, that is, replacing the two input signed graphs with switching-equivalent ones does not change the switching class of the resulting product. Although some different notions of the corona product for signed graphs have already appeared in the literature, the one we propose is the first to satisfy this property. As a consequence of our construction, we develop a method that, given a signed graph and two of its spanning trees, produces an infinite family of pairs of signed graphs that are not switching isomorphic but are cospectral with respect to the adjacency and net Laplacian matrices.

This chapter is based on [20], joint work with Francesco Belardo, Alfredo Donno, and Stefano Spessato.

4.1 Introduction and preliminaries

The corona product of two ordinary graphs Γ_1 and Γ_2 is the graph $\Gamma = \Gamma_1 \circ \Gamma_2$ formed from one copy of Γ_1 and a copy of Γ_2 for each vertex of Γ_1, where the i-th vertex of Γ_1 is adjacent to every vertex in the i-th copy of Γ_2. It should be noted that definitions of the corona product for signed graphs already exist, and excellent spectral analyses have been carried out for them. However, none of these definitions satisfies all three of the desired properties, in particular failing to maintain switching stability. To the best of our knowledge, ours is the first definition to satisfy all three.

Clearly, achieving such a result requires making some a priori choices. In our case, the corona product depends on the choice of a spanning tree $\mathcal{T}$ of Γ_2, and for this reason our corona product is denoted by $\circ_{\mathcal{T}}$. Nevertheless, the three properties are satisfied for every choice of $\mathcal{T}$, and for each we obtain a generalisation of the classical corona product.

4.1.1 Corona products

Consider two graphs $\Gamma_1 = (V_{\Gamma_1}, E_{\Gamma_1})$ and $\Gamma_2 = (V_{\Gamma_2}, E_{\Gamma_2})$. For each x in V_{Γ_1}, we denote by Γ_2^x a graph which is isomorphic to Γ_2 and, for simplicity, if v is a vertex of Γ_2, we denote by v^x its corresponding vertex in Γ_2^x. With this, the corona product of Γ_1 and Γ_2 is the graph denoted by $\Gamma_1 \circ \Gamma_2$ where

$$V_{\Gamma_1 \circ \Gamma_2} = V_{\Gamma_1} \bigcup_{x \in V_{\Gamma_1}} V_{\Gamma_2^x},$$

and there is an edge between two vertices u and v if and only if one of the following conditions is satisfied:

- edges of the first type: u and v are both in V_{Γ_1} and $\{u, v\} \in E_{\Gamma_1}$,
- edges of the second type: there is an $x \in V_{\Gamma_1}$ such that $u, v \in V_{\Gamma_2^x}$ and $\{u, v\} \in E_{\Gamma_2^x}$,
- edges of the third type: $u = x \in V_{\Gamma_1}$ and $v \in V_{\Gamma_2^x}$.

Suppose one wishes to define a corona product for signed graphs that is a good generalisation of the classical product. It is rather natural to assign to an edge of the first type the same sign that the edge has in Γ_1, and similarly for an edge of the second type, the natural choice is the sign of the corresponding edge in Γ_2. Clearly, the main issue to address is determining which value to assign to the edges of the third type. Indeed, to the best of our knowledge, all definitions of corona products for signed graphs given in the literature agree on the signs of edges of the first and second type, and differ from one another precisely in the choice of signs for the edges connecting the vertices of Γ_1 with those of each copy of Γ_2. Essentially, the definitions given are the following three.

1. The $\circ^+$ product [88]: in this product, all edges of the third type are assigned a positive sign. This definition has the advantage of being an actual generalisation of the classical corona product, by identifying unsigned graphs with all-positive graphs. Moreover, several spectral properties are generalised. However, it is not a switching-stable operation and it is not even true that the product of two balanced graphs is itself balanced.

2. The $\circ^-$ product [88]: in this case, all edges of the third type are assigned to be negative. Conceptually, it is not very different from the previous product: in certain specific contexts, unsigned graphs are identified with all-negative graphs, that is, graphs whose edges all have sign equal to -1. Clearly, by interpreting unsigned graphs in this way, this operation is a generalisation of the classical product. However, this product is also not switching stable, nor does it preserve balancedness.

3. The corona product of marked signed graphs [2]: by defining this type of product, one admits that the choice of the sign to assign to the edges of the third type is not unique and that it is necessary to make an a priori choice, a position that is shared by the authors of this chapter. Indeed, this product is not defined for signed graphs, but for marked signed graphs, which are triples (Γ, σ, μ) where (Γ, σ) is a signed graph and $\mu : V_\Gamma \to \{\pm 1\}$ is a marking function. Given two marked signed graphs $(\Gamma_1, \sigma_1, \mu_1)$ and $(\Gamma_2, \sigma_2, \mu_2)$, the corona product defined in [2] is a signed graph whose underlying graph is $\Gamma_1 \circ \Gamma_2$, and the sign of an edge $\{x, y\}$ of the third type is given by $\mu_1(x)\mu_2(y)$. The problem with this definition is that it is too general; indeed, it is neither a generalisation for every choice of the marking functions, nor is it switching stable.

This chapter is organised as follows. In Section 4.2, we introduce our corona product for signed graphs and prove that it satisfies the three desired properties. In Section 4.3, exploiting the notion of the $\mathcal{T}$-net Laplacian that we introduce, together with its properties, we present a method to construct an infinite family of pairs of signed graphs that are not switching isomorphic but are cospectral with respect to the adjacency and net Laplacian matrices. In Section 4.4, we give some concluding remarks and possible directions for future research.

4.2 A switching-stable corona product of signed graphs

4.2.1 Pseudo-potentials and the $\mathcal{T}$-corona product

Recall the concepts of the pseudo-potential from Section 1.7 of the introduction. These are fundamental ingredients for defining a switching-stable corona product of signed graphs.

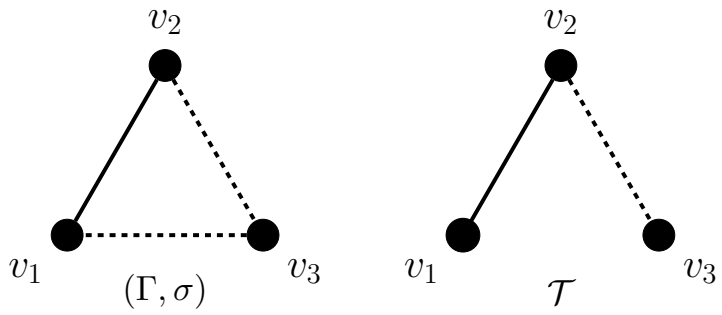


Figure 4.1: The signed graph (Γ, σ) and the spanning tree $\mathcal{T}$ with fixed vertex v_1 of Example 4.2.1.

Example 4.2.1. Let K_3 be the complete graph with vertex set $V_{K_3} = \{v_1, v_2, v_3\}$. Consider the signed

graph (K_3, σ) where $\sigma(v_1, v_2) = +1$, $\sigma(v_2, v_3) = \sigma(v_3, v_1) = -1$. Choose $\mathcal{T}$ to be the spanning tree of K_3 consisting of the two edges $\{v_1, v_2\}$ and $\{v_2, v_3\}$. Fix the vertex $v_1 \in V_{K_3}$. Then, in computing the pseudo-potentials, we have

$$U_{\mathcal{T},v_1,\sigma}(v_1) = +1, \qquad U_{\mathcal{T},v_1,\sigma}(v_2) = +1, \qquad U_{\mathcal{T},v_1,\sigma}(v_3) = -1.$$

This example is illustrated in Figure 4.1.

In the following definition and example, we use Γ_2^x to denote the x-th copy of Γ_2, according to the definition of corona product. Recall that in $\Gamma_1 \circ \Gamma_2$ the vertex $x \in V_{\Gamma_1}$ is adjacent to every vertex of Γ_2^x (edges of third type).

Definition 4.2.2. Let (Γ_1, σ_1) and (Γ_2, σ_2) be two signed graphs. Fix a spanning tree (forest) $\mathcal{T}$ of Γ_2 and a vertex v (collection of vertices $\overline{v}$) in V_{Γ_2}. The $\mathcal{T}$-corona product $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ is the signed graph with underlying graph $\Gamma_1 \circ \Gamma_2$ and signature $\sigma_1 \circ_{\mathcal{T}} \sigma_2$, defined as follows:

$$\sigma_1 \circ_{\mathcal{T}} \sigma_2(x, y) = \begin{cases} \sigma_1(x, y) & \text{if } \{x, y\} \in E_{\Gamma_1} \\ \sigma_2(x, y) & \text{if } \{x, y\} \in E_{\Gamma_2^z} \text{ for some } z \in V_{\Gamma_1} \\ U_{\mathcal{T},v,\sigma_2}(y) & \text{if } x \in V_{\Gamma_1} \text{ and } y \in V_{\Gamma_2^x}. \end{cases}$$

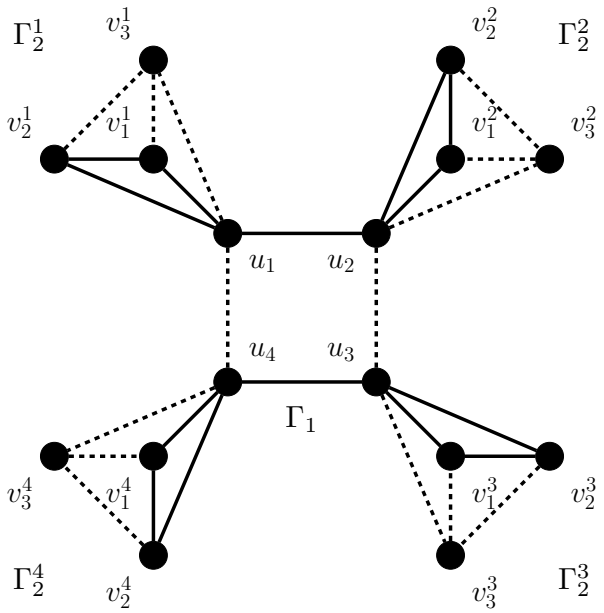


Figure 4.2: The $\mathcal{T}$-corona product $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ of Example 4.2.3.

Example 4.2.3. Consider the graph $C_4 = (V_{C_4}, E_{C_4})$ where $V_{C_4} = \{u_1, u_2, u_3, u_4\}$ and u_j is adjacent to u_k if and only if $k = j + 1$ whenever $j = 1, 2, 3$ or $k = 1$ and $j = 4$. Let (Γ_1, σ_1) be the graph where $\Gamma_1 = C_4$ and $\sigma_1(u_1, u_2) = \sigma_1(u_3, u_4) = +1$, $\sigma_1(u_1, u_4) = \sigma_1(u_2, u_3) = -1$. Take (Γ_2, σ_2) and $\mathcal{T}$ to be the signed graph (K_3, σ) and the spanning tree of Example 4.2.1, respectively, and again fix the vertex $v_1 \in V_{\Gamma_2}$.

The underlying graph of the $\mathcal{T}$-corona product is obtained by connecting each vertex of Γ_1 to every vertex of its corresponding copy of Γ_2. More precisely, for each $i = 1, 2, 3, 4$, the vertex u_i is adjacent to every vertex in $V_{\Gamma_2^i} = \{v_1^i, v_2^i, v_3^i\}$.

The edges internal to Γ_1 and the copies of Γ_2 retain the signatures induced by σ_1 and σ_2, respectively. The edges connecting u_i to v_j^i, for $j = 1, 2, 3$, take their signatures from the value of the pseudo-potential $U_{\mathcal{T},v_1^i,\sigma_2}(v_j^i)$ and we computed these values explicitly in Example 4.2.1. Hence, for $i = 1, 2, 3, 4$,

$$\sigma_1 \circ_{\mathcal{T}} \sigma_2(u_i, v_1^i) = +1, \qquad \sigma_1 \circ_{\mathcal{T}} \sigma_2(u_i, v_2^i) = +1, \qquad \sigma_1 \circ_{\mathcal{T}} \sigma_2(u_i, v_3^i) = -1.$$

This example is illustrated in Figure 4.2.

Before making an important remark, we recall what it means for a signed graph to be $\mathcal{T}$-positive, along with the definition of the $\mathcal{T}$-representative of a switching class.

Definition 4.2.4. Let (Γ, σ) be a signed graph and let $\mathcal{T}$ be a spanning tree of Γ. Then (Γ, σ) is $\mathcal{T}$-positive if $\sigma(x, y) = 1$ for each edge $\{x, y\}$ in $\mathcal{T}$.

Notice that, for each spanning tree $\mathcal{T}$ of Γ and for each switching class of signatures on Γ, there is exactly one representative of that class which is $\mathcal{T}$-positive [100, Lemma 3.1]. Therefore, it makes sense to give the following definition.

Definition 4.2.5. Let (Γ, σ) be a signed graph and let $\mathcal{T}$ be a spanning tree of Γ. The $\mathcal{T}$-representative of the switching class $[\sigma]$ is the unique $\mathcal{T}$-positive signed graph $(\Gamma, \sigma_{\mathcal{T}})$ with $\sigma_{\mathcal{T}} \in [\sigma]$.

Now we can proceed.

Remark 4.2.6. Below, we describe the connections between our definition and the other variants.

- First of all, by identifying unsigned graphs with all-positive signed graphs, one sees that the $\mathcal{T}$-corona product is indeed an extension of the classical corona product. In fact, the $\mathcal{T}$-corona product of two all-positive signed graphs is again all-positive.
- From the viewpoint of [2], the $\mathcal{T}$-corona product of two signed graphs (Γ_1, σ_1) and (Γ_2, σ_2) can also be interpreted as the product of the two marked signed graphs $(\Gamma_1, \sigma_1, \mathbf{1})$ and $(\Gamma_2, \sigma_2, U_{\mathcal{T},v,\sigma_2})$, where $\mathbf{1} : V_{\Gamma_1} \to \{\pm 1\}$ is the function assigning $+1$ to every vertex.
- Finally, observe that if (Γ_2, σ_2) is $\mathcal{T}$-positive, then

$$(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2) = (\Gamma_1, \sigma_1) \circ^{+} (\Gamma_2, \sigma_2),$$

  where $\circ^{+}$ is the corona product introduced in [88].

This final bullet point will be useful later, when we investigate the spectral properties of the $\mathcal{T}$-corona product.

4.2.2 Switching stability and spectral properties

In this subsection we prove the switching stability of the $\mathcal{T}$-corona product of signed graphs and then establish some spectral properties. To prove switching stability, it is convenient to reinterpret the corona attachment through cones over the second component of the product. Accordingly, we recall the notion of a cone over a graph.

Let $\Gamma = (V_\Gamma, E_\Gamma)$ be a graph. The cone over Γ is the graph $C\Gamma = (V_{C\Gamma}, E_{C\Gamma})$ obtained by adjoining a new vertex $w \notin V_\Gamma$, with

$$V_{C\Gamma} = V_\Gamma \cup \{w\} \quad \text{and} \quad E_{C\Gamma} = E_\Gamma \cup \{\{w, x\} : x \in V_\Gamma\}.$$

We see that, given a signature σ on Γ, if one fixes a spanning tree (forest) $\mathcal{T}$ of Γ and a vertex v (collection of vertices $\overline{v}$) in V_Γ, then it is possible to define a signature $\overline{\sigma}^{\mathcal{T}}$ on $C\Gamma$. Indeed, it is sufficient to define $\overline{\sigma}^{\mathcal{T}}(x, y) = \sigma(x, y)$ whenever x is adjacent to y in Γ and

$$\overline{\sigma}^{\mathcal{T}}(w, x) = U_{\mathcal{T},v,\sigma}(x), \tag{4.2.1}$$

otherwise. In addition, one can see that $(C\Gamma, \overline{\sigma}^{\mathcal{T}})$ may be interpreted as the corona product $(K_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma, \sigma)$, where K_1 is the graph having exactly one vertex and σ_1 is the so-called empty function: the unique signature admissible on K_1.

Notice that if $A_{(\Gamma,\sigma)}$ is the adjacency matrix of (Γ, σ) with $V_\Gamma = \{x_1, x_2, \ldots, x_n\}$, then the adjacency matrix of $(C\Gamma, \overline{\sigma}^{\mathcal{T}})$ has the following block structure after a suitable reordering of vertices:

$$A_{(C\Gamma,\overline{\sigma}^{\mathcal{T}})} = \left(\begin{array}{ccc|c} & A_{(\Gamma,\sigma)} & & \begin{matrix} U_{\mathcal{T},v,\sigma}(x_1) \\ \vdots \\ U_{\mathcal{T},v,\sigma}(x_n) \end{matrix} \\ \hline U_{\mathcal{T},v,\sigma}(x_1) & \cdots & U_{\mathcal{T},v,\sigma}(x_n) & 0 \end{array}\right). \tag{4.2.2}$$

Proposition 4.2.7. *Let (Γ_1, σ_1) and (Γ_2, σ_2) be two signed graphs of order n_1 and n_2, respectively. Let $\mathcal{T}$ be a spanning tree (forest) of Γ_2 and a vertex v (collection of vertices $\overline{v}$) in V_{Γ_2}. With respect to a suitable order of the vertices, the adjacency matrix of $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ can be expressed as follows:*

$$A_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_2)} = A_{(\Gamma_1,\sigma_1)} \otimes E_{n_2+1} + I_{n_1} \otimes A_{(C\Gamma_2,\overline{\sigma_2}^{\mathcal{T}})}, \tag{4.2.3}$$

where E_{n_2+1} denotes the $(n_2+1) \times (n_2+1)$ matrix whose only non-zero entry is a 1 in the bottom-right corner, and $\otimes$ denotes the Kronecker product of matrices.

Proof. Notice that the vertex set of $\Gamma_1 \circ \Gamma_2$ can be identified with the Cartesian product $V_{\Gamma_1} \times V_{C\Gamma_2}$: each $x \in V_{\Gamma_1}$ corresponds to the pair (x, w), where w is the newly introduced vertex of $C\Gamma_2$, and each $y \in V_{\Gamma_2^x}$ corresponds to (x, y).

Fix an order on V_{Γ_1} and an order on $C\Gamma_2$. Assume that the new vertex w in $C\Gamma_2$ is the last one with respect to the order on $C\Gamma_2$. Then, if we fix the lexicographic order on $V_{\Gamma_1\circ\Gamma_2}$, we have (x, y) adjacent to (x', y') if and only if one of the two following conditions holds:

- $x = x'$ and y is adjacent to y',
- x is adjacent to x' and $y = y' = w$.

Observe that the non-zero entries of $I_{n_1} \otimes A_{(C\Gamma_2,\overline{\sigma}^{\mathcal{T}})}$ are exactly those indexed by pairs $((x, y), (x', y'))$ satisfying the first condition. Similarly, the non-zero entries of $A_{(\Gamma_1,\sigma_1)} \otimes E_{n_2+1}$ are exactly those indexed by pairs satisfying the second. □

The following lemma establishes that the operation assigning a signed cone to each signed graph is switching stable.

Lemma 4.2.8. *Let Γ be a graph. Fix a spanning tree (forest) $\mathcal{T}$ of Γ and a vertex v (collection of vertices $\overline{v}$) in V_Γ and let σ_1 and σ_2 be two signatures on Γ. Then, if σ_1 and σ_2 are switching equivalent, so are the cones $(C\Gamma, \overline{\sigma_1}^{\mathcal{T}})$ and $(C\Gamma, \overline{\sigma_2}^{\mathcal{T}})$.*

Proof. It is sufficient to consider the case $(\Gamma, \sigma_2) = (\Gamma, \sigma_{\mathcal{T}})$, where $(\Gamma, \sigma_{\mathcal{T}})$ is the $\mathcal{T}$-representative of $[\sigma_1]$. In what follows, we denote (Γ, σ_1) simply by (Γ, σ).

As shown in [33, Remark 2.7],

$$A_{(\Gamma,\sigma_{\mathcal{T}})} = U_{\mathcal{T},v,\sigma} A_{(\Gamma,\sigma)} U_{\mathcal{T},v,\sigma},$$

where $U_{\mathcal{T},v,\sigma}$ is the diagonal matrix indexed by the vertices of Γ whose i-th entry is $U_{\mathcal{T},v,\sigma}(v_i)$. Denote by w the vertex added to V_Γ when defining the cone over Γ. Observe that the edges of the form $\{w, v_i\}$ in the cone of $(\Gamma, \sigma_{\mathcal{T}})$ with respect to $\mathcal{T}$ are positive, so by applying Equation (4.2.2) we obtain

$$A_{(C\Gamma,\overline{\sigma}^{\mathcal{T}}_{\mathcal{T}})} = \left(\begin{array}{ccc|c} & A_{(\Gamma,\sigma_{\mathcal{T}})} & & \begin{matrix}1\\ \vdots \\ 1\end{matrix} \\ \hline 1 & \cdots & 1 & 0 \end{array}\right).$$

Denote by $\overline{U}_{\mathcal{T},v,\sigma}$ the diagonal matrix indexed by the vertices of $C\Gamma$ such that $\overline{U}_{\mathcal{T},v,\sigma}(w) = 1$ and

$\overline{U}_{\mathcal{T},v,\sigma}(v_i) = U_{\mathcal{T},v,\sigma}(v_i)$. Then, by Equation (4.2.2), we have:

$$
\begin{aligned}
&\overline{U}_{\mathcal{T},v,\sigma} A_{(C\Gamma,\overline{\sigma}^{\mathcal{T}})} \overline{U}_{\mathcal{T},v,\sigma} \\
&= \left(\begin{array}{ccc|c} & U_{\mathcal{T},v,\sigma} & & \begin{matrix}0\\ \vdots \\ 0\end{matrix} \\ \hline 0 & \dots & 0 & 1 \end{array}\right)
\left(\begin{array}{ccc|c} & A_{(\Gamma,\sigma)} & & \begin{matrix}U_{\mathcal{T},v,\sigma}(x_1)\\ \vdots \\ U_{\mathcal{T},v,\sigma}(x_n)\end{matrix} \\ \hline U_{\mathcal{T},v,\sigma}(x_1) & \dots & U_{\mathcal{T},v,\sigma}(x_n) & 0 \end{array}\right)
\left(\begin{array}{ccc|c} & U_{\mathcal{T},v,\sigma} & & \begin{matrix}0\\ \vdots \\ 0\end{matrix} \\ \hline 0 & \dots & 0 & 1 \end{array}\right) \\
&= \left(\begin{array}{ccc|c} & U_{\mathcal{T},v,\sigma} & & \begin{matrix}0\\ \vdots \\ 0\end{matrix} \\ \hline 0 & \dots & 0 & 1 \end{array}\right)
\left(\begin{array}{ccc|c} & A_{(\Gamma,\sigma)} U_{\mathcal{T},v,\sigma} & & \begin{matrix}U_{\mathcal{T},v,\sigma}(x_1)\\ \vdots \\ U_{\mathcal{T},v,\sigma}(x_n)\end{matrix} \\ \hline 1 & \dots & 1 & 0 \end{array}\right) \\
&= \left(\begin{array}{ccc|c} & U_{\mathcal{T},v,\sigma} A_{(\Gamma,\sigma)} U_{\mathcal{T},v,\sigma} & & \begin{matrix}1\\ \vdots \\ 1\end{matrix} \\ \hline 1 & \dots & 1 & 0 \end{array}\right) \\
&= \left(\begin{array}{ccc|c} & A_{(\Gamma,\sigma_{\mathcal{T}})} & & \begin{matrix}1\\ \vdots \\ 1\end{matrix} \\ \hline 1 & \dots & 1 & 0 \end{array}\right) = A_{(C\Gamma,\overline{\sigma}^{\mathcal{T}}_{\mathcal{T}})}.
\end{aligned}
$$

Hence $(C\Gamma, \overline{\sigma}^{\mathcal{T}})$ and $(C\Gamma, \overline{\sigma}^{\mathcal{T}}_{\mathcal{T}})$ are switching equivalent, completing the proof. □

Remark 4.2.9. Notice that when fixing different vertices in Γ, the resulting signatures on the cone $C\Gamma$ remain switching equivalent. In particular, if Γ is connected, the resulting signature is either $\overline{\sigma}^{\mathcal{T}}_2$ itself, or the signature obtained by switching the signs of all edges of the form $\{w, x\}$.

Now we are ready to prove the switching stability of the $\mathcal{T}$-corona product of signed graphs.

Theorem 4.2.10. *Let $\Gamma_1 = (V_{\Gamma_1}, E_{\Gamma_1})$ and $\Gamma_2 = (V_{\Gamma_2}, E_{\Gamma_2})$ be two graphs of order n_1 and n_2, respectively. Let $\mathcal{T}$ be a spanning tree (forest) and v a vertex ($\overline{v}$ a collection of vertices) of Γ_2. Let σ_1 and σ_1' be two switching-equivalent signatures on Γ_1 and let σ_2 and σ_2' be two switching-equivalent signatures on Γ_2. Then the signed graphs*

$$(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2) \quad \text{and} \quad (\Gamma_1, \sigma_1') \circ_{\mathcal{T}} (\Gamma_2, \sigma_2')$$

are switching equivalent.

Proof. Since $[\sigma_1] = [\sigma_1']$ and $[\sigma_2] = [\sigma_2']$, there exist diagonal matrices S_1 and S_2 such that

$$A_{(\Gamma_1,\sigma_1)} = S_1 A_{(\Gamma_1,\sigma_1')} S_1 \quad \text{and} \quad A_{(\Gamma_2,\sigma_2)} = S_2 A_{(\Gamma_2,\sigma_2')} S_2. \tag{4.2.4}$$

Thanks to Lemma 4.2.8, we also know that there is a diagonal matrix $\overline{S}_2$ such that

$$A_{(C\Gamma_2,\overline{\sigma_2}^{\mathcal{T}})} = \overline{S}_2 A_{(C\Gamma_2,\overline{\sigma_2'}^{\mathcal{T}})} \overline{S}_2.$$

Observe that the matrix $S_1 \otimes \overline{S}_2$ is diagonal. From Equation (4.2.3), we obtain

$$
\begin{aligned}
&(S_1 \otimes \overline{S}_2) \cdot A_{(\Gamma_1,\sigma_1')\circ_{\mathcal{T}}(\Gamma_2,\sigma_2')} \cdot (S_1 \otimes \overline{S}_2) \\
&\qquad = (S_1 \otimes \overline{S}_2) \cdot (A_{(\Gamma_1,\sigma_1')} \otimes E_{n_2+1} + I_{n_1} \otimes A_{(C\Gamma_2,\overline{\sigma_2'}^{\mathcal{T}})}) \cdot (S_1 \otimes \overline{S}_2) \\
&\qquad = (S_1 A_{(\Gamma_1,\sigma_1')} S_1) \otimes (\overline{S}_2 E_{n_2+1} \overline{S}_2) + (S_1 I_{n_1} S_1) \otimes (\overline{S}_2 A_{(C\Gamma_2,\overline{\sigma_2'}^{\mathcal{T}})} \overline{S}_2) \\
&\qquad = A_{(\Gamma_1,\sigma_1)} \otimes E_{n_2+1} + I_{n_1} \otimes A_{(C\Gamma_2,\overline{\sigma_2}^{\mathcal{T}})} = A_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_2)}.
\end{aligned}
$$

□

Remark 4.2.11. As a consequence of Remark 4.2.9, any $\mathcal{T}$-corona products of the same two signed graphs (Γ_1, σ_1) and (Γ_2, σ_2), constructed using the same spanning tree but different fixed vertices, are switching equivalent. This implies that, up to switching equivalence, $\circ_{\mathcal{T}}$ depends only on the choice of spanning tree.

This leads us to the following corollaries.

Corollary 4.2.12. *Under the same assumptions as in Theorem 4.2.10, the signed graphs* (Γ_1, σ_1) *and* (Γ_2, σ_2) *are balanced if and only if the* $\mathcal{T}$*-corona product* $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ *is balanced.*

Proof. We first prove the forward implication. Recall that a signed graph (Γ, σ) is balanced if and only if it is switching equivalent to $(\Gamma, +)$, the all-positive signed graph having Γ as its underlying graph. Notice that

$$(\Gamma_1, +) \circ_{\mathcal{T}} (\Gamma_2, +) = (\Gamma_1 \circ \Gamma_2, +),$$

and we can conclude by applying Theorem 4.2.10.

We now prove the converse. Let (Γ, σ) be a signed graph, fix a non-empty subset $B \subseteq V_\Gamma$, and let $(\Gamma_B, \tilde{\sigma})$ be the subgraph of (Γ, σ) induced by B, i.e. $V_{\Gamma_B} = B$ and two vertices $x, y \in B$ are adjacent in Γ_B if and only if they are adjacent in Γ. The signature $\tilde{\sigma}$ on Γ_B is defined by $\tilde{\sigma}(x, y) = \sigma(x, y)$. Notice that if (Γ, σ) is balanced, then $(\Gamma_B, \tilde{\sigma})$ is also balanced for each subset B. Indeed, the adjacency matrix of $(\Gamma_B, \tilde{\sigma})$ can be obtained from $A_{(\Gamma,\sigma)}$ by deleting the rows and columns corresponding to the vertices in $V_\Gamma \setminus B$. Hence,

$$A_{(\Gamma,\sigma)} = S A_{(\Gamma,+)} S \implies A_{(\Gamma_B,\tilde{\sigma})} = S_B A_{(\Gamma_B,+)} S_B,$$

where S_B is obtained from S by deleting the rows and columns corresponding to the vertices in $V_\Gamma \setminus B$. In order to conclude, it is sufficient to recall that

$$V_{\Gamma_1 \circ \Gamma_2} = V_{\Gamma_1} \bigcup_{x \in V_{\Gamma_1}} V_{\Gamma_2^x}$$

and that the subgraphs induced by V_{Γ_1} and $V_{\Gamma_2^x}$ are copies of (Γ_1, σ_1) and (Γ_2, σ_2), respectively, by Definition 4.2.2. □

Corollary 4.2.13. *Under the same assumptions as in Theorem 4.2.10, denote by* $(\Gamma_2, \sigma_{(2,\mathcal{T})})$ *the* $\mathcal{T}$*-representative of the switching class of* (Γ_2, σ_2)*. Then* $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ *and* $(\Gamma_1, \sigma_1) \circ^{+} (\Gamma_2, \sigma_{(2,\mathcal{T})})$ *are adjacency and Laplacian cospectral.*

Proof. By Theorem 4.2.10, the signed graphs $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ and $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_{(2,\mathcal{T})})$ are switching equivalent and, in particular, they are adjacency and Laplacian cospectral. Moreover, as stated in the third bullet point of Remark 4.2.6, since $(\Gamma_2, \sigma_{(2,\mathcal{T})})$ is $\mathcal{T}$-positive,

$$(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_{(2,\mathcal{T})}) = (\Gamma_1, \sigma_1) \circ^{+} (\Gamma_2, \sigma_{(2,\mathcal{T})}).$$

This concludes the proof. □

The previous corollary shows that, up to a switching of the signature of Γ_2, which naturally preserves its spectrum, the spectral analysis of the adjacency matrix of the $\mathcal{T}$-corona product is essentially the same as that of the positive corona product defined in [88]. However, the most significant results, both for the adjacency spectrum and for the net Laplacian spectrum, are obtained when (Γ_2, σ_2) is net-regular. Since net-regularity is not preserved under switching equivalence, it is necessary to adopt a switching-stable notion of net-regularity (introduced in [33]) in order to apply the results of [2] and [88] in our setting. This is precisely the approach taken in the next section.

4.3 Laplacians of the corona product

4.3.1 $\mathcal{T}$-net-regularity and $\mathcal{T}$-net Laplacian

Given a graph Γ, denote by $S(v)$ the neighbourhood of v in Γ, that is, $S(v) = \{w \in V_\Gamma : \{w, v\} \in E_\Gamma\}$. The following definition is taken from [33].

Definition 4.3.1. Let (Γ, σ) be a signed graph, $\mathcal{T}$ a spanning tree (forest) of Γ and z be a vertex (collection of vertices) in V_Γ. The $\mathcal{T}$-net degree of a vertex v of Γ is the value

$$d_\sigma^{\mathcal{T}}(v) = \sum_{s \in S(v)} U_{\mathcal{T},z,\sigma}(v)\sigma(v,s)U_{\mathcal{T},z,\sigma}(s).$$

A signed graph (Γ, σ) is $\mathcal{T}$-net-regular if each vertex of (Γ, σ) has the same $\mathcal{T}$-net degree.

Remark 4.3.2. In [33, Remark 3.4], it is shown that the $\mathcal{T}$-net degree does not depend on the choice of the vertex z. Furthermore, [33, Proposition 3.5] demonstrates that, for two switching-equivalent signed graphs (Γ, σ) and (Γ, φ), it holds that $d_\sigma^{\mathcal{T}}(v) = d_\varphi^{\mathcal{T}}(v)$ for every $v \in V_\Gamma$. This implies that, unlike net-regularity, $\mathcal{T}$-net-regularity is a property of switching classes. Note that if (Γ, σ) is $\mathcal{T}$-positive, then the $\mathcal{T}$-net degree of each vertex coincides with its usual net degree, and it follows that (Γ, σ) is $\mathcal{T}$-net-regular if and only if $(\Gamma, \sigma_{\mathcal{T}})$ is net-regular.

Motivated by the notion of the net Laplacian matrix in Equation (1.4.1), we introduce a variant in the following definition.

Definition 4.3.3. Let (Γ, σ) be a signed graph and let $\mathcal{T}$ be a spanning tree of Γ. The $\mathcal{T}$-net Laplacian matrix of (Γ, σ) is the matrix indexed by V_Γ and defined as

$$L^{\mathcal{T}}_{(\Gamma,\sigma)} = D^{\mathcal{T}}_{(\Gamma,\sigma)} - A_{(\Gamma,\sigma)},$$

where $D^{\mathcal{T}}_{(\Gamma,\sigma)}$ is the diagonal matrix of $\mathcal{T}$-net degrees.

Two signed graphs (Γ, σ) and (Γ, φ) are $\mathcal{T}$-net Laplacian cospectral if and only if $\operatorname{spec}(L^{\mathcal{T}}_{(\Gamma,\sigma)}) = \operatorname{spec}(L^{\mathcal{T}}_{(\Gamma,\varphi)})$.

Example 4.3.4. Take (Γ, σ) and $\mathcal{T}$ to be those of Example 4.2.1, once again fixing vertex v_1. Then the $\mathcal{T}$-net degrees of the vertices are computed as

$$\begin{aligned}
d_\sigma^{\mathcal{T}}(v_1) &= U_{\mathcal{T},v_1,\sigma}(v_1)\sigma(v_1,v_2)U_{\mathcal{T},v_1,\sigma}(v_2) + U_{\mathcal{T},v_1,\sigma}(v_1)\sigma(v_1,v_3)U_{\mathcal{T},v_1,\sigma}(v_3)\\
&= (+1)(+1)(+1) + (+1)(-1)(-1) = 2,\\
d_\sigma^{\mathcal{T}}(v_2) &= U_{\mathcal{T},v_1,\sigma}(v_2)\sigma(v_2,v_1)U_{\mathcal{T},v_1,\sigma}(v_1) + U_{\mathcal{T},v_1,\sigma}(v_2)\sigma(v_2,v_3)U_{\mathcal{T},v_1,\sigma}(v_3)\\
&= (+1)(+1)(+1) + (+1)(-1)(-1) = 2,\\
d_\sigma^{\mathcal{T}}(v_3) &= U_{\mathcal{T},v_1,\sigma}(v_3)\sigma(v_3,v_1)U_{\mathcal{T},v_1,\sigma}(v_1) + U_{\mathcal{T},v_1,\sigma}(v_3)\sigma(v_3,v_2)U_{\mathcal{T},v_1,\sigma}(v_2)\\
&= (-1)(-1)(+1) + (-1)(-1)(+1) = 2.
\end{aligned}$$

Hence (Γ, σ) is $\mathcal{T}$-net-regular. One can verify that the $\mathcal{T}$-representative in this case is the graph with the all-positive signature, and that $(\Gamma, \sigma_{\mathcal{T}}) = (\Gamma, +)$ is indeed net-regular, as we would expect from Remark 4.3.2.

The Laplacian, net Laplacian, and $\mathcal{T}$-net Laplacian matrices of (Γ, σ) are

$$L_{(\Gamma,\sigma)} = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \qquad L^{\pm}_{(\Gamma,\sigma)} = \begin{pmatrix} 0 & -1 & 1 \\ -1 & 0 & 1 \\ 1 & 1 & -2 \end{pmatrix}, \qquad L^{\mathcal{T}}_{(\Gamma,\sigma)} = \begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix}.$$

Proposition 4.3.5. *Let Γ be a graph and let $\mathcal{T}$ be a spanning tree of Γ. Then for each signature σ on Γ, the signed graph (Γ, σ) is $\mathcal{T}$-net Laplacian cospectral with $(\Gamma, \sigma_{\mathcal{T}})$. Furthermore,*

$$\operatorname{spec}\left(L^{\mathcal{T}}_{(\Gamma,\sigma)}\right) = \operatorname{spec}\left(L^{\pm}_{(\Gamma,\sigma_{\mathcal{T}})}\right).$$

Proof. Fix v in V_Γ. Let $U_{\mathcal{T},v,\sigma}$ be the diagonal matrix whose w-th entry is $U_{\mathcal{T},v,\sigma}(w)$. As noted in [33, Remark 2.7],

$$A_{(\Gamma,\sigma_{\mathcal{T}})} = U_{\mathcal{T},v,\sigma} A_{(\Gamma,\sigma)} U_{\mathcal{T},v,\sigma}.$$

Moreover, $U^2_{\mathcal{T},v,\sigma}$ is equal to the identity matrix and $D^{\mathcal{T}}_{(\Gamma,\sigma)} = D^{\mathcal{T}}_{(\Gamma,\sigma_{\mathcal{T}})}$ as a consequence of Remark 4.3.2. Hence

$$\begin{aligned} L^{\mathcal{T}}_{(\Gamma,\sigma_{\mathcal{T}})} &= D^{\mathcal{T}}_{(\Gamma,\sigma_{\mathcal{T}})} - A_{(\Gamma,\sigma_{\mathcal{T}})} \\ &= D^{\mathcal{T}}_{(\Gamma,\sigma)} - U_{\mathcal{T},v,\sigma} A_{(\Gamma,\sigma)} U_{\mathcal{T},v,\sigma} \\ &= U_{\mathcal{T},v,\sigma}(D^{\mathcal{T}}_{(\Gamma,\sigma)} - A_{(\Gamma,\sigma)}) U_{\mathcal{T},v,\sigma} \\ &= U_{\mathcal{T},v,\sigma} L^{\mathcal{T}}_{(\Gamma,\sigma)} U_{\mathcal{T},v,\sigma}. \end{aligned}$$

Additionally, since $D^{\mathcal{T}}_{(\Gamma,\sigma_{\mathcal{T}})} = D^{\pm}_{(\Gamma,\sigma_{\mathcal{T}})}$, we have $L^{\mathcal{T}}_{(\Gamma,\sigma_{\mathcal{T}})} = L^{\pm}_{(\Gamma,\sigma_{\mathcal{T}})}$, which completes the proof. □

Proposition 4.3.6. *Let (Γ,σ) be a signed graph and let $\mathcal{T}$ be a spanning tree (forest) of Γ. If (Γ,σ) is $\mathcal{T}$-net-regular with $\mathcal{T}$-net degree d, then $d \in \operatorname{spec}(A_{(\Gamma,\sigma)})$ and $0 \in \operatorname{spec}(L^{\mathcal{T}}_{(\Gamma,\sigma)})$.*

Proof. Observe that $\operatorname{spec}(A_{(\Gamma,\sigma)}) = \operatorname{spec}(A_{(\Gamma,\sigma_{\mathcal{T}})})$ since they are switching equivalent. Moreover $\operatorname{spec}(L^{\mathcal{T}}_{(\Gamma,\sigma)}) = \operatorname{spec}(L^{\pm}_{(\Gamma,\sigma_{\mathcal{T}})})$ as a consequence of Proposition 4.3.5. Since (Γ,σ) is $\mathcal{T}$-net-regular, the signed graph $(\Gamma,\sigma_{\mathcal{T}})$ is net-regular as a consequence of Remark 4.3.2. In particular, the net degree of a vertex v in $(\Gamma,\sigma_{\mathcal{T}})$ is d and this implies that $d \in \operatorname{spec}(A_{(\Gamma,\sigma_{\mathcal{T}})})$ and $0 \in \operatorname{spec}(L^{\pm}_{(\Gamma,\sigma_{\mathcal{T}})})$, giving $d \in \operatorname{spec}(A_{(\Gamma,\sigma)})$ and $0 \in \operatorname{spec}(L^{\mathcal{T}}_{(\Gamma,\sigma)})$, as required. □

The previous proposition allows us to describe the adjacency spectrum and the $\mathcal{T}$-net Laplacian spectrum of the $\mathcal{T}$-corona product of two signed graphs.

Proposition 4.3.7. *Let (Γ_1,σ_1) and (Γ_2,σ_2) be two signed graphs of order n_1 and n_2, respectively, and let $\mathcal{T}$ be a spanning tree of Γ_2. Assume that (Γ_2,σ_2) is $\mathcal{T}$-net-regular with $\mathcal{T}$-net degree d, and that $\operatorname{spec}(A_{(\Gamma_1,\sigma_1)}) = \{\alpha_1,\alpha_2,\dots,\alpha_{n_1}\}$ and $\operatorname{spec}(A_{(\Gamma_2,\sigma_2)}) = \{\beta_1 = d,\beta_2,\dots,\beta_{n_2}\}$. Then the adjacency spectrum of the $\mathcal{T}$-corona product $(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_2)$ is*

$$\left\{\beta_2^{(n_1)},\beta_3^{(n_1)},\dots,\beta_{n_2}^{(n_1)},\gamma_{(1,+)},\gamma_{(1,-)},\dots,\gamma_{(n_1,+)},\gamma_{(n_1,-)}\right\},$$

where

$$\gamma_{(i,\pm)} = \frac{1}{2}\left(d+\alpha_i \pm \sqrt{(d-\alpha_i)^2+4n_2}\right).$$

Proof. From Corollary 4.2.13, we know that $(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_2)$ and $(\Gamma_1,\sigma_1)\circ^{+}(\Gamma_2,\sigma_{(2,\mathcal{T})})$ are adjacency cospectral. Notice that $(\Gamma_1,\sigma_1)\circ^{+}(\Gamma_2,\sigma_{(2,\mathcal{T})})$ can be viewed as the corona product of the marked signed graphs $(\Gamma_1,\sigma_1,\mathbf{1})$ and $(\Gamma_2,\sigma_{(2,\mathcal{T})},\mathbf{1})$, where $\mathbf{1}$ is the function assigning $+1$ to every vertex. From here, since $(\Gamma_2,\sigma_{(2,\mathcal{T})})$ is net-regular with net degree d by Remark 4.3.2, it suffices to apply [2, Theorem 2.3]. □

Proposition 4.3.8. *Let (Γ_1,σ_1) and (Γ_2,σ_2) be two signed graphs of order n_1 and n_2, respectively, and let $\mathcal{T}_1$ and $\mathcal{T}_2$ be spanning trees of Γ_1 and Γ_2, respectively. Assume that $\operatorname{spec}(L^{\mathcal{T}_1}_{(\Gamma_1,\sigma_1)}) = \{\alpha_1,\alpha_2,\dots,\alpha_{n_1}\}$, $\operatorname{spec}(L^{\mathcal{T}_2}_{(\Gamma_2,\sigma_2)}) = \{\beta_1,\beta_2,\dots,\beta_{n_2}\}$, and that there exists $k \in \{1,2,\dots,n_2\}$ such that $\beta_k = 0$. Denote by $\mathcal{T}_{1,2}$ any spanning tree of $\Gamma_1\circ\Gamma_2$ obtained by taking the disjoint union of one copy of $\mathcal{T}_1$ and n_1 copies of $\mathcal{T}_2$, and adding, for each $x \in V_{\Gamma_1}$, exactly one edge connecting x with a vertex of Γ_2^x. Then the $\mathcal{T}_{1,2}$-net Laplacian spectrum of $(\Gamma_1,\sigma_1)\circ_{\mathcal{T}_2}(\Gamma_2,\sigma_2)$ is*

$$\{(\beta_j+1)^{(n_1)} : 1 \le j \le n_2,\ j \ne k\} \cup \{\gamma_{(i,+)},\gamma_{(i,-)} : 1 \le i \le n_1\},$$

where

$$\gamma_{(i,\pm)} = \frac{1}{2}\left(\alpha_i+n_2+1 \pm \sqrt{(\alpha_i+n_2+1)^2-4\alpha_i}\right).$$

Proof. As a consequence of Theorem 4.2.10 and Proposition 4.3.5, we can assume that (Γ_1,σ_1) and (Γ_2,σ_2) are $\mathcal{T}_1$-positive and $\mathcal{T}_2$-positive, respectively. Observe that

$$(\Gamma_1,\sigma_{(1,\mathcal{T}_1)})\circ_{\mathcal{T}_2}(\Gamma_2,\sigma_{(2,\mathcal{T}_2)}) = (\Gamma_1,\sigma_{(1,\mathcal{T}_1)})\circ^{+}(\Gamma_2,\sigma_{(2,\mathcal{T}_2)})$$

is $\mathcal{T}_{1,2}$-positive since the pseudo-potential of $(\Gamma_2, \sigma_{(2,\mathcal{T}_2)})$ with respect to $\mathcal{T}_2$ takes value $+1$ for each vertex in V_{Γ_2}. So its $\mathcal{T}_{1,2}$-net Laplacian spectrum coincides with the standard net Laplacian spectrum of $(\Gamma_1, \sigma_{(1,\mathcal{T}_1)}) \circ^+ (\Gamma_2, \sigma_{(2,\mathcal{T}_2)})$.

The result follows as a direct consequence of [88, Theorem 3.2]. □

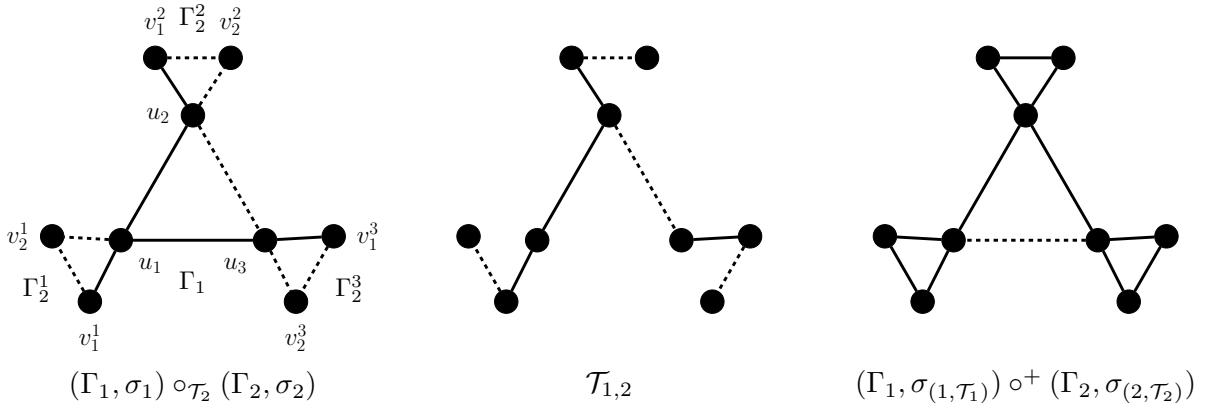


Figure 4.3: The $\mathcal{T}_2$-corona product $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}_2} (\Gamma_2, \sigma_2)$, the spanning tree $\mathcal{T}_{1,2}$, and the $\mathcal{T}_{1,2}$-positive product $(\Gamma_1, \sigma_{(1,\mathcal{T}_1)}) \circ^+ (\Gamma_2, \sigma_{(2,\mathcal{T}_2)})$ of Example 4.3.9.

Example 4.3.9. Let $(\Gamma_1, \sigma_1) = (K_3, \sigma_1)$ with $V_{\Gamma_1} = \{u_1, u_2, u_3\}$ and $\sigma_1(u_1, u_2) = \sigma_1(u_1, u_3) = 1$ and $\sigma_1(u_2, u_3) = -1$. Let $(\Gamma_2, \sigma_2) = (K_2, -)$ with $V_{\Gamma_2} = \{v_1, v_2\}$. For Γ_1, take $\mathcal{T}_1 = \{\{u_1, u_2\}, \{u_2, u_3\}\}$ as the spanning tree and u_1 as the fixed vertex. For Γ_2, take $\mathcal{T}_2 = \{\{v_1, v_2\}\}$ as the spanning tree and v_1 as the fixed vertex. Then

$$U_{\mathcal{T}_1,u_1,\sigma_1}(u_1) = +1, \quad U_{\mathcal{T}_1,u_1,\sigma_1}(u_2) = +1, \quad U_{\mathcal{T}_1,u_1,\sigma_1}(u_3) = -1,$$
$$U_{\mathcal{T}_2,v_1,\sigma_2}(v_1) = +1, \quad U_{\mathcal{T}_2,v_1,\sigma_2}(v_2) = -1.$$

With this information, the $\mathcal{T}_1$-net and $\mathcal{T}_2$-net degrees can be calculated as

$$d^{\mathcal{T}_1}_{\sigma_1}(u_1) = 0, \quad d^{\mathcal{T}_1}_{\sigma_1}(u_2) = 2, \quad d^{\mathcal{T}_1}_{\sigma_1}(u_3) = 0, \quad d^{\mathcal{T}_2}_{\sigma_2}(v_1) = 1, \quad d^{\mathcal{T}_2}_{\sigma_2}(v_2) = 1.$$

Hence the $\mathcal{T}_1$-net and $\mathcal{T}_2$-net Laplacian matrices and spectra are

$$L^{\mathcal{T}_1}_{(\Gamma_1,\sigma_1)} = \begin{pmatrix} 0 & -1 & -1 \\ -1 & 2 & 1 \\ -1 & 1 & 0 \end{pmatrix}, \qquad \operatorname{spec}(L^{\mathcal{T}_1}_{(\Gamma_1,\sigma_1)}) = \{3, 0, -1\},$$
$$L^{\mathcal{T}_2}_{(\Gamma_2,\sigma_2)} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \operatorname{spec}(L^{\mathcal{T}_2}_{(\Gamma_2,\sigma_2)}) = \{2, 0\}.$$

For the spanning tree $\mathcal{T}_{1,2}$ of $\Gamma_1 \circ \Gamma_2$, we add the edges $\{u_1, v_1^1\}$, $\{u_2, v_1^2\}$, and $\{u_3, v_1^3\}$. Now, fixing the vertex u_1, we can compute the $\mathcal{T}_{1,2}$-net degrees of the vertices in the product to obtain

$$d^{\mathcal{T}_{1,2}}_{(\sigma_1\circ_{\mathcal{T}_2}\sigma_2)}(u_1) = 2, \quad d^{\mathcal{T}_{1,2}}_{(\sigma_1\circ_{\mathcal{T}_2}\sigma_2)}(u_2) = 4, \quad d^{\mathcal{T}_{1,2}}_{(\sigma_1\circ_{\mathcal{T}_2}\sigma_2)}(u_3) = 2, \quad d^{\mathcal{T}_{1,2}}_{(\sigma_1\circ_{\mathcal{T}_2}\sigma_2)}(v_i^j) = 2,$$

for each $i = 1, 2$, and $j = 1, 2, 3$.

Therefore, the $\mathcal{T}_{1,2}$-net Laplacian matrix and spectrum of the product are

$$L^{\mathcal{T}_{1,2}}_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}_2}(\Gamma_2,\sigma_2)} = \left(\begin{array}{ccc|ccccc} 2 & -1 & -1 & -1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 4 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ -1 & 1 & 2 & 0 & 0 & 0 & 0 & -1 & 1 \\ \hline -1 & 0 & 0 & 2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 2 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 1 & 2 \end{array}\right)$$

and

$$\operatorname{spec}\left(L^{\mathcal{T}_{1,2}}_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}_2}(\Gamma_2,\sigma_2)}\right) = \{5.449, 3, 3, 3, 3, 2.414, 0.551, 0, -0.414\}.$$

One can check that the eigenvalues here match precisely those obtained from the expression in Proposition 4.3.8. Furthermore, it can be seen that they coincide with the standard net Laplacian eigenvalues of the $\mathcal{T}_{1,2}$-positive product $(\Gamma_1, \sigma_{(1,\mathcal{T}_1)}) \circ^+ (\Gamma_2, \sigma_{(2,\mathcal{T}_2)})$, as expected.

This example is illustrated in Figure 4.3.

4.3.2 Standard Laplacian of the $\mathcal{T}$-corona product

We now provide the Laplacian characteristic polynomial of the $\mathcal{T}$-corona product of two arbitrary signed graphs.

Proposition 4.3.10. *Suppose that (Γ_1, σ_1) and (Γ_2, σ_2) are signed graphs of order n_1 and n_2, respectively. Fix a spanning tree (forest) $\mathcal{T}$ of Γ_2 and a vertex v (collection of vertices $\overline{v}$) in V_{Γ_2}. Then the Laplacian characteristic polynomial of the $\mathcal{T}$-corona product $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ is*

$$P(L_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_2)}; \lambda) = \big(P(L_{(\Gamma_2,\sigma_2)}; \lambda - 1)\big)^{n_1} \cdot P(L_{(\Gamma_1,\sigma_1)}; \lambda - n_2 - \chi_{L_{(\Gamma_2,\sigma_{(2,\mathcal{T})})}}(\lambda - 1)).$$

In particular, the Laplacian spectrum of $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ is completely determined by the Laplacian characteristic polynomials of (Γ_1, σ_1) and (Γ_2, σ_2) along with the coronal of $L_{(\Gamma_2,\sigma_{(2,\mathcal{T})})}$.

Proof. By Theorem 4.2.10, $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_2)$ is switching equivalent to $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_{(2,\mathcal{T})})$ and from the third bullet point of Remark 4.2.6, we have

$$(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_{(2,\mathcal{T})}) = (\Gamma_1, \sigma_1) \circ^+ (\Gamma_2, \sigma_{(2,\mathcal{T})}).$$

So, similarly to the unsigned case [72, Theorem 3.1], the Laplacian matrix of $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_{(2,\mathcal{T})})$ can be written as

$$L_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_{(2,\mathcal{T})})} = \begin{pmatrix} L_{(\Gamma_1,\sigma_1)} + n_2 I_{n_1} & -\mathbf{1}^T_{n_2} \otimes I_{n_1} \\ -(\mathbf{1}^T_{n_2} \otimes I_{n_1})^T & (L_{(\Gamma_2,\sigma_{(2,\mathcal{T})})} + I_{n_2}) \otimes I_{n_1} \end{pmatrix}.$$

Since the Laplacian characteristic polynomials of switching-equivalent signed graphs coincide, it suffices to consider the Laplacian characteristic polynomial of $(\Gamma_1, \sigma_1) \circ_{\mathcal{T}} (\Gamma_2, \sigma_{(2,\mathcal{T})})$. Since (Γ_2, σ_2) and $(\Gamma_2, \sigma_{(2,\mathcal{T})})$ are switching equivalent, by Lemma 1.8.1 we have:

$$\begin{aligned} P(L_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_2)}; \lambda) &= P(L_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_{(2,\mathcal{T})})}; \lambda) \\ &= \det\Big(\lambda I_{n_1(n_2+1)} - L_{(\Gamma_1,\sigma_1)\circ_{\mathcal{T}}(\Gamma_2,\sigma_{(2,\mathcal{T})})}\Big). \end{aligned}$$

The result now follows by expanding this determinant using Lemma 1.8.1 and simplifying via Definition 1.8.3, exactly as carried out in [72, Theorem 3.1]. □

4.3.3 Infinite pairs of adjacency and net Laplacian cospectral signed graphs

We now detail how the results obtained so far allow us to construct infinite families of pairs of signed graphs that share the same underlying graph, adjacency spectrum, and net Laplacian spectrum.

Before proceeding, let (Γ, σ) be a signed graph and let $\mathcal{T}$ be a spanning tree of Γ. We will use the following notation:

$$\bigcirc^{\mathcal{T}}_0 (\Gamma, \sigma) = (\Gamma, \sigma)$$

and, for each $k \geq 1$,

$$\bigcirc^{\mathcal{T}}_k (\Gamma, \sigma) = \big(\bigcirc^{\mathcal{T}}_{k-1}(\Gamma, \sigma)\big) \circ_{\mathcal{T}} (\Gamma, \sigma).$$

Similarly, for each $k \geq 0$, we denote the k-th iterated corona product of the unsigned graph Γ with itself by $\bigcirc_k \Gamma$, and the iterated $\circ^+$ corona product of the signed graph (Γ, σ) with itself by $\bigcirc^+_k (\Gamma, \sigma)$.

Moreover, put $\mathcal{T}^0 = \mathcal{T}$ and, for each $k \geq 1$, let $\mathcal{T}^k$ be a spanning tree of $\bigcirc_k \Gamma$, recursively defined as follows: $\mathcal{T}^k$ is obtained as the union of $\mathcal{T}^{k-1}$, a copy $\mathcal{T}^x$ of $\mathcal{T}$ for each vertex x of $\bigcirc_{k-1}\Gamma$, and exactly one edge $\{x, v\}$ with $x \in \mathcal{T}^{k-1}$ and $v \in \mathcal{T}^x$.

Theorem 4.3.11. *Let (Γ,σ) be a signed graph and let $\mathcal{T}_1$ and $\mathcal{T}_2$ be two spanning trees of Γ. Assume that (Γ,σ) is both $\mathcal{T}_1$-net-regular and $\mathcal{T}_2$-net-regular with the same $\mathcal{T}$-net degree d. Then, for each $k\geq 1$, the signed graphs*

$$\bigcirc_k^{\mathcal{T}_1}(\Gamma,\sigma) \quad \text{and} \quad \bigcirc_k^{\mathcal{T}_2}(\Gamma,\sigma)$$

are adjacency cospectral and the $\mathcal{T}_1^k$-net Laplacian spectrum of $\bigcirc_k^{\mathcal{T}_1}(\Gamma,\sigma)$ coincides with the $\mathcal{T}_2^k$-net Laplacian spectrum of $\bigcirc_k^{\mathcal{T}_2}(\Gamma,\sigma)$.

Proof. This is a direct consequence of Propositions 4.3.6, 4.3.7 and 4.3.8. □

Remark 4.3.12. Notice that, in general, $\bigcirc_k^{\mathcal{T}_1}(\Gamma,\sigma)$ and $\bigcirc_k^{\mathcal{T}_2}(\Gamma,\sigma)$ are not switching isomorphic. An example is provided by the Mickey Mouse signed graph $(\mathcal{M}_{n,s},\Sigma)$ defined in Example 4.3.16 together with the two spanning trees $\mathcal{T}_a$ and $\mathcal{T}_b$.

Remark 4.3.13. Note that Proposition 4.3.10 expresses the Laplacian characteristic polynomial of the $\mathcal{T}$-corona product in terms of the Laplacian characteristic polynomials of (Γ_1,σ_1) and (Γ_2,σ_2), together with the coronal of $(\Gamma_2,\sigma_{(2,\mathcal{T})})$. While the Laplacian characteristic polynomial of a signed graph is invariant under switching equivalence, the coronal is not: two switching-equivalent signed graphs may have different coronals. As a consequence, the pairs of signed graphs constructed in Theorem 4.3.11 are not in general Laplacian cospectral. This is precisely the kind of obstruction that switching-stable notions are designed to avoid, and it underscores a central idea of this chapter.

Corollary 4.3.14. *Assume that (Γ,σ), $\mathcal{T}_1$, and $\mathcal{T}_2$ satisfy the same assumptions as in Theorem 4.3.11 and that Γ is connected. Then the signed graphs*

$$\bigcirc_k^{+}(\Gamma,\sigma_{\mathcal{T}_1}) \quad \text{and} \quad \bigcirc_k^{+}(\Gamma,\sigma_{\mathcal{T}_2})$$

share the same underlying graph and they are cospectral with respect to both the adjacency and net Laplacian matrices. Additionally, if the signed graph $(C\Gamma,\overline{\sigma}^{\mathcal{T}_1})$ is not switching isomorphic to $(C\Gamma,\overline{\sigma}^{\mathcal{T}_2})$, then for each $k\geq 1$, the signed graph $\bigcirc_k^{+}(\Gamma,\sigma_{\mathcal{T}_1})$ is not switching isomorphic to $\bigcirc_k^{+}(\Gamma,\sigma_{\mathcal{T}_2})$.

Proof. The two signed graphs share the same underlying graph. Moreover, they share the same adjacency spectrum and the same net Laplacian spectrum as direct consequences of Theorems 4.2.10 and 4.3.11.

For $k\geq 1$, $\bigcirc_k\Gamma$ is obtained from $\bigcirc_{k-1}\Gamma$ by attaching, to each vertex of $\bigcirc_{k-1}\Gamma$, a copy of Γ via the corona product. If $k=1$ and $|V_\Gamma|=1$, the conclusion is trivial, so assume not. A vertex in a newly attached copy has degree at most $|V_\Gamma|$, while a vertex in the base copy has degree at least $|V_\Gamma|+1$. Hence any isomorphism between two such graphs must preserve the partition of new and base vertices, and consequently maps the family of cones added at the final step to the corresponding family in the other graph. Each such cone is switching equivalent to $(C\Gamma,\overline{\sigma_{\mathcal{T}_1}}^{\mathcal{T}_1})$ or $(C\Gamma,\overline{\sigma_{\mathcal{T}_2}}^{\mathcal{T}_2})$ by Lemma 4.2.8, and so the isomorphism would induce a switching isomorphism between these cones, contradicting the hypothesis. □

Remark 4.3.15. Observe that, given two non-switching-isomorphic signed graphs (Γ,σ_1) and (Γ,σ_2) on the same underlying graph Γ, in which no vertex is adjacent to all the others, the corresponding cones constructed with respect to two spanning trees $\mathcal{T}_1$ and $\mathcal{T}_2$ are also not switching isomorphic. Indeed, writing $n=|V_\Gamma|$, the added vertex w has degree n in the cone, while every $x\in V_\Gamma$ has degree $d_\Gamma(x)+1\leq n-1$; hence w is the unique vertex of maximum degree in both cones, and every switching isomorphism between the cones must fix w. Since the two cone signatures agree with σ_1 and σ_2 on E_Γ, such a switching isomorphism would restrict to a switching isomorphism between (Γ,σ_1) and (Γ,σ_2), contradicting the assumption. The converse, however, does not hold: in Example 4.3.16 we will see that there exist signed graphs that are switching equivalent yet whose cones are not switching isomorphic. This is the reason why Corollary 4.3.14 is stated in terms of the cones rather than the signed graphs themselves, yielding a more general result than one might initially expect.

In general, it is very difficult to construct infinite families of non-switching-isomorphic signed graphs which share the same underlying graph and are simultaneously adjacency and net Laplacian cospectral. Thanks to Corollary 4.3.14, however, it becomes easy to construct several examples of such families. While Proposition 4.3.10 also determines the Laplacian spectrum of the $\mathcal{T}$-corona product, its dependence on the coronal prevents this construction from extending to Laplacian cospectrality in general. This is illustrated in the following example.

Example 4.3.16. Let $n > 3$ be an even integer, and let $1 < s < n-1$ be an integer which is coprime with n. Let $\Gamma_{n,s} = (V_{n,s}, E_{n,s})$ be the graph with $V_{n,s} = \{v_1, v_2, \ldots, v_n\}$ and v_i adjacent to v_j if and only if $[j \pm 1]_n = [i]_n$ or $[j \pm s]_n = [i]_n$, where $[x]_n$ is the class modulo n of x. Define the signature σ on $\Gamma_{n,s}$ by

$$\sigma(v_i, v_j) = \begin{cases} +1, & \text{if } [j \pm 1]_n = [i]_n, \\ -1, & \text{otherwise.} \end{cases}$$

Denote by $\mathcal{T}_1$ the spanning tree with edges $\{v_i, v_{i+1}\}$ for $1 \leq i \leq n-1$. Next, let $\mathcal{T}_2$ be the spanning tree formed by the negative path starting at v_1, proceeding to each subsequent vertex by adding s modulo n, and ending at $v_{[1+(n-1)s]_n}$. For example, when $(n, s) = (8, 3)$, the tree $\mathcal{T}_2$ has edges $\{v_1, v_4\}$, $\{v_4, v_7\}$, $\{v_7, v_2\}$, $\{v_2, v_5\}$, $\{v_5, v_8\}$, $\{v_8, v_3\}$, $\{v_3, v_6\}$.

The graph $(\Gamma_{n,s}, \sigma)$ is simultaneously $\mathcal{T}_1$-net and $\mathcal{T}_2$-net-regular with $d^{\sigma}_{\mathcal{T}_1}(v) = d^{\sigma}_{\mathcal{T}_2}(v) = 0$ for all vertices v. This graph does not satisfy the assumptions in Corollary 4.3.14. Indeed, $(\Gamma_{n,s}, \sigma) = (\Gamma_{n,s}, \sigma_{\mathcal{T}_1})$ and $(\Gamma_{n,s}, \sigma_{\mathcal{T}_2})$ are isomorphic and the same holds for their cones with respect to $\mathcal{T}_1$ and $\mathcal{T}_2$. However, this graph can be used to construct another that does satisfy the assumptions in Corollary 4.3.14.

Consider three copies (Γ_1, σ), (Γ_2, σ), and (Γ_3, σ) of $(\Gamma_{n,s}, \sigma)$. Let $V_{\Gamma_1} = \{x_1, x_2, \ldots, x_n\}$, $V_{\Gamma_2} = \{y_1, y_2, \ldots, y_n\}$ and $V_{\Gamma_3} = \{z_1, z_2, \ldots, z_n\}$. The Mickey Mouse signed graph of size (n, s) is the signed graph $(\mathcal{M}_{n,s}, \Sigma)$ obtained by identifying $x_1 \in V_{\Gamma_1}$ with $y_1 \in V_{\Gamma_2}$, and $y_2 \in V_{\Gamma_2}$ with $z_1 \in V_{\Gamma_3}$. The signature Σ agrees with σ on the edges of each Γ_i, $i = 1, 2, 3$. Then let $\mathcal{T}_a$ be the union of the three copies of $\mathcal{T}_1$ in Γ_1, Γ_2, and Γ_3, and let $\mathcal{T}_b$ be the union of the copies of $\mathcal{T}_1$ from Γ_1 and Γ_3 along with $\mathcal{T}_2$ from Γ_2.

It is easy to check that $(\mathcal{M}_{n,s}, \Sigma)$ is both $\mathcal{T}_a$-net and $\mathcal{T}_b$-net-regular with $d^{\Sigma}_{\mathcal{T}_a}(v) = d^{\Sigma}_{\mathcal{T}_b}(v) = 0$. This time, however, $(\mathcal{M}_{n,s}, \Sigma)$ and the trees $\mathcal{T}_a$ and $\mathcal{T}_b$ satisfy the condition of Corollary 4.3.14, which we shall now prove.

Let w be the new vertex in the cone $C\mathcal{M}_{n,s}$, and consider the triangle wy_1y_2, where y_1 and y_2 are vertices of Γ_2. The cone apex w is the unique vertex of maximum degree in $C\mathcal{M}_{n,s}$; indeed, w has degree $3n-2$ while every base vertex has degree at most 9. Moreover, y_1 and y_2 are the unique vertices of degree 8 in the base $\mathcal{M}_{n,s}$, while every other base vertex has degree 4. Hence any automorphism of $C\mathcal{M}_{n,s}$ must fix w and preserve the set $\{y_1, y_2\}$. Since wy_1y_2 is the only triangle involving w and both y_1 and y_2, the triangle is preserved as a set. However, as a consequence of Equation (4.2.1), the sign of this triangle with respect to the signature induced by $\mathcal{T}_a$ is 1 and the sign of the same triangle with respect to the signature induced by $\mathcal{T}_b$ is -1.

Indeed, the pseudo-potential induced by $\mathcal{T}_a$ with respect to any root is $+1$ for each vertex in $\mathcal{M}_{n,s}$, since all edges of $\mathcal{T}_a$ are positive. For $\mathcal{T}_b$, taking y_1 as the root, the pseudo-potential on each $y_i \in V_{\Gamma_2}$ is computed by traversing the path $y_1, y_{1+s}, y_{1+2s}, \ldots$ (indices modulo n), along which every edge is negative. Since each step multiplies the pseudo-potential by -1, the pseudo-potential of y_{1+js} is $(-1)^j$, that is, $+1$ on $y_i \in V_{\Gamma_2}$ if and only if i is odd. Since $\mathcal{T}_b$ uses $\mathcal{T}_1$ in Γ_1, all edges of which are positive, and x_1 is identified with y_1, the pseudo-potential is $+1$ on each $x_i \in V_{\Gamma_1}$. Finally, since $\mathcal{T}_b$ also uses $\mathcal{T}_1$ in Γ_3 and z_1 is identified with y_2, whose pseudo-potential is -1, the pseudo-potential is -1 on each $z_i \in V_{\Gamma_3}$.

This implies that the two cones are not switching isomorphic. Hence, as a consequence of Corollary 4.3.14, for each $k \geq 1$, the signed graphs

$$\bigcirc_k^+(\mathcal{M}_{n,s}, \Sigma_{\mathcal{T}_a}) \quad \text{and} \quad \bigcirc_k^+(\mathcal{M}_{n,s}, \Sigma_{\mathcal{T}_b})$$

are non-switching-isomorphic signed graphs sharing the same underlying graph, the same adjacency spectrum, and the same net Laplacian spectrum for each pair (n, s) where n is even and s is coprime with n and $1 < s < n-1$.

Furthermore, a direct computation shows that $\bigcirc_1^+(\mathcal{M}_{8,3}, \Sigma_{\mathcal{T}_a})$ and $\bigcirc_1^+(\mathcal{M}_{8,3}, \Sigma_{\mathcal{T}_b})$ are adjacency cospectral but not Laplacian cospectral, confirming that Corollary 4.3.14 cannot be extended to include Laplacian cospectrality in general.

The Mickey Mouse graph $(\mathcal{M}_{8,3}, \Sigma)$ and its spanning trees $\mathcal{T}_a$ and $\mathcal{T}_b$ are presented in Figure 4.4.

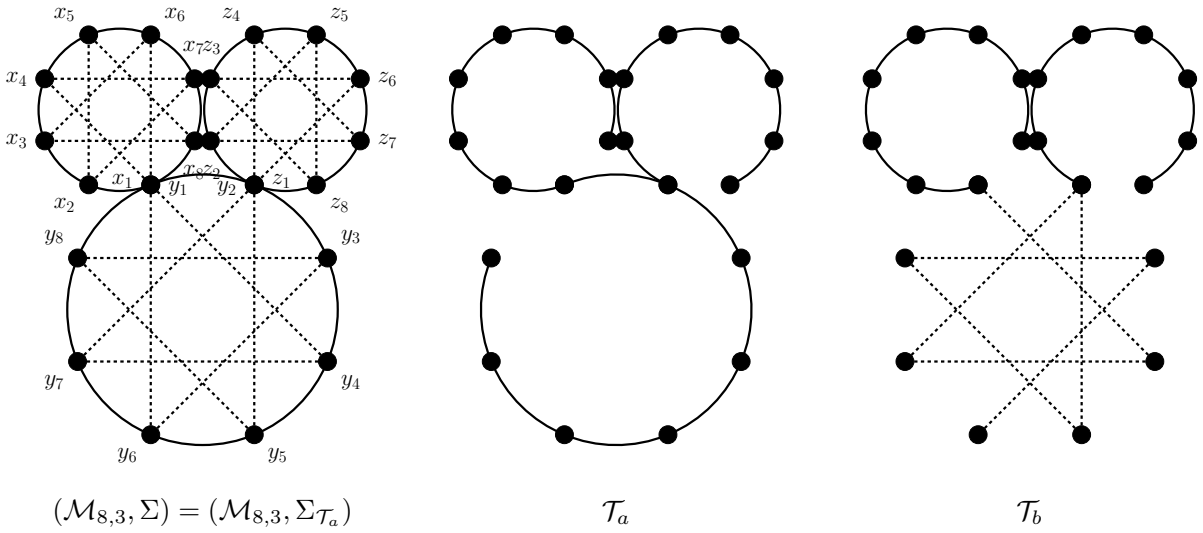


Figure 4.4: The Mickey Mouse graph $(\mathcal{M}_{8,3}, \Sigma)$ and its spanning trees $\mathcal{T}_a$ and $\mathcal{T}_b$.

4.4 Summary and outlook

We conclude by observing that the switching-stable corona product defined in this chapter extends naturally to the more general setting of gain graphs; indeed, such extensions for other operations are developed in Chapters 5 and 7.

Exactly as in the case of signed graphs, there is a line of research devoted to extending classical graph products to products of gain graphs. Some constructions have been generalised to gain graphs with values in arbitrary groups [32, 34, 35], some have been defined for complex unit gain graphs [23, 61], and others have been developed exclusively for the signed case.

Our intuition is that, by suitably extending the pseudo-potential machinery in the broader context of gain graphs, it should be possible to provide a switching-stable definition for gain graphs analogous to the one given here. The question that remains, at least in the case of complex unit gain graphs, is whether the good spectral properties of the (net) Laplacian of the corona product are preserved under the various generalisations of the Laplacian.

Chapter 5

On H-joins of complex unit gain graphs and their stability

Let H be a graph of order k and let $\mathcal{F} = \{\Gamma_1, \Gamma_2, \ldots, \Gamma_k\}$ be a family of vertex-disjoint graphs. Then the H-join of the family $\mathcal{F}$ is obtained by replacing each vertex v_i of H with the graph Γ_i of $\mathcal{F}$ and preserving the adjacencies existing in H. This chapter presents two distinct definitions of the H-join of complex unit gain graphs. The first is a direct extension of the H-join of signed graphs and, as in that setting, is switching stable only with respect to H. The second, on the other hand, requires the fixing of a spanning forest in order to be defined but has the advantage of being stable under switchings of both H and the component Γ_i graphs. When applied to signed graphs this provides a new definition of the H-join which does not coincide with the usual version and has the stronger property of being switching stable with respect to all of the graphs involved. A spectral analysis is provided for both constructions.

This chapter is based on [61], joint work with Stefano Spessato and Francesco Belardo.

5.1 Introduction and preliminaries

In this chapter, we focus on the H-join of complex unit gain graphs. The first iteration of the operation we are studying was introduced as the generalised composition of graphs by Schwenk in 1974 in [87]. It was then rediscovered in 2011 by Cardoso et al. in [28] as an application of generalising Fiedler's lemma. Since then there has been much work in the theory of both ordinary graphs and signed graphs. For more information, see [41, 26, 27, 29, 57, 71, 85, 99, 104].

In [33], the authors introduce the notion of pseudo-potential functions on signed graphs and give a solution to an open problem concerning the switching stability of the lexicographic product of signed graphs. Following the approach of works such as [26] and [104], we begin by providing a direct extension of the H-join to the setting of complex unit gain graphs. In the second half of this chapter, we extend the pseudo-potential functions of [33] to complex unit gain graphs and, together with the direct extension of the H-join from the first half, define a distinct and more robust construction that avoids the stability and balance issues present in the original setting. Since many join operations and families of graphs can be recovered as special cases or subgraphs of the H-join, H-generalised-join, and H_m-join operations, see [41], our novel approach offers a unified solution to a wide range of potential stability and balance problems.

The remainder of this chapter is structured as follows. In Section 5.2, we define the H-join operation of complex unit gain graphs, determine the characteristic polynomial and spectrum, show that the operation is stable under switching of (H, ψ), and give a characterisation of the balancedness of the produced graph. In Section 5.3, we define pseudo-potential functions on complex unit gain graphs which enable us to provide an alternative definition for the gain function of an H-join, one that is stable under switchings of both (H, ψ) and the component (Γ_i, φ_i) graphs and consequently relaxes the requirements for balancedness. In Section 5.4, we present a chronological survey of signatures and gain functions that have been given for the lexicographic product of two graphs, an operation that can be viewed as a special

case of the H-join, in which each Γ_i is isomorphic to the others, along with some discussion of how these relate to the constructions that we have proposed. In Section 5.5, we provide some concluding remarks.

5.2 The H-join operation

We shall begin with the definition of the H-join of complex unit gain graphs and then detail the structuring of its adjacency matrix along with the computations of its characteristic polynomial and spectrum. We conclude the section by establishing the stability of the H-join under switching of (H, ψ) and characterising when the join is balanced.

5.2.1 Definition

First we describe how this operation works with ordinary graphs. Let H be a graph of order k and let $\mathcal{F} = \{\Gamma_1, \Gamma_2, \ldots, \Gamma_k\}$ be a family of vertex-disjoint graphs. The H-join of $\mathcal{F}$ is obtained by first replacing each vertex v_i of H with the graph Γ_i of $\mathcal{F}$ and then, for each edge between vertices v_i and v_j in E_H, introducing new edges between every vertex in Γ_i and every vertex in Γ_j.

This procedure can be taken to the realm of complex unit gain graphs with the following definition.

Definition 5.2.1. Let (H, ψ) be a complex unit gain graph with $V_H = \{v_1, v_2, \ldots, v_k\}$ and let $\mathcal{F} = \{(\Gamma_1, \varphi_1), (\Gamma_2, \varphi_2), \ldots, (\Gamma_k, \varphi_k)\}$ be a family of vertex-disjoint complex unit gain graphs where (Γ_i, φ_i) is a complex unit gain graph of order n_i for each $1 \leq i \leq k$. Then the H-join of this family is the complex unit gain graph $(\Gamma, \Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1, \varphi_1), (\Gamma_2, \varphi_2), \ldots, (\Gamma_k, \varphi_k)\}$, sometimes denoted $(\Gamma, \Phi) = \bigvee^{\mathcal{F}}_{(H,\psi)}$ when the family is clear from context, with vertex set $V_\Gamma = \bigcup_{i=1}^{k} V_{\Gamma_i}$, edge set

$$E_\Gamma = \left(\bigcup_{i=1}^{k} E_{\Gamma_i}\right) \cup \left(\bigcup_{v_iv_j \in E_H} \{uv : u \in V_{\Gamma_i}, v \in V_{\Gamma_j}\}\right),$$

and gain function Φ satisfying

$$\Phi(uv) = \begin{cases} \varphi_i(uv), & \text{if } u, v \in V_{\Gamma_i} \text{ and } uv \in E_{\Gamma_i}, \\ \psi(v_iv_j), & \text{if } u \in V_{\Gamma_i},\ v \in V_{\Gamma_j}, \text{ and } v_iv_j \in E_H. \end{cases}$$

Note that the gain function Φ is well defined since $\Phi(uv) = \Phi(vu)^{-1}$ follows immediately from $\varphi_i(uv) = \varphi_i(vu)^{-1}$ and $\psi(v_iv_j) = \psi(v_jv_i)^{-1}$.

5.2.2 Adjacency matrix and characteristic polynomial

We can now describe the adjacency matrix of a complex unit gain graph obtained via the H-join procedure and, consequently, determine its characteristic polynomial.

Let (H, ψ) be a complex unit gain graph with $V_H = \{v_1, v_2, \ldots, v_k\}$. Consider the H-join graph $(\Gamma, \Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1, \varphi_1), (\Gamma_2, \varphi_2), \ldots, (\Gamma_k, \varphi_k)\}$ where (Γ_i, φ_i) is a complex unit gain graph of order n_i for each $1 \leq i \leq k$. Then, with consistent vertex labellings, the adjacency matrix of (Γ, Φ) can be expressed as the $k \times k$ block matrix $A_{(\Gamma,\Phi)} = (A_{ij})$ with

$$A_{ij} = \begin{cases} A_{(\Gamma_i,\varphi_i)}, & \text{if } i = j, \\ \psi(v_iv_j)J_{n_i \times n_j}, & \text{if } i \neq j \text{ and } v_iv_j \in E_H, \\ O_{n_i \times n_j}, & \text{otherwise.} \end{cases}$$

An example, with the blocks clearly depicted, is given in Example 5.2.2.

Example 5.2.2. Take $(H, \psi) = (P_3, \psi)$, $(\Gamma_1, \varphi_1) = (K_3, \varphi_1)$, $(\Gamma_2, \varphi_2) = (K_2, \varphi_2)$, and $(\Gamma_3, \varphi_3) = (C_4, \varphi_3)$ with gain functions and vertex labellings as shown in Figure 5.1. Then for the adjacency matrix of the

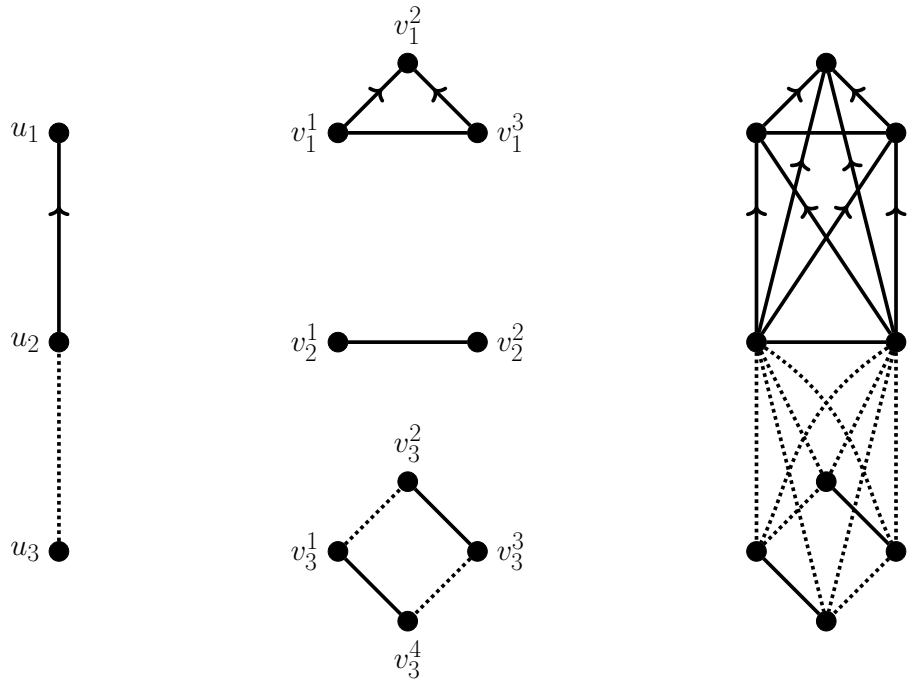


Figure 5.1: Take $(H,\psi) = (P_3,\psi)$, $(\Gamma_1,\varphi_1) = (K_3,\varphi_1)$, $(\Gamma_2,\varphi_2) = (K_2,\varphi_2)$, and $(\Gamma_3,\varphi_3) = (C_4,\varphi_3)$ with gain functions and vertex labellings as shown. Then the right-hand side of the diagram depicts the H-join of $\mathcal{F} = \{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),(\Gamma_3,\varphi_3)\}$. Here and throughout this chapter, solid lines represent positive edges, dashed lines represent negative edges, and lines directed from a to b represent edges with gain functions $\varphi(ab) = i$ and $\varphi(ba) = -i$.

H-join of $\mathcal{F} = \{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),(\Gamma_3,\varphi_3)\}$ we have

$$A_{(\Gamma,\Phi)} = \left(\begin{array}{ccc|cc|cccc} 0 & i & 1 & -i & -i & 0 & 0 & 0 & 0 \\ -i & 0 & -i & -i & -i & 0 & 0 & 0 & 0 \\ 1 & i & 0 & -i & -i & 0 & 0 & 0 & 0 \\ \hline i & i & i & 0 & 1 & -1 & -1 & -1 & -1 \\ i & i & i & 1 & 0 & -1 & -1 & -1 & -1 \\ \hline 0 & 0 & 0 & -1 & -1 & 0 & -1 & 0 & 1 \\ 0 & 0 & 0 & -1 & -1 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & -1 & -1 & 1 & 0 & -1 & 0 \end{array}\right).$$

Theorem 5.2.3. *Consider a complex unit gain graph* $(\Gamma,\Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\ldots,(\Gamma_k,\varphi_k)\}$. *Define the characteristic polynomial of* (Γ_i,φ_i) *for each* $1 \le i \le k$ *as* $p_i = \det\big(\lambda I_{n_i} - A_{(\Gamma_i,\varphi_i)}\big)$ *and denote* $\chi_i(\lambda) = \mathbf{1}_{n_i}^T(\lambda I_{n_i} - A_{(\Gamma_i,\varphi_i)})^{-1}\mathbf{1}_{n_i}$. *Then, the characteristic polynomial of* (Γ,Φ) *is*

$$P(A_{(\Gamma,\Phi)};\,\lambda) = \left(\prod_{i=1}^{k} p_i\right)\det(M),$$

where $M = (m_{ij})$ *is the* $k \times k$ *matrix with*

$$m_{ij} = \begin{cases} 1, & \text{if } i = j, \\ -\psi(v_iv_j)\chi_i, & \text{if } v_iv_j \in E_H, \\ 0, & \text{otherwise.} \end{cases}$$

Proof. Within this proof we shall use ψ_{ij} to denote $\psi(v_iv_j)$. Thus, we have

$$A_{(\Gamma,\Phi)} = \begin{pmatrix} A_{(\Gamma_1,\varphi_1)} & \psi_{12}J_{n_1\times n_2} & \dots & \psi_{1k}J_{n_1\times n_k} \\ \psi_{21}J_{n_2\times n_1} & A_{(\Gamma_2,\varphi_2)} & \dots & \psi_{2k}J_{n_2\times n_k} \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{k1}J_{n_k\times n_1} & \psi_{k2}J_{n_k\times n_2} & \dots & A_{(\Gamma_k,\varphi_k)} \end{pmatrix}.$$

Since

$$P(A_{(\Gamma,\Phi)};\,\lambda) = \det \underbrace{\begin{pmatrix} \lambda I_{n_1} - A_{(\Gamma_1,\varphi_1)} & -\psi_{12}J_{n_1\times n_2} & \dots & -\psi_{1,k-1}J_{n_1\times n_{k-1}} & -\psi_{1k}J_{n_1\times n_k} \\ -\psi_{21}J_{n_2\times n_1} & \lambda I_{n_2} - A_{(\Gamma_2,\varphi_2)} & \dots & -\psi_{2,k-1}J_{n_2\times n_{k-1}} & -\psi_{2k}J_{n_2\times n_k} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -\psi_{k-1,1}J_{n_{k-1}\times n_1} & -\psi_{k-1,2}J_{n_{k-1}\times n_2} & \dots & \lambda I_{n_{k-1}} - A_{(\Gamma_{k-1},\varphi_{k-1})} & -\psi_{k-1,k}J_{n_{k-1}\times n_k} \\ -\psi_{k1}J_{n_k\times n_1} & -\psi_{k2}J_{n_k\times n_2} & \dots & -\psi_{k,k-1}J_{n_k\times n_{k-1}} & \lambda I_{n_k} - A_{(\Gamma_k,\varphi_k)} \end{pmatrix}}_{Q},$$

partitioning this matrix as the block matrix $Q = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$ such that

$$\begin{aligned} A &= \begin{pmatrix} \lambda I_{n_1} - A_{(\Gamma_1,\varphi_1)} & -\psi_{12}J_{n_1\times n_2} & \dots & -\psi_{1,k-1}J_{n_1\times n_{k-1}} \\ -\psi_{21}J_{n_2\times n_1} & \lambda I_{n_2} - A_{(\Gamma_2,\varphi_2)} & \dots & -\psi_{2,k-1}J_{n_2\times n_{k-1}} \\ \vdots & \vdots & \ddots & \vdots \\ -\psi_{k-1,1}J_{n_{k-1}\times n_1} & -\psi_{k-1,2}J_{n_{k-1}\times n_2} & \dots & \lambda I_{n_{k-1}} - A_{(\Gamma_{k-1},\varphi_{k-1})} \end{pmatrix}, \\ B &= \begin{pmatrix} -\psi_{1k}J_{n_1\times n_k} \\ -\psi_{2k}J_{n_2\times n_k} \\ \vdots \\ -\psi_{k-1,k}J_{n_{k-1}\times n_k} \end{pmatrix}, \\ C &= \begin{pmatrix} -\psi_{k1}J_{n_k\times n_1} & -\psi_{k2}J_{n_k\times n_2} & \dots & -\psi_{k,k-1}J_{n_k\times n_{k-1}} \end{pmatrix}, \\ D &= \begin{pmatrix} \lambda I_{n_k} - A_{(\Gamma_k,\varphi_k)} \end{pmatrix}, \end{aligned}$$

and applying Schur's determinant formula, it follows that

$$P(A_{(\Gamma,\Phi)};\,\lambda) = \det(Q) = \det(D)\det\big(A - BD^{-1}C\big) = \det(D)\det(Q/D), \tag{5.2.1}$$

where $\det(Q/D)$ is the Schur complement of the matrix D in Q.

Let us compute the determinant of Q/D, taking into account that

$$\begin{aligned} \psi_{ij}\psi_{ji} &= \psi_{ij}\psi_{ij}^{-1} = 1, \\ J_{n_i\times n_j} &= \mathbf{1}_{n_i}\mathbf{1}_{n_j}^T, \\ \chi_i &= \mathbf{1}_{n_i}^T(\lambda I_{n_i} - A_{(\Gamma_i,\varphi_i)})^{-1}\mathbf{1}_{n_i}. \end{aligned}$$

First of all, we have

$$\begin{aligned} \det(Q/D) &= \det\left(A - \underbrace{\begin{pmatrix} -\psi_{1k}J_{n_1\times n_k} \\ -\psi_{2k}J_{n_2\times n_k} \\ \vdots \\ -\psi_{k-1,k}J_{n_{k-1}\times n_k} \end{pmatrix}}_{B} \underbrace{\big(\lambda I_{n_k} - A_{(\Gamma_k,\varphi_k)}\big)^{-1}}_{D^{-1}} \underbrace{\begin{pmatrix} -\psi_{k1}J_{n_k\times n_1} \\ -\psi_{k2}J_{n_k\times n_2} \\ \vdots \\ -\psi_{k,k-1}J_{n_k\times n_{k-1}} \end{pmatrix}^T}_{C}\right) \\ &= \det\left(A - \begin{pmatrix} \psi_{1k}\psi_{k1}\mathbf{1}_{n_1}\chi_k\mathbf{1}_{n_1}^T & \psi_{1k}\psi_{k2}\mathbf{1}_{n_1}\chi_k\mathbf{1}_{n_2}^T & \dots & \psi_{1k}\psi_{k,k-1}\mathbf{1}_{n_1}\chi_k\mathbf{1}_{n_{k-1}}^T \\ \psi_{2k}\psi_{k1}\mathbf{1}_{n_2}\chi_k\mathbf{1}_{n_1}^T & \psi_{2k}\psi_{k2}\mathbf{1}_{n_2}\chi_k\mathbf{1}_{n_2}^T & \dots & \psi_{2k}\psi_{k,k-1}\mathbf{1}_{n_2}\chi_k\mathbf{1}_{n_{k-1}}^T \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{k-1,k}\psi_{k1}\mathbf{1}_{n_{k-1}}\chi_k\mathbf{1}_{n_1}^T & \psi_{k-1,k}\psi_{k2}\mathbf{1}_{n_{k-1}}\chi_k\mathbf{1}_{n_2}^T & \dots & \psi_{k-1,k}\psi_{k,k-1}\mathbf{1}_{n_{k-1}}\chi_k\mathbf{1}_{n_{k-1}}^T \end{pmatrix}\right). \end{aligned}$$

Substituting the block form of A and collecting terms, the diagonal blocks simplify as noted above, giving

$$\begin{aligned}
\det(Q/D) &= \begin{pmatrix} \lambda I_{n_1} - A_{(\Gamma_1,\varphi_1)} & O & \dots & O \\ O & \lambda I_{n_2} - A_{(\Gamma_2,\varphi_2)} & \dots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \dots & \lambda I_{n_{k-1}} - A_{(\Gamma_{k-1},\varphi_{k-1})} \end{pmatrix} \\
&\quad - \begin{pmatrix} \mathbf{1}_{n_1}\chi_k\mathbf{1}_{n_1}^T & \mathbf{1}_{n_1}(\psi_{1k}\psi_{k2}\chi_k + \psi_{12})\mathbf{1}_{n_2}^T & \dots & \mathbf{1}_{n_1}(\psi_{1k}\psi_{k,k-1}\chi_k + \psi_{1,k-1})\mathbf{1}_{n_{k-1}}^T \\ \mathbf{1}_{n_2}(\psi_{2k}\psi_{k1}\chi_k + \psi_{21})\mathbf{1}_{n_1}^T & \mathbf{1}_{n_2}\chi_k\mathbf{1}_{n_2}^T & \dots & \mathbf{1}_{n_2}(\psi_{2k}\psi_{k,k-1}\chi_k + \psi_{2,k-1})\mathbf{1}_{n_{k-1}}^T \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1}_{n_{k-1}}(\psi_{k-1,k}\psi_{k1}\chi_k + \psi_{k-1,1})\mathbf{1}_{n_1}^T & \mathbf{1}_{n_{k-1}}(\psi_{k-1,k}\psi_{k2}\chi_k + \psi_{k-1,2})\mathbf{1}_{n_2}^T & \dots & \mathbf{1}_{n_{k-1}}\chi_k\mathbf{1}_{n_{k-1}}^T \end{pmatrix} \\
&= \begin{pmatrix} \lambda I_{n_1} - A_{(\Gamma_1,\varphi_1)} & O & \dots & O \\ O & \lambda I_{n_2} - A_{(\Gamma_2,\varphi_2)} & \dots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \dots & \lambda I_{n_{k-1}} - A_{(\Gamma_{k-1},\varphi_{k-1})} \end{pmatrix} \\
&\quad - \left(\begin{pmatrix} \mathbf{1}_{n_1}\chi_k & \mathbf{1}_{n_1}(\psi_{1k}\psi_{k2}\chi_k + \psi_{12}) & \dots & \mathbf{1}_{n_1}(\psi_{1k}\psi_{k,k-1}\chi_k + \psi_{1,k-1}) \\ \mathbf{1}_{n_2}(\psi_{2k}\psi_{k1}\chi_k + \psi_{21}) & \mathbf{1}_{n_2}\chi_k & \dots & \mathbf{1}_{n_2}(\psi_{2k}\psi_{k,k-1}\chi_k + \psi_{2,k-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1}_{n_{k-1}}(\psi_{k-1,k}\psi_{k1}\chi_k + \psi_{k-1,1}) & \mathbf{1}_{n_{k-1}}(\psi_{k-1,k}\psi_{k2}\chi_k + \psi_{k-1,2}) & \dots & \mathbf{1}_{n_{k-1}}\chi_k \end{pmatrix} \right. \\
&\quad \left. \cdot \begin{pmatrix} \mathbf{1}_{n_1}^T & O & \dots & O \\ O & \mathbf{1}_{n_2}^T & \dots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \dots & \mathbf{1}_{n_{k-1}}^T \end{pmatrix} \right).
\end{aligned}$$

At this point we can apply Lemma 1.8.2(b) to obtain the determinant of Q/D as

$$\begin{aligned}
\det(Q/D) &= \left(\prod_{i=1}^{k-1} p_i\right) \det\left(I_{k-1} - \begin{pmatrix} \mathbf{1}_{n_1}^T & O & \dots & O \\ O & \mathbf{1}_{n_2}^T & \dots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \dots & \mathbf{1}_{n_{k-1}}^T \end{pmatrix} \begin{pmatrix} \lambda I_{n_1} - A_{(\Gamma_1,\varphi_1)} & O & \dots & O \\ O & \lambda I_{n_2} - A_{(\Gamma_2,\varphi_2)} & \dots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \dots & \lambda I_{n_{k-1}} - A_{(\Gamma_{k-1},\varphi_{k-1})} \end{pmatrix}^{-1} \right. \\
&\quad \left. \cdot \begin{pmatrix} \mathbf{1}_{n_1}\chi_k & \mathbf{1}_{n_1}(\psi_{1k}\psi_{k2}\chi_k + \psi_{12}) & \dots & \mathbf{1}_{n_1}(\psi_{1k}\psi_{k,k-1}\chi_k + \psi_{1,k-1}) \\ \mathbf{1}_{n_2}(\psi_{2k}\psi_{k1}\chi_k + \psi_{21}) & \mathbf{1}_{n_2}\chi_k & \dots & \mathbf{1}_{n_2}(\psi_{2k}\psi_{k,k-1}\chi_k + \psi_{2,k-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{1}_{n_{k-1}}(\psi_{k-1,k}\psi_{k1}\chi_k + \psi_{k-1,1}) & \mathbf{1}_{n_{k-1}}(\psi_{k-1,k}\psi_{k2}\chi_k + \psi_{k-1,2}) & \dots & \mathbf{1}_{n_{k-1}}\chi_k \end{pmatrix} \right) \\
&= \left(\prod_{i=1}^{k-1} p_i\right) \det\left(I_{k-1} \right. \\
&\quad \left. - \begin{pmatrix} \chi_1\chi_k & \chi_1(\psi_{1k}\psi_{k2}\chi_k + \psi_{12}) & \dots & \chi_1(\psi_{1k}\psi_{k,k-1}\chi_k + \psi_{1,k-1}) \\ \chi_2(\psi_{2k}\psi_{k1}\chi_k + \psi_{21}) & \chi_2\chi_k & \dots & \chi_2(\psi_{2k}\psi_{k,k-1}\chi_k + \psi_{2,k-1}) \\ \vdots & \vdots & \ddots & \vdots \\ \chi_{k-1}(\psi_{k-1,k}\psi_{k1}\chi_k + \psi_{k-1,1}) & \chi_{k-1}(\psi_{k-1,k}\psi_{k2}\chi_k + \psi_{k-1,2}) & \dots & \chi_{k-1}\chi_k \end{pmatrix} \right) \\
&= \left(\prod_{i=1}^{k-1} p_i\right) \det\left(\begin{pmatrix} 1 & -\psi_{12}\chi_1 & \dots & -\psi_{1,k-1}\chi_1 \\ -\psi_{21}\chi_2 & 1 & \dots & -\psi_{2,k-1}\chi_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\psi_{k-1,1}\chi_{k-1} & -\psi_{k-1,2}\chi_{k-1} & \dots & 1 \end{pmatrix} \right. \\
&\quad \left. - \begin{pmatrix} \chi_1\chi_k & \psi_{1k}\psi_{k2}\chi_1\chi_k & \dots & \psi_{1k}\psi_{k,k-1}\chi_1\chi_k \\ \psi_{2k}\psi_{k1}\chi_2\chi_k & \chi_2\chi_k & \dots & \psi_{2k}\psi_{k,k-1}\chi_2\chi_k \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{k-1,k}\psi_{k1}\chi_{k-1}\chi_k & \psi_{k-1,k}\psi_{k2}\chi_{k-1}\chi_k & \dots & \chi_{k-1}\chi_k \end{pmatrix} \right) \\
&= \left(\prod_{i=1}^{k-1} p_i\right) \det\left(\begin{pmatrix} 1 & -\psi_{12}\chi_1 & \dots & -\psi_{1,k-1}\chi_1 \\ -\psi_{21}\chi_2 & 1 & \dots & -\psi_{2,k-1}\chi_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\psi_{k-1,1}\chi_{k-1} & -\psi_{k-1,2}\chi_{k-1} & \dots & 1 \end{pmatrix} - \begin{pmatrix} \psi_{1k}\chi_1 \\ \psi_{2k}\chi_2 \\ \vdots \\ \psi_{k-1,k}\chi_{k-1} \end{pmatrix} (1) \begin{pmatrix} \psi_{k1}\chi_k \\ \psi_{k2}\chi_k \\ \vdots \\ \psi_{k,k-1}\chi_k \end{pmatrix}^T \right).
\end{aligned}$$

Employing a reversal of Schur's complement formula here gives us

$$\det(Q/D) = \left(\prod_{i=1}^{k-1} p_i\right) \det\begin{pmatrix} 1 & -\psi_{12}\chi_1 & \dots & -\psi_{1k}\chi_1 \\ -\psi_{21}\chi_2 & 1 & \dots & -\psi_{2k}\chi_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\psi_{k1}\chi_k & -\psi_{k2}\chi_k & \dots & 1 \end{pmatrix}.$$

Finally, from Equation (5.2.1) and substituting $\psi(v_iv_j)$ back in for ψ_{ij}, we have

$$P(A_{(\Gamma,\Phi)};\,\lambda) = \left(\prod_{i=1}^{k} p_i\right) \det \begin{pmatrix} 1 & -\psi(v_1v_2)\chi_1 & \dots & -\psi(v_1v_k)\chi_1 \\ -\psi(v_2v_1)\chi_2 & 1 & \dots & -\psi(v_2v_k)\chi_2 \\ \vdots & \vdots & \ddots & \vdots \\ -\psi(v_kv_1)\chi_k & -\psi(v_kv_2)\chi_k & \dots & 1 \end{pmatrix} = \left(\prod_{i=1}^{k} p_i\right) \det(M).$$

This concludes the proof. □

5.2.3 Spectrum

Here, we adapt ideas presented by the authors in [85] to our context so that we can derive a spectral description of the H-join in terms of the eigenvalues of the component (Γ_i, φ_i) graphs.

First we recall the definition of what it means for an eigenvalue to be main or non-main.

Definition 5.2.4 ([85, Definition 2])**.** Let M be an $n \times n$ normal complex matrix and let $\mathbf{u}$ be an $n \times 1$ complex vector. An eigenvalue λ of M is called a $\mathbf{u}$-main eigenvalue if the corresponding eigenspace $\mathcal{E}_M(\lambda)$ is not orthogonal to $\mathbf{u}$. If an eigenvalue of M is not $\mathbf{u}$-main, we say it is $\mathbf{u}$-non-main. In the case of $\mathbf{u} = \mathbf{1}_n$ we do not specify the vector and simply call the eigenvalue λ a main or non-main eigenvalue of M.

Now we shall state a lemma from [85] along with its proof when $\mathbf{u} = \mathbf{1}_n$, as this will be useful later on.

Lemma 5.2.5 ([85, Lemma 2])**.** *Let M be an $n \times n$ normal complex matrix and let $\mathbf{u}$ be an $n \times 1$ complex vector. Then the singularities of $\mathbf{u}^*(\lambda I - M)^{-1}\mathbf{u}$ are simple and correspond to the $\mathbf{u}$-main eigenvalues of M.*

Proof. We consider only the case when $\mathbf{u} = \mathbf{1}_n$. Let $\{\theta_1, \theta_2, \dots, \theta_k\}$ be the distinct eigenvalues of M ordered so that $\{\theta_1, \theta_2, \dots, \theta_m\}$, with $m \leq k$, is the set of main eigenvalues of M. Suppose that the spectral decomposition of M is $M = \sum_{i=1}^{k} \theta_i E_{\theta_i}$, where E_{θ_i} is the orthogonal projection on the eigenspace of θ_i. Then $(\lambda I - M)^{-1} = \sum_{i=1}^{k} \frac{E_{\theta_i}}{\lambda - \theta_i}$ and $\chi_M(\lambda) = \mathbf{1}_n^T(\lambda I - M)^{-1}\mathbf{1}_n = \sum_{i=1}^{k} \frac{\mathbf{1}_n^T E_{\theta_i}\mathbf{1}_n}{\lambda - \theta_i}$. Note that $\mathbf{1}_n^T E_{\theta_i}\mathbf{1}_n \neq 0$ if and only if θ_i is a main eigenvalue of M, so $\chi_M(\lambda) = \sum_{i=1}^{m} \frac{\mathbf{1}_n^T E_{\theta_i}\mathbf{1}_n}{\lambda - \theta_i}$, and the result follows. □

In [85, Theorem 2], with $\rho_{i,j}$ being arbitrary complex numbers for $1 \leq i, j \leq k$ and $i \neq j$, they define the function $\gamma_i(\lambda) = \gamma_{M_i}(u_i, v_i; \lambda) = v_i^T(\lambda I - M_i)^{-1}u_i$ and two matrices

$$A(\mathbf{M}, \mathbf{u}, \rho) := \begin{pmatrix} M_1 & \rho_{1,2}u_1v_2^T & \dots & \rho_{1,k}u_1v_k^T \\ \rho_{2,1}u_2v_1^T & M_2 & \dots & \rho_{2,k}u_2v_k^T \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{k,1}u_kv_1^T & \rho_{k,2}u_kv_2^T & \dots & M_k \end{pmatrix}, \quad \widetilde{A}(\mathbf{M}, \mathbf{u}, \rho) := \begin{pmatrix} \frac{1}{\gamma_1(\lambda)} & -\rho_{1,2} & \dots & -\rho_{1,k} \\ -\rho_{2,1} & \frac{1}{\gamma_2(\lambda)} & \dots & -\rho_{2,k} \\ \vdots & \vdots & \ddots & \vdots \\ -\rho_{k,1} & -\rho_{k,2} & \dots & \frac{1}{\gamma_k(\lambda)} \end{pmatrix}.$$

For full details, the reader is directed to that paper. For our purposes, we can see that when $M_i = A_{(\Gamma_i,\varphi_i)}$, $u_i = v_i = \mathbf{1}_{n_i}$, and $\rho_{i,j} = \varphi(v_iv_j)$, we have $\gamma_i(\lambda) = \chi_i(\lambda)$ and $A(\mathbf{M}, \mathbf{u}, \rho) = A_{(\Gamma,\Phi)}$.

Since the adjacency matrices of the complex unit gain graphs (Γ_i, φ_i) are normal, we can suppose that $\{\theta_1, \theta_2, \dots, \theta_{m_i}\}$ is the set of distinct main eigenvalues of (Γ_i, φ_i) for each $1 \leq i \leq k$ and then, as discussed in the proof of Lemma 5.2.5, we can write $\chi_i = \frac{f_i}{g_i}$ where $g_i = \prod_{j=1}^{m_i}(\lambda - \theta_j)$. Hence, by Theorem 5.2.3, we have

$$P(A_{(\Gamma,\Phi)};\,\lambda) = \left(\prod_{i=1}^{k} \frac{p_i}{g_i}\right) \det\left(\Omega(\lambda)\right), \tag{5.2.2}$$

where $\Omega(\lambda) = (\omega_{ij}(\lambda))$ is the $k \times k$ matrix with

$$\omega_{ij}(\lambda) = \begin{cases} g_i(\lambda), & \text{if } i = j, \\ -\psi(v_iv_j)f_i(\lambda), & \text{if } i \neq j. \end{cases}$$

Consequently, we can describe the spectrum of the H-join as follows.

Theorem 5.2.6. *Let (H,ψ) be a complex unit gain graph with $V_H = \{v_1, v_2, \ldots, v_k\}$ and let $\mathcal{F} = \{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\ldots,(\Gamma_k,\varphi_k)\}$ be a family of vertex-disjoint graphs where (Γ_i,φ_i) is a complex unit gain graph of order n_i for each $1 \leq i \leq k$. Consider the H-join of this family, $(\Gamma,\Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\ldots,(\Gamma_k,\varphi_k)\}$. Then*

(a) if λ is a non-main eigenvalue of (Γ_i,φ_i) of multiplicity μ, it follows that λ is an eigenvalue of (Γ,Φ) of multiplicity at least μ,

(b) if λ is a main eigenvalue of (Γ_i,φ_i) of multiplicity μ, it follows that λ is an eigenvalue of (Γ,Φ) of multiplicity at least $\mu-1$,

(c) the remaining eigenvalues of (Γ,Φ) are precisely the roots of the polynomial $\det(\Omega(\lambda))$.

Proof. By Lemma 5.2.5, the singularities of χ_i are main eigenvalues and they are simple. Now the proof easily follows from Equation (5.2.2). □

Recall the notions of outgain and ingain for gain graphs from Section 1.5. Restricting these to the group of complex units, we have the following.

Definition 5.2.7. Let u be a vertex of a complex unit gain graph (Γ,φ). The numbers

$$d^{\rightarrow}_{(\Gamma,\varphi)}(u) = \sum_{uv\in E_\Gamma} \varphi(uv) \qquad \text{and} \qquad d^{\leftarrow}_{(\Gamma,\varphi)}(u) = \sum_{uv\in E_\Gamma} \varphi(vu)$$

are, respectively, called the $\mathbb{T}$-outgain and the $\mathbb{T}$-ingain of the vertex u.

If u is an isolated vertex, then $d^{\rightarrow}_{(\Gamma,\varphi)}(u)$ and $d^{\leftarrow}_{(\Gamma,\varphi)}(u)$ are assumed to be equal to zero. In all cases, $d^{\leftarrow}_{(\Gamma,\varphi)}(u)$ is the complex conjugate of $d^{\rightarrow}_{(\Gamma,\varphi)}(u)$. It is worth noting that this holds specifically because the adjacency matrices of complex unit gain graphs are Hermitian, and may not hold for arbitrary gain groups.

Moreover, Brunetti provided the following definition of gain regularity for the complex unit case.

Definition 5.2.8 ([23, Definition 1.2])**.** An r-$\mathbb{T}$-regular graph is a complex unit gain graph (Γ,φ) such that $d^{\rightarrow}_{(\Gamma,\varphi)}(u) = r$ for all $u \in V_\Gamma$.

By Lemma 1.8.4 and Definition 5.2.8, we see that for an r-$\mathbb{T}$-regular graph (Γ,φ) on n vertices, we have $\chi_{A_{(\Gamma,\varphi)}}(\lambda) = \frac{n}{\lambda - r}$.

Consider the H-join of a family of vertex-disjoint graphs $(\Gamma,\Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\ldots,(\Gamma_k,\varphi_k)\}$ where for each $1 \leq i \leq k$, the graph (Γ_i,φ_i) has order n_i and is r_i-$\mathbb{T}$-regular. Since $A_{(\Gamma_i,\varphi_i)}$ has constant row sum r_i, the all-one vector $\mathbf{1}_{n_i}$ will be an eigenvector with eigenvalue r_i, and so r_i will clearly be a main eigenvalue of (Γ_i,φ_i). Furthermore, since $A_{(\Gamma_i,\varphi_i)}$ is Hermitian, all of the other eigenvectors will be orthogonal to $\mathbf{1}_{n_i}$, meaning all of the other eigenvalues of (Γ_i,φ_i) will be non-main.

We have the following corollary.

Corollary 5.2.9. *Let $(\Gamma,\Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\ldots,(\Gamma_k,\varphi_k)\}$ be the H-join of a family of vertex-disjoint graphs $\mathcal{F} = \{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\ldots,(\Gamma_k,\varphi_k)\}$ in which (Γ_i,φ_i) is an r_i-$\mathbb{T}$-regular graph of order n_i for each $1 \leq i \leq k$. Then*

$$\operatorname{spec}(\Gamma,\Phi) = \left(\bigcup_{i=1}^{k} (\operatorname{spec}(\Gamma_i,\varphi_i) \setminus \{r_i\})\right) \cup \operatorname{spec}(\Psi),$$

where $\Psi = (\psi_{ij})$ is the $k \times k$ matrix with

$$\psi_{ij} = \begin{cases} r_i, & \text{if } i = j, \\ \psi(v_iv_j)n_j, & \text{if } i \neq j \text{ and } v_iv_j \in E_H, \\ 0, & \text{otherwise.} \end{cases}$$

Here $\operatorname{spec}(\Gamma_i,\varphi_i) \setminus \{r_i\}$ denotes the multiset obtained by removing exactly one occurrence of r_i.

5.2.4 A detailed example

In this section we present a fully worked example to illustrate the results we have given so far.

Example 5.2.10. Let $(H,\psi) = (P_3,\psi)$, $(\Gamma_1,\varphi_1) = (K_3,\varphi_1)$, $(\Gamma_2,\varphi_2) = (K_2,\varphi_2)$, and $(\Gamma_3,\varphi_3) = (C_4,\varphi_3)$ with gain functions and vertex labellings as shown in Figure 5.1. Then if we take the adjacency matrix of $(\Gamma,\Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),(\Gamma_3,\varphi_3)\}$, which is given explicitly in Example 5.2.2, and compute the characteristic polynomial directly we have

$$P(A_{(\Gamma,\Phi)};\,\lambda) = \lambda^9 - 22\lambda^7 - 20\lambda^6 + 97\lambda^5 + 124\lambda^4 - 84\lambda^3 - 176\lambda^2 - 64\lambda.$$

Taking the roots of this, the spectrum is

$$\operatorname{spec}(\Gamma,\Phi) = \{4.548, 2, 1.456, 0, -0.741, -1, -1, -2, -3.263\}.$$

Now let us gather all of the components needed to make use of the formula in Theorem 5.2.3. Firstly we have

$$\psi(v_1v_2) = -i,\ \psi(v_2v_1) = i,\ \psi(v_2v_3) = -1,\ \psi(v_3v_2) = -1.$$

Then we have

$$p_1 = \lambda^3 - 3\lambda - 2,\ p_2 = \lambda^2 - 1,\ p_3 = \lambda^4 - 4\lambda^2,$$

along with

$$\chi_1 = \frac{3\lambda - 1}{(\lambda-2)(\lambda+1)},\ \chi_2 = \frac{2}{\lambda - 1},\ \chi_3 = \frac{4}{\lambda}.$$

Hence

$$\det(M) = \det\begin{pmatrix} 1 & \frac{(3\lambda-1)i}{(\lambda-2)(\lambda+1)} & 0 \\ -\frac{2i}{\lambda-1} & 1 & \frac{2}{\lambda-1} \\ 0 & \frac{4}{\lambda} & 1 \end{pmatrix} = \frac{\lambda^4 - 2\lambda^3 - 15\lambda^2 + 12\lambda + 16}{\lambda(\lambda-2)(\lambda-1)(\lambda+1)}.$$

Putting all of this together gives us exactly the desired characteristic polynomial

$$\begin{aligned} P(A_{(\Gamma,\Phi)};\,\lambda) &= (\lambda^3 - 3\lambda - 2)(\lambda^2 - 1)(\lambda^4 - 4\lambda^2)\left(\frac{\lambda^4 - 2\lambda^3 - 15\lambda^2 + 12\lambda + 16}{\lambda(\lambda-2)(\lambda-1)(\lambda+1)}\right) \\ &= \lambda^9 - 22\lambda^7 - 20\lambda^6 + 97\lambda^5 + 124\lambda^4 - 84\lambda^3 - 176\lambda^2 - 64\lambda. \end{aligned}$$

Now we can look to the spectral relations given in Theorem 5.2.6. For (Γ_1,φ_1) the spectrum is $\{2,-1,-1\}$ and we have

$$\mathcal{E}_{A_{(\Gamma_1,\varphi_1)}}(2) = \operatorname{span}\left\{\begin{pmatrix}1\\-i\\1\end{pmatrix}\right\},\quad \mathcal{E}_{A_{(\Gamma_1,\varphi_1)}}(-1) = \operatorname{span}\left\{\begin{pmatrix}-i\\1\\0\end{pmatrix},\begin{pmatrix}-1\\0\\1\end{pmatrix}\right\}.$$

Hence both 2 and -1 are main eigenvalues, of multiplicity one and two respectively. Thus -1 will be an eigenvalue of $A_{(\Gamma,\Phi)}$ at least once.

For (Γ_2,φ_2) the spectrum is $\{1,-1\}$ and we have

$$\mathcal{E}_{A_{(\Gamma_2,\varphi_2)}}(1) = \operatorname{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\},\quad \mathcal{E}_{A_{(\Gamma_2,\varphi_2)}}(-1) = \operatorname{span}\left\{\begin{pmatrix}-1\\1\end{pmatrix}\right\}.$$

Hence 1 is a main eigenvalue and -1 is a non-main eigenvalue, each of multiplicity one. Therefore -1 will be an eigenvalue of $A_{(\Gamma,\Phi)}$ at least once. Combining this with what we have for (Γ_1,φ_1), we see that -1 will actually be an eigenvalue of $A_{(\Gamma,\Phi)}$ at least twice.

For (Γ_3,φ_3) the spectrum is $\{2,0,0,-2\}$ and we have

$$\mathcal{E}_{A_{(\Gamma_3,\varphi_3)}}(2) = \operatorname{span}\left\{\begin{pmatrix}1\\-1\\-1\\1\end{pmatrix}\right\},\ \mathcal{E}_{A_{(\Gamma_3,\varphi_3)}}(0) = \operatorname{span}\left\{\begin{pmatrix}1\\0\\1\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\\1\end{pmatrix}\right\},\ \mathcal{E}_{A_{(\Gamma_3,\varphi_3)}}(-2) = \operatorname{span}\left\{\begin{pmatrix}-1\\-1\\1\\1\end{pmatrix}\right\}.$$

Hence 0 is a main eigenvalue of multiplicity two, while 2 and -2 are both non-main eigenvalues of multiplicity one. Therefore each of 2, 0, and -2 will appear as an eigenvalue of $A_{(\Gamma,\Phi)}$ at least once.

So far we have collected the eigenvalues $\{2, 0, -1, -1, -2\}$. For those that remain, we need to turn our attention to the polynomial $\det(\Omega(\lambda))$. We have

$$\det(\Omega(\lambda)) = \det \begin{pmatrix} (\lambda-2)(\lambda+1) & (3\lambda-1)i & 0 \\ -2i & \lambda-1 & 2 \\ 0 & 4 & \lambda \end{pmatrix} = \lambda^4 - 2\lambda^3 - 15\lambda^2 + 12\lambda + 16,$$

which has roots $\lambda = 4.548, 1.456, -0.741, -3.263$.

Again, putting all of this together gives us precisely the desired result

$$\operatorname{spec}(\Gamma, \Phi) = \{4.548, 2, 1.456, 0, -0.741, -1, -1, -2, -3.263\}.$$

5.2.5 Stability under switching of (H, ψ)

It is shown in [57] that the H_m-join of signed graphs remains stable under switching of (H, σ). This means that if we perform the operation on any two switching-equivalent signed graphs (H, σ) and (H, σ'), the two signed graphs that are produced are also switching equivalent.

Now we can show that there is an analogous result for the H-join operation of complex unit gain graphs.

Proposition 5.2.11. *Let (H, ψ) and (H, ψ') be switching-equivalent complex unit gain graphs of order k and let $\mathcal{F} = \{(\Gamma_1, \varphi_1), (\Gamma_2, \varphi_2), \ldots, (\Gamma_k, \varphi_k)\}$. Then $(\Gamma, \Phi) = \bigvee_{(H,\psi)}^{\mathcal{F}}$ and $(\Gamma, \Phi') = \bigvee_{(H,\psi')}^{\mathcal{F}}$ are also switching equivalent.*

Proof. Switching equivalence between (H, ψ) and (H, ψ') implies that for some diagonal matrix S with entries $s_1, s_2, \ldots, s_k$ in $\mathbb{T}$, we have $S^* A_{(H,\psi)} S = A_{(H,\psi')}$. Equivalently, in terms of the gain functions,

$$s_i^* \psi(v_i v_j) s_j = \psi'(v_i v_j) \quad \text{for all } v_i v_j \in E_H.$$

Let $(\Gamma, \Phi) = \bigvee_{(H,\psi)}^{\mathcal{F}}$ and $(\Gamma, \Phi') = \bigvee_{(H,\psi')}^{\mathcal{F}}$. As in a previous proof, from here we shall use ψ_{ij} to denote $\psi(v_i v_j)$ and ψ'_{ij} to denote $\psi'(v_i v_j)$, with the convention that $\psi_{ij} = 0$ and $\psi'_{ij} = 0$ when $v_i v_j \notin E_H$. By considering the adjacency matrix of (Γ, Φ') we can see that

$$\begin{aligned} A_{(\Gamma,\Phi')} &= \begin{pmatrix} A_{(\Gamma_1,\varphi_1)} & \psi'_{12} J_{n_1 \times n_2} & \cdots & \psi'_{1k} J_{n_1 \times n_k} \\ \psi'_{21} J_{n_2 \times n_1} & A_{(\Gamma_2,\varphi_2)} & \cdots & \psi'_{2k} J_{n_2 \times n_k} \\ \vdots & \vdots & \ddots & \vdots \\ \psi'_{k1} J_{n_k \times n_1} & \psi'_{k2} J_{n_k \times n_2} & \cdots & A_{(\Gamma_k,\varphi_k)} \end{pmatrix} \\ &= \begin{pmatrix} A_{(\Gamma_1,\varphi_1)} & s_1^* \psi_{12} s_2 J_{n_1 \times n_2} & \cdots & s_1^* \psi_{1k} s_k J_{n_1 \times n_k} \\ s_2^* \psi_{21} s_1 J_{n_2 \times n_1} & A_{(\Gamma_2,\varphi_2)} & \cdots & s_2^* \psi_{2k} s_k J_{n_2 \times n_k} \\ \vdots & \vdots & \ddots & \vdots \\ s_k^* \psi_{k1} s_1 J_{n_k \times n_1} & s_k^* \psi_{k2} s_2 J_{n_k \times n_2} & \cdots & A_{(\Gamma_k,\varphi_k)} \end{pmatrix} \\ &= \begin{pmatrix} s_1 I_{n_1} & O & \cdots & O \\ O & s_2 I_{n_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & s_k I_{n_k} \end{pmatrix}^* \begin{pmatrix} A_{(\Gamma_1,\varphi_1)} & \psi_{12} J_{n_1 \times n_2} & \cdots & \psi_{1k} J_{n_1 \times n_k} \\ \psi_{21} J_{n_2 \times n_1} & A_{(\Gamma_2,\varphi_2)} & \cdots & \psi_{2k} J_{n_2 \times n_k} \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{k1} J_{n_k \times n_1} & \psi_{k2} J_{n_k \times n_2} & \cdots & A_{(\Gamma_k,\varphi_k)} \end{pmatrix} \begin{pmatrix} s_1 I_{n_1} & O & \cdots & O \\ O & s_2 I_{n_2} & \cdots & O \\ \vdots & \vdots & \ddots & \vdots \\ O & O & \cdots & s_k I_{n_k} \end{pmatrix}. \end{aligned}$$

Hence $A_{(\Gamma,\Phi')} = S^* A_{(\Gamma,\Phi)} S$ for some diagonal matrix S and so by definition $(\Gamma, \Phi) = \bigvee_{(H,\psi)}^{\mathcal{F}}$ is switching equivalent to $(\Gamma, \Phi') = \bigvee_{(H,\psi')}^{\mathcal{F}}$ and we are done. □

One way to think of the switching here is as follows: whenever we perform switching on a vertex $v_i \in V_H$, it yields the same outcome as performing switchings with the same element on every vertex of V_{Γ_i} in the compound graph produced by the H-join operation.

5.2.6 Balancedness of H-join

In [104, Theorem 1], the authors give conditions under which a graph produced by the signed H-join operation is balanced. We can state a similar theorem for the complex unit H-join.

Theorem 5.2.12. *Let (H,ψ) be a complex unit gain graph of order k with no isolated vertex and let $\mathcal{F}=\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\dots,(\Gamma_k,\varphi_k)\}$ be a family of vertex-disjoint complex unit gain graphs such that the union $E_{\Gamma_1}\cup E_{\Gamma_2}\cup\cdots\cup E_{\Gamma_k}$ is non-empty. If $(\Gamma,\Phi)=\bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\dots,(\Gamma_k,\varphi_k)\}$, then (Γ,Φ) is balanced if and only if (H,ψ) is balanced and (Γ_i,φ_i) is all-positive for each $1\le i\le k$.*

Proof. If (H,ψ) is balanced then it can be switched to the all-positive graph. From Proposition 5.2.11 we know that for switching-equivalent (H,ψ) and (H,ψ'), the join graphs $(\Gamma,\Phi)=\bigvee^{\mathcal{F}}_{(H,\psi)}$ and $(\Gamma,\Phi')=\bigvee^{\mathcal{F}}_{(H,\psi')}$ are also switching equivalent. If each $(\Gamma_i,\varphi_i)\in\mathcal{F}$ is all-positive, it is easy to see that for a balanced (H,ψ), the join graph (Γ,Φ) is switching equivalent to the all-positive graph, and so it is balanced.

Conversely, suppose that (Γ,Φ) is a balanced complex unit gain graph. For $1\le i\le k$, we can take a single vertex from each (Γ_i,φ_i) and see that the induced subgraph is balanced. Since this subgraph is isomorphic to (H,ψ), it is the case that (H,ψ) must be balanced. Additionally, since (Γ,Φ) is balanced, the product of the edge gains in every closed walk is equal to 1. Suppose we have $u,v\in V_{\Gamma_i}$ with $\varphi_i(uv)=\alpha$. Let $w\in V_{\Gamma_j}$ and let $v_iv_j\in E_H$. Then in (Γ,Φ) we will have the triangle uvw and this will have gain $\varphi_i(uv)\psi(v_iv_j)\psi(v_jv_i)=\varphi_i(uv)=\alpha$. Clearly, for this triangle to be balanced we need $\alpha=1$, hence $\varphi_i(uv)=1$ for all edges in each (Γ_i,φ_i), making each (Γ_i,φ_i) all-positive. □

5.3 The stable H-join operation

As demonstrated, the graph produced by the H-join operation is stable under switching of (H,ψ). However, the gain function in Definition 5.2.1 does not maintain stability under switchings of the individual component (Γ_i,φ_i) graphs. For instance, consider $(H,\psi)=(\Gamma_1,\varphi_1)=(\Gamma_2,\varphi_2)=(K_2,+)$. Then the graph obtained will be an all-positive, hence balanced, complete graph on four vertices. However, taking $(\Gamma_2,\varphi_2)=(K_2,-)$, that is, the complete graph on two vertices joined by an edge having gain function equal to -1, which is clearly switching equivalent to $(K_2,+)$, means that the operation now produces an unbalanced complete graph on four vertices, and so is evidently not stable.

In this section we will show that by imposing certain conditions, it is possible to define gain functions on H-joins such that they are also stable under switchings of their constituents. Furthermore, we can show that this process eases the constraints on which graphs will return a balanced product and note that the spectral analysis we have just done can also be applied in this case.

5.3.1 Pseudo-potentials

Recall Section 1.7 of the introduction. We now extend several properties of pseudo-potentials to the complex unit gain setting.

Lemma 5.3.1. *Let (Γ,φ) be a complex unit gain graph and let $\mathcal{T}$ be a spanning forest of Γ. Then*

(a) if Γ is connected, then for any pair $v,v'\in V_\Gamma$, there exists an element $z=z(v,v')\in\mathbb{T}$ (depending on v and v') such that

$$U_{\mathcal{T},v,\varphi}=zU_{\mathcal{T},v',\varphi},$$

(b) if (Γ,φ) is balanced, for any spanning tree $\mathcal{T}'$ of Γ we have

$$U_{\mathcal{T},\overline{v},\varphi}=U_{\mathcal{T}',\overline{v},\varphi},$$

(c) if φ and φ' are two switching-equivalent complex unit gain functions on Γ and the switching function s satisfies $s(v)=1$ for each $v\in\overline{v}$, there exists a diagonal matrix S with diagonal entries in $\mathbb{T}$ such that

$$A_{(\Gamma,\varphi')}=S^*A_{(\Gamma,\varphi)}S\quad\text{and}\quad U_{\mathcal{T},\overline{v},\varphi'}=U_{\mathcal{T},\overline{v},\varphi}S=SU_{\mathcal{T},\overline{v},\varphi}.$$

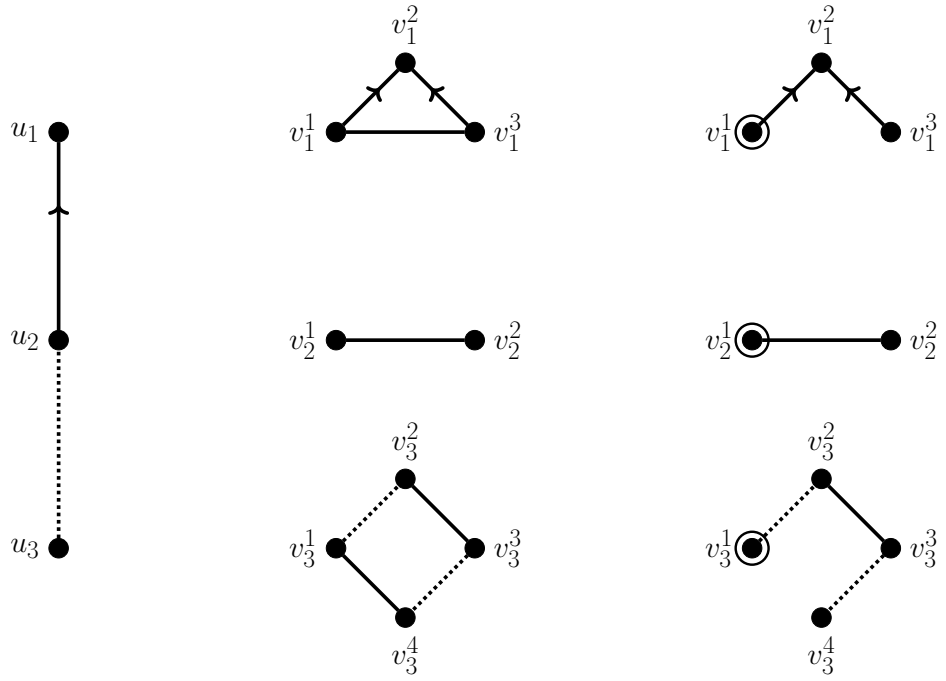


Figure 5.2: The complex unit gain graphs (H, ψ), (Γ_1, φ_1), (Γ_2, φ_2), and (Γ_3, φ_3) from Figure 5.1 and the spanning trees $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$ with fixed vertices v_1^1, v_2^1, and v_3^1.

Proof. The element z in the first point is given by $\varphi(\gamma_{vv'})$. The remainder of the proof is essentially the same as is given in [33, Lemma 2.3]. □

Definition 5.3.2. Let (Γ, φ) be a complex unit gain graph of order n and let $\mathcal{T}$ be a spanning forest of Γ. Fix a vertex v_i in each connected component of Γ and denote this collection by $\overline{v} = \{v_1, v_2, \ldots, v_r\}$. Then the complete graph induced by (Γ, φ) and $\mathcal{T}$ is the complex unit gain graph $(K_n, \overline{\varphi}_{\mathcal{T},\overline{v}})$ whose vertex set is equal to V_Γ and

$$\overline{\varphi}_{\mathcal{T},\overline{v}}(xy) = \begin{cases} \varphi(xy), & \text{if } xy \in E_\Gamma, \\ U_{\mathcal{T},\overline{v},\varphi}(x)^{-1} U_{\mathcal{T},\overline{v},\varphi}(y), & \text{otherwise.} \end{cases}$$

Remark 5.3.3. Notice that the adjacency matrix of $(K_n, \overline{\varphi}_{\mathcal{T},\overline{v}})$ is given by

$$A_{(K_n,\overline{\varphi}_{\mathcal{T},\overline{v}})} = A_{(\Gamma,\varphi)} + U^{-1}_{\mathcal{T},\overline{v},\varphi} A_{\Gamma^c} U_{\mathcal{T},\overline{v},\varphi},$$

where A_{Γ^c} is the adjacency matrix of the (unsigned) graph Γ^c, the complement of Γ.

Proposition 5.3.4. *Let (Γ, φ) be a complex unit gain graph and let $\mathcal{T}$ be a spanning forest of Γ. Then the switching class of $(K_n, \overline{\varphi}_{\mathcal{T},\overline{v}})$ depends on neither the choice of fixed vertices $\overline{v}$ nor on the choice of the representative for the switching class of (Γ, φ). Finally, (Γ, φ) is balanced if and only if $(K_n, \overline{\varphi}_{\mathcal{T},\overline{v}})$ is balanced.*

Proof. We prove the two independence claims separately.

Independence of fixed vertices. Suppose (Γ, φ) has exactly r connected components and let $\overline{v} = \{v_1, \ldots, v_r\}$ and $\overline{v}' = \{v_1', \ldots, v_r'\}$ be two lists of vertices, one from each connected component. By Lemma 5.3.1(a), for each connected component there exists $z_i \in \mathbb{T}$ such that $U_{\mathcal{T},v_i',\varphi}(w) = z_i\, U_{\mathcal{T},v_i,\varphi}(w)$ for all w in that component. Let S be the diagonal matrix with $S_{ww} = z_i$ for vertices w in the i-th component. Then S is constant on each connected component, so S commutes with $A_{(\Gamma,\varphi)}$. Moreover, the gain functions $\overline{\varphi}_{\mathcal{T},\overline{v}}$ and $\overline{\varphi}_{\mathcal{T},\overline{v}'}$ satisfy

$$A_{(K_n,\overline{\varphi}_{\mathcal{T},\overline{v}'})} = S^{-1} A_{(K_n,\overline{\varphi}_{\mathcal{T},\overline{v}})} S,$$

as can be verified directly from the definition. Hence the switching class is independent of the choice of fixed vertices.

Independence of the switching-equivalent representative. Let φ_1 and φ_2 be switching-equivalent gain functions on Γ, with $\varphi_2(u,w) = s(u)^{-1}\varphi_1(u,w)s(w)$ for some switching function $s : V_\Gamma \to \mathbb{T}$. We may assume without loss of generality that $s(v_i) = 1$ for each chosen root v_i; indeed, since $\mathbb{T}$ is abelian, replacing s by $\tilde{s}(x) = s(v_i)^{-1}s(x)$ for x in component i leaves the switched gain function φ_2 unchanged and normalises every root to 1. Let S be the diagonal matrix indexed by V_Γ with $S_{xx} = s(x)$ for the normalised function s. Then $A_{(\Gamma,\varphi_2)} = S^{-1}A_{(\Gamma,\varphi_1)}S$ and we claim that

$$A_{(K_n,\overline{\varphi_2}_{\mathcal{T},\overline{v}})} = S^{-1}A_{(K_n,\overline{\varphi_1}_{\mathcal{T},\overline{v}})}S.$$

This follows directly from the definition: for an edge xy of Γ, the gain is $\varphi_2(x,y) = s(x)^{-1}\varphi_1(x,y)s(y)$; for an edge xy of the complement, the gain is $U_{\mathcal{T},\overline{v},\varphi_2}(x)^{-1}U_{\mathcal{T},\overline{v},\varphi_2}(y)$, and after normalising the roots as above, Lemma 1.7.5 applied componentwise gives $s(x)^{-1}U_{\mathcal{T},\overline{v},\varphi_1}(x)^{-1}U_{\mathcal{T},\overline{v},\varphi_1}(y)s(y)$. Hence the gain on every edge is conjugated by S. Thus the switching class is independent of the representative.

For the final claim, note that (Γ,φ) is a subgraph of $(K_n,\overline{\varphi}_{\mathcal{T},\overline{v}})$ with the same gains on E_Γ, so if the latter is balanced then so is the former. Conversely, if (Γ,φ) is balanced then, by the independence just established together with Proposition 1.6.2, we may assume $\varphi \equiv 1$; then $U_{\mathcal{T},\overline{v},\varphi} \equiv 1$, so every gain of $(K_n,\overline{\varphi}_{\mathcal{T},\overline{v}})$ is equal to 1 and it is balanced. □

5.3.2 Switching-stable definition

Before providing the definition for the $H_{\mathcal{T},\overline{v}}$-join, also referred to as the stable H-join, we illustrate a scheme for labelling the vertices. Let (H,ψ) be a complex unit gain graph of order k with vertex set $V_H = \{u_1,u_2,\dots,u_k\}$ and let (Γ_i,φ_i) be a complex unit gain graph of order n_i with vertex set $V_{\Gamma_i} = \{v_i^1,v_i^2,\dots,v_i^{n_i}\}$ for each $1 \le i \le k$. Then we can take the vertex set of $(\Gamma,\Phi) = \bigvee_{(H,\psi)}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\dots,(\Gamma_k,\varphi_k)\}$ as

$$V_\Gamma = \bigcup_{i=1}^{k}(\{u_i\} \times V_{\Gamma_i}). \tag{5.3.1}$$

Example 5.3.5. Let $H = P_3$ and $V_H = \{u_1,u_2,u_3\}$. Let $\Gamma_1 = K_3$ and $V_{\Gamma_1} = \{v_1^1,v_1^2,v_1^3\}$, $\Gamma_2 = K_2$ and $V_{\Gamma_2} = \{v_2^1,v_2^2\}$, $\Gamma_3 = C_4$ and $V_{\Gamma_3} = \{v_3^1,v_3^2,v_3^3,v_3^4\}$. Then, according to Equation (5.3.1), we have

$$V_\Gamma = \{(u_1,v_1^1),(u_1,v_1^2),(u_1,v_1^3),(u_2,v_2^1),(u_2,v_2^2),(u_3,v_3^1),(u_3,v_3^2),(u_3,v_3^3),(u_3,v_3^4)\}.$$

With this labelling we can proceed to defining the stable H-join.

Definition 5.3.6. Let (H,ψ) be a complex unit gain graph with $V_H = \{u_1,u_2,\dots,u_k\}$ and let $\mathcal{F} = \{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\dots,(\Gamma_k,\varphi_k)\}$ be a family of vertex-disjoint graphs where (Γ_i,φ_i) is a complex unit gain graph of order n_i with vertex set $V_{\Gamma_i} = \{v_i^1,v_i^2,\dots,v_i^{n_i}\}$ for each $1 \le i \le k$.

Fix a spanning tree $\mathcal{T}_i$ of each Γ_i (with the usual convention that if Γ_i is disconnected, $\mathcal{T}_i$ denotes a spanning forest with one root per connected component), fix a vertex v_i^a in each connected component of Γ_i, and take the pseudo-potential functions $U_i = U_{\mathcal{T}_i,v_i^a,\varphi_i} \colon V_{\Gamma_i} \to \mathbb{T}$ as $U_i(w_i^b) = \varphi_i(\gamma_{v_i^a w_i^b})$ for each w_i^b in V_{Γ_i}.

Then the $H_{\mathcal{T},\overline{v}}$-join of this family is the complex unit gain graph $(\Gamma,\Phi) = \bigvee_{(H,\psi)}^{\mathcal{T}}\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),\dots,(\Gamma_k,\varphi_k)\}$, sometimes denoted $(\Gamma,\Phi) = \bigvee_{(H,\psi)}^{\mathcal{FT}}$, with vertex set $V_\Gamma = \bigcup_{i=1}^{k}(\{u_i\} \times V_{\Gamma_i})$, edge set

$$E_\Gamma = \left(\bigcup_{i=1}^{k}\left\{\{(u_i,v_i^s),(u_i,v_i^t)\} : \{v_i^s,v_i^t\} \in E_{\Gamma_i}\right\}\right) \cup \left(\bigcup_{u_iu_j \in E_H}\left\{\{(u_i,v_i^s),(u_j,v_j^t)\} : v_i^s \in V_{\Gamma_i},\, v_j^t \in V_{\Gamma_j}\right\}\right),$$

and gain function Φ satisfying

$$\Phi((u_i,v_i^s),(u_j,v_j^t)) = \begin{cases} \varphi_i(v_i^sv_i^t), & \text{if } i = j \text{ and } v_i^sv_i^t \in E_{\Gamma_i}, \\ \psi(u_iu_j)U_i(v_i^s)^{-1}U_j(v_j^t), & \text{if } u_iu_j \in E_H. \end{cases}$$

As before, note that the gain function Φ is well defined since

$$\Phi((u_i, v_i^s), (u_j, v_j^t)) = \Phi((u_j, v_j^t), (u_i, v_i^s))^{-1}$$

follows immediately from

$$\varphi_i(v_i^s v_i^t) = \varphi_i(v_i^t v_i^s)^{-1} \quad \text{and} \quad \psi(u_i u_j) U_i(v_i^s)^{-1} U_j(v_j^t) = \left(\psi(u_i u_j)^{-1} U_i(v_i^s) U_j(v_j^t)^{-1}\right)^{-1}.$$

We can see that by this definition, the $H_{\mathcal{T},\overline{v}}$-join of a family which is $\mathcal{T}$-positive coincides with the basic definition from the previous section.

5.3.3 Stability under switchings of (Γ_i, φ_i)

With this new definition, the operation remains stable under switching of (H, ψ) as before and is also stable under switchings of each component graph (Γ_i, φ_i), as we shall prove in this section.

Remark 5.3.7. While considering only the underlying graphs, without applying any gain functions, notice that each closed walk W on the H-join of a family $\mathcal{F} = \{\Gamma_1, \Gamma_2, \ldots, \Gamma_k\}$ has the form

$$(u_{i_1}, v_{i_1}^{1_1}) \ldots (u_{i_1}, v_{i_1}^{k_1})(u_{i_2}, v_{i_2}^{1_2}) \ldots (u_{i_2}, v_{i_2}^{k_2})(u_{i_3}, v_{i_3}^{1_3}) \ldots (u_{i_j}, v_{i_j}^{k_j})(u_{i_1}, v_{i_1}^{1^{(j+1)}}) \ldots (u_{i_1}, v_{i_1}^{1_1})$$

where $u_{i_s} \sim u_{i_{s+1}}$ for each $1 \leq s \leq j-1$, $u_{i_j} \sim u_{i_1}$, and $v_{i_s}^{r_s} \sim v_{i_s}^{(r+1)_s}$ for each $1 \leq s \leq j$. In the following proofs we use W_H to denote the closed walk on H induced by W, which is defined as

$$W_H = u_{i_1} u_{i_2} \ldots u_{i_j} u_{i_1}.$$

Moreover, W can be viewed as a closed walk in the complete graph K_Γ, obtained by taking the union of $\Gamma = \sqcup_{i=1}^k \Gamma_i$ with its complement.

Lemma 5.3.8. *Let (H, ψ) be a complex unit gain graph of order k and let $\mathcal{F} = \{(\Gamma_1, \varphi_1), (\Gamma_2, \varphi_2), \ldots, (\Gamma_k, \varphi_k)\}$ be a family of vertex-disjoint complex unit gain graphs. Fix a spanning forest $\mathcal{T}$ of $\Gamma = \sqcup_{i=1}^k \Gamma_i$, and choose a vertex v_i in each connected component of Γ. Let $\overline{v} = \{v_1, v_2, \ldots, v_r\}$ and denote by U_i the pseudo-potential on (Γ_i, φ_i) induced by $\mathcal{T}$ and $\overline{v}$. Let W be a closed walk in the $H_{\mathcal{T},\overline{v}}$-join of the family $\mathcal{F}$. Then*

$$\Phi(W) = \psi(W_H) \overline{\varphi}_{\mathcal{T},\overline{v}}(W),$$

where Φ is the gain function in the $H_{\mathcal{T},\overline{v}}$-join and $\overline{\varphi}_{\mathcal{T},\overline{v}}$ is the gain function induced by the (Γ_i, φ_i) graphs, $\mathcal{T}$, and $\overline{v}$ in the complete graph K_Γ.

Proof. It follows by an easy computation. We have

$$\begin{aligned}
\Phi(W) &= \Phi((u_{i_1}, v_{i_1}^{1_1}) \ldots (u_{i_1}, v_{i_1}^{k_1})(u_{i_2}, v_{i_2}^{1_2}) \ldots (u_{i_j}, v_{i_j}^{k_j})(u_{i_1}, v_{i_1}^{1^{(j+1)}}) \ldots (u_{i_1}, v_{i_1}^{1_1})) \\
&= \varphi_{i_1}(v_{i_1}^{1_1}, v_{i_1}^{2_1}) \ldots \varphi_{i_1}(v_{i_1}^{(k-1)_1}, v_{i_1}^{k_1}) \cdot \psi(u_{i_1}, u_{i_2}) U_{i_1}(v_{i_1}^{k_1})^{-1} U_{i_2}(v_{i_2}^{1_2}) \\
&\qquad \ldots \varphi_{i_1}(v_{i_1}^{1^{(j+1)}}, v_{i_1}^{2^{(j+1)}}) \ldots \varphi_{i_1}(v_{i_1}^{(k-1)^{(j+1)}}, v_{i_1}^{1_1}) \\
&= [\psi(u_{i_1}, u_{i_2}) \ldots \psi(u_{i_j}, u_{i_1})] \\
&\qquad \cdot [\varphi_{i_1}(v_{i_1}^{1_1}, v_{i_1}^{2_1}) \ldots \varphi_{i_1}(v_{i_1}^{(k-1)_1}, v_{i_1}^{k_1}) \cdot U_{i_1}(v_{i_1}^{k_1})^{-1} U_{i_2}(v_{i_2}^{1_2}) \ldots \varphi_{i_1}(v_{i_1}^{(k-1)^{(j+1)}}, v_{i_1}^{1_1})] \\
&= \psi(W_H) \overline{\varphi}_{\mathcal{T},\overline{v}}(W).
\end{aligned}$$

□

The following theorem presents the main result of this section, establishing the stability of the $H_{\mathcal{T},\overline{v}}$-join.

Theorem 5.3.9. *Let (H, ψ) be a complex unit gain graph of order k and let $\mathcal{F} = \{(\Gamma_1, \varphi_1), (\Gamma_2, \varphi_2), \ldots, (\Gamma_k, \varphi_k)\}$ be a family of vertex-disjoint complex unit gain graphs. Fix a spanning forest $\mathcal{T}$ of $\Gamma = \sqcup_{i=1}^k \Gamma_i$ and choose a vertex v_i in each connected component of Γ. Let $\overline{v} = \{v_1, v_2, \ldots, v_r\}$. Then*

- *the switching class of the $H_{\mathcal{T},\overline{v}}$-join of $\mathcal{F}$ depends only on $\mathcal{T}$ and is independent of the choice of vertices,*
- *the $H_{\mathcal{T},\overline{v}}$-join operation is switching stable on both (H,ψ) and on each member (Γ_i,φ_i) of the family $\mathcal{F}$, that is, replacing any (Γ_i,φ_i) with some switching-equivalent (Γ_i,φ_i') leaves the switching class of the $H_{\mathcal{T},\overline{v}}$-join unchanged.*

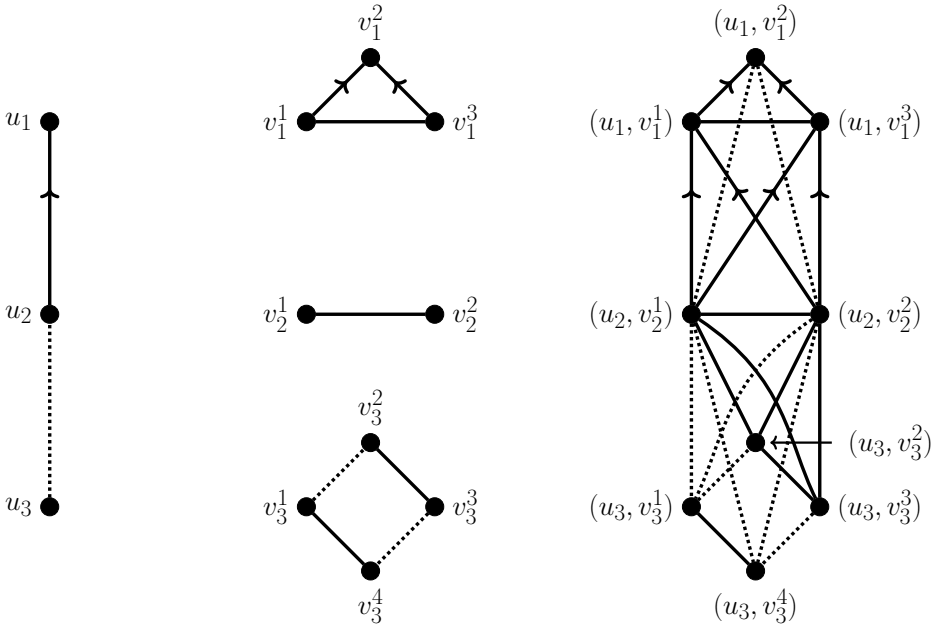
Proof. This is an immediate consequence of Proposition 5.3.4 and Lemma 5.3.8. □

With this we can directly give a characterisation of the balancedness of a stable H-join.

Corollary 5.3.10. *The $H_{\mathcal{T},\overline{v}}$-join is a balanced complex unit gain graph if and only if (H,ψ) and all component graphs (Γ_i,φ_i) are balanced. Moreover, when the join is balanced, its switching class is independent of the choice of $\mathcal{T}$.*

Remark 5.3.11. It is clear from Theorem 5.3.9 and Corollary 5.3.10 that, when applied to the setting of signed graphs, Definition 5.3.6 will provide a switching-stable signature for the signed H-join (see [104]), the signed H-generalised-join (see [71]), and the signed H_m-join (see [57]) operations.

Since the operation is switching stable, to compute the spectrum of the stable H-join, one can, without loss of generality, consider the stable H-join obtained by replacing each (Γ_i,φ_i) with the $\mathcal{T}$-positive representatives of each switching class involved. In this specific case the standard H-join and the stable H-join coincide, and so the spectral analysis done in the previous section also applies.



(H,ψ) $\qquad \mathcal{F}=\{(\Gamma_1,\varphi_1),(\Gamma_2,\varphi_2),(\Gamma_3,\varphi_3)\}$ $\qquad (\Gamma,\Phi)=\bigvee_{(H,\psi)}^{\mathcal{F}\mathcal{T}}$

Figure 5.3: The stable H-join of the graphs, spanning trees, and fixed vertices presented in Figure 5.2. The vertices are labelled according to Equation (5.3.1) and the graph produced is balanced.

Now we present an example to illustrate explicitly how to compute this new gain function and to show that from balanced components we obtain a balanced product.

Example 5.3.12. Take (H,ψ), (Γ_1,φ_1), (Γ_2,φ_2), and (Γ_3,φ_3) from Example 5.2.10 and note that each of these graphs is balanced. Let $\mathcal{T}_1$, $\mathcal{T}_2$, and $\mathcal{T}_3$ be the spanning trees depicted in Figure 5.2. Fix the vertex v_i^1 in each $\mathcal{T}_i$ and denote $U_i = U_{\mathcal{T}_i,v_i^1,\varphi_i}$. Then for the pseudo-potentials we have

$$U_1(v_1^1)=1, \quad U_1(v_1^2)=i, \quad U_1(v_1^3)=1,$$

$$U_2(v_2^1) = 1, \quad U_2(v_2^2) = 1,$$
$$U_3(v_3^1) = 1, \quad U_3(v_3^2) = -1, \quad U_3(v_3^3) = -1, \quad U_3(v_3^4) = 1.$$

Using these we can compute the gain function according to Definition 5.3.6. We know that the edges already present in the (Γ_i, φ_i) graphs will remain the same so let us look at (u_2, v_2^1) and see how we create the remaining adjacencies with respect to the gains and pseudo-potentials. We have

$$\begin{aligned}
\varphi((u_2, v_2^1), (u_1, v_1^1)) &= \psi(u_2u_1)U_2(v_2^1)^{-1}U_1(v_1^1) = i \cdot 1 \cdot 1 = i,\\
\varphi((u_2, v_2^1), (u_1, v_1^2)) &= \psi(u_2u_1)U_2(v_2^1)^{-1}U_1(v_1^2) = i \cdot 1 \cdot i = -1,\\
\varphi((u_2, v_2^1), (u_1, v_1^3)) &= \psi(u_2u_1)U_2(v_2^1)^{-1}U_1(v_1^3) = i \cdot 1 \cdot 1 = i,\\
\varphi((u_2, v_2^1), (u_3, v_3^1)) &= \psi(u_2u_3)U_2(v_2^1)^{-1}U_3(v_3^1) = -1 \cdot 1 \cdot 1 = -1,\\
\varphi((u_2, v_2^1), (u_3, v_3^2)) &= \psi(u_2u_3)U_2(v_2^1)^{-1}U_3(v_3^2) = -1 \cdot 1 \cdot -1 = 1,\\
\varphi((u_2, v_2^1), (u_3, v_3^3)) &= \psi(u_2u_3)U_2(v_2^1)^{-1}U_3(v_3^3) = -1 \cdot 1 \cdot -1 = 1,\\
\varphi((u_2, v_2^1), (u_3, v_3^4)) &= \psi(u_2u_3)U_2(v_2^1)^{-1}U_3(v_3^4) = -1 \cdot 1 \cdot 1 = -1.
\end{aligned}$$

Doing this again for (u_2, v_2^2) completes the gain function and gives us the complex unit gain graph shown in Figure 5.3, which is balanced.

5.4 Lexicographic product

Let H and Γ be two ordinary graphs. The lexicographic product $H[\Gamma]$ was introduced by Harary in 1959 [50] and is the graph whose set of vertices is $V_H \times V_\Gamma$, with (u, v) adjacent to (u', v') whenever u is adjacent to u' (edges of type I) or $u = u'$ and v is adjacent to v' (edges of type II). It can be interpreted as the H-join of a homogeneous family, that is, a family of graphs consisting of copies of the same graph Γ.

In this section, we provide a survey of all of the definitions of the lexicographic product that have been introduced so far for signed graphs and complex unit gain graphs, along with some characterisations of when they are balanced and how these relate to the gain functions we have defined for the H-join.

In 2012, Hameed and Germina [45] defined an initial lexicographic product (let us call it HG) for signed graphs and provided a characterisation of its balancedness. In particular, given two signatures σ_H on H and σ_Γ on Γ, the HG-lexicographic product assigns the value $\sigma_H(uu')$ to edges of type I and $\sigma_\Gamma(vv')$ to edges of type II.

In 2019, Brunetti, Cavaleri, and Donno [24] defined another lexicographic product (we will refer to it as BCD) for signed graphs. They claimed that, differently from HG, their operation was switching stable. However, they later disproved this claim by giving a counterexample in an erratum [25]. The BCD-lexicographic product divides the edges of type I into two families: edges of type Ia, where $u \sim u'$ and $v \sim v'$, and the remainder, where $u \sim u'$ and $v \nsim v'$, which we shall call edges of type Ib. The signature of an edge of type Ia is given by $\sigma_H(uu')\sigma_\Gamma(vv')$, while for those of type Ib the signature coincides with HG.

In 2021, Zhang, Wu, and He [104] defined the H-join on signed graphs (let us denote it ZWH) and provided a characterisation of its balancedness. This definition coincides with the HG-lexicographic product when all of the signed graphs (Γ_i, σ_i) in $\mathcal{F} = \{(\Gamma_1, \sigma_1), (\Gamma_2, \sigma_2), \dots, (\Gamma_k, \sigma_k)\}$ are identical, say, when $\mathcal{F}$ is homogeneous.

In 2024, Brunetti [23], in a paper on complex unit gain graphs, introduced the complex versions of both HG and BCD. The complex gain functions here are defined by simply replacing the signatures σ_H and σ_Γ with the corresponding complex gain functions φ_H and φ_Γ.

In 2025, Cavaleri, Donno, and Spessato [33] defined a switching-stable version of the lexicographic product (which we shall call CDS) for signed graphs. This definition requires fixing a spanning tree and divides the edges of type Ib into a further two families, which we denote with Ib_1 and Ib_2. The edges of type Ib_1 are the edges where $u \sim u'$, $v \nsim v'$, and $v \neq v'$. For these edges, the signature is $\sigma_H(uu')\sigma^c_{\Gamma\mathcal{T}}(vv')$. Those edges left over from Ib, where $u \sim u'$ and $v = v'$, are the edges of type Ib_2, and their signature is equal to that of BCD.

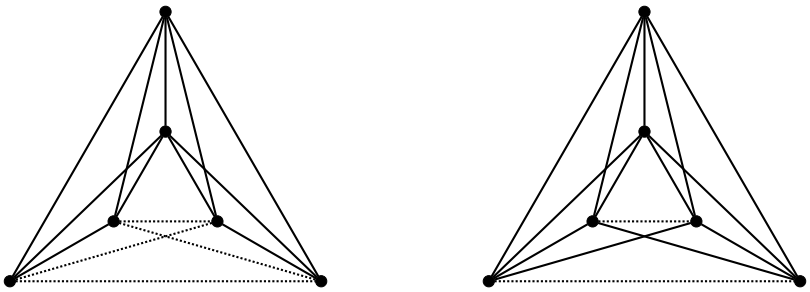


Figure 5.4: On the left, the CDS-lexicographic product of a positive $(K_2, +)$ and (C_3, σ), a signed 3-cycle with exactly two positive edges. In this specific case the CDS product does not depend on the choice of the spanning tree $\mathcal{T}$. On the right, the stable K_2-join of $\mathcal{F} = \{(C_3, \sigma), (C_3, \sigma)\}$ where the spanning trees are, in each copy of (C_3, σ), the positive subgraph of (C_3, σ).

In the present chapter, we have introduced two complex gain functions on the H-join of a family of vertex-disjoint graphs, in Definitions 5.2.1 and 5.3.6. The first is the complex version of ZWH, which, in the case of a homogeneous $\mathcal{F}$, becomes the complex version of HG defined by Brunetti in 2024. The second, which we call the stable H-join, is a switching-stable definition that depends on the choice of a spanning forest.

Although they may appear similar, the stable H-join and CDS do not generally coincide, even when restricted to the subcase of signed graphs with a homogeneous family $\mathcal{F}$. In fact, they are not even switching equivalent. An example of this dissimilarity is depicted in Figure 5.4.

On the other hand, these two operations do coincide when all of the graphs involved are balanced and the same fixed vertices are chosen in each copy of the relevant components.

Regardless, we have discovered a new switching-stable method for defining the lexicographic product of two signed (and complex unit gain) graphs, thus giving another answer to the question posed in the erratum of Brunetti, Cavaleri, and Donno [25].

This confirms, as already shown in [33], that there are several non-trivial switching-stable ways to define such stable products. Furthermore, we have discovered that these methods need not stem from the same underlying technique described in [33].

5.5 Summary and outlook

In this chapter we have extended the notion of the H-join to the realm of complex unit gain graphs and provided two distinct definitions: a direct extension that is switching stable with respect to the root graph (H, ψ), and a more refined version, the $H_{\mathcal{T},\overline{v}}$-join, that achieves full switching stability with respect to all components through the use of pseudo-potentials. We have computed the characteristic polynomial and spectrum of the resulting graphs and characterised precisely when each construction yields a balanced gain graph. These constructions unify and extend several previously known operations, including the lexicographic product and the signed H-join, and offer a new switching-stable approach to the lexicographic product of signed and complex unit gain graphs.

Clearly, a natural next step would be to extend this further to work with any arbitrary group G, not just the group of complex units. In order to do this, we would likely need to make use of G-phases and group representations, which shall be introduced in the subsequent chapter.

Chapter 6

Spectra of gain Q and R graphs via G-phases and group representations

Dragoš M. Cvetković introduced the Q and R graphs as natural intermediate constructions between the subdivision graph and the total graph. The Q graph is obtained from the subdivision graph by joining subdivision vertices whenever their corresponding edges are adjacent, while the R graph is obtained from a root graph by introducing a new vertex for each edge and joining it to both endpoints of that edge. We continue this by introducing their gain graph analogues, constructed via G-phases and group representations, and show that switching-equivalent gain graphs give rise to switching-equivalent gain Q and R graphs. For an r-regular G-gain graph (Γ, ψ), and an irreducible unitary representation π of G, we determine the π-spectra of the represented adjacency and Laplacian matrices of the associated gain Q and R graphs, expressed in terms of the π-spectrum of (Γ, ψ). As special cases, our results recover the known spectra for the classical Q and R graphs and the signed R graph, and provide a new result in the case of the signed Q graph.

This chapter is based on [63], joint work with Olwethu Waka.

6.1 Introduction and preliminaries

The total graph was first introduced by Behzad [14], and Cvetković [37] later introduced the Q and R graphs as natural intermediate constructions between the subdivision graph and the total graph, obtaining their spectra alongside those of the line graph and subdivision graph; these results are collected in [38]. These constructions are illustrated in Figure 6.1.

We briefly recall the relevant classical constructions. For an undirected simple graph Γ with vertex set V_Γ of order n and edge set E_Γ of size m, the vertex-edge incidence matrix of Γ is the $n \times m$ matrix B defined by $B_{ij} = 1$ if vertex v_i is incident with edge e_j and $B_{ij} = 0$ otherwise. The line graph $L(\Gamma)$ of Γ is the graph whose vertices are in one-to-one correspondence with the edges of Γ, with two vertices of $L(\Gamma)$ being adjacent if and only if the corresponding edges in Γ have a vertex in common. The subdivision graph $S(\Gamma)$ of a graph Γ is the graph obtained by inserting a new vertex into every edge of Γ, or equivalently, by replacing each edge with a path of length 2. Recall from, say, [38] that any subdivision graph $S(\Gamma)$ is bipartite.

The Q graph $Q(\Gamma)$ of Γ is obtained from the subdivision by additionally joining two subdivision vertices whenever their corresponding edges are adjacent in Γ. The R graph $R(\Gamma)$ of Γ is obtained from Γ by introducing a new vertex for each edge of Γ and joining each such vertex to both of the endpoints of the corresponding edge. Equivalently, each edge of Γ is supplemented by an additional path of length 2 between its endpoints. Further to this, the total graph $T(\Gamma)$ of Γ is obtained from the subdivision by adding all of the edges of the line graph onto the newly introduced subdivision vertices.

The adjacency matrices of the subdivision graph, the Q graph, the R graph, and the total graph are

$$A_{S(\Gamma)} = \begin{pmatrix} O & B \\ B^T & O \end{pmatrix}, \quad A_{Q(\Gamma)} = \begin{pmatrix} O & B \\ B^T & A_{L(\Gamma)} \end{pmatrix}, \quad A_{R(\Gamma)} = \begin{pmatrix} A_\Gamma & B \\ B^T & O \end{pmatrix}, \quad A_{T(\Gamma)} = \begin{pmatrix} A_\Gamma & B \\ B^T & A_{L(\Gamma)} \end{pmatrix}.$$

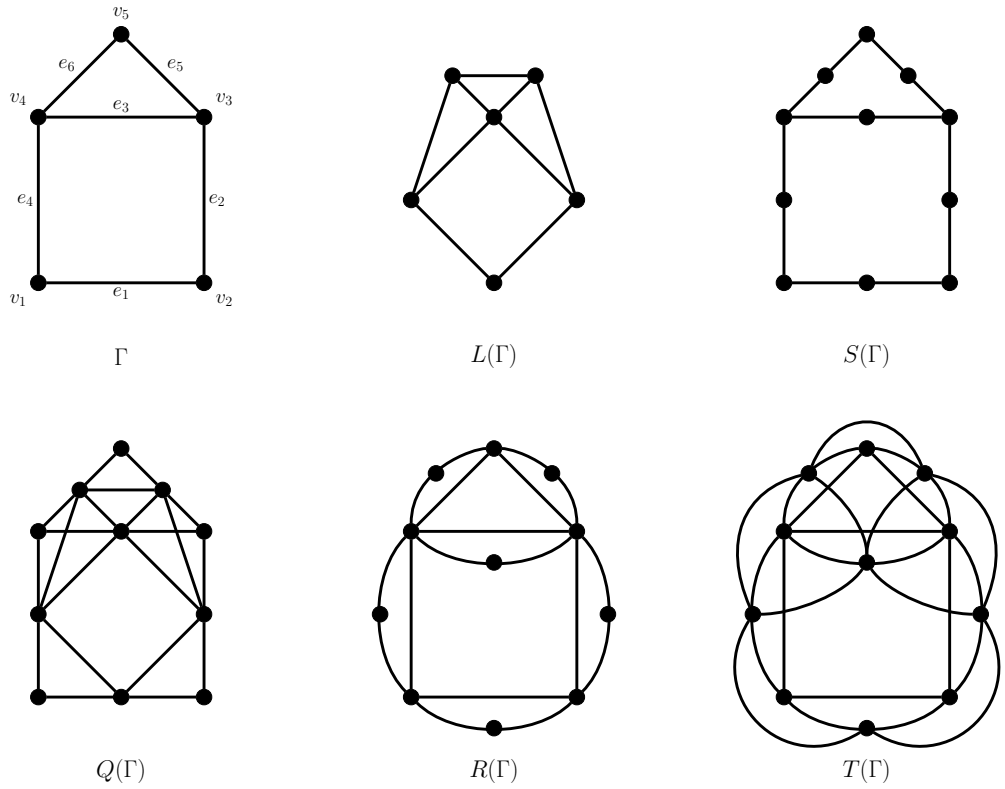


Figure 6.1: A graph Γ along with its line graph $L(\Gamma)$, its subdivision graph $S(\Gamma)$, the related graphs $Q(\Gamma)$ and $R(\Gamma)$, and its total graph $T(\Gamma)$.

The generalisation of such constructions to signed graphs, where edges can be either positive or negative, has received attention in recent years [21, 22, 80], and the further generalisation to gain graphs, in which edge labels are drawn from an arbitrary group G, has been pursued via the framework of G-phases and group representations developed in [30, 34]. Within this setting, the π-spectrum of a gain graph, defined via an irreducible unitary representation π of G, provides a natural notion of a spectrum that generalises the classical and signed cases. The gain analogues of the line graph, subdivision graph, and total graph have been studied in such works as [6, 30, 34], and the present chapter completes this picture by introducing and studying the gain Q and R graphs.

6.1.1 Group representations and G-phases

In group theory, an element $s \in G$ is called an involution if $s^2 = 1_G$ and $s \neq 1_G$; equivalently, s is a non-identity element of order 2. We denote by $Z(G) = \{g \in G : gh = hg \text{ for all } h \in G\}$ the centre of G, and say that an element $g \in G$ is central if $g \in Z(G)$. Thus s is a central involution if $s \in Z(G)$, $s^2 = 1_G$ and $s \neq 1_G$.

Throughout this chapter and the next we fix a central element $s \in G$ with $s^2 = 1_G$. Following [34], we allow the degenerate case $s = 1_G$, which recovers the classical unsigned constructions. When $s \neq 1_G$, such an s is a central involution in the above sense. For brevity we refer to s as the fixed central involution in both cases.

For example, consider the multiplicative group of complex units, $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$. We see that the only involution is -1, and since the group is abelian, its centre is $Z(\mathbb{T}) = \mathbb{T}$. In particular, -1 is a central involution of $\mathbb{T}$.

The group ring $\mathbb{C}G$ is the set of all formal linear combinations $\sum_{x \in G} f_x x$, with $f_x \in \mathbb{C}$, equipped with the natural addition and multiplication inherited from G. Furthermore, there is an involution * on $\mathbb{C}G$ defined as $f^* := \sum_{x \in G} \overline{f_x} x^{-1}$, where $\overline{f_x}$ denotes the complex conjugate of f_x. There exists a natural embedding of G in $\mathbb{C}G$, and we simply write 0 to indicate the zero-vector of $\mathbb{C}G$.

A representation π of G of degree k (we write $\deg \pi = k$) is a group homomorphism $\pi : G \to GL_k(\mathbb{C})$

where $GL_k(\mathbb{C})$ is the group of all invertible matrices in $M_k(\mathbb{C})$. That is, each group element $g \in G$ is assigned an invertible $k \times k$ complex matrix $\pi(g)$, in a way that respects the group structure, i.e. $\pi(gh) = \pi(g)\pi(h)$ for all $g, h \in G$.

Such a representation $\pi : G \to GL_k(\mathbb{C})$ can be linearly extended to $M_{n,m}(\mathbb{C}G)$, the set of $n \times m$ matrices whose entries are elements of the group ring $\mathbb{C}G$. We keep the notation

$$\pi : M_{n,m}(\mathbb{C}G) \to M_{nk,mk}(\mathbb{C}).$$

For $A \in M_{n,m}(\mathbb{C}G)$, the element $\pi(A) \in M_{nk,mk}(\mathbb{C})$ is often called the Fourier transform of A at π. The matrix $\pi(A)$ is obtained from A by replacing each occurrence of an element $g \in G$ with the block $\pi(g) \in GL_k(\mathbb{C})$, and each 0 with the $k \times k$ zero matrix.

For instance, we may have

$$A = \begin{pmatrix} g & h \\ 0 & g^{-1} \end{pmatrix}, \qquad \pi(A) = \begin{pmatrix} \pi(g) & \pi(h) \\ O & \pi(g)^{-1} \end{pmatrix}.$$

Consider a G-gain graph (Γ, φ) with adjacency matrix $A_{(\Gamma,\varphi)} \in M_n(\mathbb{C}G)$. Then the matrix $\pi(A_{(\Gamma,\varphi)}) \in M_{nk}(\mathbb{C})$ is called the represented adjacency matrix of (Γ, φ) with respect to the representation π. Note that if π is unitary, then $\pi(A_{(\Gamma,\varphi)})$ is Hermitian and we say that its (real) spectrum is the π-spectrum of (Γ, φ).

Definition 6.1.1 ([34, Definition 2.4])**.** For a graph $\Gamma = (V_\Gamma, E_\Gamma)$ and a group G, with $|V_\Gamma| = n$ and $|E_\Gamma| = m$, a matrix $H \in M_{n,m}(\mathbb{C}G)$ is said to be a G-phase of Γ if $H_{i,j} \in G$ when $v_i \in e_j$ and $H_{i,j} = 0$ when $v_i \notin e_j$. We denote by $\mathcal{H}_\Gamma \subset M_{n,m}(\mathbb{C}G)$ the set of all G-phases of Γ.

Let G^n be the n-th direct power of the group G, where the group product is defined coordinate-wise. Notice that, when an ordering of V_Γ is fixed, an element $g \in G^n$ can be regarded as a map $g\colon V_\Gamma \to G$ such that $g(v_i) = g_i$. Moreover, the group G^n is canonically isomorphic to the group of all diagonal matrices of $M_n(\mathbb{C}G)$ with diagonal entries in G, via the bijection

$$g = (g_1, g_2, \ldots, g_n) \mapsto \operatorname{diag}(g_1, g_2, \ldots, g_n) = \begin{pmatrix} g_1 & & \\ & \ddots & \\ & & g_n \end{pmatrix} \in M_n(\mathbb{C}G).$$

In particular, we can multiply a matrix in $M_{n,m}(\mathbb{C}G)$ by an element of G^n on the left or by an element of G^m on the right. These actions can be restricted to $\mathcal{H}_\Gamma$ (see [30, Lemma 4.4]). For $f = (f_1, f_2, \ldots, f_n) \in G^n$, we write f^* for the conjugate transpose of the associated diagonal matrix, that is, $f^* = \operatorname{diag}(f_1^{-1}, f_2^{-1}, \ldots, f_n^{-1})$.

In the following definition, we set up how a G-phase plays the role of the classical incidence matrix of Γ.

Definition 6.1.2 ([34, Definition 2.5])**.** Let Γ be a graph. Then three actions on the set $\mathcal{H}_\Gamma$ of G-phases of Γ can be defined as follows. For every $f \in G^n$, $g \in G^m$, $H \in \mathcal{H}_\Gamma$, we put:

$$\begin{aligned} r\colon G^m \times \mathcal{H}_\Gamma \to \mathcal{H}_\Gamma, &\qquad r(g, H) = Hg, \\ l\colon G^n \times \mathcal{H}_\Gamma \to \mathcal{H}_\Gamma, &\qquad l(f, H) = f^*H, \\ (l \times r)\colon (G^n \times G^m) \times \mathcal{H}_\Gamma \to \mathcal{H}_\Gamma, &\qquad (l \times r)((f, g), H) = f^*Hg. \end{aligned}$$

Now, we set up how we obtain the gain equivalents of the graph and the line graph. In each case, we need the freedom to choose a parameter: a (possibly trivial) central involution s of G.

Definition 6.1.3 ([34, Definition 2.6])**.** Let s be a fixed central involution of the group G. Let Γ be a graph with $V_\Gamma = \{v_1, v_2, \ldots, v_n\}$ and $E_\Gamma = \{e_1, e_2, \ldots, e_m\}$. We have the maps

$$\begin{aligned} \Phi\colon \mathcal{H}_\Gamma \to G(\Gamma), &\qquad \Phi(H)(v_i, v_j) = sH_{i,k}(H_{j,k})^{-1}, \\ \Phi_L\colon \mathcal{H}_\Gamma \to G(L(\Gamma)), &\qquad \Phi_L(H)(e_p, e_q) = s(H_{r,p})^{-1}H_{r,q}, \end{aligned}$$

where $e_k = \{v_i, v_j\}$ and $v_r = e_p \cap e_q$.

Once orderings for V_Γ and E_Γ are fixed, we will write, with a slight abuse of notation, $\Phi(H)$ and $\Phi_L(H)$ for the adjacency matrices of the gain graphs $(\Gamma, \Phi(H))$ and $(L(\Gamma), \Phi_L(H))$, respectively. Similarly, the elements $f \in G^n$ and $g \in G^m$ can be interpreted as switching functions as in Equation (1.6.1). In summary, $f = (f_1, f_2, \ldots, f_n) \in G^n$ can be considered both as the matrix $\mathrm{diag}(f_1, f_2, \ldots, f_n) \in M_n(\mathbb{C}G)$ and as a map $f\colon V_\Gamma \to G$. Likewise, $g = (g_1, g_2, \ldots, g_m) \in G^m$ can be considered both as the matrix $\mathrm{diag}(g_1, g_2, \ldots, g_m) \in M_m(\mathbb{C}G)$ and as a map $g\colon V_{L(\Gamma)} \to G$. As a consequence, the gain functions $\Phi(H)^f$ and $\Phi_L(H)^g$ correspond to the adjacency matrices $f^* A_{(\Gamma,\Phi(H))} f$ and $g^* A_{(L(\Gamma),\Phi_L(H))} g$.

Notice that, with the notations just introduced and computing the diagonal entries of HH^* and H^*H, where s acts entry-wise, we can rewrite Definition 6.1.3 as follows:

$$\Phi(H) = s(HH^* - \deg(\Gamma)), \qquad \Phi_L(H) = s(H^*H - 2I), \tag{6.1.1}$$

where $\deg(\Gamma) = \mathrm{diag}(\deg(v_1)1_G, \deg(v_2)1_G, \ldots, \deg(v_n)1_G)$ and I is the identity matrix of $M_m(\mathbb{C}G)$.

Lemma 6.1.4 ([30, Lemma 4.24]). *Let $H \in \mathcal{H}_\Gamma$, $f \in G^n$, $g \in G^m$. We have*

$$\Phi(f^*Hg) = \Phi(H)^f,$$
$$\Phi_L(f^*Hg) = \Phi_L(H)^g.$$

The remainder of this chapter is structured as follows. In Section 6.2, we give our definitions of gain Q and R graphs and prove that switching-equivalent gain graphs yield switching-equivalent gain Q and R graphs. In Section 6.3, we determine the π-spectra of the represented adjacency and Laplacian matrices of the gain Q and R graphs in terms of the π-spectrum of (Γ, φ), and show that known classical and signed spectral results are recovered as special cases.

6.2 Gain Q and R graphs

6.2.1 Maps and matrices

In this section we shall provide our definitions of gain Q and R graphs, give an example using quaternions, and prove that two switching-equivalent root gain graphs yield two switching-equivalent gain Q and R graphs.

Definition 6.2.1. Let s be a fixed central involution of the group G. Let Γ be a graph with $V_\Gamma = \{v_1, v_2, \ldots, v_n\}$ and $E_\Gamma = \{e_1, e_2, \ldots, e_m\}$. We define the map

$$\Phi_Q\colon \mathcal{H}_\Gamma \to G(Q(\Gamma)), \quad \Phi_Q(H)(x,y) = \begin{cases} 0 & \text{if } x = v_i, y = v_j, \\ H_{i,k} & \text{if } x = v_i, y = e_k, v_i \in e_k, \\ (H_{i,k})^* & \text{if } x = e_k, y = v_i, v_i \in e_k, \\ s(H_{r,p})^{-1} H_{r,q} & \text{if } x = e_p, y = e_q, v_r = e_p \cap e_q. \end{cases}$$

We call $(Q(\Gamma), \Phi_Q(H))$ a gain Q graph.

For the adjacency matrix $A_{(Q(\Gamma),\Phi_Q(H))}$ we write $\Phi_Q(H)$, and for every $H \in \mathcal{H}_\Gamma$ we have

$$\Phi_Q(H) = \begin{pmatrix} O_n & H \\ H^* & \Phi_L(H) \end{pmatrix} \in M_{n+m}(\mathbb{C}G).$$

Definition 6.2.2. Let s be a fixed central involution of the group G. Let Γ be a graph with $V_\Gamma = \{v_1, v_2, \ldots, v_n\}$ and $E_\Gamma = \{e_1, e_2, \ldots, e_m\}$. We define the map

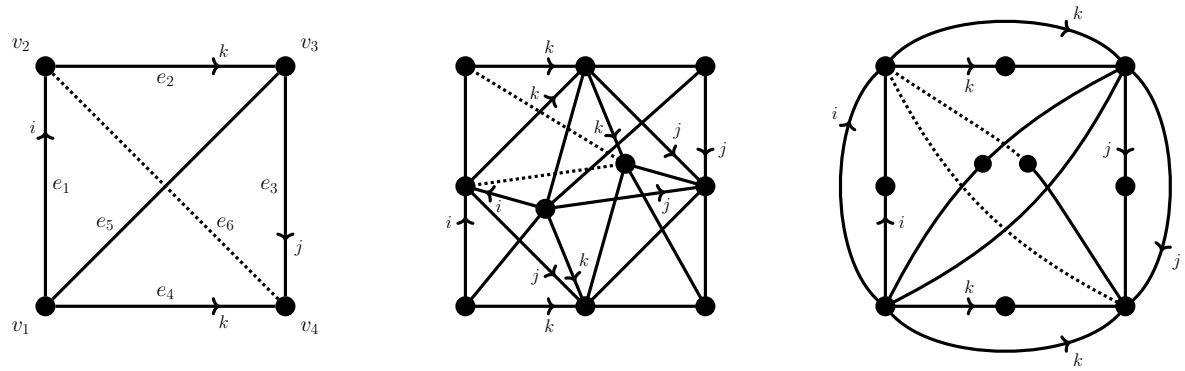
$$\Phi_R\colon \mathcal{H}_\Gamma \to G(R(\Gamma)), \quad \Phi_R(H)(x,y) = \begin{cases} sH_{i,k}(H_{j,k})^{-1} & \text{if } x = v_i, y = v_j, e_k = \{v_i, v_j\}, \\ H_{i,k} & \text{if } x = v_i, y = e_k, v_i \in e_k, \\ (H_{i,k})^* & \text{if } x = e_k, y = v_i, v_i \in e_k, \\ 0 & \text{if } x = e_p, y = e_q. \end{cases}$$

We call $(R(\Gamma), \Phi_R(H))$ a gain R graph.

For the adjacency matrix $A_{(R(\Gamma),\Phi_R(H))}$ we write $\Phi_R(H)$, and for every $H \in \mathcal{H}_\Gamma$ we have

$$\Phi_R(H) = \begin{pmatrix} \Phi(H) & H \\ H^* & O_m \end{pmatrix} \in M_{n+m}(\mathbb{C}G).$$

Remark 6.2.3. The gain subgraph of $(Q(\Gamma), \Phi_Q(H))$ induced by the subset of vertices E_Γ (regarded as a subset of the vertices of $Q(\Gamma)$) is a copy of the gain line graph $(L(\Gamma), \Phi_L(H))$. Similarly, the gain subgraph of $(R(\Gamma), \Phi_R(H))$ induced by the subset of vertices V_Γ is a copy of the original gain graph $(\Gamma, \Phi(H))$.



(Γ, φ) $(Q(\Gamma), \kappa)$ $(R(\Gamma), \rho)$

Figure 6.2: The $\mathbb{H}$-gain graphs (Γ, φ), $(Q(\Gamma), \kappa)$, and $(R(\Gamma), \rho)$ of Example 6.2.4.

Example 6.2.4. Recall the group of quaternions, defined by the presentation

$$\mathbb{H} = \langle i, j, k \mid i^2 = j^2 = k^2 = ijk = -1 \rangle,$$

where -1 is central and $(-1)^2 = 1$. Explicitly, $\mathbb{H} = \{\pm 1, \pm i, \pm j, \pm k\}$, with multiplication determined by $ij = k$, $jk = i$, $ki = j$, and $ji = -k$, $kj = -i$, $ik = -j$.

Let (Γ, φ) be the $\mathbb{H}$-gain graph with the following adjacency matrix $A_{(\Gamma,\varphi)}$, and consider the following $\mathbb{H}$-phase H:

$$A_{(\Gamma,\varphi)} = \begin{pmatrix} 0 & i & 1 & k \\ -i & 0 & k & -1 \\ 1 & -k & 0 & j \\ -k & -1 & -j & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} i & 0 & 0 & k & 1 & 0 \\ 1 & k & 0 & 0 & 0 & -1 \\ 0 & 1 & j & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{pmatrix} \in \mathcal{H}_\Gamma.$$

With $s = 1$, it can be verified that $s(HH^* - \deg(\Gamma)) = A_{(\Gamma,\varphi)}$, as expected from Equation (6.1.1).

Then, applying the map given in Definition 6.2.1, we have

$$\Phi_Q(H) = \left(\begin{array}{cccc|cccccc} 0 & 0 & 0 & 0 & i & 0 & 0 & k & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & k & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 1 & j & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ \hline -i & 1 & 0 & 0 & 0 & k & 0 & j & -i & -1 \\ 0 & -k & 1 & 0 & -k & 0 & j & 0 & 1 & k \\ 0 & 0 & -j & 1 & 0 & -j & 0 & 1 & -j & 1 \\ -k & 0 & 0 & 1 & -j & 0 & 1 & 0 & -k & 1 \\ 1 & 0 & 1 & 0 & i & 1 & j & k & 0 & 0 \\ 0 & -1 & 0 & 1 & -1 & -k & 1 & 1 & 0 & 0 \end{array}\right).$$

Applying the map given in Definition 6.2.2 instead, we get

$$\Phi_R(H) = \left(\begin{array}{cccc|cccccc} 0 & i & 1 & k & i & 0 & 0 & k & 1 & 0 \\ -i & 0 & k & -1 & 1 & k & 0 & 0 & 0 & -1 \\ 1 & -k & 0 & j & 0 & 1 & j & 0 & 1 & 0 \\ -k & -1 & -j & 0 & 0 & 0 & 1 & 1 & 0 & 1 \\ \hline -i & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -k & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -j & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ -k & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right).$$

The $\mathbb{H}$-gain graphs (Γ, φ), $(Q(\Gamma), \kappa)$, and $(R(\Gamma), \rho)$ are displayed in Figure 6.2.

6.2.2 Switching equivalences

The following theorems demonstrate that if a representative in a switching class of gain functions on $Q(\Gamma)$ or $R(\Gamma)$ is realised as a gain Q or R graph by a G-phase of Γ, then all the other gain functions in the same class are realised as such.

Theorem 6.2.5. *Let $\kappa \in G(Q(\Gamma))$ be a gain function on the Q graph $Q(\Gamma)$ of Γ. If $\kappa \sim \Phi_Q(H)$ for some $H \in \mathcal{H}_\Gamma$, then there exists $H' \in \mathcal{H}_\Gamma$, with $H' \sim_{l\times r} H$, such that $\kappa = \Phi_Q(H')$.*

Proof. Let us denote by $\kappa_{L(\Gamma)}$ the gain function κ restricted to the vertices associated with E_Γ. In matrix form, we have

$$\kappa = \begin{pmatrix} O & H' \\ H'^* & \kappa_{L(\Gamma)} \end{pmatrix},$$

for some $H' \in \mathcal{H}_\Gamma$ and we have to show that $\kappa = \Phi_Q(H')$. That is, that $\kappa_{L(\Gamma)} = \Phi_L(H')$.

We know that $\kappa \sim \Phi_Q(H)$ for some $H \in \mathcal{H}_\Gamma$, and this implies that there exist $f \in G^n$ and $g \in G^m$ such that

$$\kappa = \begin{pmatrix} O & H' \\ H'^* & \kappa_{L(\Gamma)} \end{pmatrix} = \begin{pmatrix} f^* & O \\ O & g^* \end{pmatrix} \begin{pmatrix} O & H \\ H^* & \Phi_L(H) \end{pmatrix} \begin{pmatrix} f & O \\ O & g \end{pmatrix} = \begin{pmatrix} O & f^*Hg \\ g^*H^*f & g^*\Phi_L(H)g \end{pmatrix}.$$

In particular, we have $\kappa_{L(\Gamma)} = \Phi_L(H)^g$ and $H' = f^*Hg$. Applying Lemma 6.1.4, we conclude that

$$\Phi_L(H') = \Phi_L(f^*Hg) = \Phi_L(H)^g = \kappa_{L(\Gamma)}.$$

□

Theorem 6.2.6. *Let $\rho \in G(R(\Gamma))$ be a gain function on the R graph $R(\Gamma)$ of Γ. If $\rho \sim \Phi_R(H)$ for some $H \in \mathcal{H}_\Gamma$, then there exists $H' \in \mathcal{H}_\Gamma$, with $H' \sim_{l\times r} H$, such that $\rho = \Phi_R(H')$.*

Proof. We perform an analogous proof as in Theorem 6.2.5, instead denoting by ρ_Γ the gain function ρ restricted to the vertices associated with V_Γ and considering the matrix form

$$\rho = \begin{pmatrix} \rho_\Gamma & H' \\ H'^* & O \end{pmatrix}.$$

for some $H' \in \mathcal{H}_\Gamma$, now needing to show that $\rho = \Phi_R(H')$. That is, that $\rho_\Gamma = \Phi(H')$. After following the same steps we obtain $\rho_\Gamma = f^*\Phi(H)f = \Phi(H)^f$, and applying Lemma 6.1.4 again, we arrive at

$$\Phi(H') = \Phi(f^*Hg) = \Phi(H)^f = \rho_\Gamma.$$

□

The subsequent theorem shows that switching-equivalent gain graphs give rise to switching-equivalent gain Q and R graphs.

Theorem 6.2.7. *For any two G-phases $H, H' \in \mathcal{H}_\Gamma$, the following statements are equivalent:*

(1) $H \sim_{l\times r} H'$,

(2) $\Phi(H) \sim \Phi(H')$,

(3) $\Phi_L(H) \sim \Phi_L(H')$,

(4) $\Phi_Q(H) \sim \Phi_Q(H')$,

(5) $\Phi_R(H) \sim \Phi_R(H')$,

(6) $\Phi_T(H) \sim \Phi_T(H')$.

Proof. The relations (1) $\iff$ (2) $\iff$ (3) $\iff$ (6) are given by Cavaleri et al. in [30] and [34].

(1) $\implies$ (4): Suppose that $H' = f^*Hg$ with $f \in G^n$ and $g \in G^m$. Then by using Lemma 6.1.4 we have

$$\Phi_Q(H') = \begin{pmatrix} O_n & H' \\ H'^* & \Phi_L(H') \end{pmatrix} = \begin{pmatrix} O_n & f^*Hg \\ (f^*Hg)^* & \Phi_L(f^*Hg) \end{pmatrix} = \begin{pmatrix} O_n & f^*Hg \\ g^*H^*f & g^*\Phi_L(H)g \end{pmatrix}.$$

Now take $(f, g) \in G^{n+m}$ and the associated $(n+m)\times(n+m)$ diagonal matrix with f in the first n positions and g in the final m positions. Then we have

$$\begin{aligned}(f,g)^*\Phi_Q(H)(f,g) &= \begin{pmatrix} f^* & O_{n\times m} \\ O_{m\times n} & g^* \end{pmatrix}\begin{pmatrix} O_n & H \\ H^* & \Phi_L(H) \end{pmatrix}\begin{pmatrix} f & O_{n\times m} \\ O_{m\times n} & g \end{pmatrix} \\ &= \begin{pmatrix} O_n & f^*Hg \\ g^*H^*f & g^*\Phi_L(H)g \end{pmatrix}.\end{aligned}$$

Now comparing this with the previous equation for $\Phi_Q(H')$ and considering the switching function $(f, g)\colon V_{Q(\Gamma)} \to G$, we have $\Phi_Q(H') = \Phi_Q(H)^{(f,g)}$. Hence $\Phi_Q(H) \sim \Phi_Q(H')$.

(1) $\implies$ (5): In taking $H' = f^*Hg$ as before and following the same steps with the adjacency matrix for R we will arrive at $\Phi_R(H) \sim \Phi_R(H')$.

(4) $\implies$ (3): If we assume that $\Phi_Q(H) \sim \Phi_Q(H')$ then in particular these gain functions restricted on the vertices of E_Γ are switching equivalent. Then by Remark 6.2.3 it must be so that $\Phi_L(H) \sim \Phi_L(H')$.

(5) $\implies$ (2): If we assume that $\Phi_R(H) \sim \Phi_R(H')$ then in particular these gain functions restricted on the vertices of V_Γ are switching equivalent. Then, again by Remark 6.2.3, it must be so that $\Phi(H) \sim \Phi(H')$. □

6.3 Spectra from regular root graphs

Let $A \in M_n(\mathbb{C}G)$ and let π be a representation of G. We denote by

$$|A|_\pi = \det(\pi(A))$$

the determinant of the Fourier transform of A at π, or the π-determinant of A.

Furthermore, for $A, B \in M_n(\mathbb{C}G)$, we write $|A| = |B|$ to mean that $|A|_\pi = |B|_\pi$ for every irreducible representation π of G. Equivalently, $|A|$ denotes the indexed family $(|A|_\pi)_{\pi\in\mathrm{Irr}(G)}$, where $\mathrm{Irr}(G)$ denotes the set of all irreducible representations of G.

A matrix $T \in M_n(\mathbb{C}G)$ is triangular if, for all $1 \le i, j \le n$ with $i < j$, one has $T_{i,j} = 0$ (lower triangular) or $T_{j,i} = 0$ (upper triangular). If $T \in M_n(\mathbb{C}G)$ is triangular and $T_{i,i} = 1_G$ for all $1 \le i \le n$, then, since the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks, $\det(\pi(T)) = 1$ for every representation π.

Furthermore, applying the Cauchy-Binet theorem, which generalises the multiplicativity of determinants, one obtains

$$|TA|_\pi = \det(\pi(TA)) = \det(\pi(T))\det(\pi(A)) = 1 \cdot |A|_\pi = |A|_\pi,$$

and since this holds for every irreducible representation π, we get

$$|TA| = |A|, \quad \forall\ A \in M_n(\mathbb{C}G). \tag{6.3.1}$$

Let s be a central involution of G. Then Schur's Lemma implies that $\pi(s) = \lambda I$ for some $\lambda \in \mathbb{C}$ for any irreducible representation π. Moreover, since $s^2 = 1_G$, we have $\lambda^2 = 1$, and so $\pi(s) = \pm I$. In the following, we write $\sigma = \pm 1$ to indicate which case holds.

6.3.1 The π-spectrum of a gain Q graph

Theorem 6.3.1. *Let Γ be an r-regular graph with n vertices, where $r \geq 2$, and let G be a group with fixed central involution s. Let π be an irreducible unitary representation of G of degree d. Suppose that $(Q(\Gamma), \kappa)$ is a gain Q graph of a G-gain graph (Γ, φ), and let $\{\lambda_1, \lambda_2, \ldots, \lambda_{nd}\}$ be the π-spectrum of (Γ, φ). Then the π-spectrum of $(Q(\Gamma), \kappa)$ consists of -2σ with multiplicity $d(m-n)$ and, for each $1 \leq i \leq nd$, the two eigenvalues*

$$\frac{1}{2}\left(\lambda_i + (r-2)\sigma \pm \sqrt{(\lambda_i + r\sigma)^2 + 4}\right),$$

where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. Since $(Q(\Gamma), \kappa)$ is a gain Q graph, by definition there exists $H \in \mathcal{H}_\Gamma$ such that $\Phi_Q(H) = \kappa$, and in particular $\Phi(H) = \varphi$. By combining Equation (6.1.1) with Definition 6.2.1, we get

$$\Phi_Q(H) = \begin{pmatrix} O & H \\ H^* & s(H^*H - 2I) \end{pmatrix}$$

and then

$$xI - \Phi_Q(H) = \begin{pmatrix} xI & -H \\ -H^* & xI - s(H^*H - 2I) \end{pmatrix}.$$

Making use of Equation (6.3.1) we will multiply this matrix by a triangular matrix whose diagonal entries are all equal to 1_G. In the following computations, we will simply write A instead of $A_{(\Gamma,\varphi)}$ to denote the adjacency matrix of (Γ, φ). Also, since Γ is r-regular and s is a central involution of G, we have that $HH^* = rI + sA$ and $s^2 = 1_G$. Therefore,

$$\begin{aligned}
|xI - \Phi_Q(H)| &= \left| \begin{pmatrix} I & O \\ -sH^* & I \end{pmatrix} \begin{pmatrix} xI & -H \\ -H^* & xI - s(H^*H - 2I) \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} xI & -H \\ -H^* - sxH^* & xI - sH^*H + sH^*H + 2sI \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} I & \frac{1}{x+2s}H \\ O & I \end{pmatrix} \begin{pmatrix} xI & -H \\ -(1+sx)H^* & (x+2s)I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} xI - \frac{1+sx}{x+2s}HH^* & O \\ -(1+sx)H^* & (x+2s)I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x+2s}I & O \\ O & (x+2s)I \end{pmatrix} \begin{pmatrix} (x^2+2sx)I - (1+sx)HH^* & O \\ -\frac{1+sx}{x+2s}H^* & I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x+2s}I & O \\ O & (x+2s)I \end{pmatrix} \begin{pmatrix} (x^2+2sx)I - (1+sx)(rI+sA) & O \\ -\frac{1+sx}{x+2s}H^* & I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x+2s}I & O \\ O & (x+2s)I \end{pmatrix} \begin{pmatrix} (x^2+(2s-rs)x - r)I - (x+s)A & O \\ -\frac{1+sx}{x+2s}H^* & I \end{pmatrix} \right|.
\end{aligned}$$

Let π be the irreducible representation of the claim and let us introduce $\sigma \in \{\pm 1\}$ such that $\sigma = \pm 1$ if $\pi(s) = \pm I$. Then we have

$$|xI - \Phi_Q(H)|_\pi = (x+2\sigma)^{d(m-n)} \prod_{i=1}^{nd} (x^2 + (2\sigma - r\sigma)x - r - (x+\sigma)\lambda_i).$$

Thus -2σ is an eigenvalue repeated $d(m-n)$ times and the remaining eigenvalues are given by

$$x = \frac{1}{2}\left(\lambda_i + (r-2)\sigma \pm \sqrt{(\lambda_i + r\sigma)^2 + 4}\right),$$

for each $1 \leq i \leq nd$. The desired result follows. □

Remark 6.3.2. Taking $G = \{1\}$ with $s = 1_G$, so that $d = 1$ and $\sigma = 1$, the gain Q graph reduces to the classical unsigned Q graph, and Theorem 6.3.1 recovers the characteristic polynomial result given in [37, Theorem 5]. To the best of our knowledge, no analogous spectral result for the signed Q graph currently exists in the literature, and Theorem 6.3.1 thus provides a new result in this setting.

6.3.2 The π-spectrum of a gain R graph

Theorem 6.3.3. *Let Γ be an r-regular graph with n vertices, where $r \geq 2$, and let G be a group with fixed central involution s. Let π be an irreducible unitary representation of G of degree d. Suppose that $(R(\Gamma), \rho)$ is a gain R graph of a G-gain graph (Γ, φ), and let $\{\lambda_1, \lambda_2, \ldots, \lambda_{nd}\}$ be the π-spectrum of (Γ, φ). Then the π-spectrum of $(R(\Gamma), \rho)$ consists of 0 with multiplicity $d(m-n)$ and, for each $1 \leq i \leq nd$, the two eigenvalues*

$$\frac{1}{2}\left(\lambda_i \pm \sqrt{\lambda_i^2 + 4\sigma\lambda_i + 4r}\right),$$

where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. Since $(R(\Gamma), \rho)$ is a gain R graph, by definition there exists $H \in \mathcal{H}_\Gamma$ such that $\Phi_R(H) = \rho$, and in particular $\Phi(H) = \varphi$. By combining Equation (6.1.1) with Definition 6.2.2, we get

$$\Phi_R(H) = \begin{pmatrix} s(HH^* - \deg(\Gamma)) & H \\ H^* & O \end{pmatrix}$$

and then

$$xI - \Phi_R(H) = \begin{pmatrix} xI - s(HH^* - rI) & -H \\ -H^* & xI \end{pmatrix}.$$

As in the proof of Theorem 6.3.1, we shall use Equation (6.3.1) and write A instead of $A_{(\Gamma,\psi)}$ to denote the adjacency matrix of (Γ, ψ). Hence,

$$\begin{aligned}
|xI - \Phi_R(H)| &= \left| \begin{pmatrix} I & \frac{1}{x}H \\ O & I \end{pmatrix} \begin{pmatrix} xI - s(HH^* - rI) & -H \\ -H^* & xI \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} xI - s(HH^* - rI) - \frac{1}{x}HH^* & O \\ -H^* & xI \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x}I & O \\ O & xI \end{pmatrix} \begin{pmatrix} x^2I - sx(HH^* - rI) - HH^* & O \\ -\frac{1}{x}H^* & I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x}I & O \\ O & xI \end{pmatrix} \begin{pmatrix} x^2I - sx(rI + sA - rI) - rI - sA & O \\ -\frac{1}{x}H^* & I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x}I & O \\ O & xI \end{pmatrix} \begin{pmatrix} (x^2 - r)I - (x+s)A & O \\ -\frac{1}{x}H^* & I \end{pmatrix} \right|.
\end{aligned}$$

Let π be the irreducible representation of the claim and let us introduce $\sigma \in \{\pm 1\}$ such that $\sigma = \pm 1$ if $\pi(s) = \pm I$. Then we have

$$|xI - \Phi_R(H)|_\pi = x^{d(m-n)} \prod_{i=1}^{nd} (x^2 - r - (x + \sigma)\lambda_i).$$

Thus 0 is an eigenvalue repeated $d(m-n)$ times and the remaining eigenvalues are given by

$$x = \frac{1}{2}\left(\lambda_i \pm \sqrt{\lambda_i^2 + 4\sigma\lambda_i + 4r}\right),$$

for each $1 \leq i \leq nd$. The desired result follows. □

Remark 6.3.4. Taking $G = \{1\}$ with $s = 1_G$, so that $d = 1$ and $\sigma = 1$, as in Remark 6.3.2, the gain R graph reduces to the classical unsigned R graph, and Theorem 6.3.3 recovers the characteristic polynomial results given in [37, Theorem 4]. Taking $G = \{1, -1\}$ with $s = -1$ and the representation $\pi(-1) = -I$, so that $d = 1$ and $\sigma = -1$, the gain R graph reduces to its signed counterpart. In this case, Theorem 6.3.3 recovers [80, Theorem 3.6], which gives the spectrum of the signed R graph of an r-regular signed graph.

Example 6.3.5. Consider the following irreducible representation π of $\mathbb{H}$ of degree 2:

$$\pi(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \pi(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \pi(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \pi(k) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$

For the $\mathbb{H}$-gain graph (Γ, φ) of Example 6.2.4, the represented adjacency matrix is

$$\pi(A_{(\Gamma,\varphi)}) = \left(\begin{array}{cc|cc|cc|cc} 0 & 0 & i & 0 & 1 & 0 & 0 & i \\ 0 & 0 & 0 & -i & 0 & 1 & i & 0 \\ \hline -i & 0 & 0 & 0 & 0 & i & -1 & 0 \\ 0 & i & 0 & 0 & i & 0 & 0 & -1 \\ \hline 1 & 0 & 0 & -i & 0 & 0 & 0 & 1 \\ 0 & 1 & -i & 0 & 0 & 0 & -1 & 0 \\ \hline 0 & -i & -1 & 0 & 0 & -1 & 0 & 0 \\ -i & 0 & 0 & -1 & 1 & 0 & 0 & 0 \end{array}\right) \in M_8(\mathbb{C}).$$

Thus, the π-spectrum of (Γ, φ) is

$$\pi\text{-spec}(\Gamma, \varphi) = \{2.414^{(2)}, 0.414^{(2)}, -0.414^{(2)}, -2.414^{(2)}\}. \tag{6.3.2}$$

Let $(Q(\Gamma), \kappa)$ be the gain graph determined by $\Phi_Q(H)$ of Example 6.2.4. After the appropriate substitutions, the π-spectrum of $(Q(\Gamma), \kappa)$ can be directly computed as

$$\pi\text{-spec}(Q(\Gamma), \kappa) = \{4.593^{(2)}, 2.686^{(2)}, 1.927^{(2)}, 0.335^{(2)}, -1.179^{(2)}, -1.271^{(2)}, -1.342^{(2)}, -1.749^{(2)}, -2^{(4)}\},$$

which matches the values obtained from implementing (6.3.2) and Theorem 6.3.1.

Similarly, let $(R(\Gamma), \rho)$ be the gain graph determined by $\Phi_R(H)$ of Example 6.2.4. After the analogous substitutions, the π-spectrum of $(R(\Gamma), \rho)$ can be found immediately as

$$\pi\text{-spec}(R(\Gamma), \rho) = \{3.828^{(2)}, 2.066^{(2)}, 1.414^{(2)}, 0.222^{(2)}, 0^{(4)}, -1.414^{(2)}, -1.652^{(2)}, -1.828^{(2)}, -2.636^{(2)}\},$$

which, likewise, matches the values obtained from combining (6.3.2) and Theorem 6.3.3.

6.3.3 The Laplacian π-spectrum of a gain Q graph

Let $L_{(Q(\Gamma),\kappa)} = \deg(Q(\Gamma)) - A_{(Q(\Gamma),\kappa)}$ be denoted by $L(\Phi_Q(H))$.

Since Γ is r-regular, we have $\deg(Q(\Gamma)) = \operatorname{diag}(rI_n, 2rI_m)$. Indeed, each vertex in $Q(\Gamma)$ corresponding to some $v_i \in V_\Gamma$ is adjacent only to the edge-vertices corresponding to edges incident with v_i, of which there are r. Each vertex in $Q(\Gamma)$ corresponding to some $e \in E_\Gamma$ is adjacent to its two endpoints in V_Γ, and additionally to every edge-vertex corresponding to an edge sharing a vertex with e. Since each of the two endpoints of e has degree r in Γ, there are $r-1$ other edges incident with each endpoint, contributing $2(r-1)$ further edge-vertex neighbours. Thus each edge-vertex has degree $2 + 2(r-1) = 2r$ in $Q(\Gamma)$.

Theorem 6.3.6. *Let Γ be an r-regular graph with n vertices, where $r \geq 2$, and let G be a group with fixed central involution s. Let π be an irreducible unitary representation of G of degree d. Suppose that $(Q(\Gamma), \kappa)$ is a gain Q graph of a G-gain graph (Γ, φ), and let $\{\lambda_1, \lambda_2, \ldots, \lambda_{nd}\}$ be the π-spectrum of (Γ, φ). Then the Laplacian π-spectrum of $(Q(\Gamma), \kappa)$ consists of $2r + 2\sigma$ with multiplicity $d(m-n)$ and, for each $1 \leq i \leq nd$, the two eigenvalues*

$$\frac{1}{2}\left(3r + 2\sigma - r\sigma - \lambda_i \pm \sqrt{\tau(\lambda_i, \sigma)}\right),$$

where

$$\tau(\lambda_i, \sigma) = \begin{cases} \lambda_i^2 + 4(r+1), & \text{if } \sigma = 1, \\ (\lambda_i - 2r)^2 + 4(1-r), & \text{if } \sigma = -1, \end{cases}$$

and $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. Since $(Q(\Gamma), \kappa)$ is a gain Q graph, by definition there exists $H \in \mathcal{H}_\Gamma$ such that $\Phi_Q(H) = \kappa$, and in particular $\Phi(H) = \varphi$. By combining Equation (6.1.1) with Definition 6.2.1, we get

$$L(\Phi_Q(H)) = \begin{pmatrix} \deg(\Gamma) & -H \\ -H^* & 2\deg(\Gamma) - s(H^*H - 2I) \end{pmatrix}$$

and then

$$xI - L(\Phi_Q(H)) = \begin{pmatrix} xI - \deg(\Gamma) & H \\ H^* & xI - 2\deg(\Gamma) + s(H^*H - 2I) \end{pmatrix}.$$

We again let A denote the adjacency matrix of (Γ, ψ) and use a similar strategy as in the proofs for the adjacency π-spectra. Consequently,

$$\begin{aligned}
|xI - L(\Phi_Q(H))| &= \left| \begin{pmatrix} xI - rI & H \\ H^* & xI - 2rI + s(H^*H - 2I) \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} I & O \\ -sH^* & I \end{pmatrix} \begin{pmatrix} (x-r)I & H \\ H^* & (x-2r-2s)I + sH^*H \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} (x-r)I & H \\ H^* - s(x-r)H^* & (x-2r-2s)I + sH^*H - sH^*H \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} I & -\frac{1}{x-2r-2s}H \\ O & I \end{pmatrix} \begin{pmatrix} (x-r)I & H \\ (1+rs-sx)H^* & (x-2r-2s)I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} (x-r)I - \frac{1+rs-sx}{x-2r-2s}HH^* & O \\ (1+rs-sx)H^* & (x-2r-2s)I \end{pmatrix} \right| \\
&= \left| \begin{pmatrix} \frac{1}{x-2r-2s}I & O \\ O & (x-2r-2s)I \end{pmatrix} \right. \\
&\qquad \left. \begin{pmatrix} (x-2r-2s)(x-r)I - (1+rs-sx)HH^* & O \\ \frac{1+rs-sx}{x-2r-2s}H^* & I \end{pmatrix} \right|.
\end{aligned}$$

Let $X = (x-2r-2s)(x-r)I - (1+rs-sx)HH^*$. Then, after substituting $HH^* = rI + sA$, we have

$$\begin{aligned}
X &= (x^2 - 2rx - 2sx - rx + 2r^2 + 2rs)I - (1 + rs - sx)(rI + sA) \\
&= (x^2 - (3r+2s)x + 2r^2 + 2rs)I - (r + r^2s - rsx)I - (s + rs^2 - s^2x)A \\
&= (x^2 + (rs - 3r - 2s)x + 2r^2 - r^2s - r + 2rs)I + (x - r - s)A.
\end{aligned}$$

Now let π be the irreducible representation of the claim and let us introduce $\sigma \in \{\pm 1\}$ such that $\sigma = \pm 1$ if $\pi(s) = \pm I$. Then we have

$$|xI - L(\Phi_Q(H))|_\pi = (x - 2r - 2\sigma)^{d(m-n)} \prod_{i=1}^{nd} (x^2 + (r\sigma - 3r - 2\sigma)x + 2r^2 - r^2\sigma - r + 2r\sigma + (x - r - \sigma)\lambda_i).$$

Thus $2r + 2\sigma$ is an eigenvalue repeated $d(m-n)$ times and the remaining eigenvalues are given by

$$x = \frac{1}{2}\left(3r + 2\sigma - r\sigma - \lambda_i \pm \sqrt{(\lambda_i - 3r - 2\sigma + r\sigma)^2 - 4(2r^2 - r^2\sigma - r + 2r\sigma - r\lambda_i - \sigma\lambda_i)}\right),$$

for each $1 \leq i \leq nd$. The desired result follows. □

6.3.4 The Laplacian π-spectrum of a gain R graph

Let $L_{(R(\Gamma),\rho)} = \deg(R(\Gamma)) - A_{(R(\Gamma),\rho)}$ be denoted by $L(\Phi_R(H))$.

Since Γ is r-regular, we have $\deg(R(\Gamma)) = \operatorname{diag}(2rI_n, 2I_m)$. Indeed, each vertex in $R(\Gamma)$ corresponding to some $v_i \in V_\Gamma$ retains its r neighbours in Γ, and is additionally adjacent to the edge-vertex corresponding to each edge incident with v_i, of which there are r. Thus each original vertex has degree $2r$ in $R(\Gamma)$. Each vertex in $R(\Gamma)$ corresponding to some $e \in E_\Gamma$ is adjacent only to the two endpoints of e in V_Γ, and so has degree 2.

Theorem 6.3.7. *Let Γ be an r-regular graph with n vertices, where $r \geq 2$, and let G be a group with fixed central involution s. Let π be an irreducible unitary representation of G of degree d. Suppose that $(R(\Gamma), \rho)$ is a gain R graph of a G-gain graph (Γ, φ), and let $\{\lambda_1, \lambda_2, \ldots, \lambda_{nd}\}$ be the π-spectrum of (Γ, φ). Then the Laplacian π-spectrum of $(R(\Gamma), \rho)$ consists of 2 with multiplicity $d(m-n)$ and, for each $1 \leq i \leq nd$, the two eigenvalues*

$$\frac{1}{2}\left(2r + 2 - \lambda_i \pm \sqrt{\tau(\lambda_i, \sigma)}\right),$$

where

$$\tau(\lambda_i, \sigma) = \begin{cases} (\lambda_i - 2r)^2 + 8\lambda_i - 4r + 4, & \text{if } \sigma = 1, \\ (\lambda_i - 2r)^2 + 4(1 - r), & \text{if } \sigma = -1, \end{cases}$$

and $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. Since $(R(\Gamma), \rho)$ is a gain R graph, by definition there exists $H \in \mathcal{H}_\Gamma$ such that $\Phi_R(H) = \rho$, and in particular $\Phi(H) = \varphi$. By combining Equation (6.1.1) with Definition 6.2.2, we get

$$L(\Phi_R(H)) = \begin{pmatrix} 2\deg(\Gamma) - s(HH^* - \deg(\Gamma)) & -H \\ -H^* & 2I \end{pmatrix}$$

and then

$$xI - L(\Phi_R(H)) = \begin{pmatrix} xI - 2\deg(\Gamma) + s(HH^* - \deg(\Gamma)) & H \\ H^* & xI - 2I \end{pmatrix}.$$

Once more denoting the adjacency matrix of (Γ, ψ) by A, we have

$$\begin{aligned} |xI - L(\Phi_R(H))| &= \left| \begin{pmatrix} xI - 2rI + s(HH^* - rI) & H \\ H^* & xI - 2I \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} I & -\frac{1}{x-2}H \\ O & I \end{pmatrix} \begin{pmatrix} (x - 2r - rs)I + sHH^* & H \\ H^* & (x-2)I \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} (x - 2r - rs)I + sHH^* - \frac{1}{x-2}HH^* & O \\ H^* & (x-2)I \end{pmatrix} \right| \\ &= \left| \begin{pmatrix} \frac{1}{x-2}I & O \\ O & (x-2)I \end{pmatrix} \begin{pmatrix} (x-2)(x-2r-rs)I + s(x-2)HH^* - HH^* & O \\ \frac{1}{x-2}H^* & I \end{pmatrix} \right|. \end{aligned}$$

Let $X = (x-2)(x-2r-rs)I + s(x-2)HH^* - HH^*$. Then, after substituting $HH^* = rI + sA$, we have

$$\begin{aligned} X &= (x^2 - 2rx - rsx - 2x + 4r + 2rs)I + s(x-2)(rI + sA) - rI - sA \\ &= (x^2 - (2r + rs + 2)x + 4r + 2rs)I + (rsx - 2rs - r)I + (s^2x - 2s^2 - s)A \\ &= (x^2 - (2r+2)x + 3r)I + (x - 2 - s)A. \end{aligned}$$

Now let π be the irreducible representation of the claim and let us introduce $\sigma \in \{\pm 1\}$ such that $\sigma = \pm 1$ if $\pi(s) = \pm I$. Then we have

$$|xI - L(\Phi_R(H))|_\pi = (x-2)^{d(m-n)} \prod_{i=1}^{nd} (x^2 - (2r+2)x + 3r + (x - 2 - \sigma)\lambda_i).$$

Thus 2 is an eigenvalue repeated $d(m-n)$ times and the remaining eigenvalues are given by

$$x = \frac{1}{2}\left(2r + 2 - \lambda_i \pm \sqrt{(\lambda_i - 2r - 2)^2 - 4(3r - 2\lambda_i - \sigma\lambda_i)}\right),$$

for each $1 \leq i \leq nd$. The desired result follows. □

Example 6.3.8. Consider the irreducible representation π of $\mathbb{H}$ and the $\mathbb{H}$-gain graph (Γ, φ) of Example 6.3.5. Recall from Equation (6.3.2) that the π-spectrum of (Γ, φ) is

$$\pi\text{-spec}(\Gamma, \varphi) = \{2.414^{(2)}, 0.414^{(2)}, -0.414^{(2)}, -2.414^{(2)}\}.$$

Let $(Q(\Gamma), \kappa)$ be the gain graph determined by $\Phi_Q(H)$ of Example 6.2.4. After the appropriate substitutions, the Laplacian π-spectrum of $(Q(\Gamma), \kappa)$ can be directly computed as

$$\pi\text{-spec}(L_{(Q(\Gamma),\kappa)}) = \{8^{(4)}, 7.543^{(2)}, 6.218^{(2)}, 5.804^{(2)}, 5.129^{(2)}, 2.871^{(2)}, 2.196^{(2)}, 1.782^{(2)}, 0.457^{(2)}\},$$

which matches the values obtained from implementing (6.3.2) and Theorem 6.3.6.

Similarly, let $(R(\Gamma), \rho)$ be the gain graph determined by $\Phi_R(H)$ of Example 6.2.4. After the analogous substitutions, the π-spectrum of $(R(\Gamma), \rho)$ can be found immediately as

$$\pi\text{-spec}(L_{(R(\Gamma),\rho)}) = \{8.504^{(2)}, 6.938^{(2)}, 6.368^{(2)}, 5.251^{(2)}, 2^{(4)}, 1.910^{(2)}, 1.476^{(2)}, 1.218^{(2)}, 0.335^{(2)}\},$$

which, in the same way, matches the values obtained from combining (6.3.2) and Theorem 6.3.7.

6.4 Summary and outlook

In this chapter we have introduced gain Q and R graphs in the framework of G-phases and group representations, proved that switching-equivalent gain graphs yield switching-equivalent gain Q and R graphs, and determined the π-spectra of their represented adjacency and Laplacian matrices for an r-regular root gain graph (Γ, φ). The eigenvalue expressions obtained are given entirely in terms of the π-spectrum of the root gain graph, and recover the known classical results of [37] as special cases. In the signed setting, Theorem 6.3.3 recovers the signed R graph result of [80], while Theorem 6.3.1 provides a new spectral result for signed Q graphs, for which no analogue previously existed in the literature.

When computing the π-spectra in Theorems 6.3.1, 6.3.3, 6.3.6, and 6.3.7, the assumption of r-regularity on the underlying graph Γ is essential to the method, as it allows the product of a G-phase H with its adjoint to be expressed cleanly in terms of the adjacency matrix since, in this case, $HH^* = rI + sA$. Extending the spectral results to irregular underlying graphs remains an open problem. Natural further directions include the study of corona products and other graph operations in the G-gain setting.

Chapter 7

Products derived from gain subdivision graphs via group representations

We introduce gain variants of the subdivision-vertex, Q-vertex, R-vertex, subdivision-edge, Q-edge, and R-edge corona products. These constructions are defined via G-phases and group representations, with each attachment edge assigned the identity gain 1_G. For an r-regular G-gain graph (Γ_1, φ_1) and a G-gain graph (Γ_2, φ_2), together with an irreducible unitary representation π of G, we determine the π-characteristic polynomials of the represented adjacency matrices for each product, in terms of the represented adjacency matrices of the component graphs and a matrix-valued coronal. When (Γ_2, φ_2) is g-G-regular and $\pi(g)$ is scalar, these expressions simplify to explicit descriptions of the π-spectra in terms of the π-spectra of the component graphs. As special cases, our results recover the known spectra for these products in the classical setting, as well as for the R-vertex and R-edge corona products in the signed setting, while also providing new results for the signed subdivision and Q cases.

We then develop the notion of pseudo-potentials for G-gain graphs, which enables us to define switching-stable versions of all six products, in which the gains assigned to the attachment edges depend on a choice of spanning tree and a fixed vertex of (Γ_2, φ_2). In particular, we show that under this modified gain assignment, balanced components yield a balanced product.

This chapter is based on [59].

7.1 Gain subdivision, Q, and R graphs

7.1.1 Maps and matrices

Definition 7.1.1. Let s be a fixed central involution of the group G. Let Γ be a connected graph with $V_\Gamma = \{v_1, v_2, \dots, v_n\}$ and $E_\Gamma = \{e_1, e_2, \dots, e_m\}$. We define the map

$$\Phi_S \colon \mathcal{H}_\Gamma \to G(S(\Gamma)), \quad \Phi_S(H)(x,y) = \begin{cases} 0 & \text{if } x = v_i, y = v_j, \\ H_{i,k} & \text{if } x = v_i, y = e_k, v_i \in e_k, \\ (H_{i,k})^* & \text{if } x = e_k, y = v_i, v_i \in e_k, \\ 0 & \text{if } x = e_p, y = e_q. \end{cases}$$

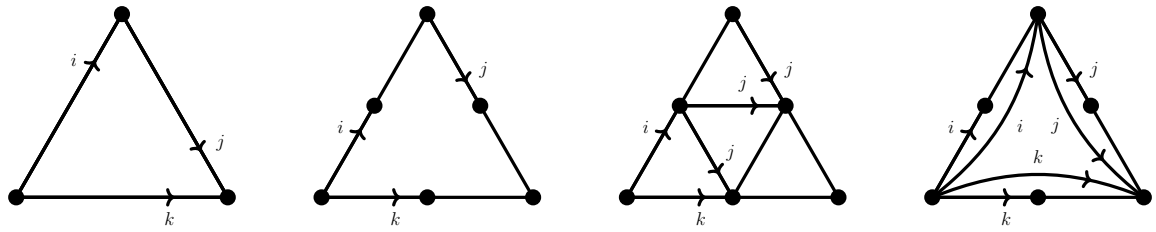
We call $(S(\Gamma), \Phi_S(H))$ a *gain subdivision graph*.

For the adjacency matrix $A_{(S(\Gamma),\Phi_S(H))}$ we write $\Phi_S(H)$, and for every $H \in \mathcal{H}_\Gamma$ we have

$$\Phi_S(H) = \begin{pmatrix} O_n & H \\ H^* & O_m \end{pmatrix} \in M_{n+m}(\mathbb{C}G).$$

Example 7.1.2. Consider the following irreducible representation π of $\mathbb{H}$ of degree 2:

$$\pi(1) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \pi(i) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}, \quad \pi(j) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \pi(k) = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}.$$



(Γ, φ) $(S(\Gamma), \Phi_S(H))$ $(Q(\Gamma), \Phi_Q(H))$ $(R(\Gamma), \Phi_R(H))$

Figure 7.1: The $\mathbb{H}$-gain graphs (Γ, φ), $(S(\Gamma), \Phi_S(H))$, $(Q(\Gamma), \Phi_Q(H))$, and $(R(\Gamma), \Phi_R(H))$ of Example 7.1.2.

Let (Γ, φ) be the $\mathbb{H}$-gain graph with the following adjacency matrix $A_{(\Gamma,\varphi)}$, and consider the following $\mathbb{H}$-phase H:

$$A_{(\Gamma,\varphi)} = \begin{pmatrix} 0 & i & k \\ -i & 0 & j \\ -k & -j & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} i & 0 & k \\ 1 & j & 0 \\ 0 & 1 & 1 \end{pmatrix} \in \mathcal{H}_\Gamma.$$

g

With $s = 1_G$, it can be verified that $s(HH^* - \deg(\Gamma)) = A_{(\Gamma,\varphi)}$, as expected from Equation (6.1.1).

The represented adjacency matrix of (Γ, φ) is

$$\pi(A_{(\Gamma,\varphi)}) = \left(\begin{array}{cc|cc|cc} 0 & 0 & i & 0 & 0 & i \\ 0 & 0 & 0 & -i & i & 0 \\ \hline -i & 0 & 0 & 0 & 0 & 1 \\ 0 & i & 0 & 0 & -1 & 0 \\ \hline 0 & -i & 0 & -1 & 0 & 0 \\ -i & 0 & 1 & 0 & 0 & 0 \end{array}\right) \in M_6(\mathbb{C}).$$

Thus, the π-spectrum of (Γ, φ) is

$$\pi\text{-spec}(\Gamma, \varphi) = \{2^{(2)}, -1^{(4)}\}. \tag{7.1.1}$$

The $\mathbb{H}$-gain graphs (Γ, φ), $(S(\Gamma), \Phi_S(H))$, $(Q(\Gamma), \Phi_Q(H))$, and $(R(\Gamma), \Phi_R(H))$ are displayed in Figure 7.1.

7.1.2 Switching equivalences

In the previous chapter, Theorems 6.2.5 and 6.2.6 demonstrated that if a representative in a switching class of gain functions on $Q(\Gamma)$ or $R(\Gamma)$ is realised as a gain Q or R graph by a G-phase of Γ, then all the other gain functions in the same class are realised as such. We can provide an analogous result for the subdivision graph.

Theorem 7.1.3. *Let $\varsigma \in G(S(\Gamma))$ be a gain function on the subdivision graph $S(\Gamma)$ of Γ. If $\varsigma \sim \Phi_S(H)$ for some $H \in \mathcal{H}_\Gamma$, then there exists $H' \in \mathcal{H}_\Gamma$, with $H' \sim_{l\times r} H$, such that $\varsigma = \Phi_S(H')$.*

Proof. We know that $\varsigma \sim \Phi_S(H)$ for some $H \in \mathcal{H}_\Gamma$. Setting $H' = f^*Hg \sim_{l\times r} H$, we have

$$\varsigma = \begin{pmatrix} f^* & O \\ O & g^* \end{pmatrix} \begin{pmatrix} O & H \\ H^* & O \end{pmatrix} \begin{pmatrix} f & O \\ O & g \end{pmatrix} = \begin{pmatrix} O & f^*Hg \\ g^*H^*f & O \end{pmatrix} = \begin{pmatrix} O & H' \\ H'^* & O \end{pmatrix} = \Phi_S(H').$$

□

7.2 Products derived from gain subdivision graphs

In the subsequent constructions, we shall use the following labelling convention. For each $i \in \{1, 2, \ldots, n_1\}$ (or $i \in \{1, 2, \ldots, m_1\}$ in the edge corona cases), let $(\Gamma_2^i, \varphi_2^i)$ denote the i-th copy of (Γ_2, φ_2).

If $y \in V_{\Gamma_2}$, we write y^i for the corresponding vertex in the i-th copy. Similarly, when fixing a spanning tree $\mathcal{T}$ of Γ_2 and a root vertex $v \in V_{\Gamma_2}$, we let $\mathcal{T}^i$ denote the corresponding spanning tree in the i-th copy and v^i the corresponding root vertex. For disconnected Γ_2, the usual convention applies: replace each spanning tree by a spanning forest with one root per connected component.

For an $m \times n$ matrix P and a $p \times q$ matrix Q, the Kronecker product $P \otimes Q$ is the $mp \times nq$ matrix obtained from P by replacing each element p_{ij} with $p_{ij}Q$. This is an associative operation with the property that $(P \otimes Q)^T = P^T \otimes Q^T$ and $(P \otimes Q)(R \otimes S) = PR \otimes QS$ whenever the products PR and QS exist. The latter implies that $(P \otimes Q)^{-1} = P^{-1} \otimes Q^{-1}$ whenever P and Q are invertible. For further details, see [38, 55].

Recall the notions of outgain, ingain, and regularity for gain graphs from Section 1.5 and the coronal and its regular form from Section 1.8.

Let $A \in M_n(\mathbb{C}G)$ and let π be a representation of G. We denote by

$$|A|_\pi = \det(\pi(A))$$

the determinant of the Fourier transform of A at π, or the π-determinant of A.

Furthermore, for $A, B \in M_n(\mathbb{C}G)$ we write $|A| = |B|$, where

$$|A| = \{|A|_\pi : \ \pi \text{ is an irreducible representation of } G\},$$

if and only if $|A|_\pi = |B|_\pi$ for all irreducible representations π of G.

A matrix $T \in M_n(\mathbb{C}G)$ is triangular if, for all $1 \le i, j \le n$ with $i < j$, one has $T_{i,j} = 0$ (lower triangular) or $T_{j,i} = 0$ (upper triangular). If $T \in M_n(\mathbb{C}G)$ is triangular and $T_{i,i} = 1_G$ for all $1 \le i \le n$, then, since the determinant of a block-triangular matrix is the product of the determinants of its diagonal blocks, $\det(\pi(T)) = 1$ for every representation π.

Furthermore, applying the Cauchy-Binet theorem, which generalises the multiplicativity of determinants, one obtains

$$|TA|_\pi = \det(\pi(TA)) = \det(\pi(T))\det(\pi(A)) = 1 \cdot |A|_\pi = |A|_\pi,$$

and since this holds for every irreducible representation π, we get

$$|TA| = |A|, \quad \forall\ A \in M_n(\mathbb{C}G).$$

Let s be a central involution of G. Then Schur's lemma implies that $\pi(s) = \lambda I$ for some $\lambda \in \mathbb{C}$ for any irreducible representation π. Moreover, since $s^2 = 1_G$, we have $\lambda^2 = 1$, and so $\pi(s) = \pm I$. In the following, we write $\sigma = \pm 1$ to indicate which case holds.

Remark 7.2.1. Throughout the following two subsections we shall have the following standing assumptions. Let G be a group with fixed central involution s and let π be an irreducible unitary representation of G of degree d. Let (Γ_1, φ_1) be an r-regular G-gain graph with n_1 vertices and m_1 edges, where $r \ge 2$, let (Γ_2, φ_2) be a G-gain graph with n_2 vertices and m_2 edges, let A_1 and A_2 denote their respective G-valued adjacency matrices, and let $\{\lambda_1, \lambda_2, \dots, \lambda_{n_1 d}\}$ and $\{\mu_1, \mu_2, \dots, \mu_{n_2 d}\}$ be the π-spectra of (Γ_1, φ_1) and (Γ_2, φ_2), respectively.

Define the matrix-valued coronal

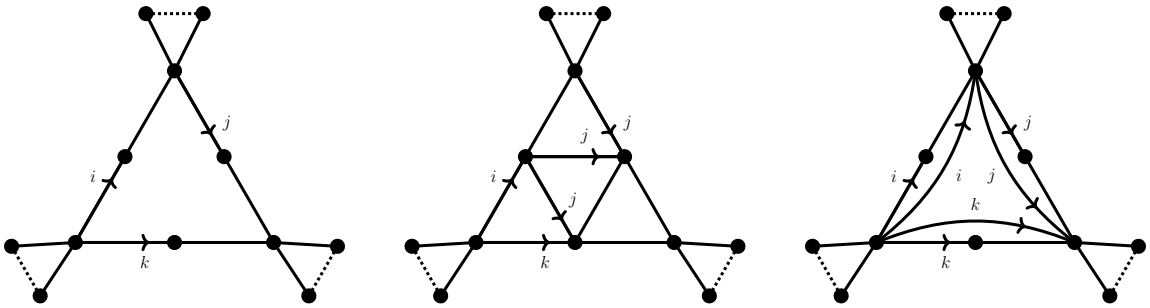
$$C_\pi(x) = (\mathbf{1}_{n_2}^T \otimes I_d)(xI_{n_2 d} - \pi(A_2))^{-1}(\mathbf{1}_{n_2} \otimes I_d),$$

for $x \notin \operatorname{spec}(\pi(A_2))$. In the following theorems, the stated identities are to be read as identities of rational functions in x; in each case both sides are polynomials.

7.2.1 Vertex corona products

Subdivision-vertex corona

Definition 7.2.2. Let $H_1 \in \mathcal{H}_{\Gamma_1}$ and let (Γ_2, φ_2) be a G-gain graph. The gain subdivision-vertex corona $(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)$ is the gain graph obtained from $(S(\Gamma_1), \Phi_S(H_1))$ and n_1 copies of (Γ_2, φ_2) by joining the i-th vertex of V_{Γ_1} by an edge with gain 1_G to every vertex in the i-th copy $(\Gamma_2^i, \varphi_2^i)$ of (Γ_2, φ_2).



$(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)$ $(\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2)$ $(\Gamma_1, \varphi_1) \odot_R (\Gamma_2, \varphi_2)$

Figure 7.2: The gain subdivision-vertex, gain Q-vertex, and gain R-vertex corona graphs of (Γ_1, φ_1) and (Γ_2, φ_2).

For the adjacency matrix of $(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)$ we write $\Phi_{\odot_S}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\odot_S}(H_1, H_2) = \begin{pmatrix} O_{n_1} & H_1 & I_{n_1} \otimes \mathbf{1}_{n_2}^T \\ H_1^* & O_{m_1} & O_{m_1 \times n_1 n_2} \\ I_{n_1} \otimes \mathbf{1}_{n_2} & O_{n_1 n_2 \times m_1} & I_{n_1} \otimes \Phi(H_2) \end{pmatrix}.$$

Theorem 7.2.3. *Under the assumptions of Remark 7.2.1, suppose that $((\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2), \Upsilon)$ is a gain subdivision-vertex corona graph of (Γ_1, φ_1) and (Γ_2, φ_2). The π-characteristic polynomial of $((\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2), \Upsilon)$ is*

$$\phi^{\pi}_{(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)} = x^{d(m_1 - n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \cdot \det\left(x^2 I_{n_1 d} - x(I_{n_1} \otimes C_\pi(x)) - r I_{n_1 d} - \sigma \pi(A_1)\right),$$

where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

In the special case where $C_\pi(x) = c_\pi(x) I_d$ (for example, when $d = 1$), this reduces to

$$\phi^{\pi}_{(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)} = x^{d(m_1 - n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \prod_{i=1}^{n_1 d} \left(x^2 - c_\pi(x) x - r - \sigma \lambda_i\right).$$

Proof. Since (Γ_1, φ_1) and (Γ_2, φ_2) are G-gain graphs, there exist $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1) = \varphi_1$ and $\Phi(H_2) = \varphi_2$, and, by definition of the gain subdivision-vertex corona, $\Upsilon = \Phi_{\odot_S}(H_1, H_2)$.

Apply the representation π to the characteristic matrix of the adjacency matrix. From the adjacency matrix of the gain subdivision-vertex corona $(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)$, we have

$$xI - \pi(\Phi_{\odot_S}(H_1, H_2)) = \begin{pmatrix} x I_{n_1 d} & -\pi(H_1) & -I_{n_1} \otimes (\mathbf{1}_{n_2}^T \otimes I_d) \\ -\pi(H_1)^* & x I_{m_1 d} & O \\ -I_{n_1} \otimes (\mathbf{1}_{n_2} \otimes I_d) & O & I_{n_1} \otimes (x I_{n_2 d} - \pi(A_2)) \end{pmatrix}.$$

Taking the Schur complement with respect to the $(3,3)$ block gives

$$\det(xI - \pi(\Phi_{\odot_S}(H_1, H_2))) = \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \cdot \det \begin{pmatrix} x I_{n_1 d} - I_{n_1} \otimes C_\pi(x) & -\pi(H_1) \\ -\pi(H_1)^* & x I_{m_1 d} \end{pmatrix},$$

where the block $(1,1)$ correction is $I_{n_1} \otimes C_\pi(x)$.

Now take the Schur complement with respect to the $(2,2)$ block. For $x \neq 0$,

$$\det \begin{pmatrix} x I_{n_1 d} - I_{n_1} \otimes C_\pi(x) & -\pi(H_1) \\ -\pi(H_1)^* & x I_{m_1 d} \end{pmatrix} = \det(x I_{m_1 d}) \cdot \det\left(x I_{n_1 d} - I_{n_1} \otimes C_\pi(x) - \frac{1}{x} \pi(H_1 H_1^*)\right).$$

Since $\deg(\Gamma_1) = rI_{n_1}$ by r-regularity, $H_1H_1^* = rI_{n_1} + sA_1$, and s is a central involution, applying π gives

$$\pi(H_1H_1^*) = rI_{n_1d} + \pi(s)\pi(A_1) = rI_{n_1d} + \sigma\pi(A_1),$$

where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Thus,

$$\det\begin{pmatrix} xI_{n_1d} - I_{n_1} \otimes C_\pi(x) & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1d} \end{pmatrix} = x^{dm_1} \det\left(xI_{n_1d} - I_{n_1} \otimes C_\pi(x) - \frac{1}{x}(rI_{n_1d} + \sigma\pi(A_1))\right).$$

Factoring x^{-1} out of the $n_1d \times n_1d$ matrix multiplies the determinant by x^{-n_1d}, giving

$$= x^{d(m_1-n_1)} \det\left(x^2I_{n_1d} - x(I_{n_1} \otimes C_\pi(x)) - rI_{n_1d} - \sigma\pi(A_1)\right).$$

Therefore,

$$\phi^\pi_{(\Gamma_1,\varphi_1)\odot_S(\Gamma_2,\varphi_2)} = x^{d(m_1-n_1)} \prod_{i=1}^{n_2d} (x-\mu_i)^{n_1} \cdot \det\left(x^2I_{n_1d} - x(I_{n_1} \otimes C_\pi(x)) - rI_{n_1d} - \sigma\pi(A_1)\right).$$

The intermediate determinant identities were derived for $x \neq 0$ and $x \notin \operatorname{spec}(\pi(A_2))$; since both sides are rational functions in x, the identity extends to all x by clearing denominators (cf. Section 1.8).

The scalar case $C_\pi(x) = c_\pi(x)I_d$ follows immediately, since then $I_{n_1} \otimes C_\pi(x) = c_\pi(x)I_{n_1d}$ commutes with $\pi(A_1)$, and the determinant factorises over the eigenvalues λ_i. □

If (Γ_2, φ_2) is g-G-regular and $\pi(g) = \gamma_\pi I_d$, then $C_\pi(x)$ is scalar and the special case of Theorem 7.2.3 applies, so we can say more.

Corollary 7.2.4. *Suppose that, under the assumptions of Theorem 7.2.3, (Γ_2, φ_2) is g-G-regular, with $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$ (automatic when g lies in the centre of $\mathbb{C}G$, by Schur's lemma, or when $d = 1$), and $\mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1} = \gamma_\pi$ for some fixed k $(1 \le k \le n_2d - d + 1)$. The π-spectrum of $((\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2), \Upsilon)$ consists of*

(i) 0 *with multiplicity* $d(m_1 - n_1)$,

(ii) μ_i *with multiplicity* n_1 *for each* $i \in \{1, 2, \ldots, k-1, k+d, \ldots, n_2d\}$,

(iii) the roots of $x^3 - \gamma_\pi x^2 - (n_2 + r + \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i)$ *for each* $1 \le i \le n_1d$, *with* $\sigma = \pm 1$ *when* $\pi(s) = \pm I$.

Proof. Item (i) follows directly from the prefactor $x^{d(m_1-n_1)}$ in Theorem 7.2.3: the eigenvalue 0 appears with multiplicity $d(m_1 - n_1)$.

Since (Γ_2, φ_2) is g-G-regular, we have $A_2\mathbf{1}_{n_2} = g\mathbf{1}_{n_2}$. Applying π gives

$$\pi(A_2)(\mathbf{1}_{n_2} \otimes I_d) = (\mathbf{1}_{n_2} \otimes I_d)\pi(g) = \gamma_\pi(\mathbf{1}_{n_2} \otimes I_d).$$

Thus,

$$C_\pi(x) = (\mathbf{1}_{n_2}^T \otimes I_d)(xI_{n_2d} - \pi(A_2))^{-1}(\mathbf{1}_{n_2} \otimes I_d) = n_2(xI_d - \gamma_\pi I_d)^{-1} = \frac{n_2}{x - \gamma_\pi} I_d.$$

So the scalar case of Theorem 7.2.3 applies with $c_\pi(x) = \frac{n_2}{x-\gamma_\pi}$.

Furthermore, $x = \gamma_\pi = \mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1}$ is the only pole of $c_\pi(x)$, and Theorem 7.2.3 implies that μ_i is a π-eigenvalue of $(\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)$ for each i specified in item (ii).

The remaining π-eigenvalues satisfy

$$(x - \gamma_\pi)\left(x^2 - \frac{n_2}{x - \gamma_\pi}x - r - \sigma\lambda_i\right) = 0.$$

Expanding and simplifying, this becomes

$$x^3 - \gamma_\pi x^2 - (n_2 + r + \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i) = 0,$$

and, as before, we have $\sigma = \pm 1$ when $\pi(s) = \pm I$. This gives us item (iii). □

Q-vertex corona

Definition 7.2.5. Let $H_1 \in \mathcal{H}_{\Gamma_1}$ and let (Γ_2, φ_2) be a G-gain graph. The gain Q-vertex corona $(\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2)$ is the gain graph obtained from $(Q(\Gamma_1), \Phi_Q(H_1))$ and n_1 copies of (Γ_2, φ_2) by joining the i-th vertex of V_{Γ_1} by an edge with gain 1_G to every vertex in the i-th copy $(\Gamma_2^i, \varphi_2^i)$ of (Γ_2, φ_2).

For the adjacency matrix of $(\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2)$ we write $\Phi_{\odot_Q}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\odot_Q}(H_1, H_2) = \begin{pmatrix} O_{n_1} & H_1 & I_{n_1} \otimes \mathbf{1}_{n_2}^T \\ H_1^* & s(H_1^* H_1 - 2I_{m_1}) & O_{m_1 \times n_1 n_2} \\ I_{n_1} \otimes \mathbf{1}_{n_2} & O_{n_1 n_2 \times m_1} & I_{n_1} \otimes \Phi(H_2) \end{pmatrix}.$$

Theorem 7.2.6. *Under the assumptions of Remark 7.2.1, suppose that $((\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2), \Upsilon)$ is a gain Q-vertex corona graph of (Γ_1, φ_1) and (Γ_2, φ_2). The π-characteristic polynomial of $((\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2), \Upsilon)$ is*

$$\phi^{\pi}_{(\Gamma_1,\varphi_1)\odot_Q(\Gamma_2,\varphi_2)} = (x + 2\sigma)^{d(m_1 - n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \cdot \det\Big(\big((x + (2-r)\sigma) I_{n_1 d} - \pi(A_1)\big)\big(x I_{n_1 d} - I_{n_1} \otimes C_\pi(x)\big) - r I_{n_1 d} - \sigma \pi(A_1) \Big),$$

where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

In the special case where $C_\pi(x) = c_\pi(x) I_d$ (for example, when $d = 1$), this reduces to

$$\phi^{\pi}_{(\Gamma_1,\varphi_1)\odot_Q(\Gamma_2,\varphi_2)} = (x + 2\sigma)^{d(m_1 - n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \cdot \prod_{i=1}^{n_1 d} \big(x^2 + ((2-r)\sigma - \lambda_i - c_\pi(x))x - r - \sigma\lambda_i + \sigma(r + \sigma\lambda_i - 2) c_\pi(x) \big).$$

Proof. Since (Γ_1, φ_1) and (Γ_2, φ_2) are G-gain graphs, there exist $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1) = \varphi_1$ and $\Phi(H_2) = \varphi_2$, and, by definition of the gain Q-vertex corona, $\Upsilon = \Phi_{\odot_Q}(H_1, H_2)$.

Apply the representation π to the characteristic matrix of the adjacency matrix. From the adjacency matrix of the gain Q-vertex corona $(\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2)$, we have

$$xI - \pi(\Phi_{\odot_Q}(H_1, H_2)) = \begin{pmatrix} x I_{n_1 d} & -\pi(H_1) & -I_{n_1} \otimes (\mathbf{1}_{n_2}^T \otimes I_d) \\ -\pi(H_1)^* & x I_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) & O \\ -I_{n_1} \otimes (\mathbf{1}_{n_2} \otimes I_d) & O & I_{n_1} \otimes (x I_{n_2 d} - \pi(A_2)) \end{pmatrix}.$$

Note that $s(H_1^* H_1 - 2I_{m_1})$ becomes $\sigma(\pi(H_1^* H_1) - 2I_{m_1 d})$, where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Taking the Schur complement with respect to the $(3,3)$ block gives

$$\det\big(xI - \pi(\Phi_{\odot_Q}(H_1, H_2))\big) = \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \cdot \det \begin{pmatrix} x I_{n_1 d} - I_{n_1} \otimes C_\pi(x) & -\pi(H_1) \\ -\pi(H_1)^* & x I_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) \end{pmatrix},$$

where the block $(1,1)$ correction is $I_{n_1} \otimes C_\pi(x)$.

Now take the Schur complement with respect to the $(2,2)$ block. Let

$$D = x I_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) = (x + 2\sigma) I_{m_1 d} - \sigma \pi(H_1^* H_1).$$

For x such that D is invertible,

$$\det \begin{pmatrix} x I_{n_1 d} - I_{n_1} \otimes C_\pi(x) & -\pi(H_1) \\ -\pi(H_1)^* & D \end{pmatrix} = \det(D) \cdot \det\big(x I_{n_1 d} - I_{n_1} \otimes C_\pi(x) - \pi(H_1) D^{-1} \pi(H_1)^* \big).$$

Since $\pi(H_1)f(\pi(H_1^*H_1)) = f(\pi(H_1)\pi(H_1)^*)\pi(H_1)$ for any rational function f defined on the relevant spectra, we have

$$\pi(H_1)D^{-1}\pi(H_1)^* = \pi(H_1H_1^*)\big((x+2\sigma)I_{n_1d} - \sigma\pi(H_1H_1^*)\big)^{-1}.$$

Moreover, since the non-zero spectra of $\pi(H_1^*H_1)$ and $\pi(H_1H_1^*)$ coincide,

$$\det(D) = (x+2\sigma)^{d(m_1-n_1)}\det\big((x+(2-r)\sigma)I_{n_1d} - \pi(A_1)\big).$$

Using $\pi(H_1H_1^*) = rI_{n_1d} + \sigma\pi(A_1)$, the determinant becomes

$$\begin{aligned}&\det\begin{pmatrix} xI_{n_1d} - I_{n_1}\otimes C_\pi(x) & -\pi(H_1) \\ -\pi(H_1)^* & D\end{pmatrix}\\ &\quad= (x+2\sigma)^{d(m_1-n_1)}\cdot\det\Big(\big((x+(2-r)\sigma)I_{n_1d} - \pi(A_1)\big)\big(xI_{n_1d} - I_{n_1}\otimes C_\pi(x)\big) - rI_{n_1d} - \sigma\pi(A_1)\Big).\end{aligned}$$

Therefore,

$$\begin{aligned}\phi^\pi_{(\Gamma_1,\varphi_1)\odot_Q(\Gamma_2,\varphi_2)} &= (x+2\sigma)^{d(m_1-n_1)}\prod_{i=1}^{n_2d}(x-\mu_i)^{n_1}\\ &\quad\cdot\det\Big(\big((x+(2-r)\sigma)I_{n_1d} - \pi(A_1)\big)\big(xI_{n_1d} - I_{n_1}\otimes C_\pi(x)\big) - rI_{n_1d} - \sigma\pi(A_1)\Big).\end{aligned}$$

The intermediate determinant identities were derived for $x \notin \operatorname{spec}(\pi(A_2))$ and for x such that D is invertible; since both sides are rational functions in x, the identity extends to all x by clearing denominators (cf. Section 1.8).

The scalar case $C_\pi(x) = c_\pi(x)I_d$ follows immediately: since $I_{n_1}\otimes C_\pi(x)$ becomes scalar and commutes with $\pi(A_1)$, the determinant factorises over the eigenvalues λ_i. For each i, the factor is

$$\big(x+(2-r)\sigma-\lambda_i\big)\big(x-c_\pi(x)\big) - r - \sigma\lambda_i = x^2 + ((2-r)\sigma - \lambda_i - c_\pi(x))x - r - \sigma\lambda_i + \sigma(r+\sigma\lambda_i-2)c_\pi(x),$$

which matches the stated scalar formula. □

If (Γ_2,φ_2) is g-G-regular and $\pi(g) = \gamma_\pi I_d$, then $C_\pi(x)$ is scalar and the special case of Theorem 7.2.6 applies, so we can say more.

Corollary 7.2.7. *Suppose that, under the assumptions of Theorem 7.2.6, (Γ_2,φ_2) is g-G-regular, with $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$ (automatic when g lies in the centre of $\mathbb{C}G$, by Schur's lemma, or when $d = 1$), and $\mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1} = \gamma_\pi$ for some fixed k $(1 \le k \le n_2d - d + 1)$. The π-spectrum of $((\Gamma_1,\varphi_1)\odot_Q(\Gamma_2,\varphi_2),\Upsilon)$ consists of*

(i) -2σ with multiplicity $d(m_1-n_1)$,

(ii) μ_i with multiplicity n_1 for each $i \in \{1,2,\dots,k-1,k+d,\dots,n_2d\}$,

(iii) the roots of

$$x^3 - (\gamma_\pi+\lambda_i+(r-2)\sigma)x^2 + (\gamma_\pi\lambda_i + (\gamma_\pi r - 2\gamma_\pi - \lambda_i)\sigma - n_2 - r)x + \lambda_i n_2 + \gamma_\pi r + (\gamma_\pi\lambda_i + n_2 r - 2n_2)\sigma$$

for each $1 \le i \le n_1d$, with $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. Item (i) follows directly from the prefactor $(x+2\sigma)^{d(m_1-n_1)}$ in Theorem 7.2.6: the eigenvalue -2σ appears with multiplicity $d(m_1-n_1)$.

As in Corollary 7.2.4, since (Γ_2,φ_2) is g-G-regular, we have

$$C_\pi(x) = \frac{n_2}{x-\gamma_\pi}I_d,$$

so the scalar case of Theorem 7.2.6 applies with $c_\pi(x) = \frac{n_2}{x-\gamma_\pi}$.

Furthermore, $x = \gamma_\pi = \mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1}$ is the only pole of $c_\pi(x)$, and Theorem 7.2.6 implies that μ_i is a π-eigenvalue of $(\Gamma_1,\varphi_1)\odot_Q(\Gamma_2,\varphi_2)$ for each i specified in item (ii).

The remaining π-eigenvalues are the roots of

$$(x-\gamma_\pi)\left(x^2+\left((2-r)\sigma-\lambda_i-\frac{n_2}{x-\gamma_\pi}\right)x-r-\sigma\lambda_i+\sigma(r+\sigma\lambda_i-2)\frac{n_2}{x-\gamma_\pi}\right)=0$$

for each $1\le i\le n_1 d$. Expanding and simplifying, this becomes

$$x^3-(\gamma_\pi+\lambda_i+(r-2)\sigma)x^2+(\gamma_\pi\lambda_i+(\gamma_\pi r-2\gamma_\pi-\lambda_i)\sigma-n_2-r)x+\lambda_i n_2+\gamma_\pi r+(\gamma_\pi\lambda_i+n_2 r-2n_2)\sigma=0,$$

and, as before, we have $\sigma=\pm1$ when $\pi(s)=\pm I$. This gives us item (iii). □

R-vertex corona

Definition 7.2.8. Let $H_1\in\mathcal{H}_{\Gamma_1}$ and let (Γ_2,φ_2) be a G-gain graph. The gain R-vertex corona $(\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2)$ is the gain graph obtained from $(R(\Gamma_1),\Phi_R(H_1))$ and n_1 copies of (Γ_2,φ_2) by joining the i-th vertex of V_{Γ_1} by an edge with gain 1_G to every vertex in the i-th copy (Γ_2^i,φ_2^i) of (Γ_2,φ_2).

For the adjacency matrix of $(\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2)$ we write $\Phi_{\odot_R}(H_1,H_2)$ and for each $H_1\in\mathcal{H}_{\Gamma_1}$ and $H_2\in\mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\odot_R}(H_1,H_2)=\begin{pmatrix}s(H_1H_1^*-\deg(\Gamma_1)) & H_1 & I_{n_1}\otimes\mathbf{1}_{n_2}^T\\ H_1^* & O_{m_1} & O_{m_1\times n_1n_2}\\ I_{n_1}\otimes\mathbf{1}_{n_2} & O_{n_1n_2\times m_1} & I_{n_1}\otimes\Phi(H_2)\end{pmatrix}.$$

Theorem 7.2.9. *Under the assumptions of Remark 7.2.1, suppose that $((\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2),\Upsilon)$ is a gain R-vertex corona graph of (Γ_1,φ_1) and (Γ_2,φ_2). The π-characteristic polynomial of $((\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2),\Upsilon)$ is*

$$\phi^\pi_{(\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2)}=x^{d(m_1-n_1)}\prod_{i=1}^{n_2d}(x-\mu_i)^{n_1}\cdot\det\Big(x^2I_{n_1d}-x(I_{n_1}\otimes C_\pi(x))-rI_{n_1d}-(x+\sigma)\pi(A_1)\Big),$$

where $\sigma=\pm1$ when $\pi(s)=\pm I$.

In the special case where $C_\pi(x)=c_\pi(x)I_d$ (for example, when $d=1$), this reduces to

$$\phi^\pi_{(\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2)}=x^{d(m_1-n_1)}\prod_{i=1}^{n_2d}(x-\mu_i)^{n_1}\prod_{i=1}^{n_1d}\left(x^2-(c_\pi(x)+\lambda_i)x-r-\sigma\lambda_i\right).$$

Proof. Since (Γ_1,φ_1) and (Γ_2,φ_2) are G-gain graphs, there exist $H_1\in\mathcal{H}_{\Gamma_1}$ and $H_2\in\mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1)=\varphi_1$ and $\Phi(H_2)=\varphi_2$, and, by definition of the gain R-vertex corona, $\Upsilon=\Phi_{\odot_R}(H_1,H_2)$.

We first note that, at the group-ring level,

$$s(H_1H_1^*-rI_{n_1})=s(rI_{n_1}+sA_1-rI_{n_1})=s^2A_1=A_1,$$

since Γ_1 is r-regular, $\deg(\Gamma_1)=rI_{n_1}$, and s is a central involution with $H_1H_1^*=rI_{n_1}+sA_1$. Thus the $(1,1)$ block of $\Phi_{\odot_R}(H_1,H_2)$ is precisely A_1.

Apply the representation π to the characteristic matrix of the adjacency matrix. From the adjacency matrix of the gain R-vertex corona $(\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2)$, we have

$$xI-\pi(\Phi_{\odot_R}(H_1,H_2))=\begin{pmatrix}xI_{n_1d}-\pi(A_1) & -\pi(H_1) & -I_{n_1}\otimes(\mathbf{1}_{n_2}^T\otimes I_d)\\ -\pi(H_1)^* & xI_{m_1d} & O\\ -I_{n_1}\otimes(\mathbf{1}_{n_2}\otimes I_d) & O & I_{n_1}\otimes(xI_{n_2d}-\pi(A_2))\end{pmatrix}.$$

Taking the Schur complement with respect to the $(3,3)$ block gives

$$\det(xI-\pi(\Phi_{\odot_R}(H_1,H_2)))=\prod_{i=1}^{n_2d}(x-\mu_i)^{n_1}\cdot\det\begin{pmatrix}xI_{n_1d}-I_{n_1}\otimes C_\pi(x)-\pi(A_1) & -\pi(H_1)\\ -\pi(H_1)^* & xI_{m_1d}\end{pmatrix},$$

where the block $(1,1)$ correction is $I_{n_1}\otimes C_\pi(x)$.

Now take the Schur complement with respect to the $(2,2)$ block. For $x \neq 0$,

$$\det \begin{pmatrix} xI_{n_1 d} - I_{n_1} \otimes C_\pi(x) - \pi(A_1) & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1 d} \end{pmatrix} = \det(xI_{m_1 d}) \det \left(xI_{n_1 d} - I_{n_1} \otimes C_\pi(x) - \pi(A_1) - \frac{1}{x}\pi(H_1 H_1^*) \right).$$

Since $\pi(H_1 H_1^*) = rI_{n_1 d} + \sigma\pi(A_1)$, this becomes

$$= x^{dm_1} \det \left(xI_{n_1 d} - I_{n_1} \otimes C_\pi(x) - \pi(A_1) - \frac{1}{x}(rI_{n_1 d} + \sigma\pi(A_1)) \right).$$

Factoring x^{-1} out of the $n_1 d \times n_1 d$ matrix multiplies the determinant by $x^{-n_1 d}$, giving

$$= x^{d(m_1 - n_1)} \det \left(x^2 I_{n_1 d} - x(I_{n_1} \otimes C_\pi(x)) - x\pi(A_1) - rI_{n_1 d} - \sigma\pi(A_1) \right).$$

Grouping the $\pi(A_1)$ terms:

$$= x^{d(m_1 - n_1)} \det \left(x^2 I_{n_1 d} - x(I_{n_1} \otimes C_\pi(x)) - rI_{n_1 d} - (x + \sigma)\pi(A_1) \right).$$

Therefore,

$$\phi^\pi_{(\Gamma_1,\varphi_1)\odot_R(\Gamma_2,\varphi_2)} = x^{d(m_1 - n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{n_1} \cdot \det \Big(x^2 I_{n_1 d} - x(I_{n_1} \otimes C_\pi(x)) - rI_{n_1 d} - (x + \sigma)\pi(A_1) \Big).$$

The intermediate determinant identities were derived for $x \neq 0$ and $x \notin \operatorname{spec}(\pi(A_2))$; since both sides are rational functions in x, the identity extends to all x by clearing denominators (cf. Section 1.8).

The scalar case $C_\pi(x) = c_\pi(x) I_d$ follows immediately: since $I_{n_1} \otimes C_\pi(x)$ becomes scalar and commutes with $\pi(A_1)$, the determinant factorises over the eigenvalues λ_i. For each i, the factor is

$$x^2 - c_\pi(x)x - r - (x + \sigma)\lambda_i = x^2 - (c_\pi(x) + \lambda_i)x - r - \sigma\lambda_i,$$

which matches the stated scalar formula. □

If (Γ_2, φ_2) is g-G-regular and $\pi(g) = \gamma_\pi I_d$, then $C_\pi(x)$ is scalar and the special case of Theorem 7.2.9 applies, so we can say more.

Corollary 7.2.10. *Suppose that, under the assumptions of Theorem 7.2.9, (Γ_2, φ_2) is g-G-regular, with $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$ (automatic when g lies in the centre of $\mathbb{C}G$, by Schur's lemma, or when $d = 1$), and $\mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1} = \gamma_\pi$ for some fixed k $(1 \leq k \leq n_2 d - d + 1)$. The π-spectrum of $((\Gamma_1, \varphi_1) \odot_R (\Gamma_2, \varphi_2), \Upsilon)$ consists of*

(i) 0 *with multiplicity* $d(m_1 - n_1)$,

(ii) μ_i *with multiplicity* n_1 *for each* $i \in \{1, 2, \ldots, k-1, k+d, \ldots, n_2 d\}$,

(iii) the roots of

$$x^3 - (\gamma_\pi + \lambda_i)x^2 + (\gamma_\pi \lambda_i - n_2 - r - \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i)$$

for each $1 \leq i \leq n_1 d$, *with* $\sigma = \pm 1$ *when* $\pi(s) = \pm I$.

Proof. Item (i) follows directly from the prefactor $x^{d(m_1 - n_1)}$ in Theorem 7.2.9: the eigenvalue 0 appears with multiplicity $d(m_1 - n_1)$.

As in Corollary 7.2.4, since (Γ_2, φ_2) is g-G-regular, we have

$$C_\pi(x) = \frac{n_2}{x - \gamma_\pi} I_d,$$

so the scalar case of Theorem 7.2.9 applies with $c_\pi(x) = \frac{n_2}{x - \gamma_\pi}$.

Furthermore, $x = \gamma_\pi = \mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1}$ is the only pole of $c_\pi(x)$, and Theorem 7.2.9 implies that μ_i is a π-eigenvalue of $(\Gamma_1, \varphi_1) \odot_R (\Gamma_2, \varphi_2)$ for each i specified in item (ii).

The remaining π-eigenvalues satisfy

$$(x - \gamma_\pi)\left(x^2 - \left(\frac{n_2}{x - \gamma_\pi} + \lambda_i\right)x - r - \sigma\lambda_i\right) = 0$$

for each $1 \leq i \leq n_1 d$. Expanding and simplifying, this becomes

$$x^3 - (\gamma_\pi + \lambda_i)x^2 + (\gamma_\pi\lambda_i - n_2 - r - \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i) = 0,$$

and, as before, we have $\sigma = \pm 1$ when $\pi(s) = \pm I$. This gives us item (iii). □

Vertex corona example

Example 7.2.11. We shall consider the products shown in Figure 7.2 along with the irreducible representation π of $\mathbb{H}$ from Example 7.1.2.

Recall from Equation (7.1.1) that the π-spectrum of (Γ_1, φ_1) is

$$\pi\text{-spec}(\Gamma_1, \varphi_1) = \{2^{(2)}, -1^{(4)}\}.$$

Let (Γ_2, φ_2) be the $\mathbb{H}$-gain graph consisting of a single edge with gain $\varphi_2(v_1, v_2) = -1$. Since $\pi(-1) = -I_2$, the represented adjacency matrix of (Γ_2, φ_2) is

$$\pi(A_{(\Gamma_2,\varphi_2)}) = \left(\begin{array}{cc|cc} 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \\ \hline -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right) \in M_4(\mathbb{C}),$$

and the π-spectrum of (Γ_2, φ_2) is

$$\pi\text{-spec}(\Gamma_2, \varphi_2) = \{1^{(2)}, -1^{(2)}\}. \tag{7.2.1}$$

After the appropriate substitutions, the π-spectra of the gain subdivision-vertex, Q-vertex, and R-vertex corona products can be directly computed as

$$\begin{aligned}
\pi\text{-spec}((\Gamma_1, \varphi_1) \odot_S (\Gamma_2, \varphi_2)) &= \{2.323^{(2)}, 1.481^{(4)}, 1^{(6)}, -0.311^{(4)}, -0.642^{(2)}, -2.170^{(4)}, -2.681^{(2)}\}, \\
\pi\text{-spec}((\Gamma_1, \varphi_1) \odot_Q (\Gamma_2, \varphi_2)) &= \{3.372^{(2)}, 1.303^{(4)}, 1^{(6)}, 0^{(2)}, -1^{(4)}, -2.303^{(4)}, -2.372^{(2)}\}, \\
\pi\text{-spec}((\Gamma_1, \varphi_1) \odot_R (\Gamma_2, \varphi_2)) &= \{3.562^{(2)}, 1^{(10)}, -0.382^{(4)}, -0.562^{(2)}, -2^{(2)}, -2.618^{(4)}\}.
\end{aligned}$$

These match the values obtained from implementing Equations (7.1.1) and (7.2.1) with Corollaries 7.2.4, 7.2.7, and 7.2.10.

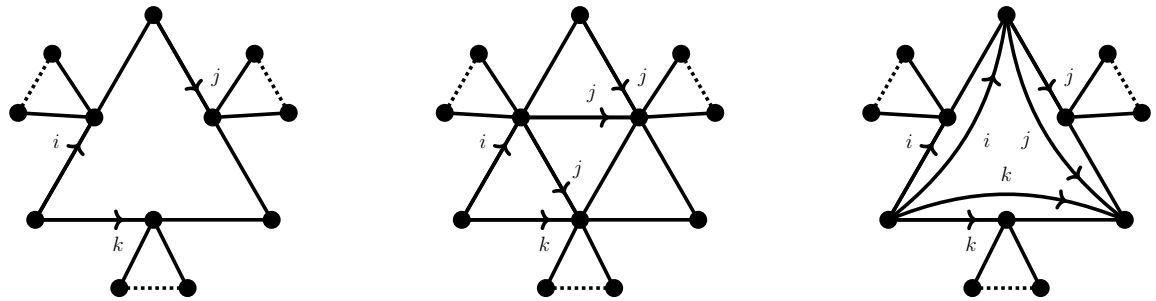
7.2.2 Edge corona products

Subdivision-edge corona

Definition 7.2.12. Let $H_1 \in \mathcal{H}_{\Gamma_1}$ and let (Γ_2, φ_2) be a G-gain graph. The gain subdivision-edge corona $(\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2)$ is the gain graph obtained from $(S(\Gamma_1), \Phi_S(H_1))$ and m_1 copies of (Γ_2, φ_2) by joining the i-th vertex of $V_{S(\Gamma_1)} \setminus V_{\Gamma_1}$ by an edge with gain 1_G to every vertex in the i-th copy $(\Gamma_2^i, \varphi_2^i)$ of (Γ_2, φ_2).

For the adjacency matrix of $(\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2)$ we write $\Phi_{\ominus_S}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\ominus_S}(H_1, H_2) = \begin{pmatrix} O_{n_1} & H_1 & O_{n_1 \times m_1 n_2} \\ H_1^* & O_{m_1} & I_{m_1} \otimes \mathbf{1}_{n_2}^T \\ O_{m_1 n_2 \times n_1} & I_{m_1} \otimes \mathbf{1}_{n_2} & I_{m_1} \otimes \Phi(H_2) \end{pmatrix}.$$



$(\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2)$ $(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)$ $(\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2)$

Figure 7.3: The gain subdivision-edge, gain Q-edge, and gain R-edge corona graphs of (Γ_1, φ_1) and (Γ_2, φ_2).

Theorem 7.2.13. *Under the assumptions of Remark 7.2.1, suppose that* $((\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2), \Upsilon)$ *is a gain subdivision-edge corona graph of* (Γ_1, φ_1) *and* (Γ_2, φ_2)*. The* π*-characteristic polynomial of* $((\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2), \Upsilon)$ *is*

$$\phi^{\pi}_{(\Gamma_1,\varphi_1)\ominus_S(\Gamma_2,\varphi_2)} = x^{d(n_1-m_1)} \prod_{i=1}^{n_2 d} (x-\mu_i)^{m_1} \cdot \det\Big(x^2 I_{m_1 d} - x(I_{m_1} \otimes C_\pi(x)) - \pi(H_1^* H_1)\Big),$$

where $H_1 \in \mathcal{H}_{\Gamma_1}$ *and* $H_2 \in \mathcal{H}_{\Gamma_2}$ *are the* G*-phases with* $\Upsilon = \Phi_{\ominus_S}(H_1, H_2)$*, and* $\sigma = \pm 1$ *when* $\pi(s) = \pm I$*.*

In the special case where $C_\pi(x) = c_\pi(x) I_d$ *(for example, when* $d = 1$*), this reduces to*

$$\phi^{\pi}_{(\Gamma_1,\varphi_1)\ominus_S(\Gamma_2,\varphi_2)} = (x - c_\pi(x))^{d(m_1-n_1)} \prod_{i=1}^{n_2 d} (x-\mu_i)^{m_1} \prod_{i=1}^{n_1 d} \left(x^2 - c_\pi(x) x - r - \sigma\lambda_i\right).$$

Proof. Since (Γ_1, φ_1) and (Γ_2, φ_2) are G-gain graphs, there exist $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1) = \varphi_1$ and $\Phi(H_2) = \varphi_2$, and, by definition of the gain subdivision-edge corona, $\Upsilon = \Phi_{\ominus_S}(H_1, H_2)$.

Apply the representation π to the characteristic matrix of the adjacency matrix. From the adjacency matrix of the gain subdivision-edge corona $(\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2)$, we have

$$xI - \pi(\Phi_{\ominus_S}(H_1, H_2)) = \begin{pmatrix} xI_{n_1 d} & -\pi(H_1) & O \\ -\pi(H_1)^* & xI_{m_1 d} & -I_{m_1} \otimes (\mathbf{1}_{n_2}^T \otimes I_d) \\ O & -I_{m_1} \otimes (\mathbf{1}_{n_2} \otimes I_d) & I_{m_1} \otimes (xI_{n_2 d} - \pi(A_2)) \end{pmatrix}.$$

Taking the Schur complement with respect to the $(3,3)$ block gives

$$\det(xI - \pi(\Phi_{\ominus_S}(H_1, H_2))) = \prod_{i=1}^{n_2 d} (x-\mu_i)^{m_1} \cdot \det\begin{pmatrix} xI_{n_1 d} & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1 d} - I_{m_1} \otimes C_\pi(x) \end{pmatrix},$$

where the block $(2,2)$ correction is $I_{m_1} \otimes C_\pi(x)$.

Now take the Schur complement with respect to the $(1,1)$ block. For $x \neq 0$,

$$\det\begin{pmatrix} xI_{n_1 d} & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1 d} - I_{m_1} \otimes C_\pi(x) \end{pmatrix} = x^{n_1 d} \det\left(xI_{m_1 d} - I_{m_1} \otimes C_\pi(x) - \frac{1}{x}\pi(H_1^* H_1)\right).$$

Factoring x^{-1} out of the $m_1 d \times m_1 d$ matrix multiplies the determinant by $x^{-m_1 d}$, giving

$$= x^{d(n_1-m_1)} \det\Big(x^2 I_{m_1 d} - x(I_{m_1} \otimes C_\pi(x)) - \pi(H_1^* H_1)\Big).$$

Therefore,

$$\phi^{\pi}_{(\Gamma_1,\varphi_1)\ominus_S(\Gamma_2,\varphi_2)} = x^{d(n_1-m_1)} \prod_{i=1}^{n_2 d} (x-\mu_i)^{m_1} \cdot \det\Big(x^2 I_{m_1 d} - x(I_{m_1} \otimes C_\pi(x)) - \pi(H_1^* H_1)\Big).$$

The intermediate determinant identities were derived for $x \neq 0$ and $x \notin \operatorname{spec}(\pi(A_2))$; since both sides are rational functions in x, the identity extends to all x by clearing denominators (cf. Section 1.8).

The scalar case $C_\pi(x) = c_\pi(x) I_d$ follows immediately: since $I_{m_1} \otimes C_\pi(x) = c_\pi(x) I_{m_1 d}$, the determinant factorises over the eigenvalues of $\pi(H_1^* H_1)$, which are the same as those of $r I_{n_1 d} + \sigma \pi(A_1)$ together with $d(m_1 - n_1)$ zero eigenvalues. Noting that

$$x^{d(n_1-m_1)} \left(x^2 - c_\pi(x) x\right)^{d(m_1-n_1)} = \left(x - c_\pi(x)\right)^{d(m_1-n_1)},$$

we obtain

$$\left(x - c_\pi(x)\right)^{d(m_1-n_1)} \prod_{i=1}^{n_1 d} \left(x^2 - c_\pi(x) x - r - \sigma \lambda_i\right),$$

so that the final determinant factor becomes the stated scalar formula when combined with $\prod_{i=1}^{n_2 d}(x - \mu_i)^{m_1}$. □

If (Γ_2, φ_2) is g-G-regular and $\pi(g) = \gamma_\pi I_d$, then $C_\pi(x)$ is scalar and the special case of Theorem 7.2.13 applies, so we can say more.

Corollary 7.2.14. *Suppose that, under the assumptions of Theorem 7.2.13, (Γ_2, φ_2) is g-G-regular, with $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$ (automatic when g lies in the centre of $\mathbb{C}G$, by Schur's lemma, or when $d = 1$), and $\mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1} = \gamma_\pi$ for some fixed k $(1 \leq k \leq n_2 d - d + 1)$. The π-spectrum of $((\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2), \Upsilon)$ consists of*

(i) the roots of $x^2 - \gamma_\pi x - n_2$, each with multiplicity $d(m_1 - n_1)$,

(ii) μ_i with multiplicity m_1 for each $i \in \{1, 2, \ldots, k-1, k+d, \ldots, n_2 d\}$,

(iii) the roots of $x^3 - \gamma_\pi x^2 - (n_2 + r + \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i)$ for each $1 \leq i \leq n_1 d$, with $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. As in Corollary 7.2.4, since (Γ_2, φ_2) is g-G-regular, we have

$$C_\pi(x) = \frac{n_2}{x - \gamma_\pi} I_d,$$

so the scalar case of Theorem 7.2.13 applies with $c_\pi(x) = \frac{n_2}{x - \gamma_\pi}$.

The scalar case gives the characteristic polynomial

$$\left(x - c_\pi(x)\right)^{d(m_1-n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{m_1} \prod_{i=1}^{n_1 d} \left(x^2 - c_\pi(x) x - r - \sigma \lambda_i\right).$$

Since $x = \gamma_\pi = \mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1}$ is the only pole of $c_\pi(x)$, Theorem 7.2.13 implies that μ_i is a π-eigenvalue of $(\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2)$ for each i specified in item (ii).

The remaining π-eigenvalues are the roots of

$$(x - \gamma_\pi)\left(x - \frac{n_2}{x - \gamma_\pi}\right) = 0$$

with multiplicity $d(m_1 - n_1)$, and

$$(x - \gamma_\pi)\left(x^2 - \frac{n_2}{x - \gamma_\pi} x - r - \sigma\lambda_i\right) = 0$$

for each $1 \leq i \leq n_1 d$. Expanding and simplifying, these become

$$x^2 - \gamma_\pi x - n_2 = 0$$

and

$$x^3 - \gamma_\pi x^2 - (n_2 + r + \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i) = 0,$$

respectively. This gives us items (i) and (iii). □

Q-edge corona

Definition 7.2.15. Let $H_1 \in \mathcal{H}_{\Gamma_1}$ and let (Γ_2, φ_2) be a G-gain graph. The gain Q-edge corona $(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)$ is the gain graph obtained from $(Q(\Gamma_1), \Phi_Q(H_1))$ and m_1 copies of (Γ_2, φ_2) by joining the i-th vertex of $V_{Q(\Gamma_1)} \setminus V_{\Gamma_1}$ by an edge with gain 1_G to every vertex in the i-th copy $(\Gamma_2^i, \varphi_2^i)$ of (Γ_2, φ_2).

For the adjacency matrix of $(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)$ we write $\Phi_{\ominus_Q}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\ominus_Q}(H_1, H_2) = \begin{pmatrix} O_{n_1} & H_1 & O_{n_1 \times m_1 n_2} \\ H_1^* & s(H_1^* H_1 - 2I_{m_1}) & I_{m_1} \otimes \mathbf{1}_{n_2}^T \\ O_{m_1 n_2 \times n_1} & I_{m_1} \otimes \mathbf{1}_{n_2} & I_{m_1} \otimes \Phi(H_2) \end{pmatrix}.$$

Theorem 7.2.16. *Under the assumptions of Remark 7.2.1, suppose that $((\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2), \Upsilon)$ is a gain Q-edge corona graph of (Γ_1, φ_1) and (Γ_2, φ_2). The π-characteristic polynomial of $((\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2), \Upsilon)$ is*

$$\phi^{\pi}_{(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)} = x^{d(n_1 - m_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{m_1} \cdot \det\Big((x^2 + 2\sigma x) I_{m_1 d} - x(I_{m_1} \otimes C_\pi(x)) - (\sigma x + 1)\pi(H_1^* H_1) \Big),$$

where $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ are the G-phases with $\Upsilon = \Phi_{\ominus_Q}(H_1, H_2)$, and $\sigma = \pm 1$ when $\pi(s) = \pm I$.

In the special case where $C_\pi(x) = c_\pi(x) I_d$ (for example, when $d = 1$), this reduces to

$$\phi^{\pi}_{(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)} = (x - c_\pi(x) + 2\sigma)^{d(m_1 - n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{m_1} \prod_{i=1}^{n_1 d} \left(x^2 + (2\sigma - r\sigma - c_\pi(x) - \lambda_i)x - r - \sigma\lambda_i \right).$$

Proof. Since (Γ_1, φ_1) and (Γ_2, φ_2) are G-gain graphs, there exist $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1) = \varphi_1$ and $\Phi(H_2) = \varphi_2$, and, by definition of the gain Q-edge corona, $\Upsilon = \Phi_{\ominus_Q}(H_1, H_2)$.

Apply the representation π to the characteristic matrix of the adjacency matrix. From the adjacency matrix of the gain Q-edge corona $(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)$, we have

$$xI - \pi(\Phi_{\ominus_Q}(H_1, H_2)) = \begin{pmatrix} xI_{n_1 d} & -\pi(H_1) & O \\ -\pi(H_1)^* & xI_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) & -I_{m_1} \otimes (\mathbf{1}_{n_2}^T \otimes I_d) \\ O & -I_{m_1} \otimes (\mathbf{1}_{n_2} \otimes I_d) & I_{m_1} \otimes (xI_{n_2 d} - \pi(A_2)) \end{pmatrix}.$$

Note that $s(H_1^* H_1 - 2I_{m_1})$ becomes $\sigma(\pi(H_1^* H_1) - 2I_{m_1 d})$, where $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Taking the Schur complement with respect to the $(3,3)$ block gives

$$\det\big(xI - \pi(\Phi_{\ominus_Q}(H_1, H_2))\big) = \prod_{i=1}^{n_2 d} (x - \mu_i)^{m_1} \cdot \det \begin{pmatrix} xI_{n_1 d} & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) - I_{m_1} \otimes C_\pi(x) \end{pmatrix},$$

where the block $(2,2)$ correction is $I_{m_1} \otimes C_\pi(x)$.

Now take the Schur complement with respect to the $(1,1)$ block. For $x \neq 0$,

$$\det \begin{pmatrix} xI_{n_1 d} & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) - I_{m_1} \otimes C_\pi(x) \end{pmatrix}$$
$$= x^{n_1 d} \det\left(xI_{m_1 d} - \sigma(\pi(H_1^* H_1) - 2I_{m_1 d}) - I_{m_1} \otimes C_\pi(x) - \frac{1}{x}\pi(H_1^* H_1) \right).$$

Collecting terms,

$$= x^{n_1 d} \det\left((x + 2\sigma) I_{m_1 d} - I_{m_1} \otimes C_\pi(x) - \left(\sigma + \frac{1}{x}\right) \pi(H_1^* H_1) \right).$$

Factoring x^{-1} out of the $m_1 d \times m_1 d$ matrix multiplies the determinant by $x^{-m_1 d}$, giving

$$= x^{d(n_1-m_1)} \det\Big((x^2+2\sigma x)I_{m_1 d} - x(I_{m_1} \otimes C_\pi(x)) - (\sigma x+1)\pi(H_1^* H_1)\Big).$$

Therefore,

$$\phi^{\pi}_{(\Gamma_1,\varphi_1)\ominus_Q(\Gamma_2,\varphi_2)} = x^{d(n_1-m_1)} \prod_{i=1}^{n_2 d}(x-\mu_i)^{m_1} \cdot \det\Big((x^2+2\sigma x)I_{m_1 d} - x(I_{m_1} \otimes C_\pi(x)) - (\sigma x+1)\pi(H_1^* H_1)\Big).$$

The intermediate determinant identities were derived for $x \neq 0$ and $x \notin \operatorname{spec}(\pi(A_2))$; since both sides are rational functions in x, the identity extends to all x by clearing denominators (cf. Section 1.8).

The scalar case $C_\pi(x) = c_\pi(x) I_d$ follows immediately: since $I_{m_1} \otimes C_\pi(x) = c_\pi(x) I_{m_1 d}$, the determinant factorises over the eigenvalues of $\pi(H_1^* H_1)$, which are the same as those of $r I_{n_1 d} + \sigma\pi(A_1)$ together with $d(m_1 - n_1)$ zero eigenvalues. The zero eigenvalue contributes $x(x + 2\sigma - c_\pi(x))$ per copy. Noting that

$$x^{d(n_1-m_1)}\big[x(x+2\sigma-c_\pi(x))\big]^{d(m_1-n_1)} = (x - c_\pi(x) + 2\sigma)^{d(m_1-n_1)},$$

we obtain

$$(x - c_\pi(x) + 2\sigma)^{d(m_1-n_1)} \prod_{i=1}^{n_1 d}\left(x^2 + (2\sigma - r\sigma - c_\pi(x) - \lambda_i)x - r - \sigma\lambda_i\right),$$

so that the final determinant factor becomes the stated scalar formula when combined with $\prod_{i=1}^{n_2 d}(x - \mu_i)^{m_1}$. □

If (Γ_2, φ_2) is g-G-regular and $\pi(g) = \gamma_\pi I_d$, then $C_\pi(x)$ is scalar and the special case of Theorem 7.2.16 applies, so we can say more.

Corollary 7.2.17. *Suppose that, under the assumptions of Theorem 7.2.16, (Γ_2, φ_2) is g-G-regular, with $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$ (automatic when g lies in the centre of $\mathbb{C}G$, by Schur's lemma, or when $d = 1$), and $\mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1} = \gamma_\pi$ for some fixed k $(1 \le k \le n_2 d - d + 1)$. The π-spectrum of $((\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2), \Upsilon)$ consists of*

(i) the roots of $x^2 + (2\sigma - \gamma_\pi)x - 2\gamma_\pi\sigma - n_2$, each with multiplicity $d(m_1 - n_1)$,

(ii) μ_i with multiplicity m_1 for each $i \in \{1, 2, \ldots, k-1, k+d, \ldots, n_2 d\}$,

(iii) the roots of

$$x^3 + (2\sigma - r\sigma - \lambda_i - \gamma_\pi)x^2 - \big(\gamma_\pi(2\sigma - r\sigma - \lambda_i) + n_2 + r + \sigma\lambda_i\big)x + \gamma_\pi(r + \sigma\lambda_i)$$

for each $1 \le i \le n_1 d$, with $\sigma = \pm 1$ when $\pi(s) = \pm I$.

Proof. As in Corollary 7.2.4, since (Γ_2, φ_2) is g-G-regular, we have

$$C_\pi(x) = \frac{n_2}{x - \gamma_\pi} I_d,$$

so the scalar case of Theorem 7.2.16 applies with $c_\pi(x) = \frac{n_2}{x - \gamma_\pi}$.

The scalar case gives the characteristic polynomial

$$(x - c_\pi(x) + 2\sigma)^{d(m_1-n_1)} \prod_{i=1}^{n_2 d}(x-\mu_i)^{m_1} \prod_{i=1}^{n_1 d}\left(x^2 + (2\sigma - r\sigma - c_\pi(x) - \lambda_i)x - r - \sigma\lambda_i\right).$$

Since $x = \gamma_\pi = \mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1}$ is the only pole of $c_\pi(x)$, Theorem 7.2.16 implies that μ_i is a π-eigenvalue of $(\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)$ for each i specified in item (ii).

The remaining π-eigenvalues are the roots of

$$(x - \gamma_\pi)\left(x - \frac{n_2}{x - \gamma_\pi} + 2\sigma\right) = 0$$

with multiplicity $d(m_1 - n_1)$, and

$$(x-\gamma_\pi)\left(x^2+(2\sigma-r\sigma-\frac{n_2}{x-\gamma_\pi}-\lambda_i)x-r-\sigma\lambda_i\right)=0$$

for each $1 \le i \le n_1 d$. Expanding and simplifying, these become

$$x^2+(2\sigma-\gamma_\pi)x-2\gamma_\pi\sigma-n_2=0$$

and

$$x^3+(2\sigma-r\sigma-\lambda_i-\gamma_\pi)x^2-\big(\gamma_\pi(2\sigma-r\sigma-\lambda_i)+n_2+r+\sigma\lambda_i\big)x+\gamma_\pi(r+\sigma\lambda_i)=0,$$

respectively. This gives us items (i) and (iii). □

R-edge corona

Definition 7.2.18. Let $H_1 \in \mathcal{H}_{\Gamma_1}$ and let (Γ_2, φ_2) be a G-gain graph. The gain R-edge corona $(\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2)$ is the gain graph obtained from $(R(\Gamma_1), \Phi_R(H_1))$ and m_1 copies of (Γ_2, φ_2) by joining the i-th vertex of $V_{R(\Gamma_1)} \setminus V_{\Gamma_1}$ by an edge with gain 1_G to every vertex in the i-th copy $(\Gamma_2^i, \varphi_2^i)$ of (Γ_2, φ_2).

For the adjacency matrix of $(\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2)$ we write $\Phi_{\ominus_R}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\ominus_R}(H_1,H_2)=\begin{pmatrix} s(H_1H_1^*-\deg(\Gamma_1)) & H_1 & O_{n_1\times m_1n_2}\\ H_1^* & O_{m_1} & I_{m_1}\otimes \mathbf{1}_{n_2}^T\\ O_{m_1n_2\times n_1} & I_{m_1}\otimes \mathbf{1}_{n_2} & I_{m_1}\otimes \Phi(H_2)\end{pmatrix}.$$

Theorem 7.2.19. *Under the assumptions of Remark 7.2.1, suppose that $((\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2), \Upsilon)$ is a gain R-edge corona graph of (Γ_1, φ_1) and (Γ_2, φ_2). The π-characteristic polynomial of $((\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2), \Upsilon)$ is*

$$\phi^\pi_{(\Gamma_1,\varphi_1)\ominus_R(\Gamma_2,\varphi_2)} = \prod_{i=1}^{n_2d}(x-\mu_i)^{m_1}\prod_{i=1}^{n_1d}(x-\lambda_i) \cdot \det\Big(xI_{m_1d}-I_{m_1}\otimes C_\pi(x)-\pi(H_1^*H_1)\big((x+\sigma r)I_{m_1d}-\sigma\pi(H_1^*H_1)\big)^{-1}\Big),$$

where $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ are the G-phases with $\Upsilon = \Phi_{\ominus_R}(H_1, H_2)$, and $\sigma = \pm 1$ when $\pi(s) = \pm I$.

In the special case where $C_\pi(x) = c_\pi(x) I_d$ (for example, when $d = 1$), this reduces to

$$\phi^\pi_{(\Gamma_1,\varphi_1)\ominus_R(\Gamma_2,\varphi_2)} = (x-c_\pi(x))^{d(m_1-n_1)}\prod_{i=1}^{n_2d}(x-\mu_i)^{m_1}\prod_{i=1}^{n_1d}\big(x^2-(c_\pi(x)+\lambda_i)x-r+(c_\pi(x)-\sigma)\lambda_i\big).$$

Proof. Since (Γ_1, φ_1) and (Γ_2, φ_2) are G-gain graphs, there exist $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1) = \varphi_1$ and $\Phi(H_2) = \varphi_2$, and, by definition of the gain R-edge corona, $\Upsilon = \Phi_{\ominus_R}(H_1, H_2)$.

We first note that, at the group-ring level,

$$s(H_1H_1^*-rI_{n_1})=s(rI_{n_1}+sA_1-rI_{n_1})=s^2A_1=A_1,$$

since Γ_1 is r-regular, $\deg(\Gamma_1) = rI_{n_1}$, and s is a central involution with $H_1H_1^* = rI_{n_1} + sA_1$. Thus the $(1,1)$ block of $\Phi_{\ominus_R}(H_1, H_2)$ is precisely A_1.

Apply the representation π to the characteristic matrix of the adjacency matrix. From the adjacency matrix of the gain R-edge corona $(\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2)$, we have

$$xI-\pi(\Phi_{\ominus_R}(H_1,H_2))=\begin{pmatrix} xI_{n_1d}-\pi(A_1) & -\pi(H_1) & O\\ -\pi(H_1)^* & xI_{m_1d} & -I_{m_1}\otimes(\mathbf{1}_{n_2}^T\otimes I_d)\\ O & -I_{m_1}\otimes(\mathbf{1}_{n_2}\otimes I_d) & I_{m_1}\otimes(xI_{n_2d}-\pi(A_2))\end{pmatrix}.$$

Taking the Schur complement with respect to the $(3,3)$ block gives

$$\det(xI - \pi(\Phi_{\ominus_R}(H_1,H_2))) = \prod_{i=1}^{n_2 d}(x-\mu_i)^{m_1} \cdot \det\begin{pmatrix} xI_{n_1 d} - \pi(A_1) & -\pi(H_1) \\ -\pi(H_1)^* & xI_{m_1 d} - I_{m_1} \otimes C_\pi(x) \end{pmatrix},$$

where the block $(2,2)$ correction is $I_{m_1} \otimes C_\pi(x)$.

Now take the Schur complement with respect to the $(1,1)$ block. Since

$$xI_{n_1 d} - \pi(A_1) = (x+\sigma r)I_{n_1 d} - \sigma\pi(H_1 H_1^*),$$

and $\pi(H_1 H_1^*) = rI_{n_1 d} + \sigma\pi(A_1)$, the intertwining identity applies. Let $W(x) = (x+\sigma r)I_{m_1 d} - \sigma\pi(H_1^* H_1)$. Then

$$\pi(H_1)^*(xI_{n_1 d} - \pi(A_1))^{-1}\pi(H_1) = \pi(H_1^* H_1)\, W(x)^{-1}.$$

Thus the 2×2 determinant becomes

$$\det(xI_{n_1 d} - \pi(A_1)) \cdot \det\left(xI_{m_1 d} - I_{m_1} \otimes C_\pi(x) - \pi(H_1^* H_1)\, W(x)^{-1}\right).$$

Since $\det(xI_{n_1 d} - \pi(A_1)) = \prod_{i=1}^{n_1 d}(x-\lambda_i)$, we obtain

$$\phi^\pi_{(\Gamma_1,\varphi_1)\ominus_R(\Gamma_2,\varphi_2)} = \prod_{i=1}^{n_2 d}(x-\mu_i)^{m_1} \prod_{i=1}^{n_1 d}(x-\lambda_i) \cdot \det\left(xI_{m_1 d} - I_{m_1} \otimes C_\pi(x) - \pi(H_1^* H_1)\, W(x)^{-1}\right),$$

where $W(x) = (x+\sigma r)I_{m_1 d} - \sigma\pi(H_1^* H_1)$.

The intermediate determinant identities were derived for $x \notin \operatorname{spec}(\pi(A_2))$ and for x such that the relevant inverses exist; since both sides are rational functions in x, the identity extends to all x by clearing denominators (cf. Section 1.8).

The scalar case $C_\pi(x) = c_\pi(x) I_d$ follows immediately. Let ν be an eigenvalue of $\pi(H_1^* H_1)$, so that the corresponding factor of the determinant is

$$x - c_\pi(x) - \frac{\nu}{x + \sigma r - \sigma\nu}.$$

For $\nu = 0$, with multiplicity $d(m_1 - n_1)$, this is $x - c_\pi(x)$. For $\nu = r + \sigma\lambda_i$ we have $x + \sigma r - \sigma\nu = x - \lambda_i$, and the factor is

$$x - c_\pi(x) - \frac{r + \sigma\lambda_i}{x - \lambda_i} = \frac{x^2 - (c_\pi(x) + \lambda_i)x - r + (c_\pi(x) - \sigma)\lambda_i}{x - \lambda_i}.$$

Multiplying by $\prod_{i=1}^{n_1 d}(x-\lambda_i)$ clears these denominators, giving

$$(x - c_\pi(x))^{d(m_1 - n_1)} \prod_{i=1}^{n_1 d}\left(x^2 - (c_\pi(x) + \lambda_i)x - r + (c_\pi(x) - \sigma)\lambda_i\right),$$

which together with $\prod_{i=1}^{n_2 d}(x-\mu_i)^{m_1}$ is the stated scalar formula. □

If (Γ_2, φ_2) is g-G-regular and $\pi(g) = \gamma_\pi I_d$, then $C_\pi(x)$ is scalar and the special case of Theorem 7.2.19 applies, so we can say more.

Corollary 7.2.20. *Suppose that, under the assumptions of Theorem 7.2.19, (Γ_2, φ_2) is g-G-regular, with $\pi(g) = \gamma_\pi I_d$ for some $\gamma_\pi \in \mathbb{C}$ (automatic when g lies in the centre of $\mathbb{C}G$, by Schur's lemma, or when $d = 1$), and $\mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1} = \gamma_\pi$ for some fixed k $(1 \le k \le n_2 d - d + 1)$. The π-spectrum of $((\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2), \Upsilon)$ consists of*

(i) the roots of $x^2 - \gamma_\pi x - n_2$, each with multiplicity $d(m_1 - n_1)$,

(ii) μ_i with multiplicity m_1 for each $i \in \{1, 2, \ldots, k-1, k+d, \ldots, n_2 d\}$,

(iii) the roots of

$$x^3 - (\gamma_\pi + \lambda_i)x^2 + (\gamma_\pi\lambda_i - n_2 - r - \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i) + n_2\lambda_i$$

for each $1 \le i \le n_1 d$*, with* $\sigma = \pm 1$ *when* $\pi(s) = \pm I$*.*

Proof. As in Corollary 7.2.4, since (Γ_2, φ_2) is g-G-regular, we have

$$C_\pi(x) = \frac{n_2}{x - \gamma_\pi} I_d,$$

so the scalar case of Theorem 7.2.19 applies with $c_\pi(x) = \frac{n_2}{x-\gamma_\pi}$.

The scalar case gives the characteristic polynomial

$$(x - c_\pi(x))^{d(m_1-n_1)} \prod_{i=1}^{n_2 d} (x - \mu_i)^{m_1} \prod_{i=1}^{n_1 d} \left(x^2 - (c_\pi(x) + \lambda_i)x - r + (c_\pi(x) - \sigma)\lambda_i\right).$$

Since $x = \gamma_\pi = \mu_k = \mu_{k+1} = \cdots = \mu_{k+d-1}$ is the only pole of $c_\pi(x)$, Theorem 7.2.19 implies that μ_i is a π-eigenvalue of $(\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2)$ for each i specified in item (ii).

The remaining π-eigenvalues are the roots of

$$(x - \gamma_\pi)\left(x - \frac{n_2}{x - \gamma_\pi}\right) = 0$$

with multiplicity $d(m_1 - n_1)$, and

$$(x - \gamma_\pi)\left(x^2 - \left(\frac{n_2}{x - \gamma_\pi} + \lambda_i\right)x - r + \left(\frac{n_2}{x - \gamma_\pi} - \sigma\right)\lambda_i\right) = 0$$

for each $1 \le i \le n_1 d$. Expanding and simplifying, these become

$$x^2 - \gamma_\pi x - n_2 = 0$$

and

$$x^3 - (\gamma_\pi + \lambda_i)x^2 + (\gamma_\pi\lambda_i - n_2 - r - \sigma\lambda_i)x + \gamma_\pi(r + \sigma\lambda_i) + n_2\lambda_i = 0,$$

respectively. This gives us items (i) and (iii). □

Edge corona example

Example 7.2.21. We shall consider the products shown in Figure 7.3 along with the irreducible representation π of $\mathbb{H}$ from Example 7.1.2.

Recall from Equation (7.1.1) that the π-spectrum of (Γ_1, φ_1) is

$$\pi\text{-spec}(\Gamma_1, \varphi_1) = \{2^{(2)}, -1^{(4)}\},$$

and from Equation (7.2.1) that the π-spectrum of (Γ_2, φ_2) is

$$\pi\text{-spec}(\Gamma_2, \varphi_2) = \{1^{(2)}, -1^{(2)}\}.$$

After the appropriate substitutions, the π-spectra of the gain subdivision-edge, Q-edge, and R-edge corona products can be directly computed as

$$\pi\text{-spec}((\Gamma_1, \varphi_1) \ominus_S (\Gamma_2, \varphi_2)) = \{2.323^{(2)}, 1.481^{(4)}, 1^{(6)}, -0.311^{(4)}, -0.642^{(2)}, -2.170^{(4)}, -2.681^{(2)}\},$$
$$\pi\text{-spec}((\Gamma_1, \varphi_1) \ominus_Q (\Gamma_2, \varphi_2)) = \{3.562^{(2)}, 1^{(10)}, -0.382^{(4)}, -0.562^{(2)}, -2^{(2)}, -2.618^{(4)}\},$$
$$\pi\text{-spec}((\Gamma_1, \varphi_1) \ominus_R (\Gamma_2, \varphi_2)) = \{3.372^{(2)}, 1.303^{(4)}, 1^{(6)}, 0^{(2)}, -1^{(4)}, -2.303^{(4)}, -2.372^{(2)}\}.$$

These match the values obtained from implementing Equations (7.1.1) and (7.2.1) with Corollaries 7.2.14, 7.2.17, and 7.2.20.

Remark 7.2.22. It is notable that in this example the π-spectra of the gain subdivision-vertex and gain subdivision-edge corona products coincide, and that the π-spectra of the gain Q-vertex and gain R-edge corona products coincide, as do those of the gain R-vertex and gain Q-edge corona products. These coincidences admit the following explanation.

For the subdivision pair, the coincidence is a direct consequence of the characteristic polynomial formulas: the second factor here satisfies the scalar-coronal hypothesis $C_\pi(x) = c_\pi(x) I_d$, and $\Gamma_1 = C_3$ satisfies $n_1 = m_1$, so the special cases of Theorems 7.2.3 and 7.2.13 reduce to the same polynomial $\prod_{i=1}^{n_2 d}(x-\mu_i)^{n_1} \prod_{i=1}^{n_1 d}\left(x^2 - c_\pi(x)x - r - \sigma\lambda_i\right)$, with both prefactors trivial. The same argument shows that the gain subdivision-vertex and subdivision-edge corona products are π-cospectral for any cycle $\Gamma_1 = C_n$ and any second factor with scalar represented coronal. The Q/R interchanges may similarly be attributed to the isomorphism between the underlying graphs $Q(C_n)$ and $R(C_n)$ swapping original and edge-vertices, which in this example is compatible with the gain assignments, and the computed spectra above are consistent with this. For second factors whose represented coronal is not scalar, these coincidences need not hold, even for $\Gamma_1 = C_3$.

Remark 7.2.23. Taking $G = \{1\}$ with the trivial representation π, the π-spectrum reduces to the ordinary adjacency spectrum, and our results recover known classical results. Specifically, Corollaries 7.2.4 and 7.2.14 recover the subdivision-vertex and subdivision-edge corona spectral results of Lu and Miao [74, Theorems 2.1, 3.1 and Corollaries 2.2, 3.2], and Corollaries 7.2.10 and 7.2.20 recover the R-vertex and R-edge corona spectral results of Lan and Zhou [69, Theorems 3.1, 4.1 and Corollaries 3.1, 4.1]. Corollaries 7.2.7 and 7.2.17 recover the Q-vertex and Q-edge corona spectral results of Najiya and Chithra [76, Propositions 5.1, 5.2 and Corollaries 5.1, 5.3] in the case where both constituent graphs are regular; our results are more general, requiring regularity only on the first factor, not on both.

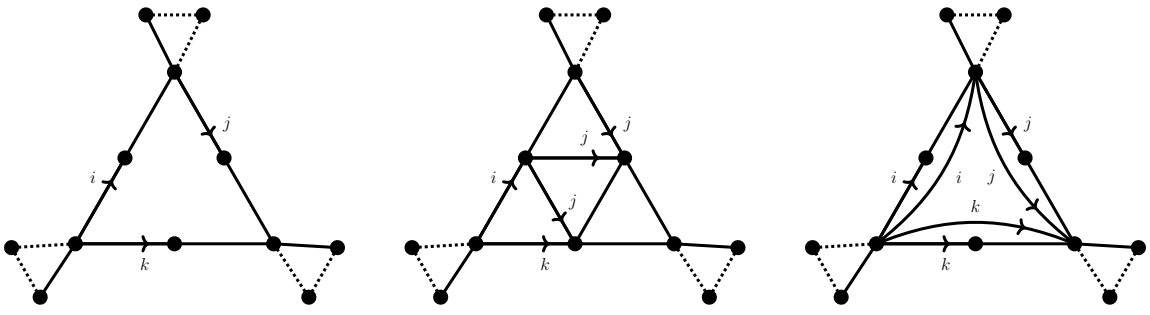
Taking $G = \{+1, -1\}$ with the representation $\pi(-1) = -1$, the π-spectrum reduces to the signed adjacency spectrum, and Corollaries 7.2.10 and 7.2.20 recover the signed R-vertex and R-edge corona spectral results of Rashid et al. [80, Theorems 5.5, 5.6, 5.11, 5.12]. The signed subdivision-vertex, Q-vertex, subdivision-edge, and Q-edge corona results of Corollaries 7.2.4, 7.2.7, 7.2.14, and 7.2.17 appear to be new in the signed setting.

7.3 Switching stability via pseudo-potentials

Once again we shall be making use of the pseudo-potentials as introduced in Section 1.7. The reader is referred there for the definition and basic properties.

In the remainder of this section we assume that the group G is abelian. Consequently the group ring $\mathbb{C}G$ is commutative, and the Kronecker product of matrices over $\mathbb{C}G$ satisfies the mixed-product property $(A \otimes B)(C \otimes D) = AC \otimes BD$ used in the proofs below.

7.3.1 Switching stability of the corona products



$(\Gamma_1, \varphi_1) \odot_S^{\mathcal{T}} (\Gamma_2, \varphi_2)$ $(\Gamma_1, \varphi_1) \odot_Q^{\mathcal{T}} (\Gamma_2, \varphi_2)$ $(\Gamma_1, \varphi_1) \odot_R^{\mathcal{T}} (\Gamma_2, \varphi_2)$

Figure 7.4: The switching-stable gain $\mathcal{T}$-subdivision-vertex, $\mathcal{T}$-Q-vertex, and $\mathcal{T}$-R-vertex corona graphs of (Γ_1, φ_1) and (Γ_2, φ_2).

In the standard case, all attachment edges are given a gain of 1_G in each direction. To provide switching-stable products, we make use of the pseudo-potentials in the following way.

For $M \in \{S, Q, R\}$, the gain function of the gain $\mathcal{T}$-M-vertex corona product is given as

$$\Phi_{\odot_M^{\mathcal{T}}}(H_1, H_2)(x, y) = \begin{cases} \Phi_M(H_1)(x, y), & \text{if } x, y \in V_{\Gamma_1} \cup E_{\Gamma_1}, \\ \Phi(H_2)(x, y), & \text{if } x, y \in V_{\Gamma_2}, \\ U_{\mathcal{T},v,\varphi_2}(y), & \text{if } x \in V_{\Gamma_1} \text{ and } y \in V_{\Gamma_2^x}. \end{cases}$$

Similarly, the gain function of the gain $\mathcal{T}$-M-edge corona product is given as

$$\Phi_{\ominus_M^{\mathcal{T}}}(H_1, H_2)(x, y) = \begin{cases} \Phi_M(H_1)(x, y), & \text{if } x, y \in V_{\Gamma_1} \cup E_{\Gamma_1}, \\ \Phi(H_2)(x, y), & \text{if } x, y \in V_{\Gamma_2}, \\ U_{\mathcal{T},v,\varphi_2}(y), & \text{if } x \in E_{\Gamma_1} \text{ and } y \in V_{\Gamma_2^x}. \end{cases}$$

In each case, the gain on the reverse arc is given by the inverse of the forward arc gain, as required in a gain graph.

Theorem 7.3.1. *Let (Γ_1, φ_1) and (Γ_1, ω_1) be switching-equivalent G-gain graphs, and let (Γ_2, φ_2) and (Γ_2, ω_2) be switching-equivalent G-gain graphs. Fix any spanning tree $\mathcal{T}$ of Γ_2 and any vertex $v \in V_{\Gamma_2}$. Then the gain subdivision-vertex, Q-vertex, and R-vertex corona products $(\Gamma_1, \varphi_1) \odot_M^{\mathcal{T}} (\Gamma_2, \varphi_2)$ and $(\Gamma_1, \omega_1) \odot_M^{\mathcal{T}} (\Gamma_2, \omega_2)$, with $M \in \{S, Q, R\}$, are switching equivalent.*

Proof. Owing to the similarities in structure, we treat the subdivision case in full and indicate the modifications required for the Q and R cases.

Since (Γ_1, φ_1) and (Γ_2, φ_2) are G-gain graphs, by definition we know that there exist $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ such that $\Phi(H_1) = \varphi_1$ and $\Phi(H_2) = \varphi_2$. Now since $\omega_1 \sim \Phi(H_1)$ and $\omega_2 \sim \Phi(H_2)$, there also exist $H_1' \in \mathcal{H}_{\Gamma_1}$ and $H_2' \in \mathcal{H}_{\Gamma_2}$ such that $\omega_1 = \Phi(H_1')$ and $\omega_2 = \Phi(H_2')$.

Therefore $\Phi(H_1) = \varphi_1 \sim \omega_1 = \Phi(H_1')$, and by Theorem 6.2.7, $H_1' = f_1^* H_1 g_1$ for some $f_1 \in G^{n_1}$ and $g_1 \in G^{m_1}$. Likewise, $H_2' = f_2^* H_2 g_2$ for some $f_2 \in G^{n_2}$ and $g_2 \in G^{m_2}$.

Let $\mathbf{U}$ be the vector of length n_2 such that the i-th entry is equal to $U_{\mathcal{T},v,\varphi_2}(y_i)$. Then for the adjacency matrix of $(\Gamma_1, \varphi_1) \odot_S^{\mathcal{T}} (\Gamma_2, \varphi_2)$ we write $\Phi_{\odot_S^{\mathcal{T}}}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\odot_S^{\mathcal{T}}}(H_1, H_2) = \begin{pmatrix} O_{n_1} & H_1 & I_{n_1} \otimes \mathbf{U}^T \\ H_1^* & O_{m_1} & O_{m_1 \times n_1 n_2} \\ I_{n_1} \otimes \mathbf{U}^* & O_{n_1 n_2 \times m_1} & I_{n_1} \otimes \Phi(H_2) \end{pmatrix}.$$

Similarly, let $\mathbf{U}'$ be the vector of length n_2 such that the i-th entry is equal to $U_{\mathcal{T},v,\omega_2}(y_i)$. Then for the adjacency matrix of $(\Gamma_1, \omega_1) \odot_S^{\mathcal{T}} (\Gamma_2, \omega_2)$ we write $\Phi_{\odot_S^{\mathcal{T}}}(H_1', H_2')$ and for each $H_1' \in \mathcal{H}_{\Gamma_1}$ and $H_2' \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\odot_S^{\mathcal{T}}}(H_1', H_2') = \begin{pmatrix} O_{n_1} & H_1' & I_{n_1} \otimes \mathbf{U}'^T \\ H_1'^* & O_{m_1} & O_{m_1 \times n_1 n_2} \\ I_{n_1} \otimes \mathbf{U}'^* & O_{n_1 n_2 \times m_1} & I_{n_1} \otimes \Phi(H_2') \end{pmatrix}.$$

We claim that $\Phi_{\odot_S^{\mathcal{T}}}(H_1, H_2)$ and $\Phi_{\odot_S^{\mathcal{T}}}(H_1', H_2')$ are switching equivalent via the switching matrix

$$F = \operatorname{diag}\left(F_1,\, F_1^L,\, F_1 \otimes F_2\right),$$

where $F_1 = \operatorname{diag}(f_1) \in M_{n_1}(\mathbb{C}G)$, $F_1^L = \operatorname{diag}(g_1) \in M_{m_1}(\mathbb{C}G)$, and $F_2 = \operatorname{diag}(f_2) \in M_{n_2}(\mathbb{C}G)$, acting on V_{Γ_1}, E_{Γ_1}, and the n_1 copies of V_{Γ_2}, respectively.

We verify this claim by checking each block of $F^* \Phi_{\odot_S^{\mathcal{T}}}(H_1, H_2) F$ and comparing the result against $\Phi_{\odot_S^{\mathcal{T}}}(H_1', H_2')$.

The $(V_{\Gamma_1}, V_{\Gamma_1})$ block: $F_1^* O_{n_1} F_1 = O_{n_1}$.

The $(V_{\Gamma_1}, E_{\Gamma_1})$ block: $F_1^* H_1 F_1^L = f_1^* H_1 g_1 = H_1'$, by Lemma 6.1.4 and Theorem 6.2.7.

The $(V_{\Gamma_1}, V_{\Gamma_2})$ block: Using the Kronecker product identity $(P \otimes Q)(R \otimes S) = PR \otimes QS$, we obtain

$$F_1^*(I_{n_1} \otimes \mathbf{U}^T)(F_1 \otimes F_2) = F_1^*(F_1 \otimes \mathbf{U}^T F_2)$$

$$\begin{aligned}&= (F_1^* F_1) \otimes (\mathbf{U}^T F_2)\\ &= I_{n_1} \otimes (\mathbf{U}^T F_2),\end{aligned}$$

since $F_1^* F_1 = \operatorname{diag}(f_1^{-1})\operatorname{diag}(f_1) = \operatorname{diag}(1_G, 1_G, \ldots, 1_G) = I_{n_1}$. Now $(\mathbf{U}^T F_2)_i = U_{\mathcal{T},v,\varphi_2}(y_i) f_2(y_i) = U_{\mathcal{T},v,\omega_2}(y_i) = (\mathbf{U}'^T)_i$ by Corollary 1.7.6(i), where we assume $f_2(v) = 1_G$ without loss of generality by Remark 1.7.7. Hence $F_1^*(I_{n_1} \otimes \mathbf{U}^T)(F_1 \otimes F_2) = I_{n_1} \otimes \mathbf{U}'^T$.

The $(E_{\Gamma_1}, V_{\Gamma_1})$ **block:** $F_1^{L*} H_1^* F_1 = (F_1^* H_1 F_1^L)^* = H_1'^*$, by Lemma 6.1.4 and Theorem 6.2.7.

The $(E_{\Gamma_1}, E_{\Gamma_1})$ **block:** $F_1^{L*} O_{m_1} F_1^L = O_{m_1}$.

The $(E_{\Gamma_1}, V_{\Gamma_2})$ **block:** $F_1^{L*} O_{m_1 \times n_1 n_2}(F_1 \otimes F_2) = O_{m_1 \times n_1 n_2}$.

The $(V_{\Gamma_2}, V_{\Gamma_1})$ **block:** Using the Kronecker product identity:

$$\begin{aligned}(F_1 \otimes F_2)^*(I_{n_1} \otimes \mathbf{U}^*)F_1 &= (F_1^* \otimes F_2^*)(I_{n_1} \otimes \mathbf{U}^*)F_1\\ &= (F_1^* \otimes F_2^* \mathbf{U}^*)F_1\\ &= (F_1^* F_1) \otimes (F_2^* \mathbf{U}^*)\\ &= I_{n_1} \otimes (F_2^* \mathbf{U}^*),\end{aligned}$$

where $(F_2^* \mathbf{U}^*)_i = f_2(y_i)^{-1} U^{-1}_{\mathcal{T},v,\varphi_2}(y_i) = (U_{\mathcal{T},v,\varphi_2}(y_i) f_2(y_i))^{-1} = U^{-1}_{\mathcal{T},v,\omega_2}(y_i) = (\mathbf{U}'^*)_i$ by Corollary 1.7.6(i) and $f_2(v) = 1_G$. Hence $(F_1 \otimes F_2)^*(I_{n_1} \otimes \mathbf{U}^*)F_1 = I_{n_1} \otimes \mathbf{U}'^*$.

The $(V_{\Gamma_2}, E_{\Gamma_1})$ **block:** $(F_1 \otimes F_2)^* O_{n_1 n_2 \times m_1} F_1^L = O_{n_1 n_2 \times m_1}$.

The $(V_{\Gamma_2}, V_{\Gamma_2})$ **block:** Using the Kronecker product identity:

$$\begin{aligned}(F_1 \otimes F_2)^*(I_{n_1} \otimes \Phi(H_2))(F_1 \otimes F_2) &= (F_1^* \otimes F_2^*)(I_{n_1} \otimes \Phi(H_2))(F_1 \otimes F_2)\\ &= (F_1^* I_{n_1} F_1) \otimes (F_2^* \Phi(H_2) F_2)\\ &= I_{n_1} \otimes \Phi(H_2'),\end{aligned}$$

since $F_2^* \Phi(H_2) F_2 = \Phi(H_2)^{f_2} = \omega_2 = \Phi(H_2')$ by Lemma 6.1.4.

Since each of the nine blocks of $F^* \Phi_{\odot_S^{\mathcal{T}}}(H_1, H_2) F$ agrees with the corresponding block of $\Phi_{\odot_S^{\mathcal{T}}}(H_1', H_2')$, we conclude that $(\Gamma_1, \varphi_1) \odot_S^{\mathcal{T}} (\Gamma_2, \varphi_2)$ and $(\Gamma_1, \omega_1) \odot_S^{\mathcal{T}} (\Gamma_2, \omega_2)$ are switching equivalent.

For Q, the $(E_{\Gamma_1}, E_{\Gamma_1})$ block contains $\Phi_L(H_1)$ instead of O_{m_1}, and then $F_1^{L*} \Phi_L(H_1) F_1^L = \Phi_L(H_1)^{g_1} = \Phi_L(H_1')$ by Lemma 6.1.4 and Theorem 6.2.7. Every other block is identical to the subdivision case, thus $(\Gamma_1, \varphi_1) \odot_Q^{\mathcal{T}} (\Gamma_2, \varphi_2)$ and $(\Gamma_1, \omega_1) \odot_Q^{\mathcal{T}} (\Gamma_2, \omega_2)$ are switching equivalent.

For R, the $(V_{\Gamma_1}, V_{\Gamma_1})$ block contains $\Phi(H_1)$ instead of O_{n_1}, and then $F_1^* \Phi(H_1) F_1 = \Phi(H_1)^{f_1} = \Phi(H_1')$ by Lemma 6.1.4 and Theorem 6.2.7. Again every other block is identical to the subdivision case, thus $(\Gamma_1, \varphi_1) \odot_R^{\mathcal{T}} (\Gamma_2, \varphi_2)$ and $(\Gamma_1, \omega_1) \odot_R^{\mathcal{T}} (\Gamma_2, \omega_2)$ are switching equivalent. □

Theorem 7.3.2. *Let* (Γ_1, φ_1) *and* (Γ_1, ω_1) *be switching-equivalent* G*-gain graphs, and let* (Γ_2, φ_2) *and* (Γ_2, ω_2) *be switching-equivalent* G*-gain graphs. Fix any spanning tree* $\mathcal{T}$ *of* Γ_2 *and any vertex* $v \in V_{\Gamma_2}$*. Then the gain subdivision-edge,* Q*-edge, and* R*-edge corona products* $(\Gamma_1, \varphi_1) \ominus_M^{\mathcal{T}} (\Gamma_2, \varphi_2)$ *and* $(\Gamma_1, \omega_1) \ominus_M^{\mathcal{T}} (\Gamma_2, \omega_2)$*, with* $M \in \{S, Q, R\}$*, are switching equivalent.*

Proof. We proceed in a very similar manner to that of the proof of Theorem 7.3.1, treating the subdivision case in full and highlighting the modifications required for the Q and R cases.

Let $\mathbf{U}$ be the vector of length n_2 such that the i-th entry is equal to $U_{\mathcal{T},v,\varphi_2}(y_i)$. Then for the adjacency matrix of $(\Gamma_1, \varphi_1) \ominus_S^{\mathcal{T}} (\Gamma_2, \varphi_2)$ we write $\Phi_{\ominus_S^{\mathcal{T}}}(H_1, H_2)$ and for each $H_1 \in \mathcal{H}_{\Gamma_1}$ and $H_2 \in \mathcal{H}_{\Gamma_2}$ we have

$$\Phi_{\ominus_S^{\mathcal{T}}}(H_1, H_2) = \begin{pmatrix} O_{n_1} & H_1 & O_{n_1 \times m_1 n_2} \\ H_1^* & O_{m_1} & I_{m_1} \otimes \mathbf{U}^T \\ O_{m_1 n_2 \times n_1} & I_{m_1} \otimes \mathbf{U}^* & I_{m_1} \otimes \Phi(H_2) \end{pmatrix}.$$

Similarly, let $\mathbf{U}'$ be the vector of length n_2 such that the i-th entry is equal to $U_{\mathcal{T},v,\omega_2}(y_i)$. Then for the adjacency matrix of $(\Gamma_1, \omega_1) \ominus_S^{\mathcal{T}} (\Gamma_2, \omega_2)$ we write $\Phi_{\ominus_S^{\mathcal{T}}}(H_1', H_2')$ and for each $H_1' \in \mathcal{H}_{\Gamma_1}$ and $H_2' \in \mathcal{H}_{\Gamma_2}$

we have

$$\Phi_{\ominus_S^{\mathcal{T}}}(H_1', H_2') = \begin{pmatrix} O_{n_1} & H_1' & O_{n_1 \times m_1 n_2} \\ H_1'^* & O_{m_1} & I_{m_1} \otimes \mathbf{U}'^T \\ O_{m_1 n_2 \times n_1} & I_{m_1} \otimes \mathbf{U}'^* & I_{m_1} \otimes \Phi(H_2') \end{pmatrix}.$$

We claim that $\Phi_{\ominus_S^{\mathcal{T}}}(H_1, H_2)$ and $\Phi_{\ominus_S^{\mathcal{T}}}(H_1', H_2')$ are switching equivalent via the switching matrix

$$F = \operatorname{diag}\left(F_1,\, F_1^L,\, F_1^L \otimes F_2\right),$$

where $F_1 = \operatorname{diag}(f_1) \in M_{n_1}(\mathbb{C}G)$, $F_1^L = \operatorname{diag}(g_1) \in M_{m_1}(\mathbb{C}G)$, and $F_2 = \operatorname{diag}(f_2) \in M_{n_2}(\mathbb{C}G)$, acting on V_{Γ_1}, E_{Γ_1}, and the m_1 copies of V_{Γ_2}, respectively.

We verify this claim by checking each block of $F^*\Phi_{\ominus_S^{\mathcal{T}}}(H_1, H_2)F$ and comparing the result against $\Phi_{\ominus_S^{\mathcal{T}}}(H_1', H_2')$.

The $(V_{\Gamma_1}, V_{\Gamma_1})$ **block:** $F_1^* O_{n_1} F_1 = O_{n_1}$.

The $(V_{\Gamma_1}, E_{\Gamma_1})$ **block:** $F_1^* H_1 F_1^L = f_1^* H_1 g_1 = H_1'$, by Lemma 6.1.4 and Theorem 6.2.7.

The $(V_{\Gamma_1}, V_{\Gamma_2})$ **block:** $F_1^* O_{n_1 \times m_1 n_2}(F_1^L \otimes F_2) = O_{n_1 \times m_1 n_2}$.

The $(E_{\Gamma_1}, V_{\Gamma_1})$ **block:** $F_1^{L*} H_1^* F_1 = (F_1^* H_1 F_1^L)^* = H_1'^*$, by Lemma 6.1.4 and Theorem 6.2.7.

The $(E_{\Gamma_1}, E_{\Gamma_1})$ **block:** $F_1^{L*} O_{m_1} F_1^L = O_{m_1}$.

The $(E_{\Gamma_1}, V_{\Gamma_2})$ **block:** Using the Kronecker product identity $(P \otimes Q)(R \otimes S) = PR \otimes QS$, we obtain

$$\begin{aligned} F_1^{L*}(I_{m_1} \otimes \mathbf{U}^T)(F_1^L \otimes F_2) &= F_1^{L*}(F_1^L \otimes \mathbf{U}^T F_2) \\ &= (F_1^{L*} F_1^L) \otimes (\mathbf{U}^T F_2) \\ &= I_{m_1} \otimes (\mathbf{U}^T F_2), \end{aligned}$$

since $F_1^{L*} F_1^L = \operatorname{diag}(g_1^{-1}) \operatorname{diag}(g_1) = \operatorname{diag}(1_G, 1_G, \ldots, 1_G) = I_{m_1}$. We know that $\mathbf{U}^T F_2 = \mathbf{U}'^T$ from Theorem 7.3.1, and so $F_1^{L*}(I_{m_1} \otimes \mathbf{U}^T)(F_1^L \otimes F_2) = I_{m_1} \otimes \mathbf{U}'^T$.

The $(V_{\Gamma_2}, V_{\Gamma_1})$ **block:** $(F_1^L \otimes F_2)^* O_{m_1 n_2 \times n_1} F_1 = O_{m_1 n_2 \times n_1}$.

The $(V_{\Gamma_2}, E_{\Gamma_1})$ **block:** Using the Kronecker product identity:

$$\begin{aligned} (F_1^L \otimes F_2)^*(I_{m_1} \otimes \mathbf{U}^*)F_1^L &= (F_1^{L*} \otimes F_2^*)(I_{m_1} \otimes \mathbf{U}^*)F_1^L \\ &= (F_1^{L*} \otimes F_2^* \mathbf{U}^*)F_1^L \\ &= (F_1^{L*} F_1^L) \otimes (F_2^* \mathbf{U}^*) \\ &= I_{m_1} \otimes (F_2^* \mathbf{U}^*), \end{aligned}$$

and we know that $F_2^* \mathbf{U}^* = \mathbf{U}'^*$ from Theorem 7.3.1. Hence $(F_1^L \otimes F_2)^*(I_{m_1} \otimes \mathbf{U}^*)F_1^L = I_{m_1} \otimes \mathbf{U}'^*$.

The $(V_{\Gamma_2}, V_{\Gamma_2})$ **block:** Using the Kronecker product identity:

$$\begin{aligned} (F_1^L \otimes F_2)^*(I_{m_1} \otimes \Phi(H_2))(F_1^L \otimes F_2) &= (F_1^{L*} \otimes F_2^*)(I_{m_1} \otimes \Phi(H_2))(F_1^L \otimes F_2) \\ &= (F_1^{L*} I_{m_1} F_1^L) \otimes (F_2^* \Phi(H_2) F_2) \\ &= I_{m_1} \otimes \Phi(H_2'), \end{aligned}$$

since $F_2^* \Phi(H_2) F_2 = \Phi(H_2)^{f_2} = \omega_2 = \Phi(H_2')$ by Lemma 6.1.4.

Since each of the nine blocks of $F^*\Phi_{\ominus_S^{\mathcal{T}}}(H_1, H_2)F$ agrees with the corresponding block of $\Phi_{\ominus_S^{\mathcal{T}}}(H_1', H_2')$, we conclude that $(\Gamma_1, \varphi_1) \ominus_S^{\mathcal{T}} (\Gamma_2, \varphi_2)$ and $(\Gamma_1, \omega_1) \ominus_S^{\mathcal{T}} (\Gamma_2, \omega_2)$ are switching equivalent.

For Q and R, the $(E_{\Gamma_1}, E_{\Gamma_1})$ and $(V_{\Gamma_1}, V_{\Gamma_1})$ blocks differ from the subdivision case in exactly the same way as in the proof of Theorem 7.3.1. So, we conclude that $(\Gamma_1, \varphi_1) \ominus_M^{\mathcal{T}} (\Gamma_2, \varphi_2)$ and $(\Gamma_1, \omega_1) \ominus_M^{\mathcal{T}} (\Gamma_2, \omega_2)$ are switching equivalent for $M \in \{Q, R\}$, and we are done. □

7.3.2 Balancedness of the products

Theorem 7.3.3. *Let $s = 1_G$. A G-gain graph (Γ, φ) is balanced if and only if each of its gain line, subdivision, Q, R, and total graphs are balanced.*

Proof. The line graph is dealt with in [30, Corollary 4.27] and the total graph is handled by [34, Corollary 3.7].

We begin with the subdivision graph. Every cycle in $(S(\Gamma), \Phi_S(H))$ must alternate between vertex-vertices from V_Γ and edge-vertices from E_Γ, since there are no edges between two vertices of the same type in the subdivision graph. So every cycle has even length, say $2k$, and passes through vertices v_{i_1}, e_{j_1}, v_{i_2}, e_{j_2}, $\ldots$, v_{i_k}, e_{j_k} in order, where each v_{i_l} is an endpoint of e_{j_l} and $e_{j_{l-1}}$. The gain of such a cycle is

$$H_{i_1,j_1}(H_{i_2,j_1})^{-1}H_{i_2,j_2}(H_{i_3,j_2})^{-1}\cdots H_{i_k,j_k}(H_{i_1,j_k})^{-1},$$

which is precisely the gain of the closed walk $v_{i_1}, v_{i_2}, \ldots, v_{i_k}, v_{i_1}$ in (Γ, φ) via Definition 6.1.3. Conversely, every closed walk in (Γ, φ) gives rise to a closed walk of this form in $(S(\Gamma), \Phi_S(H))$. Hence (Γ, φ) is balanced if and only if $(S(\Gamma), \Phi_S(H))$ is balanced.

Now we consider the Q graph. Note that $(S(\Gamma), \Phi_S(H))$ is an induced subgraph of $(Q(\Gamma), \Phi_Q(H))$, so a balanced Q graph implies a balanced subdivision graph, which by the above implies a balanced (Γ, φ). Every arc in $(Q(\Gamma), \Phi_Q(H))$ is of one of two types: a subdivision arc between a vertex-vertex v_i and an edge-vertex e_k with gain $H_{i,k}$ or $(H_{i,k})^{-1}$, or a line arc between two edge-vertices e_p and e_q sharing a common vertex $v_r = e_p \cap e_q$ with gain $(H_{r,p})^{-1}H_{r,q}$. Since there are no arcs between two vertex-vertices, every cycle consists of subdivision arcs and line arcs in some combination. In all cases, the gain of the cycle is a product of terms $H_{i,k}$ and $(H_{i,k})^{-1}$ in which consecutive terms share an index, and the product telescopes to

$$H_{i_1,j_1}(H_{i_2,j_1})^{-1}H_{i_2,j_2}(H_{i_3,j_2})^{-1}\cdots H_{i_k,j_k}(H_{i_1,j_k})^{-1},$$

which, as in the subdivision case, is the gain of the closed walk $v_{i_1}, v_{i_2}, \ldots, v_{i_k}, v_{i_1}$ in (Γ, φ). Hence (Γ, φ) is balanced if and only if $(Q(\Gamma), \Phi_Q(H))$ is balanced.

Finally, we consider the R graph. Here we note that (Γ, φ) itself is an induced subgraph of $(R(\Gamma), \Phi_R(H))$, and so a balanced R graph immediately implies a balanced (Γ, φ). Every arc in $(R(\Gamma), \Phi_R(H))$ is of one of two types: an arc between two vertex-vertices v_i and v_j with gain $\varphi(v_i, v_j)$, or an arc between a vertex-vertex v_i and an edge-vertex e_k with gain $H_{i,k}$ or $(H_{i,k})^{-1}$. Every cycle in $(R(\Gamma), \Phi_R(H))$ through an edge-vertex $e_k = \{v_i, v_j\}$ must enter and exit via v_i and v_j, since e_k is only adjacent to v_i and v_j in $R(\Gamma)$, contributing gain $H_{i,k}(H_{j,k})^{-1} = \varphi(v_i, v_j)$ by Definition 6.1.3 with $s = 1_G$, which is precisely the gain of the arc from v_i to v_j in (Γ, φ). Hence every closed walk in $(R(\Gamma), \Phi_R(H))$ has the same gain as a corresponding closed walk in (Γ, φ), thus is balanced if (Γ, φ) is balanced. Therefore (Γ, φ) is balanced if and only if $(R(\Gamma), \Phi_R(H))$ is balanced. □

Remark 7.3.4. We expect that the result extends to general s-balance in the sense of [34, Corollary 3.7], whereby an s-balanced (Γ, φ) gives rise to s-balanced gain line, subdivision, Q, R, and total graphs, and vice versa. We leave this generalisation for future work.

Theorem 7.3.5. *Let $s = 1_G$. The standard gain subdivision-vertex, Q-vertex, R-vertex, subdivision-edge, Q-edge, and R-edge corona products of two G-gain graphs (Γ_1, φ_1) and (Γ_2, φ_2) are balanced if and only if $\varphi_1 \in [1_G]$ and $\varphi_2 = 1_G$, provided that in the edge-corona cases $m_1 > 0$.*

Proof. Suppose that $\varphi_1 \in [1_G]$ and $\varphi_2 = 1_G$. Then the derived graphs $(S(\Gamma_1), \Phi_S(H_1))$, $(Q(\Gamma_1), \Phi_Q(H_1))$, and $(R(\Gamma_1), \Phi_R(H_1))$ are balanced by Theorem 7.3.3. Since every edge in (Γ_2, φ_2) has gain 1_G and every attachment edge in the standard vertex and edge corona operations has gain 1_G, every closed walk in the product has gain 1_G, and the new graph is balanced.

Conversely, suppose that the product is balanced. Since the graphs $(S(\Gamma_1), \Phi_S(H_1))$, $(Q(\Gamma_1), \Phi_Q(H_1))$, and $(R(\Gamma_1), \Phi_R(H_1))$ each appear as an induced subgraph in the relevant product, each must be balanced, hence (Γ_1, φ_1) must be balanced by Theorem 7.3.3. The second component, (Γ_2, φ_2), also appears as an induced subgraph in the product, there being n_1 copies in each vertex corona and m_1 copies in each edge corona. Take any edge $v_iv_j \in E_{\Gamma_2}$ and any vertex u corresponding to either a vertex or edge of Γ_1

as appropriate. Then the product contains the triangle uv_iv_j, which (denoting the gain function of the product by Υ) has gain

$$\Upsilon(uv_i)\Upsilon(v_iv_j)\Upsilon(v_ju) = 1_G \cdot \varphi_2(v_iv_j) \cdot 1_G = \varphi_2(v_iv_j).$$

Therefore, to have a balanced gain subdivision-vertex, Q-vertex, R-vertex, subdivision-edge, Q-edge, or R-edge corona product, it is necessary that $\varphi_2(v_iv_j) = 1_G$ for all edges in (Γ_2, φ_2). □

Theorem 7.3.6. *Let $s = 1_G$. The switching-stable gain $\mathcal{T}$-subdivision-vertex, $\mathcal{T}$-Q-vertex, $\mathcal{T}$-R-vertex, $\mathcal{T}$-subdivision-edge, $\mathcal{T}$-Q-edge, and $\mathcal{T}$-R-edge corona products of two G-gain graphs (Γ_1, φ_1) and (Γ_2, φ_2) are balanced if and only if $\varphi_1 \in [1_G]$ and $\varphi_2 \in [1_G]$, independent of the choice of spanning tree.*

Proof. If (Γ_1, φ_1) is unbalanced, then so is each of its gain subdivision, Q, and R graphs by Theorem 7.3.3, and so are the relevant products. Likewise, since (Γ_2, φ_2) appears as an induced subgraph in each product, an unbalanced (Γ_2, φ_2) implies an unbalanced product.

Replacing φ_1 and φ_2 with some switching-equivalent gain functions ω_1 and ω_2 leaves the switching classes of the $\mathcal{T}$-subdivision-vertex, $\mathcal{T}$-Q-vertex, $\mathcal{T}$-R-vertex, $\mathcal{T}$-subdivision-edge, $\mathcal{T}$-Q-edge, and $\mathcal{T}$-R-edge corona products unchanged, by Theorems 7.3.1 and 7.3.2. Moreover, by Lemma 1.7.8, since (Γ_2, φ_2) is balanced, different choices of spanning tree give the same values for the pseudo-potential. Accordingly, we may assume without loss of generality that $\varphi_1 = 1_G$ and $\varphi_2 = 1_G$. Under these identity gain functions, every edge within the copies of $(S(\Gamma_1), \Phi_S(H_1))$, $(Q(\Gamma_1), \Phi_Q(H_1))$, or $(R(\Gamma_1), \Phi_R(H_1))$ has gain 1_G, every edge within each copy of (Γ_2, φ_2) has gain 1_G, and every attachment edge has gain 1_G by definition of the pseudo-potential with identity gain function. Hence every closed walk in the product has gain 1_G, and the product is balanced. □

Example 7.3.7. We shall consider the products shown in Figure 7.4 along with the irreducible representation π of $\mathbb{H}$ from Example 7.1.2.

After the appropriate substitutions, the π-spectra of the gain $\mathcal{T}$-subdivision-vertex, $\mathcal{T}$-Q-vertex, and $\mathcal{T}$-R-vertex corona products can be directly computed as

$$\pi\text{-}\operatorname{spec}((\Gamma_1, \varphi_1) \odot_S^{\mathcal{T}} (\Gamma_2, \varphi_2)) = \{2.681^{(2)}, 2.170^{(4)}, 0.642^{(2)}, 0.311^{(4)}, -1^{(6)}, -1.481^{(4)}, -2.323^{(2)}\},$$
$$\pi\text{-}\operatorname{spec}((\Gamma_1, \varphi_1) \odot_Q^{\mathcal{T}} (\Gamma_2, \varphi_2)) = \{3.489^{(2)}, 2.115^{(4)}, 1.289^{(2)}, -0.254^{(4)}, -1^{(6)}, -1.779^{(2)}, -1.861^{(4)}\},$$
$$\pi\text{-}\operatorname{spec}((\Gamma_1, \varphi_1) \odot_R^{\mathcal{T}} (\Gamma_2, \varphi_2)) = \{3.779^{(2)}, 1.861^{(4)}, 0.711^{(2)}, 0.254^{(4)}, -1^{(6)}, -1.489^{(2)}, -2.115^{(4)}\}.$$

Moreover, since (Γ_1, φ_1) and (Γ_2, φ_2) are both balanced, by Theorem 7.3.6 these vertex corona products are balanced. Indeed, their π-spectra coincide with those of the adjacency matrices of the classical vertex corona products $\Gamma_1 \odot_S \Gamma_2$, $\Gamma_1 \odot_Q \Gamma_2$, and $\Gamma_1 \odot_R \Gamma_2$, with each classical eigenvalue of multiplicity k appearing with multiplicity $2k$ in the π-spectrum, as expected for a representation of degree 2.

7.4 Summary and outlook

In this chapter we have introduced gain variants of the six subdivision-based corona products and determined the π-characteristic polynomials and π-spectra of their adjacency matrices for an r-regular first component and an arbitrary second, with further simplifications when the second factor is g-G-regular. These results recover and extend a number of known classical and signed spectral formulas. We have also shown how pseudo-potentials lead to switching-stable versions of all six products, characterised their balancedness, and observed that switching-stability currently requires the group G to be abelian; removing this restriction is a natural direction for future work. Further possible developments include the study of Laplacian and normalised Laplacian spectra for these products, as well as the extension of similar techniques to other graph compositions over gain graphs.

Chapter 8

Concluding remarks and future work

This concluding chapter synthesises the contributions of the preceding chapters, reflects on the methodological approach developed throughout, and charts directions for future research. The central theme unifying this thesis has been the systematic extension of classical graph operations to signed and gain graphs in a manner that respects switching equivalence, a fundamental relation in the theory of signed and gain graphs. By developing switching-stable constructions, we have ensured that the spectral properties of the products generalise suitably, thereby providing robust extensions of classical results.

8.1 Summary of contributions

Chapter 2: Graphs with integer reciprocal eigenvalue properties

In Chapter 2, we introduced the integer anti-reciprocal eigenvalue properties m-($-$R) and m-($-$SR), complementing the earlier notions of m-(R) and m-(SR) due to Barik, Pati, and Verma [13]. We showed that the Kronecker product of signed graphs behaves naturally with respect to these properties, and established that the simple m-corona of a signed graph always satisfies m-($-$SR), with m-(SR) following when the root graph has symmetric spectrum. We provided a complete classification of hypercubes: while the classical hypercube Q_n satisfies no integer reciprocal eigenvalue property for $n \geq 5$, a suitable signing always yields n-(SR) and n-($-$SR). We also constructed explicit infinite families of graphs satisfying m-(R) but not m-(SR), and non-corona graphs satisfying both properties in both the unsigned and signed settings.

Chapter 3: Spectra and balance of signed graphs obtained from vertex splittings

In Chapter 3, we introduced the neighbours splitting (NS) and non-neighbours splitting (NNS) joins for signed graphs, generalising the classical splitting joins to the signed setting. We derived their adjacency and Laplacian characteristic polynomials, and provided explicit spectral descriptions for co-regular components. A key contribution was the development of switching-stable variants using pseudo-potential functions, which are stable under switching of either component, providing a significant improvement over the standard variants at the cost of fixing a spanning tree. We also characterised balancedness: the standard joins are balanced only when both inputs are all-positive, while the switching-stable variants are balanced precisely when both inputs are balanced.

Chapter 4: A switching-stable corona product of signed graphs

In Chapter 4, we addressed the long-standing problem of the lack of a switching-stable corona product for signed graphs. We introduced the $\mathcal{T}$-corona product of two signed graphs (Γ_1, σ_1) and (Γ_2, σ_2), where the signature on attachment edges is determined by the pseudo-potential of (Γ_2, σ_2) with respect to a fixed spanning tree $\mathcal{T}$. This construction satisfies identity preservation, structural compatibility, and switching stability. We proved that the $\mathcal{T}$-corona product is balanced if and only if both inputs are balanced, and introduced the notion of the $\mathcal{T}$-net Laplacian. A significant application was the construction of infinite families of non-switching-isomorphic cospectral signed graph pairs using the Mickey Mouse graph example.

Chapter 5: On H-joins of complex unit gain graphs and their stability

In Chapter 5, we extended the H-join operation to complex unit gain graphs. We again provided two distinct definitions: a direct extension that is switching stable only with respect to the root graph, and a stable H-join using pseudo-potentials to achieve full switching stability. As well as deriving the characteristic polynomial and spectrum, we also characterised balancedness. When restricted to the homogeneous case, the stable H-join provides a new switching-stable variant of the lexicographic product, distinct from existing constructions.

Chapter 6: Spectra of gain Q and R graphs via G-phases and group representations

In Chapter 6, we introduced gain analogues of the classical Q and R graphs of Cvetković [37] using the framework of G-phases and group representations. We proved that switching-equivalent gain graphs give rise to switching-equivalent gain Q and R graphs. For an r-regular G-gain graph and an irreducible unitary representation π, we determined the π-spectra of the represented adjacency and Laplacian matrices, recovering the classical spectra and providing new results for signed Q graphs.

Chapter 7: Products derived from gain subdivision graphs via group representations

In Chapter 7, we introduced gain variants of six subdivision-based corona products and determined their π-characteristic polynomials and π-spectra. The results recover and extend known classical and signed spectral formulas. We developed pseudo-potentials for G-gain graphs over abelian groups, enabling switching-stable variants of all six products in which balanced components yield a balanced product.

8.2 Methodological reflections

Throughout this thesis, the pseudo-potential technique emerged as a consistently effective method for constructing switching-stable graph operations. By fixing a spanning tree and defining signatures or gains on newly added edges in terms of the pseudo-potentials, we achieved switching stability for the signed NS and NNS joins, the signed corona product, the complex unit H-join, and the subdivision-based corona products over abelian groups. The success of this approach across multiple constructions suggests that pseudo-potentials capture the essential structure of switching equivalence in a way that is compatible with graph composition.

However, this approach comes at a cost: the constructions depend on the choice of a spanning tree, and different choices yield different signatures on the attachment edges. Thus, the pseudo-potential method provides a family of switching-stable constructions rather than a single canonical one. Moreover, in the gain graph setting, the approach currently requires the group G to be abelian; extending to non-abelian groups remains an open challenge.

The pseudo-potential technique is not the only possible approach to switching stability, as demonstrated by the CDS lexicographic product [33], which is not switching equivalent to the stable H-join in the homogeneous case. A deeper understanding of the space of all switching-stable generalisations of a given graph operation would be valuable.

8.3 Future work

Several avenues for further investigation arise from this work, some perhaps more immediate and others more long term.

The work on integer reciprocal eigenvalue properties raises the question of characterising which signatures yield m-(SR) or m-($-$SR) for a given underlying graph. The normalised Laplacian spectra of the NS and NNS joins also remain open for non-co-regular components.

The spectral results for gain Q and R graphs were obtained under the assumption of r-regularity; extending to irregular graphs requires a more sophisticated analysis of the matrix HH^*, and the Laplacian spectra of the gain corona products have not been fully explored.

More generally, the pseudo-potential technique raises the question: for which graph operations does this technique work, and what are its limitations? Developing a general theory of pseudo-potential-based constructions would provide a unified framework for switching-stable generalisations.

Furthermore, extending the techniques of constructing switching-stable gain graphs to non-abelian groups remains open, as the commutativity of the group ring $\mathbb{C}G$ was essential in Chapter 7. Possible routes could be to develop a non-commutative version of pseudo-potentials using left and right regular representations, or to work directly with the representation matrices. The relationship between the switching-stable constructions and other approaches deserves deeper investigation, as does the potential application of this work to network science, physics, and computer science.

Pseudo-potentials were first introduced by Cavaleri, Donno, and Spessato in their definition of the complement of a signed graph [33]. Since the signature they assign to the complement is determined by a pseudo-potential, the resulting complement is always balanced. A natural avenue for future investigation is therefore the search for an alternative notion of signed complement that need not be balanced, yet still retains interesting spectral properties. A closely related question arises for strongly regular signed graphs (see [89]): building on the combinatorial and switching-stable notions of signed strong regularity, and on the fact that the complement of a strongly regular signed graph is again strongly regular, one could ask whether such a signed complement, freed from the balance constraint, preserves strong regularity and its associated spectral structure.

Another direction concerns the spectral theory of products of oriented graphs and its relationship to the signed and complex unit gain graph settings studied here. Building on the correspondence between the skew spectrum of an oriented graph and the spectrum of an associated signed graph [62, 91, 92, 93, 94, 96], it would be natural to ask whether the products and switching-stable operations developed in this thesis admit oriented-graph analogues, and whether their skew spectra can be recovered from the corresponding signed or gain spectra through this association.

8.3.1 Open problems

1. Characterise the signatures of a given underlying graph that yield integer reciprocal eigenvalue properties.
2. Compute the characteristic polynomials and spectra of the normalised Laplacian matrices of the signed NS and NNS joins of arbitrary signed graphs.
3. Develop a general theory of pseudo-potential-based switching-stable constructions.
4. Extend the switching-stable gain corona products of Chapter 7 to non-abelian groups.
5. Explore applications of switching-stable gain graph constructions to network science and other fields.
6. Study the spectral properties of products of oriented graphs and their relationship to the corresponding signed and complex unit gain graphs.

8.4 Concluding remarks

In summary, this thesis has contributed to the theory of signed and gain graphs by developing switching-stable variants of fundamental graph operations, establishing spectral results for these operations, and demonstrating their utility through explicit constructions. The pseudo-potential technique emerges as a unifying concept that consistently provides switching-stable generalisations, albeit at the cost of fixing a spanning tree. Furthermore, we have shown that this approach is not unique, suggesting a rich landscape of possible methods yet to be explored.

The framework developed in this thesis (combining switching stability, pseudo-potentials, and group representations) provides a foundation for further advances in spectral graph theory and its applications. As the theory of signed and gain graphs continues to develop, we anticipate that the methods and results presented here will contribute to a deeper understanding of the spectral properties of these structures, and will find applications in the fields where signed and gain graphs naturally arise.

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Bibliography

[1] A. Abiad, F. Belardo, A.P. Khramova. A switching method for constructing cospectral gain graphs. Discrete Math. **347** (2024), no. 4, Paper No. 113838, 9 pp., DOI: https://doi.org/10.1016/j.disc.2023.113838. 36

[2] B. Adhikari, A. Singh, S.K. Yadav. Corona product of signed graphs and its application to modeling signed networks. Discrete Math. Algorithms Appl. **15** (2023), no. 1, Paper No. 2250062, 32 pp., DOI: https://doi.org/10.1142/S1793830922500628. 56, 58, 61, 63

[3] U. Ahmad, S. Hameed, S. Jabeen. Class of weighted graphs with strong anti-reciprocal eigenvalue property. Linear Multilinear Algebra **68** (2020), no. 6, 1129–1139, DOI: https://doi.org/10.1080/03081087.2018.1532489. 15

[4] U. Ahmad, S. Hameed, S. Jabeen. Noncorona graphs with strong anti-reciprocal eigenvalue property. Linear Multilinear Algebra **69** (2021), no. 10, 1878–1888, DOI: https://doi.org/10.1080/03081087.2019.1646204. 15

[5] S. Akhter, S. Hameed, U. Ahmad. Signed graphs with strong anti-reciprocal eigenvalue property. Comm. Algebra **51** (2023), no. 10, 4271–4279, DOI: https://doi.org/10.1080/00927872.2023.2204357. 15

[6] A. Alazemi, M. Anđelić, F. Belardo, M. Brunetti, C.M. da Fonseca. Line and subdivision graphs determined by $\mathbb{T}_4$-gain graphs. Mathematics **7** (2019), no. 10, Paper No. 926, 12 pp., DOI: https://doi.org/10.3390/math7100926. 2, 86

[7] S. Barik, S. Ghosh, D. Mondal. On graphs with strong anti-reciprocal eigenvalue property. Linear Multilinear Algebra **70** (2022), no. 21, 6698–6711, DOI: https://doi.org/10.1080/03081087.2021.1968330. 15

[8] S. Barik, D. Mondal, S. Pati, K. Sarma. Singular graphs and the reciprocal eigenvalue property. Discrete Math. **347** (2024), no. 7, Paper No. 114003, 7 pp., DOI: https://doi.org/10.1016/j.disc.2024.114003. 15

[9] S. Barik, M. Nath, S. Pati, B.K. Sarma. Unicyclic graphs with the strong reciprocal eigenvalue property. Electron. J. Linear Algebra **17** (2008), 139–153, DOI: https://doi.org/10.13001/1081-3810.1255. 15

[10] S. Barik, M. Neumann, S. Pati. On nonsingular trees and a reciprocal eigenvalue property. Linear Multilinear Algebra **54** (2006), no. 6, 453–465, DOI: https://doi.org/10.1080/03081080600792897. 15

[11] S. Barik, S. Pati. Classes of nonbipartite graphs with reciprocal eigenvalue property. Linear Algebra Appl. **612** (2021), 177–187, DOI: https://doi.org/10.1016/j.laa.2020.10.039. 15

[12] S. Barik, S. Pati, B.K. Sarma. The spectrum of the corona of two graphs. SIAM J. Discrete Math. **21** (2007), no. 1, 47–56, DOI: https://doi.org/10.1137/050624029. 15

[13] S. Barik, S. Pati, P. Verma. Trees with integer reciprocal eigenvalue property. Linear Multilinear Algebra **74** (2026), no. 10, 1298–1314, DOI: https://doi.org/10.1080/03081087.2026.2666137. 15, 16, 18, 33, 123

[14] M. Behzad. Graphs and their chromatic numbers. PhD thesis, Michigan State University, 1965. 2, 85

[15] F. Belardo, M. Brunetti, M. Cavaleri, A. Donno. Constructing cospectral signed graphs. Linear Multilinear Algebra **69** (2021), no. 14, 2717–2732, DOI: https://doi.org/10.1080/03081087.2019.1694483. 36

[16] F. Belardo, M. Brunetti, M. Cavaleri, A. Donno. Godsil-McKay switching for mixed and gain graphs over the circle group. Linear Algebra Appl. **614** (2021), 256–269, DOI: https://doi.org/10.1016/j.laa.2020.04.025. 2

[17] F. Belardo, M. Brunetti, M. Cavaleri, A. Donno. Construction of cospectral graphs, signed graphs and $\mathbb{T}$-gain graphs via partial transpose. J. Algebraic Combin. **60** (2024), no. 1, 191–224, DOI: https://doi.org/10.1007/s10801-024-01332-z. 2, 36

[18] F. Belardo, M. Brunetti, N.J. Coble, N. Reff, H. Skogman. Spectra of quaternion unit gain graphs. Linear Algebra Appl. **632** (2022), 15–49, DOI: https://doi.org/10.1016/j.laa.2021.09.009. 5

[19] F. Belardo, M. Brunetti, S. Khan. NEPS of complex unit gain graphs. Electron. J. Linear Algebra **39** (2023), 621–643, DOI: https://doi.org/10.13001/ela.2023.8015. 2

[20] F. Belardo, A. Donno, C. Huntington, S. Spessato. A switching-stable corona product of signed graphs. Under review. iii, 2, 55

[21] F. Belardo, C. Huntington. Signed graphs with strong (anti-)reciprocal eigenvalue property. Spec. Matrices **12** (2024), Paper No. 20240017, 28 pp., DOI: https://doi.org/10.1515/spma-2024-0017. 15, 25, 31, 86

[22] F. Belardo, Z. Stanić, T. Zaslavsky. Total graph of a signed graph. Ars Math. Contemp. **23** (2023), no. 1, Paper No. 2, 17 pp., DOI: https://doi.org/10.26493/1855-3974.2842.6b5. 2, 86

[23] M. Brunetti. Eigenvalues of complex unit gain graphs and gain regularity. Spec. Matrices **12** (2024), Paper No. 20240005, 15 pp., DOI: https://doi.org/10.1515/spma-2024-0005. 2, 6, 68, 75, 83

[24] M. Brunetti, M. Cavaleri, A. Donno. A lexicographic product for signed graphs. Australas. J. Combin. **74** (2019), 332–343. 2, 83

[25] M. Brunetti, M. Cavaleri, A. Donno. Erratum to the article "A lexicographic product for signed graphs". Australas. J. Combin. **75** (2019), 256–258. 83, 84

[26] D.M. Cardoso, M.A. de Freitas, E.A. Martins, M. Robbiano. Spectra of graphs obtained by a generalization of the join graph operation. Discrete Math. **313** (2013), no. 5, 733–741, DOI: https://doi.org/10.1016/j.disc.2012.10.016. 69

[27] D.M. Cardoso, H. Gomes, S.J. Pinheiro. The H-join of arbitrary families of graphs – the universal adjacency spectrum. Linear Algebra Appl. **648** (2022), 160–180, DOI: https://doi.org/10.1016/j.laa.2022.04.015. 69

[28] D.M. Cardoso, I. Gutman, E.A. Martins, M. Robbiano. A generalization of Fiedler's lemma and some applications. Linear Multilinear Algebra **59** (2011), no. 8, 929–942, DOI: https://doi.org/10.1080/03081087.2010.536982. 2, 69

[29] D.M. Cardoso, E.A. Martins, M. Robbiano, O. Rojo. Eigenvalues of a H-generalized join graph operation constrained by vertex subsets. Linear Algebra Appl. **438** (2013), no. 8, 3278–3290, DOI: https://doi.org/10.1016/j.laa.2012.12.004. 69

[30] M. Cavaleri, D. D'Angeli, A. Donno. Gain-line graphs via G-phases and group representations. Linear Algebra Appl. **613** (2021), 241–270, DOI: https://doi.org/10.1016/j.laa.2020.11.009. 86, 87, 88, 91, 120

[31] M. Cavaleri, D. D'Angeli, A. Donno. Characterizations of line graphs in signed and gain graphs. European J. Combin. **102** (2022), Paper No. 103479, 19 pp., DOI: https://doi.org/10.1016/j.ejc.2021.103479. 2

[32] M. Cavaleri, A. Donno, S. Spessato. The NEPS of gain graphs over arbitrary groups and its spectra. Adv. Group Theory Appl. **19** (2024), 105–153, DOI: https://doi.org/10.32037/agta-2024-011. 2, 4, 68

[33] M. Cavaleri, A. Donno, S. Spessato. On the complement of a signed graph. Discrete Math. **348** (2025), no. 6, Paper No. 114433, 21 pp., DOI: https://doi.org/10.1016/j.disc.2025.114433. 9, 36, 50, 59, 61, 62, 69, 79, 83, 84, 124, 125

[34] M. Cavaleri, A. Donno, S. Spessato. Gain total graphs and their spectra via G-phases and group representations. Linear Algebra Appl. **721** (2025), 484–512, DOI: https://doi.org/10.1016/j.laa.2024.03.002. 2, 68, 86, 87, 91, 120

[35] M. Cavaleri, A. Donno, S. Spessato. On the wreath product of signed and gain graphs and its spectrum. Ars Math. Contemp. **25** (2025), no. 3, Paper No. 10, 30 pp., DOI: https://doi.org/10.26493/1855-3974.3332.d9b. 2, 68

[36] F.R.K. Chung. Spectral Graph Theory. CBMS Regional Conference Series in Mathematics, No. 92, American Mathematical Society, Providence, RI, 1997. 1, 53

[37] D.M. Cvetković. Spectra of graphs formed by some unary operations. Publ. Inst. Math. (Beograd) (N.S.) **19(33)** (1975), 37–41. 2, 85, 93, 97, 124

[38] D.M. Cvetković, M. Doob, H. Sachs. Spectra of Graphs: Theory and Applications. 3rd ed., Johann Ambrosius Barth, Leipzig, 1995. 2, 3, 12, 21, 35, 36, 85, 101

[39] E.R. van Dam, W.H. Haemers. Which graphs are determined by their spectrum? Linear Algebra Appl. **373** (2003), 241–272, DOI: https://doi.org/10.1016/S0024-3795(03)00483-X. 35

[40] R. Frucht, F. Harary. On the corona of two graphs. Aequationes Math. **4** (1970), 322–325, DOI: https://doi.org/10.1007/BF01844162. 2

[41] R. Ganeshbabu, G. Arunkumar. On the spectrum of generalized H-join operation constrained by indexing maps – I. arXiv:2402.10557 [math.CO] (2024), DOI: https://doi.org/10.48550/arXiv.2402.10557. 12, 69

[42] K.A. Germina, S. Hameed K., T. Zaslavsky. On products and line graphs of signed graphs, their eigenvalues and energy. Linear Algebra Appl. **435** (2011), no. 10, 2432–2450, DOI: https://doi.org/10.1016/j.laa.2010.10.026. 2, 15, 17

[43] I. Gohberg (ed.). I. Schur Methods in Operator Theory and Signal Processing. Operator Theory: Advances and Applications, vol. 18, Birkhäuser Verlag, Basel, 1986. 131

[44] S. Hameed, U. Ahmad. Inverse of the adjacency matrices and strong anti-reciprocal eigenvalue property. Linear Multilinear Algebra **70** (2022), no. 14, 2739–2764, DOI: https://doi.org/10.1080/03081087.2020.1812495. 15

[45] S. Hameed K., K.A. Germina. On composition of signed graphs. Discuss. Math. Graph Theory **32** (2012), no. 3, 507–516, DOI: https://doi.org/10.7151/dmgt.1615. 2, 83

[46] S. Hameed K., V. Paul, K.A. Germina. On co-regular signed graphs. Australas. J. Combin. **62** (2015), 8–17. 6

[47] R. Hammack, W. Imrich, S. Klavžar. Handbook of Product Graphs. Discrete Mathematics and its Applications (Boca Raton), 2nd ed., CRC Press, Boca Raton, FL, 2011. 2

[48] S. Hamud, A. Berman. New constructions of nonregular cospectral graphs. Spec. Matrices **12** (2024), Paper No. 20230109, 21 pp., DOI: https://doi.org/10.1515/spma-2023-0109. 36, 37, 38

[49] F. Harary. On the notion of balance of a signed graph. Michigan Math. J. **2** (1953/54), 143–146. 7

[50] F. Harary. On the group of the composition of two graphs. Duke Math. J. **26** (1959), 29–34. 83

[51] F. Harary. Graph Theory. Addison-Wesley Publishing Co., Reading, MA, 1969. 37

[52] F. Harary, C. St. J.A. Nash-Williams. On Eulerian and Hamiltonian graphs and line graphs. Canad. Math. Bull. **8** (1965), 701–709, DOI: https://doi.org/10.4153/CMB-1965-051-3. 2

[53] F. Harary, R.Z. Norman. Some properties of line digraphs. Rend. Circ. Mat. Palermo (2) **9** (1960), 161–168, DOI: https://doi.org/10.1007/BF02854581. 2

[54] F. Heider. Attitudes and cognitive organization. J. Psychol. **21** (1946), no. 1, 107–112, DOI: https://doi.org/10.1080/00223980.1946.9917275. 7

[55] R.A. Horn, C.R. Johnson. Topics in Matrix Analysis. Cambridge University Press, Cambridge, 1991. 6, 37, 101

[56] H. Huang. Induced subgraphs of hypercubes and a proof of the sensitivity conjecture. Ann. of Math. (2) **190** (2019), no. 3, 949–955, DOI: https://doi.org/10.4007/annals.2019.190.3.6. 20

[57] C. Huntington. The $\dot{H}$-join operation of signed graphs constrained by indexing maps. RAIRO Oper. Res. **59** (2025), no. 5, 2501–2515, DOI: https://doi.org/10.1051/ro/2025100. 2, 69, 77, 82

[58] C. Huntington. Spectra and balance of signed graphs obtained from vertex splittings. Discuss. Math. Graph Theory, in press, DOI: https://doi.org/10.7151/dmgt.2643. iii, 2, 35

[59] C. Huntington. Products derived from gain subdivision graphs via group representations. In preparation. iii, 2, 99

[60] C. Huntington, F. Belardo, S. Barik. Graphs with integer reciprocal eigenvalue properties. Under review. iii, 2, 15

[61] C. Huntington, S. Spessato, F. Belardo. On H-joins of complex unit gain graphs and their stability. Discrete Math. **349** (2026), no. 7, Paper No. 115069, 17 pp., DOI: https://doi.org/10.1016/j.disc.2026.115069. iii, 2, 9, 36, 68, 69

[62] C. Huntington, Z. Stanić, F. Belardo. Further spectral relations between oriented graphs and corresponding signed graphs. Under review. 2, 125

[63] C. Huntington, O. Waka. Spectra of gain Q and R graphs via G-phases and group representations. Under review. iii, 2, 85

[64] D. Kőnig. Theorie der endlichen und unendlichen Graphen. Akademische Verlagsgesellschaft, Leipzig, 1936. 7

[65] J. Krausz. Démonstration nouvelle d'un théorème de Whitney sur les réseaux. Mat. Fiz. Lapok **50** (1943), 75–85. 2

[66] K. Kuratowski. Sur le problème des courbes gauches en topologie. Fund. Math. **15** (1930), 271–283, DOI: https://doi.org/10.4064/fm-15-1-271-283. 2

[67] J.D. LaGrange. Boolean rings and reciprocal eigenvalue properties. Linear Algebra Appl. **436** (2012), no. 7, 1863–1871, DOI: https://doi.org/10.1016/j.laa.2011.05.042. 15

[68] D. Lajou. On Cartesian products of signed graphs. Discrete Appl. Math. **319** (2022), 533–555, DOI: https://doi.org/10.1016/j.dam.2021.03.023. 2

[69] J. Lan, B. Zhou. Spectra of graph operations based on R-graph. Linear Multilinear Algebra **63** (2015), no. 7, 1401–1422, DOI: https://doi.org/10.1080/03081087.2014.941292. 116

[70] S. Laplante, R. Naserasr, A. Sunny, Z. Wang. Sensitivity conjecture and signed hypercubes. ACM Trans. Comput. Theory **18** (2026), no. 1, Paper No. 6, 25 pp., DOI: https://doi.org/10.1145/3777401. 15, 20

[71] Y. Li, P. Zhang, K. Xu. The spectra of signed graphs obtained by $\dot{H}$-(generalized) join operation. Comput. Appl. Math. **42** (2023), no. 2, Paper No. 77, 15 pp., DOI: https://doi.org/10.1007/s40314-023-02205-0. 69, 82

[72] Q. Liu. The Laplacian spectrum of corona of two graphs. Kragujevac J. Math. **38** (2014), no. 1, 163–170, DOI: https://doi.org/10.5937/KgJMath1401163L. 65

[73] Z. Lu, X. Ma, M. Zhang. Spectra of graph operations based on splitting graph. J. Appl. Anal. Comput. **13** (2023), no. 1, 133–155, DOI: https://doi.org/10.11948/20210446. 36, 37

[74] P. Lu, Y. Miao. Spectra of the subdivision-vertex and subdivision-edge coronae. arXiv:1302.0457 [math.CO] (2013), DOI: https://doi.org/10.48550/arXiv.1302.0457. 116

[75] C. McLeman, E. McNicholas. Spectra of coronae. Linear Algebra Appl. **435** (2011), no. 5, 998–1007, DOI: https://doi.org/10.1016/j.laa.2011.02.007. 12

[76] V.K. Najiya, A.V. Chithra. Constructions of A_α-cospectral graphs using some corona operations. arXiv:2406.07183 [math.CO] (2024), DOI: https://doi.org/10.48550/arXiv.2406.07183. 116

[77] S.K. Panda, S. Pati. Graphs with reciprocal eigenvalue properties. Electron. J. Linear Algebra **31** (2016), 511–514, DOI: https://doi.org/10.13001/1081-3810.3336. 15, 20, 21

[78] S. Pirzada, T. Shamsher, M.A. Bhat. A note on the class of weighted graphs with strong anti-reciprocal eigenvalue property. Linear Multilinear Algebra **71** (2023), no. 1, 1–4, DOI: https://doi.org/10.1080/03081087.2021.2014776. 15

[79] B.R. Rakshith, K.C. Das, B.J. Manjunatha. Some new families of noncorona graphs with strong anti-reciprocal eigenvalue property. Bull. Malays. Math. Sci. Soc. **45** (2022), no. 5, 2597–2618, DOI: https://doi.org/10.1007/s40840-022-01364-3. 15, 25, 26

[80] M.R. Ul Rashid, S. Pirzada, T. Shamsher, Z. Stanić. Spectra of subdivisions of signed graphs, signed R-graphs and related products. Rev. Un. Mat. Argentina, in press, DOI: https://doi.org/10.33044/revuma.4483. 2, 86, 93, 97, 116

[81] M.R. Ul Rashid, S. Pirzada, Z. Stanić. Spectra of the Mycielskian of a signed graph and related products. Discrete Appl. Math. **370** (2025), 124–144, DOI: https://doi.org/10.1016/j.dam.2025.03.017. 2

[82] N. Reff. Spectral properties of complex unit gain graphs. Linear Algebra Appl. **436** (2012), no. 9, 3165–3176, DOI: https://doi.org/10.1016/j.laa.2011.10.021. 36, 50

[83] N. Reff. Oriented gain graphs, line graphs and eigenvalues. Linear Algebra Appl. **506** (2016), 316–328, DOI: https://doi.org/10.1016/j.laa.2016.05.040. 2

[84] J. Richardson. A Life of Picasso, Vol. III: The Triumphant Years, 1917–1932. Alfred A. Knopf, New York, 2007. 1

[85] M. Saravanan, S.P. Murugan, G. Arunkumar. A generalization of Fiedler's lemma and the spectra of H-join of graphs. Linear Algebra Appl. **625** (2021), 20–43, DOI: https://doi.org/10.1016/j.laa.2021.04.015. 69, 74

[86] I. Schur. Über Potenzreihen, die im Innern des Einheitskreises beschränkt sind [I]. J. Reine Angew. Math. **147** (1917), 205–232, DOI: https://doi.org/10.1515/crll.1917.147.205 (translated into English in pp. 31–59 of the book by Gohberg [43]). 12

[87] A.J. Schwenk. Computing the characteristic polynomial of a graph. In: Graphs and Combinatorics (R.A. Bari, F. Harary, eds.), Lecture Notes in Math., Vol. 406, Springer, Berlin, 1974, 153–172. 2, 21, 69

[88] T. Shamsher, S. Pirzada, Z. Stanić. Net Laplacian eigenvalues of certain corona-like products of signed graphs. Bol. Soc. Mat. Mex. (3) **30** (2024), no. 3, Paper No. 80, 14 pp., DOI: https://doi.org/10.1007/s40590-024-00655-y. 2, 56, 58, 61, 64

[89] Z. Stanić. On strongly regular signed graphs. Discrete Appl. Math. **271** (2019), 184–190, DOI: https://doi.org/10.1016/j.dam.2019.06.017. 125

[90] Z. Stanić. On the spectrum of the net Laplacian matrix of a signed graph. Bull. Math. Soc. Sci. Math. Roumanie (N.S.) **63(111)** (2020), no. 2, 205–213. 5

[91] Z. Stanić. On cospectral oriented graphs and cospectral signed graphs. Linear Multilinear Algebra **70** (2022), no. 19, 3689–3701, DOI: https://doi.org/10.1080/03081087.2020.1852153. 125

[92] Z. Stanić. Some relations between the skew spectrum of an oriented graph and the spectrum of certain closely associated signed graphs. Rev. Un. Mat. Argentina **63** (2022), no. 1, 41–50, DOI: https://doi.org/10.33044/revuma.1914. 2, 125

[93] Z. Stanić. Relations between the skew spectrum of an oriented graph and the spectrum of an associated signed graph. Linear Algebra Appl. **676** (2023), 241–250, DOI: https://doi.org/10.1016/j.laa.2023.07.015. 125

[94] Z. Stanić. Spectra of signed graphs and related oriented graphs. Ars Math. Contemp. **24** (2024), no. 3, Paper No. 10, 14 pp., DOI: https://doi.org/10.26493/1855-3974.3227.fd5. 125

[95] Z. Stanić. A construction of cospectral signed line graphs. Commun. Comb. Optim. **11** (2026), no. 3, 985–993, DOI: https://doi.org/10.22049/cco.2024.30034.2284. 36

[96] Z. Stanić. Characterizations of signed and oriented graphs exhibiting a large multiplicity of one eigenvalue. Discrete Appl. Math. **393** (2026), 17–24, DOI: https://doi.org/10.1016/j.dam.2026.06.010. 125

[97] K. Wagner. Über eine Eigenschaft der ebenen Komplexe. Math. Ann. **114** (1937), no. 1, 570–590, DOI: https://doi.org/10.1007/BF01594196. 2

[98] H. Whitney. Congruent graphs and the connectivity of graphs. Amer. J. Math. **54** (1932), no. 1, 150–168, DOI: https://doi.org/10.2307/2371086. 2

[99] B. Wu, Y. Lou, C. He. Signless Laplacian and normalized Laplacian on the H-join operation of graphs. Discrete Math. Algorithms Appl. **6** (2014), no. 3, Paper No. 1450046, 13 pp., DOI: https://doi.org/10.1142/S1793830914500463. 69

[100] T. Zaslavsky. Signed graphs. Discrete Appl. Math. **4** (1982), no. 1, 47–74, DOI: https://doi.org/10.1016/0166-218X(82)90033-6. 2, 3, 11, 36, 58

[101] T. Zaslavsky. Biased graphs. I. Bias, balance, and gains. J. Combin. Theory Ser. B **47** (1989), no. 1, 32–52, DOI: https://doi.org/10.1016/0095-8956(89)90063-4. 3, 7, 9

[102] T. Zaslavsky. Biased graphs. II. The three matroids. J. Combin. Theory Ser. B **51** (1991), no. 1, 46–72, DOI: https://doi.org/10.1016/0095-8956(91)90005-5. 7

[103] F. Zhang (ed.). The Schur Complement and Its Applications. Numerical Methods and Algorithms, Vol. 4, Springer-Verlag, New York, 2005. 12

[104] P. Zhang, B. Wu, C. He. Balancedness and spectra of signed graphs obtained by $\dot{H}$-join operation. Comput. Appl. Math. **40** (2021), no. 3, Paper No. 90, 11 pp., DOI: https://doi.org/10.1007/s40314-021-01481-y. 2, 69, 78, 82, 83